CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S01	0	START OF PRIME MISSION		2004-135T18:40	May14	Fri	---	28	227	Start of prime mission (2004-2008)
S01	0	MAG	Segment	2004-135T18:40	May14	Fri	---	28	227	MAG_000_PrePhoebe (27d); Early Rev 0 prior to Phoebe
S01	0	S01 Begins		2004-135T18:40	May14	Fri	---	28	227	S01 Sequence. Duration = 36 d
S01	0	TCM-20	Phoebe targ	2004-148T16:25	Jul03	Sat	---	15	240	Phoebe targeting maneuver, 34 m/s. D/L start, burn ~6h later.
S01	0	SOST	Segment	2004-163T04:02	Jun11	Fri	---	1	14	SOST_000PH (2d); SOST Rev 00 Targeted Phoebe
S01	0	00PH (t) [P1]	PHOEBE	2004-163T19:34	Jun11	Fri	---	137	15	Inbound 2068.1 km flyby, v = 6.4 km/s, phase = 24 deg
S01	0	MAG	Segment	2004-164T22:32	Jun12	Sat	1	136	16	MAG_000_PreSOI (18d); Rev 0 after Phoebe before SOI
S01	0	TCM-21	Phoebe c/u	2004-168T15:07	Jul03	Sat	5	132	20	Phoebe cleanup maneuver, 4 m/s. D/L start, burn ~6h later.
S02	0	S02 Begins		2004-171T21:52	Jun19	Sat	8	129	3	S02 Sequence. Duration = 41 d
S02	0	00PO (nt)	POLYDEUCES	2004-182T20:57	Jun30	Wed	19	118	14	Inbound 67990.9 km flyby, v = 13.3 km/s, phase = 90 deg
S02	0	00CA (nt)	CALYPSO	2004-182T22:27	Jun30	Wed	19	118	14	Inbound 52100.1 km flyby, v = 14.1 km/s, phase = 92 deg
S02	0	00ME (nt)	METHONE	2004-183T00:16	Jul01	Thu	19	118	14	Inbound 51202.4 km flyby, v = 12.9 km/s, phase = 96 deg
S02	0	00MI (nt)	MIMAS	2004-183T00:30	Jul01	Thu	19	118	14	Inbound 76487.7 km flyby, v = 22.3 km/s, phase = 80 deg
S02	0	G_ring	DUST HAZARD	2004-183T00:46	Jul01	Thu	19	118	14	Egress at 2004-183T00:47 (0.0 min); protection = HGA to ram and MEA cover closure.
S02	0	Ring CRX	Ascending	2004-183T00:47	Jul01	Thu	19	118	14	r = 2.628 Rs (158525 km)
S02	0	JE_ring	DUST HAZARD	2004-183T00:49	Jul01	Thu	19	118	14	Egress at 2004-183T00:50 (0.0 min); protection = HGA to ram and MEA cover closure.
S02	0	SOI Start		2004-183T01:12	Jul01	Thu	19	118	14	SOI burn start, DV= 626 m/s, duration = 01:36:20 (hh:mm:ss)
S02	0	00PA (nt)	PANDORA	2004-183T01:16	Jul01	Thu	19	118	0	Inbound 89854.0 km flyby, v = 26.8 km/s, phase = 89 deg
S02	0	00JA (nt)	JANUS	2004-183T01:51	Jul01	Thu	19	118	0	Inbound 67681.8 km flyby, v = 12.8 km/s, phase = 106 deg
S02	0	Earth OCC	RING	2004-183T02:21	Jul01	Thu	19	118	0	Egress = 2004-183T04:21 (121 min)
S02	0	Sun OCC	RING	2004-183T02:21	Jul01	Thu	19	118	0	Egress = 2004-183T04:21 (120 min)
S02	0	Periapse		2004-183T02:38	Jul01	Thu	19	118	0	R = 1.330 Rs, lat = 17 deg, phase = 94 deg
S02	0	SOI End		2004-183T02:48	Jul01	Thu	19	118	0	SOI burn end
S02	0	SOI	Segment	2004-183T03:07	Jul01	Thu	19	118	0	SOI_000 (1d); SOI TWT
S02	0	Earth OCC	SATURN	2004-183T03:33	Jul01	Thu	19	117	0	Egress = 2004-183T04:09 (36 min)
S02	0	Sun OCC	SATURN	2004-183T03:36	Jul01	Thu	19	117	0	Egress = 2004-183T04:08 (32 min)
S02	0	00PM (nt)	PROMETHEUS	2004-183T03:57	Jul01	Thu	19	117	0	Outbound 107413.3 km flyby, v = 28.5 km/s, phase = 91 deg
S02	0	00PN (nt)	PAN	2004-183T04:02	Jul01	Thu	19	117	0	Outbound 57684.4 km flyby, v = 22.8 km/s, phase = 88 deg
S02	0	00AE (nt)	AEGAEON	2004-183T04:21	Jul01	Thu	19	117	0	Outbound 112909.9 km flyby, v = 26.0 km/s, phase = 94 deg
S02	0	JE_ring	DUST HAZARD	2004-183T04:30	Jul01	Thu	19	117	0	Egress at 2004-183T04:30 (0.0 min); protection = HGA to ram and MEA cover closure.
S02	0	G_ring	DUST HAZARD	2004-183T04:33	Jul01	Thu	19	117	0	Egress at 2004-183T04:34 (0.0 min); protection = HGA to ram and MEA cover closure.
S02	0	Ring CRX	Descending	2004-183T04:34	Jul01	Thu	19	117	0	r = 2.632 Rs (158780 km)
S02	0	TOST	Segment	2004-184T01:06	Jul02	Fri	20	117	1	TOST_000TI_T0 (2d); TOST 000TI T0
S02	0	00TI (nt)	TITAN	2004-184T09:29	Jul02	Fri	21	116	1	Outbound 339120.0 km flyby, v = 8.3 km/s, phase = 67 deg
S02	0	OTM-001	SOI c/u	2004-185T14:06	Jul03	Sat	22	115	3	SOI cleanup. D/L start, burn ~6h later. Backup D/L 186T14:06
S02	0	MAG	Segment	2004-185T23:06	Jul03	Sat	22	115	0	MAG_000_PostTitan (34d); Rev 0 after Titan
S02	0	SEP = 3.0 deg	Conjunction	2004-187T04:21	Jul05	Mon	23	113	2	Commanding must not be required below 3.0 deg
S02	0	SEP = 2.0 deg	Conjunction	2004-188T10:13	Jul06	Tue	25	112	3	Plan no SSR playback below 2.0 deg
S02	0	SEP = 1.0 deg	Conjunction	2004-189T16:28	Jul07	Wed	26	111	4	Require no DSN passes below 1.0 deg
S02	0	SEP = 1.0 deg	Conjunction	2004-192T01:04	Jul10	Sat	28	109	6	Require no DSN passes below 1.0 deg
S02	0	SEP = 2.0 deg	Conjunction	2004-193T07:12	Jul11	Sun	29	107	8	Plan no SSR playback below 2.0 deg
S02	0	SEP = 3.0 deg	Conjunction	2004-194T12:52	Jul12	Mon	31	106	9	Commanding must not be required below 3.0 deg
S02	0	OTM-001A	SOI c/u	2004-199T13:21	Jul17	Sat	36	101	14	SOI cleanup. D/L start, burn ~6h later. Backup D/L 200T13:21
S03	0	S03 Begins		2004-212T21:32	Jul30	Fri	49	88	13	S03 Sequence. Duration = 44 d
S03	0	SATURN	Segment	2004-219T13:37	Aug06	Fri	56	81	20	Saturn_000_00A (80d); Atmosphere Segment - Rev 0-A
S03	0	OTM-002	PRM	2004-236T09:53	Aug23	Mon	73	64	37	Periapsis raise. D/L start, burn ~6h later. Backup D/L 237T09:38
S03	A	Apoapse		2004-240T08:57	Aug27	Fri	77	60	4	Per = 124.1 d, inc = 17.6 deg, r = 150.767 Rs, phase = 87 deg
S03	A	OTM-003	PRM c/u	2004-251T10:30	Sep07	Tue	88	49	15	Periapsis raise clean up. D/L start, burn ~6h later. Backup D/L 252T10:25
S04	A	S04 Begins		2004-256T11:35	Sep12	Sun	93	44	5	S04 Sequence. Duration = 36 d
S05	A	S05 Begins		2004-292T09:30	Oct18	Mon	129	8	41	S05 Sequence. Duration = 28 d
S05	A	OTM-004	ATI-3	2004-297T00:16	Oct23	Sat	133	4	46	D/L start, burn ~6h later. Backup D/L 298T00:01
S05	A	TOST	Segment	2004-299T16:31	Oct25	Mon	136	1	3	TOST_00ATI_TA (2d); TOST Segment for 00ATI (TA)
S05	A	0ATI (t) [TA]	TITAN	2004-300T15:30	Oct26	Tue	137	48	4	Inbound 1174.0 km flyby, v = 6.1 km/s, phase = 91 deg
S05	A	Ring CRX	Ascending	2004-300T16:53	Oct26	Tue	0	48	4	r = 19.506 Rs (1176795 km)
S05	A	RINGS	Segment	2004-301T09:16	Oct27	Wed	1	47	4	RINGS_REV_A (5d); Rings REV A Segment
S05	A	Periapse		2004-302T10:20	Oct28	Thu	2	46	5	R = 6.170 Rs, lat = 12 deg, phase = 104 deg
S05	A	Ring CRX	Descending	2004-302T20:01	Oct28	Thu	2	46	6	r = 8.082 Rs (487586 km)
S05	A	OTM-005	ATI+3	2004-303T00:15	Oct29	Fri	2	45	6	D/L start, burn ~6h later. Backup D/L 304T00:30
S05	A	MAG	Segment	2004-306T10:45	Nov01	Mon	6	42	3	MAG_00A_00B (32d); Rev A/B
S06	A	S06 Begins		2004-320T07:49	Nov15	Mon	20	28	17	S06 Sequence. Duration = 31 d
S06	A	OTM-006	APO	2004-325T23:00	Nov20	Sat	25	23	23	D/L start, burn ~6h later. Backup D/L 326T23:00
S06	B	Apoapse		2004-326T08:41	Nov21	Sun	26	22	0	Per = 47.9 d, inc = 13.8 deg, r = 78.090 Rs, phase = 77 deg
S06	B	SATURN	Segment	2004-338T15:36	Dec03	Fri	38	10	13	Saturn_00Ba (7d); Saturn Rev B - a (Inbound)
S06	B	OTM-007	BTI-3	2004-344T21:06	Dec09	Thu	44	4	19	D/L start, burn ~6h later. Backup D/L 345T21:05
S06	B	TOST	Segment	2004-345T06:06	Dec10	Fri	45	3	0	TOST_00BTI_TB (4d); TOST Segment for 00BTI (TB)
S06	B	0BTI (t) [TB]	TITAN	2004-348T11:38	Dec13	Mon	48	32	4	Inbound 1192.3 km flyby, v = 6.1 km/s, phase = 102 deg
S06	B	Sun OCC	TITAN	2004-348T11:52	Dec13	Mon	0	32	4	Egress = 2004-348T12:15 (23 min)
S06	B	Ring CRX	Ascending	2004-348T14:58	Dec13	Mon	0	32	4	r = 18.860 Rs (1137811 km)
S06	B	SATURN	Segment	2004-349T07:52	Dec14	Tue	1	31	4	Saturn_00Bb (2d); Saturn Rev B - b (Periapse)
S06	B	0BDI (nt)	DIONE	2004-350T01:41	Dec15	Wed	2	30	5	Inbound 72066.8 km flyby, v = 5.3 km/s, phase = 84 deg
S06	B	0BMI (nt)	MIMAS	2004-350T05:02	Dec15	Wed	2	30	5	Inbound 106749.6 km flyby, v = 1.6 km/s, phase = 90 deg
S06	B	0BAE (nt)	AEGAEON	2004-350T05:43	Dec15	Wed	2	30	5	Inbound 121062.3 km flyby, v = 1.0 km/s, phase = 108 deg
S06	B	Periapse		2004-350T05:51	Dec15	Wed	2	30	5	R = 4.764 Rs, lat = 4 deg, phase = 108 deg
S06	B	Ring CRX	Descending	2004-350T11:03	Dec15	Wed	2	30	6	r = 5.750 Rs (346893 km)
S07	B	MAG	Segment	2004-351T13:22	Dec16	Thu	3	29	7	MAG_00B (15d); Rev B
S07	B	S07 Begins		2004-351T13:22	Dec16	Thu	3	29	7	S07 Sequence. Duration = 37 d
S07	B	OTM-008	PTM	2004-351T19:22	Dec16	Thu	3	29	7	Probe targeting. D/L start, burn ~6h later. Backup D/L 352T19:22
S07	B	OTM-009	PTM c/u	2004-357T18:52	Dec22	Wed	9	23	6	Probe targeting cleanup. D/L start, burn ~6h later. Backup D/L 358T18:52
S07	B	PROBE RELEASE		2004-360T02:00	Dec25	Sat	12	20	2	At entry interface - 20.3 d
S07	B	OTM-010	ODM	2004-362T18:37	Dec27	Mon	14	18	5	Orbit deflection. D/L start, burn ~6h later. Backup D/L 363T18:38
S07	B	SOST	Segment	2004-366T01:30	Dec31	Fri	18	14	3	SOST_00CIA (2d); SOST segment early in Rev C (Iapetus)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S07	C	**Apoapse**		2004-366T07:02	Dec31	Fri	18	14	4	Per = 31.9 d, inc = 5.3 deg, r = 59.651 Rs, phase = 72 deg
S07	C	0CIA (nt)	**IAPETUS**	2004-366T18:49	Dec31	Fri	18	14	4	Inbound 123401.4 km flyby, v = 2.0 km/s, phase = 94 deg
S07	C	**MAG**	**Segment**	2005-002T04:38	Jan02	Sun	20	12	5	MAG_00C (13d); MAG TWT Rev C
S07	C	**OTM-010A**	ODM c/u	2005-003T17:38	Jan03	Mon	21	11	7	D/L start, burn ~6h later. Backup D/L 004T17:38
S07	C	**PROBE ENTRY**		2005-014T09:06	Jan14	Fri	32	0	11	Entry interface alt=1270 km, Tc c/a -2.1 h
S07	C	**0CTI (t) [TC]**	**TITAN**	2005-014T11:12	Jan14	Fri	32	32	11	Inbound 60003.3 km flyby, v = 5.4 km/s, phase = 93 deg
S07	C	Ring CRX	Ascending	2005-014T17:05	Jan14	Fri	0	32	11	r = 18.360 Rs (1107646 km)
S07	C	**SOST**	**Segment**	2005-015T12:00	Jan15	Sat	1	31	12	SOST_00C (3d); SOST segment in Rev C (non-targeted satellites)
S07	C	**OTM-011**	CTI+2	2005-016T03:20	Jan16	Sun	2	30	12	D/L start, burn ~6h later. Backup D/L 017T03:30
S07	C	0CME (nt)	**METHONE**	2005-016T04:58	Jan16	Sun	2	30	0	Inbound 113120.3 km flyby, v = 1.8 km/s, phase = 78 deg
S07	C	0CMI (nt)	**MIMAS**	2005-016T06:08	Jan16	Sun	2	30	0	Inbound 107630.4 km flyby, v = 1.3 km/s, phase = 100 deg
S07	C	Periapse		2005-016T06:25	Jan16	Sun	2	30	0	R = 4.824 Rs, lat = 4 deg, phase = 106 deg
S07	C	0CPL (nt)	**PALLENE**	2005-016T07:26	Jan16	Sun	2	30	0	Outbound 90824.4 km flyby, v = 2.4 km/s, phase = 138 deg
S07	C	Ring CRX	Descending	2005-016T11:58	Jan16	Sun	2	30	0	r = 5.905 Rs (356223 km)
S07	C	**MAG**	**Segment**	2005-018T10:00	Jan18	Tue	4	28	2	MAG_00C_003 (17d); Mag Rev C/3
S08	C	**S08 Begins**		2005-022T10:38	Jan22	Sat	8	24	6	S08 Sequence. Duration = 36 d
S08	C	**OTM-012**	~APO	2005-028T01:08	Jan28	Fri	14	18	12	D/L start, burn ~6h later. Backup D/L 029T01:08
S08	3	**Apoapse**		2005-032T03:26	Feb01	Tue	18	14	4	Per = 31.8 d, inc = 5.2 deg, r = 59.250 Rs, phase = 74 deg
S08	3	**SATURN**	**Segment**	2005-035T11:07	Feb04	Fri	21	11	7	Saturn_003 (10d); Saturn Rev 3 Inbound
S08	3	**OTM-013**	3TI-3	2005-043T00:07	Feb12	Sat	29	3	15	D/L start, burn ~6h later. Backup D/L 043T16:37
S08	3	**TOST**	**Segment**	2005-045T09:07	Feb14	Mon	31	1	2	TOST_003TI_T3 (2d); TOST Segment 003TI (T3)
S08	3	**03TI (dt) [T3]**	**TITAN**	2005-046T06:58	Feb15	Tue	32	2	3	Inbound 1579.0 km flyby, v = 6.0 km/s, phase = 99 deg
S08	3	**SOST**	**Segment**	2005-047T08:59	Feb16	Wed	1	1	4	SOST_003 (2d); SOST segment in Rev 3 (Enceladus non-targeted)
S08	3	E_ring_lg	Dust Crossing	2005-047T22:39	Feb16	Wed	2	0	5	Egress at 2005-048T00:09 (0.1 min); protection = none.
S08	3	03PA (nt)	**PANDORA**	2005-047T23:03	Feb16	Wed	2	0	5	Inbound 102608.6 km flyby, v = 1.6 km/s, phase = 62 deg
S08	3	Pallene	Dust Crossing	2005-048T00:03	Feb17	Thu	2	0	5	Egress at 2005-048T01:51 (0.1 min); protection = none.
S08	3	03EP (nt)	**EPIMETHEUS**	2005-048T00:10	Feb17	Thu	2	0	5	Inbound 73455.4 km flyby, v = 2.9 km/s, phase = 78 deg
S08	3	03AT (nt)	**ATLAS**	2005-048T00:23	Feb17	Thu	2	0	5	Inbound 76935.0 km flyby, v = 1.6 km/s, phase = 96 deg
S08	3	03CA (nt)	**CALYPSO**	2005-048T00:35	Feb17	Thu	2	0	5	Inbound 83229.2 km flyby, v = 6.9 km/s, phase = 67 deg
S08	3	Ring CRX	Ascending	2005-048T00:50	Feb17	Thu	2	0	5	r = 3.499 Rs (211082 km)
S08	3	Periapse		2005-048T00:57	Feb17	Thu	2	0	5	R = 3.498 Rs, lat = 0 deg, phase = 115 deg
S08	3	03AE (nt)	**AEGAEON**	2005-048T01:23	Feb17	Thu	2	0	5	Outbound 107572.0 km flyby, v = 8.3 km/s, phase = 157 deg
S08	3	E_ring_lg	Dust Crossing	2005-048T01:46	Feb17	Thu	2	0	5	Egress at 2005-048T03:00 (0.1 min); protection = none.
S08	3	**03EN (dt) [E0]**	**ENCELADUS**	2005-048T03:30	Feb17	Thu	2	20	5	Outbound 1260.7 km flyby, v = 6.7 km/s, phase = 113 deg
S08	3	**Earth OCC**	ENCELADUS	2005-048T03:37	Feb17	Thu	0	20	5	Egress = 2005-048T03:38 (1 min)
S08	3	03PO (nt)	**POLYDEUCES**	2005-048T08:48	Feb17	Thu	0	20	5	Outbound 6446.7 km flyby, v = 8.6 km/s, phase = 121 deg
S08	3	**OTM-014**	3TI+3	2005-049T00:00	Feb18	Fri	1	19	6	D/L start, burn ~6h later. Backup D/L 049T23:36
S08	3	**XD**	**Segment**	2005-049T08:43	Feb18	Fri	1	19	0	XD_003_004 (18d); Cross Discipline Segment Outbound 3-Inbound 4
S08	3	Ring CRX	Descending	2005-057T04:25	Feb26	Sat	9	11	8	r = 43.975 Rs (2653035 km)
S09	3	**S09 Begins**		2005-058T00:36	Feb27	Sun	10	10	9	S09 Sequence. Duration = 41 d
S09	4	**Apoapse**		2005-058T06:20	Feb27	Sun	10	10	9	Per = 20.5 d, inc = 0.4 deg, r = 44.303 Rs, phase = 65 deg
S09	4	**OTM-015**	~APO	2005-060T22:50	Mar01	Tue	13	7	12	D/L start, burn ~6h later. Backup D/L 061T22:50
S09	4	**OTM-016**	4EN-3	2005-064T22:35	Mar05	Sat	17	3	4	D/L start, burn ~6h later. Backup D/L 065T22:20
S09	4	**SOST**	**Segment**	2005-067T07:35	Mar08	Tue	19	1	2	SOST_004EN (2d); SOST Rev 04 Targeted Enceladus
S09	4	04HE (nt)	**HELENE**	2005-068T04:33	Mar09	Wed	20	0	3	Inbound 74490.5 km flyby, v = 6.5 km/s, phase = 67 deg
S09	4	E_ring_lg	**DUST HAZARD**	2005-068T08:06	Mar09	Wed	20	0	3	Egress at 2005-068T10:53 (0.1 min); protection = HGA to ram and MEA cover closure.
S09	4	**04EN (t) [E1]**	**ENCELADUS**	2005-068T09:08	Mar09	Wed	20	22	3	Inbound 497.3 km flyby, v = 6.6 km/s, phase = 43 deg
S09	4	Pallene	**DUST HAZARD**	2005-068T10:47	Mar09	Wed	0	22	4	Egress at 2005-068T12:31 (0.1 min); protection = MEA cover closure.
S09	4	Ring CRX	Ascending	2005-068T10:48	Mar09	Wed	0	22	4	r = 3.559 Rs (214695 km)
S09	4	04AT (nt)	**ATLAS**	2005-068T11:29	Mar09	Wed	0	22	4	Inbound 73847.2 km flyby, v = 1.6 km/s, phase = 108 deg
S09	4	Periapse		2005-068T11:40	Mar09	Wed	0	22	4	R = 3.502 Rs, lat = 0 deg, phase = 115 deg
S09	4	04TE (nt)	**TETHYS**	2005-068T11:44	Mar09	Wed	0	22	4	Outbound 82863.3 km flyby, v = 6.9 km/s, phase = 64 deg
S09	4	E_ring_lg	**DUST HAZARD**	2005-068T12:26	Mar09	Wed	0	22	4	Egress at 2005-068T15:13 (0.1 min); protection = MEA cover closure.
S09	4	**XD**	**Segment**	2005-069T07:26	Mar10	Thu	1	22	4	XD_004_005 (18d); Cross Discipline Segment Outbound 4 - Inbound 5
S09	4	**OTM-017**	4EN+3	2005-070T21:20	Mar11	Fri	3	20	6	D/L start, burn ~6h later. Backup D/L 072T20:49
S09	4	Ring CRX	Descending	2005-073T16:46	Mar14	Mon	5	17	3	r = 36.913 Rs (2226935 km)
S09	4	**OTM-018**	~APO	2005-078T12:19	Mar19	Sat	10	12	8	D/L start, burn ~6h later. Backup D/L 079T12:19
S09	5	**Apoapse**		2005-078T17:35	Mar19	Sat	10	12	0	Per = 20.5 d, inc = 0.2 deg, r = 44.390 Rs, phase = 65 deg
S09	5	**OTM-019**	5TI-3	2005-086T20:00	Mar27	Sun	18	4	8	D/L start, burn ~6h later. Backup D/L 087T19:17
S09	5	**MAG**	**Segment**	2005-087T05:09	Mar28	Mon	19	4	0	MAG_005 (3d); Mag Rev 5
S09	5	05TE (nt)	**TETHYS**	2005-088T18:46	Mar29	Tue	20	2	2	Inbound 108638.1 km flyby, v = 12.2 km/s, phase = 124 deg
S09	5	E_ring_lg	**DUST HAZARD**	2005-088T20:04	Mar29	Tue	20	2	2	Egress at 2005-088T22:49 (0.1 min); protection = MEA cover closure.
S09	5	05EN (nt)	**ENCELADUS**	2005-088T20:32	Mar29	Tue	20	2	2	Inbound 55573.4 km flyby, v = 9.7 km/s, phase = 134 deg
S09	5	05PL (nt)	**PALLENE**	2005-088T21:48	Mar29	Tue	21	2	2	Inbound 30586.6 km flyby, v = 7.0 km/s, phase = 137 deg
S09	5	05AT (nt)	**ATLAS**	2005-088T22:01	Mar29	Tue	21	2	2	Inbound 97311.2 km flyby, v = 1.4 km/s, phase = 68 deg
S09	5	Pallene	**DUST HAZARD**	2005-088T22:44	Mar29	Tue	21	2	2	Egress at 2005-089T00:31 (0.1 min); protection = MEA cover closure.
S09	5	Ring CRX	Ascending	2005-088T22:59	Mar29	Tue	21	2	2	r = 3.531 Rs (213006 km)
S09	5	05EP (nt)	**EPIMETHEUS**	2005-088T23:25	Mar29	Tue	21	2	2	Inbound 62012.1 km flyby, v = 2.3 km/s, phase = 102 deg
S09	5	Periapse		2005-088T23:38	Mar29	Tue	21	2	2	R = 3.498 Rs, lat = 0 deg, phase = 114 deg
S09	5	E_ring_lg	**DUST HAZARD**	2005-089T00:26	Mar30	Wed	21	2	2	Egress at 2005-089T02:32 (0.1 min); protection = MEA cover closure.
S09	5	05ME (nt)	**METHONE**	2005-089T00:51	Mar30	Wed	21	2	2	Outbound 52681.2 km flyby, v = 7.4 km/s, phase = 82 deg
S09	5	05TL (nt)	**TELESTO**	2005-089T01:32	Mar30	Wed	21	2	2	Outbound 74531.9 km flyby, v = 6.5 km/s, phase = 61 deg
S09	5	05PO (nt)	**POLYDEUCES**	2005-089T05:59	Mar30	Wed	21	2	2	Outbound 104025.0 km flyby, v = 6.0 km/s, phase = 59 deg
S09	5	**TOST**	**Segment**	2005-090T05:50	Mar31	Thu	22	1	3	TOST_005TI_T4 (2d); TOST Segment 005TI (T4)
S09	5	**05TI (t) [T4]**	**TITAN**	2005-090T20:05	Mar31	Thu	22	16	4	Outbound 2403.5 km flyby, v = 5.9 km/s, phase = 66 deg
S09	5	Ring CRX	Descending	2005-090T22:07	Mar31	Thu	0	16	4	r = 21.276 Rs (1283557 km)
S09	5	**XD**	**Segment**	2005-092T00:22	Apr02	Sat	1	15	5	XD_005_006 (10d); Cross Discipline Segment Outbound 5 - Inbound 6
S09	5	**OTM-020**	5TI+3	2005-093T20:22	Apr03	Sun	3	13	7	D/L start, burn ~6h later. Backup D/L 094T16:22

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S09	6	**Apoapse**		2005-096T23:32	Apr06	Wed	6	10	3	Per = 16.0 d, inc = 7.4 deg, r = 37.956 Rs, phase = 72 deg
S10	6	**S10 Begins**		2005-099T05:15	Apr09	Sat	8	8	5	S10 Sequence. Duration = 35 d
S10	6	**OTM-021**	~APO	2005-099T20:00	Apr09	Sat	9	7	6	D/L start, burn ~6h later. Backup D/L 100T16:00
S10	6	**RINGS**	**Segment**	2005-102T06:30	Apr12	Tue	11	5	2	RINGS_6 (2d); RINGS Rev 6
S10	6	**OTM-022**	6TI-3	2005-103T20:40	Apr13	Wed	13	3	4	D/L start, burn ~6h later. Backup D/L 105T02:14
S10	6	**SATURN**	**Segment**	2005-104T05:55	Apr14	Thu	13	3	0	Saturn_006 (2d); Saturn Rev 6 Periapse
S10	6	06PL (nt)	**PALLENE**	2005-104T20:31	Apr14	Thu	14	2	1	Inbound 119294.2 km flyby, v = 16.6 km/s, phase = 115 deg
S10	6	06ME (nt)	**METHONE**	2005-104T20:52	Apr14	Thu	14	2	1	Inbound 50424.3 km flyby, v = 12.3 km/s, phase = 118 deg
S10	6	G_ring	**DUST HAZARD**	2005-104T22:15	Apr14	Thu	14	2	1	Egress at 2005-104T22:17 (0.0 min); protection = HGA to ram and MEA cover closure.
S10	6	Ring CRX	Ascending	2005-104T22:16	Apr14	Thu	14	2	1	r = 2.734 Rs (164959 km)
S10	6	Periapse		2005-104T23:16	Apr14	Thu	14	2	1	R = 2.594 Rs, lat = 3 deg, phase = 108 deg
S10	6	**Earth OCC**	RING	2005-104T23:49	Apr14	Thu	14	2	1	Egress = 2005-105T01:45 (117 min)
S10	6	**Sun OCC**	RING	2005-104T23:58	Apr14	Thu	14	2	1	Egress = 2005-105T02:00 (122 min)
S10	6	06EP (nt)	**EPIMETHEUS**	2005-105T00:14	Apr15	Fri	14	2	1	Outbound 46179.6 km flyby, v = 9.8 km/s, phase = 80 deg
S10	6	06MI (nt)	**MIMAS**	2005-105T01:25	Apr15	Fri	14	2	1	Outbound 82488.4 km flyby, v = 14.0 km/s, phase = 94 deg
S10	6	06CA (nt)	**CALYPSO**	2005-105T04:21	Apr15	Fri	14	2	1	Outbound 70531.7 km flyby, v = 12.5 km/s, phase = 115 deg
S10	6	**TOST**	**Segment**	2005-106T06:25	Apr16	Sat	15	1	2	TOST_006TI_T5 (2d); TOST Segment 006TI (T5)
S10	6	**Earth OCC**	TITAN	2005-106T19:12	Apr16	Sat	0	89	3	Egress = 2005-106T19:18 (7 min)
S10	6	**06TI (t) [T5]**	**TITAN**	2005-106T19:12	Apr16	Sat	16	89	3	Outbound 1027.4 km flyby, v = 6.1 km/s, phase = 127 deg
S10	6	**Sun OCC**	TITAN	2005-106T19:13	Apr16	Sat	0	89	3	Egress = 2005-106T19:20 (8 min)
S10	6	Ring CRX	Descending	2005-106T19:58	Apr16	Sat	0	89	3	r = 20.890 Rs (1260316 km)
S10	6	**XD**	**Segment**	2005-107T21:25	Apr17	Sun	1	88	4	XD_006_007 (12d); Cross Discipline Segment Outbound 6 - Inbound 7
S10	6	**OTM-023**	6TI+3	2005-109T18:59	Apr19	Tue	3	86	6	D/L start, burn ~6h later. Backup D/L 110T18:44

S10	7	**Apoapse**		2005-113T23:30	Apr23	Sat	7	82	4	Per = 18.2 d, inc = 21.7 deg, r = 40.629 Rs, phase = 65 deg
S10	7	**OTM-024**	~APO	2005-118T18:58	Apr28	Thu	12	77	9	D/L start, burn ~6h later. Backup D/L 119T18:28
S10	7	**RINGS**	**Segment**	2005-120T04:13	Apr30	Sat	13	76	1	RINGS_7 (5d); RINGS Rev 7
S10	7	07HE (nt)	**HELENE**	2005-122T19:19	May02	Mon	16	73	4	Inbound 114174.6 km flyby, v = 9.9 km/s, phase = 107 deg
S10	7	07TE (nt)	**TETHYS**	2005-122T21:48	May02	Mon	16	73	4	Inbound 51855.5 km flyby, v = 9.3 km/s, phase = 110 deg
S10	7	07EP (nt)	**EPIMETHEUS**	2005-122T23:03	May02	Mon	16	73	4	Inbound 118460.8 km flyby, v = 6.2 km/s, phase = 46 deg
S10	7	E_ring_lg	**DUST HAZARD**	2005-122T23:33	May02	Mon	16	73	4	Egress at 2005-122T23:38 (0.0 min); protection = MEA cover closure.
S10	7	Ring CRX	Ascending	2005-122T23:36	May02	Mon	16	73	4	r = 3.895 Rs (234996 km)
S10	7	Periapse		2005-123T01:43	May03	Tue	16	73	4	R = 3.590 Rs, lat = 12 deg, phase = 115 deg
S10	7	**Earth OCC**	RING	2005-123T02:56	May03	Tue	16	73	4	Egress = 2005-123T08:38 (343 min)
S10	7	**Sun OCC**	RING	2005-123T03:24	May03	Tue	16	73	4	Egress = 2005-123T09:24 (361 min)
S10	7	**Earth OCC**	SATURN	2005-123T04:40	May03	Tue	16	73	4	Egress = 2005-123T07:03 (144 min)
S10	7	**Sun OCC**	SATURN	2005-123T05:11	May03	Tue	16	73	4	Egress = 2005-123T07:43 (153 min)
S10	7	07TL (nt)	**TELESTO**	2005-123T06:15	May03	Tue	16	73	4	Outbound 108305.9 km flyby, v = 8.3 km/s, phase = 98 deg
S10	7	07TI (nt)	**TITAN**	2005-124T05:37	May04	Wed	17	72	5	Outbound 864240.3 km flyby, v = 10.3 km/s, phase = 154 deg
S10	7	**SATURN**	**Segment**	2005-124T20:30	May04	Wed	18	71	6	Saturn_007 (1d); Saturn Rev 7 Outbound
S10	7	Ring CRX	Descending	2005-125T02:07	May05	Thu	18	71	6	r = 21.422 Rs (1292379 km)
S10	7	**XD**	**Segment**	2005-126T03:40	May06	Fri	19	70	7	XD_007_008 (12d); Cross Discipline Segment Outbound 7-Inbound 8

S10	8	**Apoapse**		2005-132T03:52	May12	Thu	25	64	13	Per = 18.2 d, inc = 21.9 deg, r = 40.608 Rs, phase = 65 deg
S11	8	**S11 Begins**		2005-134T02:50	May14	Sat	27	62	15	S11 Sequence. Duration = 35 d
S11	8	**RINGS**	**Segment**	2005-138T03:00	May18	Wed	31	58	19	RINGS_Rev_8 (5d); RINGS_Rev_8
S11	8	E_ring_lg	**DUST HAZARD**	2005-141T03:51	May21	Sat	34	55	22	Egress at 2005-141T03:56 (0.0 min); protection = MEA cover closure.
S11	8	Ring CRX	Ascending	2005-141T03:54	May21	Sat	34	55	22	r = 3.903 Rs (235467 km)
S11	8	Periapse		2005-141T06:02	May21	Sat	34	55	22	R = 3.591 Rs, lat = 12 deg, phase = 114 deg
S11	8	08AT (nt)	**ATLAS**	2005-141T06:30	May21	Sat	34	55	22	Outbound 99861.1 km flyby, v = 6.7 km/s, phase = 145 deg
S11	8	08PM (nt)	**PROMETHEUS**	2005-141T06:48	May21	Sat	34	55	22	Outbound 107599.9 km flyby, v = 6.9 km/s, phase = 153 deg
S11	8	**Earth OCC**	RING	2005-141T07:21	May21	Sat	35	55	23	Egress = 2005-141T13:07 (347 min)
S11	8	**Sun OCC**	RING	2005-141T07:46	May21	Sat	35	55	23	Egress = 2005-141T13:48 (362 min)
S11	8	08EN (nt)	**ENCELADUS**	2005-141T08:13	May21	Sat	35	54	23	Outbound 101964.9 km flyby, v = 8.8 km/s, phase = 86 deg
S11	8	**Earth OCC**	SATURN	2005-141T09:06	May21	Sat	35	54	23	Egress = 2005-141T11:31 (146 min)
S11	8	**Sun OCC**	SATURN	2005-141T09:33	May21	Sat	35	54	23	Egress = 2005-141T12:06 (154 min)
S11	8	**SATURN**	**Segment**	2005-142T19:00	May22	Sun	36	53	24	Saturn_008 (1d); Saturn Rev 8 Outbound
S11	8	Ring CRX	Descending	2005-143T05:45	May23	Mon	36	53	24	r = 21.229 Rs (1280748 km)
S11	8	**XD**	**Segment**	2005-144T02:42	May24	Tue	37	52	25	XD_008_009 (12d); Cross Discipline Segment Outbound 8 - Inbound 9

S11	9	**Apoapse**		2005-150T08:21	May30	Mon	44	45	32	Per = 18.2 d, inc = 21.9 deg, r = 40.626 Rs, phase = 66 deg
S11	9	**RINGS**	**Segment**	2005-156T02:00	Jun05	Sun	49	40	37	RINGS_Rev_9 (5d); RINGS_Rev_9
S11	9	09TI (nt)	**TITAN**	2005-157T19:05	Jun06	Mon	51	38	39	Inbound 425570.4 km flyby, v = 5.8 km/s, phase = 83 deg
S11	9	E_ring_lg	**DUST HAZARD**	2005-159T08:23	Jun08	Wed	53	36	41	Egress at 2005-159T08:28 (0.0 min); protection = MEA cover closure.
S11	9	Ring CRX	Ascending	2005-159T08:26	Jun08	Wed	53	36	41	r = 3.914 Rs (236127 km)
S11	9	09PL (nt)	**PALLENE**	2005-159T09:57	Jun08	Wed	53	36	41	Inbound 64544.8 km flyby, v = 7.0 km/s, phase = 56 deg
S11	9	09CA (nt)	**CALYPSO**	2005-159T10:24	Jun08	Wed	53	36	41	Inbound 97710.2 km flyby, v = 7.8 km/s, phase = 64 deg
S11	9	Periapse		2005-159T10:36	Jun08	Wed	53	36	41	R = 3.591 Rs, lat = 12 deg, phase = 114 deg
S11	9	**Earth OCC**	RING	2005-159T12:03	Jun08	Wed	53	36	41	Egress = 2005-159T17:54 (352 min)
S11	9	**Sun OCC**	RING	2005-159T12:24	Jun08	Wed	53	36	41	Egress = 2005-159T18:26 (362 min)
S11	9	**Earth OCC**	SATURN	2005-159T13:49	Jun08	Wed	53	36	41	Egress = 2005-159T16:17 (149 min)
S11	9	**Sun OCC**	SATURN	2005-159T14:10	Jun08	Wed	53	36	41	Egress = 2005-159T16:44 (155 min)
S11	9	**SATURN**	**Segment**	2005-160T17:21	Jun09	Thu	54	35	42	Saturn_009 (2d); Saturn Rev 9 Outbound
S11	9	Ring CRX	Descending	2005-161T09:08	Jun10	Fri	55	34	43	r = 20.903 Rs (1261058 km)
S11	9	**XD**	**Segment**	2005-163T01:25	Jun12	Sun	56	33	44	XD_009_010 (11d); Cross Discipline Segment Outbound 9 - Inbound 10

S11	10	**Apoapse**		2005-168T13:12	Jun17	Fri	62	27	50	Per = 18.2 d, inc = 21.8 deg, r = 40.663 Rs, phase = 66 deg
S12	10	**S12 Begins**		2005-169T01:34	Jun18	Sat	62	27	50	S12 Sequence. Duration = 44 d
S12	10	10TI (nt)	**TITAN**	2005-173T12:37	Jun22	Wed	67	22	55	Inbound 920614.9 km flyby, v = 3.7 km/s, phase = 65 deg
S12	10	**RINGS**	**Segment**	2005-174T01:30	Jun23	Thu	67	22	55	RINGS_Rev_10 (5d); RINGS_Rev_10
S12	10	10TI (nt)	**TITAN**	2005-177T07:10	Jun26	Sun	70	19	59	Inbound 874592.4 km flyby, v = 7.7 km/s, phase = 111 deg
S12	10	E_ring_lg	**DUST HAZARD**	2005-177T13:27	Jun26	Sun	71	18	59	Egress at 2005-177T13:32 (0.0 min); protection = MEA cover closure.
S12	10	Ring CRX	Ascending	2005-177T13:29	Jun26	Sun	71	18	59	r = 3.934 Rs (237361 km)
S12	10	10PN (nt)	**PAN**	2005-177T14:23	Jun26	Sun	71	18	59	Inbound 94954.9 km flyby, v = 6.2 km/s, phase = 84 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S12	10	10TL (nt)	**TELESTO**	2005-177T14:48	Jun26	Sun	71	18	59	Inbound 87057.7 km flyby, v = 8.0 km/s, phase = 64 deg
S12	10	Periapse		2005-177T15:42	Jun26	Sun	71	18	59	R = 3.603 Rs, lat = 12 deg, phase = 113 deg
S12	10	**Earth OCC**	RING	2005-177T17:20	Jun26	Sun	71	18	59	Egress = 2005-177T23:18 (358 min)
S12	10	**Sun OCC**	RING	2005-177T17:35	Jun26	Sun	71	18	59	Egress = 2005-177T23:38 (364 min)
S12	10	**Earth OCC**	SATURN	2005-177T19:07	Jun26	Sun	71	18	59	Egress = 2005-177T21:38 (152 min)
S12	10	**Sun OCC**	SATURN	2005-177T19:21	Jun26	Sun	71	18	59	Egress = 2005-177T21:56 (156 min)
S12	10	**SATURN**	**Segment**	2005-179T00:30	Jun28	Tue	72	17	60	Saturn_010 (1d); Saturn Rev 10 Outbound
S12	10	Ring CRX	Descending	2005-179T13:44	Jun28	Tue	73	16	61	r = 20.762 Rs (1252546 km)
S12	10	**XD**	**Segment**	2005-180T01:25	Jun29	Wed	73	16	61	XD_010_011 (13d); Cross Discipline Segment Outbound 10-Inbound 11

S12	11	**Apoapse**		2005-186T18:57	Jul05	Tue	80	9	68	Per = 18.3 d, inc = 21.8 deg, r = 40.748 Rs, phase = 67 deg
S12	11	**OTM-025**	11EN-6	2005-189T14:37	Jul08	Fri	83	6	71	D/L start, burn ~6h later. Backup D/L 190T14:37
S12	11	**SATURN**	**Segment**	2005-192T23:30	Jul11	Mon	86	3	3	Saturn_011_In (1d); Saturn Rev 11 Inbound
S12	11	**SOST**	**Segment**	2005-193T23:30	Jul12	Tue	87	2	4	SOST_011EN (3d); SOST Rev 11 Targeted Enceladus
S12	11	11PM (nt)	**PROMETHEUS**	2005-195T19:31	Jul14	Thu	89	0	6	Inbound 122149.3 km flyby, v = 6.1 km/s, phase = 51 deg
S12	11	E_ring_lg	**DUST HAZARD**	2005-195T19:53	Jul14	Thu	89	0	6	Egress at 2005-195T19:58 (0.0 min); protection = MEA cover closure.
S12	11	**11EN (t) [E2]**	**ENCELADUS**	2005-195T19:55	Jul14	Thu	89	39	6	Inbound 166.3 km flyby, v = 8.2 km/s, phase = 64 deg
S12	11	Ring CRX	Ascending	2005-195T19:56	Jul14	Thu	0	39	6	r = 3.947 Rs (238115 km)
S12	11	11ME (nt)	**METHONE**	2005-195T20:48	Jul14	Thu	0	39	6	Inbound 36705.4 km flyby, v = 7.6 km/s, phase = 110 deg
S12	11	11EP (nt)	**EPIMETHEUS**	2005-195T20:55	Jul14	Thu	0	38	6	Inbound 76769.5 km flyby, v = 6.3 km/s, phase = 84 deg
S12	11	Periapse		2005-195T22:10	Jul14	Thu	0	38	6	R = 3.609 Rs, lat = 13 deg, phase = 113 deg
S12	11	11AE (nt)	**AEGAEON**	2005-195T22:11	Jul14	Thu	0	38	6	Outbound 118582.1 km flyby, v = 11.1 km/s, phase = 163 deg
S12	11	**Earth OCC**	RING	2005-196T00:02	Jul15	Fri	0	38	6	Egress = 2005-196T06:04 (363 min)
S12	11	**Sun OCC**	RING	2005-196T00:07	Jul15	Fri	0	38	6	Egress = 2005-196T06:11 (364 min)
S12	11	**Earth OCC**	SATURN	2005-196T01:48	Jul15	Fri	0	38	6	Egress = 2005-196T04:23 (155 min)
S12	11	**Sun OCC**	SATURN	2005-196T01:52	Jul15	Fri	0	38	6	Egress = 2005-196T04:28 (157 min)
S12	11	**SATURN**	**Segment**	2005-196T23:30	Jul15	Fri	1	37	7	Saturn_011_Out (2d); Saturn Rev 11 Outbound
S12	11	Ring CRX	Descending	2005-197T19:43	Jul16	Sat	2	37	8	r = 20.630 Rs (1244598 km)
S12	11	**XD**	**Segment**	2005-198T23:20	Jul17	Sun	3	35	9	XD_011_012 (12d); Cross Discipline Segment Outbound 11-Inbound 12
S12	11	SEP = 3.0 deg	Conjunction	2005-201T04:31	Jul20	Wed	5	33	12	Commanding must not be required below 3.0 deg
S12	11	SEP = 2.0 deg	Conjunction	2005-202T09:58	Jul21	Thu	7	32	13	Plan no SSR playback below 2.0 deg
S12	11	SEP = 1.0 deg	Conjunction	2005-203T15:49	Jul22	Fri	8	31	14	Require no DSN passes below 1.0 deg

S12	12	**Apoapse**		2005-205T01:57	Jul24	Sun	9	29	15	Per = 18.3 d, inc = 21.8 deg, r = 40.827 Rs, phase = 67 deg
S12	12	SEP = 1.0 deg	Conjunction	2005-205T23:08	Jul24	Sun	10	28	16	Require no DSN passes below 1.0 deg
S12	12	SEP = 2.0 deg	Conjunction	2005-207T04:45	Jul26	Tue	11	27	18	Plan no SSR playback below 2.0 deg
S12	12	SEP = 3.0 deg	Conjunction	2005-208T09:48	Jul27	Wed	13	26	19	Commanding must not be required below 3.0 deg
S12	12	**SATURN**	**Segment**	2005-210T21:30	Jul29	Fri	15	23	21	Saturn_012_In (2d); Saturn Rev 12 Inbound
S13	12	**SOST**	**Segment**	2005-212T22:00	Jul31	Sun	17	21	23	SOST_012MI (3d); SOST Rev 12 Non-targeted Mimas
S13	12	**S13 Begins**		2005-212T22:00	Jul31	Sun	17	21	23	S13 Sequence. Duration = 30 d
S13	12	E_ring_lg	**DUST HAZARD**	2005-214T03:30	Aug02	Tue	18	20	25	Egress at 2005-214T03:35 (0.0 min); protection = MEA cover closure.
S13	12	Ring CRX	Ascending	2005-214T03:33	Aug02	Tue	18	20	25	r = 3.948 Rs (238184 km)
S13	12	12DA (nt)	**DAPHNIS**	2005-214T03:58	Aug02	Tue	18	20	25	Inbound 102508.5 km flyby, v = 6.1 km/s, phase = 68 deg
S13	12	12MI (nt)	**MIMAS**	2005-214T04:21	Aug02	Tue	18	20	25	Inbound 61148.8 km flyby, v = 6.1 km/s, phase = 58 deg
S13	12	12PM (nt)	**PROMETHEUS**	2005-214T05:17	Aug02	Tue	18	20	25	Inbound 84970.7 km flyby, v = 6.3 km/s, phase = 111 deg
S13	12	Periapse		2005-214T05:48	Aug02	Tue	18	20	25	R = 3.604 Rs, lat = 13 deg, phase = 112 deg
S13	12	12CA (nt)	**CALYPSO**	2005-214T07:47	Aug02	Tue	18	20	25	Outbound 106192.2 km flyby, v = 7.1 km/s, phase = 73 deg
S13	12	**Sun OCC**	RING	2005-214T07:48	Aug02	Tue	18	20	25	Egress = 2005-214T13:52 (365 min)
S13	12	**Earth OCC**	RING	2005-214T07:54	Aug02	Tue	18	20	25	Egress = 2005-214T14:00 (367 min)
S13	12	**Sun OCC**	SATURN	2005-214T09:33	Aug02	Tue	19	20	25	Egress = 2005-214T12:09 (157 min)
S13	12	**Earth OCC**	SATURN	2005-214T09:38	Aug02	Tue	19	20	25	Egress = 2005-214T12:16 (159 min)
S13	12	12TI (nt)	**TITAN**	2005-214T15:55	Aug02	Tue	19	20	25	Outbound 939763.3 km flyby, v = 6.5 km/s, phase = 35 deg
S13	12	**OTM-026**	NEARP	2005-215T05:50	Aug03	Wed	19	19	26	D/L start, burn ~6h later. Backup D/L 216T13:20
S13	12	**SATURN**	**Segment**	2005-215T14:50	Aug03	Wed	20	19	0	Saturn_012_Out (1d); Saturn Rev 12 Outbound
S13	12	Ring CRX	Descending	2005-216T02:39	Aug04	Thu	20	18	1	r = 20.428 Rs (1232417 km)
S13	12	**XD**	**Segment**	2005-216T22:20	Aug04	Thu	21	17	2	XD_012_013 (13d); Cross Discipline Segment Outbound 12 - Inbound 13
S13	12	12TI (nt)	**TITAN**	2005-218T12:33	Aug06	Sat	23	16	3	Outbound 837687.8 km flyby, v = 3.8 km/s, phase = 62 deg
S13	12	**OTM-027**	~APO	2005-222T07:21	Aug10	Wed	26	12	7	D/L start, burn ~6h later. Backup D/L 223T12:52

S13	13	**Apoapse**		2005-223T08:29	Aug11	Thu	28	11	1	Per = 18.2 d, inc = 21.9 deg, r = 40.691 Rs, phase = 68 deg
S13	13	**RINGS**	**Segment**	2005-229T18:00	Aug17	Wed	34	5	7	RINGS_Rev_13 (4d); Rings Rev 13 segment
S13	13	**OTM-028**	13TI-3	2005-230T05:00	Aug18	Thu	34	4	8	D/L start, burn ~6h later. Backup D/L 231T05:00
S13	13	13TE (nt)	**TETHYS**	2005-232T07:02	Aug20	Sat	36	2	2	Inbound 122847.5 km flyby, v = 12.4 km/s, phase = 123 deg
S13	13	E_ring_lg	**DUST HAZARD**	2005-232T08:57	Aug20	Sat	37	2	2	Egress at 2005-232T09:02 (0.0 min); protection = MEA cover closure.
S13	13	Ring CRX	Ascending	2005-232T09:00	Aug20	Sat	37	2	2	r = 3.930 Rs (237122 km)
S13	13	Periapse		2005-232T11:15	Aug20	Sat	37	2	2	R = 3.585 Rs, lat = 13 deg, phase = 112 deg
S13	13	13TL (nt)	**TELESTO**	2005-232T12:13	Aug20	Sat	37	2	2	Outbound 105277.7 km flyby, v = 7.8 km/s, phase = 67 deg
S13	13	**Sun OCC**	RING	2005-232T13:17	Aug20	Sat	37	2	2	Egress = 2005-232T19:22 (365 min)
S13	13	**Earth OCC**	RING	2005-232T13:37	Aug20	Sat	37	2	2	Egress = 2005-232T19:45 (368 min)
S13	13	**Sun OCC**	SATURN	2005-232T15:01	Aug20	Sat	37	2	2	Egress = 2005-232T17:38 (158 min)
S13	13	**Earth OCC**	SATURN	2005-232T15:17	Aug20	Sat	37	2	2	Egress = 2005-232T17:57 (161 min)
S13	13	**TOST**	**Segment**	2005-233T21:20	Aug21	Sun	38	0	4	TOST_013TI_T6 (2d); TOST Segment 013TI (T6)
S13	13	Ring CRX	Descending	2005-234T07:31	Aug22	Mon	38	0	4	r = 20.270 Rs (1222898 km)
S13	13	**13TI (t) [T6]**	**TITAN**	2005-234T08:54	Aug22	Mon	39	16	4	Outbound 3660.2 km flyby, v = 5.9 km/s, phase = 44 deg
S13	13	**XD**	**Segment**	2005-235T13:42	Aug23	Tue	1	15	5	XD_013_014 (10d); Cross Discipline Segment Outbound 13 - Inbound 14
S13	13	**OTM-029**	13TI+3	2005-237T11:08	Aug25	Thu	3	13	7	D/L start, burn ~6h later. Backup D/L 238T10:59

S13	14	**Apoapse**		2005-240T12:02	Aug28	Sun	6	10	3	Per = 16.0 d, inc = 15.6 deg, r = 37.798 Rs, phase = 75 deg
S13	14	**OTM-030**	~APO	2005-242T12:43	Aug30	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 243T10:38
S14	14	**S14 Begins**		2005-242T21:43	Aug30	Tue	9	7	0	S14 Sequence. Duration = 39 d
S14	14	**RINGS**	**Segment**	2005-245T20:00	Sep02	Fri	11	5	3	RINGS_Rev_14 (4d); Rings Rev 14 Segment
S14	14	**OTM-031**	14TI-3	2005-246T11:30	Sep03	Sat	12	4	4	D/L start, burn ~6h later. Backup D/L 247T10:30
S14	14	G_ring	**DUST HAZARD**	2005-248T10:38	Sep05	Mon	14	2	2	Egress at 2005-248T10:39 (0.0 min); protection = MEA cover closure.
S14	14	Ring CRX	Ascending	2005-248T10:39	Sep05	Mon	14	2	2	r = 2.928 Rs (176673 km)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S14	14	14AE (nt)	**AEGAEON**	2005-248T10:48	Sep05	Mon	14	2	2	Inbound 8517.2 km flyby, v = 8.1 km/s, phase = 115 deg
S14	14	14PA (nt)	**PANDORA**	2005-248T11:31	Sep05	Mon	14	2	2	Inbound 51938.2 km flyby, v = 7.0 km/s, phase = 57 deg
S14	14	14PM (nt)	**PROMETHEUS**	2005-248T11:32	Sep05	Mon	14	2	2	Inbound 102396.9 km flyby, v = 14.6 km/s, phase = 151 deg
S14	14	Periapse		2005-248T11:52	Sep05	Mon	14	2	2	R = 2.748 Rs, lat = 8 deg, phase = 105 deg
S14	14	**Sun OCC**	RING	2005-248T12:38	Sep05	Mon	14	2	2	Egress = 2005-248T17:53 (316 min)
S14	14	**Earth OCC**	RING	2005-248T12:52	Sep05	Mon	14	2	2	Egress = 2005-248T18:22 (330 min)
S14	14	14ME (nt)	**METHONE**	2005-248T13:57	Sep05	Mon	14	2	2	Outbound 63024.5 km flyby, v = 10.2 km/s, phase = 91 deg
S14	14	**Sun OCC**	SATURN	2005-248T14:37	Sep05	Mon	14	2	2	Egress = 2005-248T16:53 (136 min)
S14	14	**Earth OCC**	SATURN	2005-248T14:53	Sep05	Mon	14	2	2	Egress = 2005-248T17:19 (147 min)
S14	14	**TOST**	**Segment**	2005-249T20:30	Sep06	Tue	15	0	3	TOST_014TI_T7 (2d); TOST Segment 014TI (T7)
S14	14	Ring CRX	Descending	2005-250T06:46	Sep07	Wed	16	0	4	r = 20.226 Rs (1220222 km)
S14	14	**14TI (dt) [T7]**	**TITAN**	2005-250T08:12	Sep07	Wed	16	17	4	Outbound 1074.8 km flyby, v = 6.1 km/s, phase = 85 deg
S14	14	**XD**	**Segment**	2005-251T12:46	Sep08	Thu	1	16	5	XD_014_015 (14d); Cross Discipline Segment Outbound 14-Inbound 15
S14	14	**OTM-032**	14TI+3	2005-253T11:09	Sep10	Sat	3	14	7	D/L start, burn ~6h later. Backup D/L 254T09:30
S14	15	**Apoapse**		2005-257T16:50	Sep14	Wed	7	9	4	Per = 18.4 d, inc = 0.3 deg, r = 41.471 Rs, phase = 74 deg
S14	15	**OTM-033**	~APO	2005-262T10:40	Sep19	Mon	12	5	9	D/L start, burn ~6h later. Backup D/L 263T10:15
S14	15	**SATURN**	**Segment**	2005-265T18:43	Sep22	Thu	15	1	3	Saturn_015 (3d); Saturn Rev 15 Periapse
S14	15	**OTM-034**	15HY-3	2005-266T01:45	Sep23	Fri	16	1	4	D/L start, burn ~6h later. Backup D/L 267T10:20
S14	15	Ring CRX	Ascending	2005-266T17:11	Sep23	Fri	16	0	1	r = 4.514 Rs (272323 km)
S14	15	E_ring_lg	**DUST HAZARD**	2005-266T17:30	Sep23	Fri	16	0	1	Egress at 2005-266T19:11 (0.1 min); protection = HGA to ram and MEA cover closure.
S14	15	15CA (nt)	**CALYPSO**	2005-266T18:51	Sep23	Fri	16	0	1	Inbound 91042.5 km flyby, v = 7.0 km/s, phase = 83 deg
S14	15	Pallene	**DUST HAZARD**	2005-266T19:10	Sep23	Fri	16	0	1	Egress at 2005-266T19:22 (0.0 min); protection = MEA cover closure.
S14	15	15MI (nt)	**MIMAS**	2005-266T19:40	Sep23	Fri	16	0	1	Inbound 70011.2 km flyby, v = 11.8 km/s, phase = 133 deg
S14	15	Periapse		2005-266T21:36	Sep23	Fri	17	0	1	R = 2.999 Rs, lat = 0 deg, phase = 107 deg
S14	15	15PM (nt)	**PROMETHEUS**	2005-266T22:26	Sep23	Fri	17	0	1	Outbound 98009.9 km flyby, v = 7.6 km/s, phase = 159 deg
S14	15	E_ring_lg	**DUST HAZARD**	2005-267T00:00	Sep24	Fri	17	0	1	Egress at 2005-267T01:40 (0.1 min); protection = MEA cover closure.
S14	15	**15TE (dt) [Te1]**	**TETHYS**	2005-267T02:42	Sep24	Sat	17	2	1	Outbound 1495.2 km flyby, v = 9.0 km/s, phase = 68 deg
S14	15	Ring CRX	Descending	2005-267T07:27	Sep24	Sat	0	2	1	r = 7.445 Rs (449149 km)
S14	15	15TI (nt)	**TITAN**	2005-267T22:47	Sep24	Sat	1	1	2	Outbound 911727.0 km flyby, v = 10.7 km/s, phase = 149 deg
S14	15	**SOST**	**Segment**	2005-268T13:40	Sep25	Sun	1	1	2	SOST_015HY (1d); SOST Rev 15 Targeted Hyperion
S14	15	**15HY (t) [H1]**	**HYPERION**	2005-269T02:25	Sep26	Mon	2	16	3	Outbound 479.0 km flyby, v = 5.6 km/s, phase = 48 deg
S14	15	**XD**	**Segment**	2005-269T19:20	Sep26	Mon	1	15	4	XD_015_016 (14d); Cross Discipline Segment Outbound 15 - Inbound 16
S14	15	**OTM-035**	15HY+3	2005-271T10:11	Sep28	Wed	2	13	5	D/L start, burn ~6h later. Backup D/L 272T10:00
S14	15	**OTM-036**	~APO	2005-274T08:26	Oct01	Sat	5	10	3	D/L start, burn ~6h later. Backup D/L 275T11:00
S14	16	**Apoapse**		2005-275T23:34	Oct02	Sun	7	9	2	Per = 18.2 d, inc = 0.3 deg, r = 41.173 Rs, phase = 74 deg
S14	16	**OTM-037**	16DI-3	2005-281T03:30	Oct08	Sat	12	4	7	D/L start, burn ~6h later. Backup D/L 282T09:12
S15	16	**S15 Begins**		2005-281T12:30	Oct08	Sat	12	3	0	S15 Sequence. Duration = 35 d
S15	16	**SOST**	**Segment**	2005-283T18:27	Oct10	Mon	15	1	3	SOST_016DI (3d); SOST Rev 16 Targeted Dione
S15	16	16TI (nt)	**TITAN**	2005-283T22:22	Oct10	Mon	15	1	3	Inbound 776295.1 km flyby, v = 9.7 km/s, phase = 65 deg
S15	16	**16DI (t) [D1]**	**DIONE**	2005-284T17:52	Oct11	Tue	16	16	4	Inbound 498.5 km flyby, v = 9.1 km/s, phase = 66 deg
S15	16	16TL (nt)	**TELESTO**	2005-284T20:26	Oct11	Tue	0	16	4	Inbound 9533.5 km flyby, v = 8.7 km/s, phase = 63 deg
S15	16	Ring CRX	Ascending	2005-284T20:38	Oct11	Tue	0	16	4	r = 4.766 Rs (287517 km)
S15	16	E_ring_lg	**DUST HAZARD**	2005-284T21:26	Oct11	Tue	0	16	4	Egress at 2005-284T23:07 (0.1 min); protection = MEA cover closure.
S15	16	16PL (nt)	**PALLENE**	2005-284T22:48	Oct11	Tue	0	16	4	Inbound 50438.7 km flyby, v = 10.8 km/s, phase = 125 deg
S15	16	Pallene	**DUST HAZARD**	2005-284T23:05	Oct11	Tue	0	16	4	Egress at 2005-284T23:05 (0.0 min); protection = MEA cover closure.
S15	16	**OTM-038**	16DI+1	2005-284T23:57	Oct11	Tue	0	16	4	D/L start, burn ~6h later. Backup D/L 286T10:40
S15	16	Periapse		2005-285T01:31	Oct12	Wed	0	16	0	R = 3.002 Rs, lat = 0 deg, phase = 106 deg
S15	16	16AT (nt)	**ATLAS**	2005-285T01:55	Oct12	Wed	0	16	0	Outbound 55661.5 km flyby, v = 4.0 km/s, phase = 143 deg
S15	16	16EN (nt)	**ENCELADUS**	2005-285T03:03	Oct12	Wed	0	16	0	Outbound 49076.4 km flyby, v = 6.8 km/s, phase = 73 deg
S15	16	E_ring_lg	**DUST HAZARD**	2005-285T04:51	Oct12	Wed	0	16	0	Egress at 2005-285T05:36 (0.0 min); protection = MEA cover closure.
S15	16	Ring CRX	Descending	2005-285T09:57	Oct12	Wed	1	16	0	r = 6.687 Rs (403427 km)
S15	16	**XD**	**Segment**	2005-286T19:55	Oct13	Thu	2	14	2	XD_016_017 (14d); Cross-Discipline Segment 16 Outbound-17 Inbound
S15	17	**Apoapse**		2005-294T00:00	Oct21	Thu	9	7	9	Per = 17.9 d, inc = 0.4 deg, r = 40.697 Rs, phase = 74 deg
S15	17	**OTM-039**	~APO	2005-294T08:58	Oct21	Fri	10	7	9	D/L start, burn ~6h later. Backup D/L 295T01:29
S15	17	**OTM-040**	17TI-3	2005-298T01:14	Oct25	Tue	13	3	4	D/L start, burn ~6h later. Backup D/L 299T01:14
S15	17	**TOST**	**Segment**	2005-300T18:29	Oct27	Thu	16	0	3	TOST_017TI_T8 (2d); TOST Segment 017TI (T8)
S15	17	**17TI (t) [T8]**	**TITAN**	2005-301T04:15	Oct28	Fri	16	30	3	Inbound 1353.0 km flyby, v = 5.9 km/s, phase = 105 deg
S15	17	**Earth OCC**	TITAN	2005-301T04:24	Oct28	Fri	0	30	3	Egress = 2005-301T04:41 (17 min)
S15	17	**SATURN**	**Segment**	2005-302T09:49	Oct29	Sat	1	29	4	Saturn_017 (2d); Saturn Rev 17 Periapse
S15	17	Ring CRX	Ascending	2005-302T21:25	Oct29	Sat	2	28	5	r = 4.699 Rs (283479 km)
S15	17	17ME (nt)	**METHONE**	2005-302T21:37	Oct29	Sat	2	28	5	Inbound 98875.3 km flyby, v = 2.1 km/s, phase = 57 deg
S15	17	Periapse		2005-302T22:56	Oct29	Sat	2	28	5	R = 4.596 Rs, lat = 0 deg, phase = 92 deg
S15	17	17CA (nt)	**CALYPSO**	2005-303T02:50	Oct30	Sun	2	28	5	Outbound 107869.7 km flyby, v = 9.4 km/s, phase = 66 deg
S15	17	**OTM-041**	17TI+3	2005-304T07:59	Oct31	Mon	3	27	6	D/L start, burn ~6h later. Backup D/L 305T08:30
S15	17	**XD**	**Segment**	2005-304T17:14	Oct31	Mon	4	26	0	XD_017_018 (25d); Cross Discipline Segment Outbound 17-Inbound 18
S15	17	Ring CRX	Descending	2005-309T17:18	Nov05	Sat	9	21	5	r = 44.355 Rs (2675932 km)
S16	17	**S16 Begins**		2005-316T17:01	Nov12	Sat	16	14	12	S16 Sequence. Duration = 35 d
S16	18	**Apoapse**		2005-317T04:57	Nov13	Sun	16	14	13	Per = 28.5 d, inc = 0.4 deg, r = 55.088 Rs, phase = 88 deg
S16	18	**OTM-042**	~APO	2005-317T08:02	Nov13	Sun	16	14	13	D/L start, burn ~6h later. Backup D/L 318T08:02
S16	18	**OTM-043**	18RH-3	2005-327T07:03	Nov23	Wed	26	4	10	D/L start, burn ~6h later. Backup D/L 328T07:03
S16	18	**SOST**	**Segment**	2005-329T15:48	Nov25	Fri	28	1	2	SOST_018RH (4d); SOST Rev 18 Targeted Rhea
S16	18	**18RH (t) [R1]**	**RHEA**	2005-330T22:38	Nov26	Sat	30	30	4	Inbound 503.6 km flyby, v = 7.3 km/s, phase = 87 deg
S16	18	Ring CRX	Ascending	2005-331T09:18	Nov27	Sun	0	29	4	r = 4.761 Rs (287256 km)
S16	18	18EN (nt)	**ENCELADUS**	2005-331T10:25	Nov27	Sun	0	29	4	Inbound 107850.8 km flyby, v = 7.5 km/s, phase = 134 deg
S16	18	Periapse		2005-331T11:21	Nov27	Sun	1	29	4	R = 4.576 Rs, lat = 0 deg, phase = 92 deg
S16	18	18HE (nt)	**HELENE**	2005-331T14:37	Nov27	Sun	1	29	4	Outbound 86435.4 km flyby, v = 5.5 km/s, phase = 87 deg
S16	18	**OTM-044**	18RH+1	2005-331T22:15	Nov27	Sun	1	29	5	D/L start, burn ~6h later. Backup D/L 333T06:33
S16	18	**XD**	**Segment**	2005-333T15:33	Nov29	Tue	3	27	2	XD_018_019 (23d); Cross Discipline Segment Outbound 18-Inbound 19
S16	18	Ring CRX	Descending	2005-336T04:53	Dec02	Fri	5	25	4	r = 36.636 Rs (2210259 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S16	19	**Apoapse**		2005-345T04:27	Dec11	Sun	14	16	13	Per = 27.4 d, inc = 0.4 deg, r = 53.563 Rs, phase = 89 deg
S16	19	**OTM-045**	~APO	2005-345T05:35	Dec11	Sun	14	16	13	D/L start, burn ~6h later. Backup D/L 346T05:35
S17	19	**S17 Begins**		2005-351T14:21	Dec17	Sat	21	9	6	S17 Sequence. Duration = 41 d
S17	19	**MAG**	**Segment**	2005-356T14:21	Dec22	Thu	26	4	11	MAG_019 (4d); Rev 19
S17	19	**OTM-046**	19TI-3	2005-357T06:25	Dec23	Fri	26	4	12	D/L start, burn ~6h later. Backup D/L 358T04:40
S17	19	Ring CRX	Ascending	2005-358T19:19	Dec24	Sat	28	2	2	r = 4.760 Rs (287193 km)
S17	19	19EN (nt)	**ENCELADUS**	2005-358T20:29	Dec24	Sat	28	2	2	Inbound 94053.5 km flyby, v = 6.7 km/s, phase = 133 deg
S17	19	Periapse		2005-358T21:23	Dec24	Sat	28	2	2	R = 4.572 Rs, lat = 0 deg, phase = 91 deg
S17	19	19PL (nt)	**PALLENE**	2005-358T22:26	Dec24	Sat	28	2	2	Outbound 107286.4 km flyby, v = 4.4 km/s, phase = 142 deg
S17	19	19TL (nt)	**TELESTO**	2005-359T00:44	Dec25	Sun	28	2	2	Outbound 19275.4 km flyby, v = 5.9 km/s, phase = 70 deg
S17	19	19HE (nt)	**HELENE**	2005-359T01:45	Dec25	Sun	28	2	2	Outbound 74718.3 km flyby, v = 5.4 km/s, phase = 90 deg
S17	19	**TOST**	**Segment**	2005-360T06:59	Dec26	Mon	29	1	3	TOST_019TI_T9 (1d); TOST Segment 019TI (T9)
S17	19	**19TI (t) [T9]**	**TITAN**	2005-360T18:59	Dec26	Mon	30	20	4	Outbound 10410.9 km flyby, v = 5.6 km/s, phase = 67 deg
S17	19	**XD**	**Segment**	2005-361T13:34	Dec27	Tue	1	19	4	XD_019_020 (18d); Cross Discipline Segment Outbound 19 - Inbound 20
S17	19	**OTM-047**	19TI+3	2005-363T20:47	Dec29	Thu	3	17	7	D/L start, burn ~6h later. Backup D/L 365T04:07
S17	19	Ring CRX	Descending	2005-363T22:02	Dec29	Thu	3	17	0	r = 37.052 Rs (2235328 km)
S17	19	**OTM-048**	~APO	2006-002T20:22	Jan02	Mon	7	13	4	D/L start, burn ~6h later. Backup D/L 004T04:17
S17	20	**Apoapse**		2006-005T14:09	Jan05	Thu	10	10	3	Per = 23.4 d, inc = 0.4 deg, r = 48.276 Rs, phase = 94 deg
S17	20	**OTM-049**	20TI-3	2006-012T03:23	Jan12	Thu	16	3	9	D/L start, burn ~6h later. Backup D/L 013T03:23
S17	20	**TOST**	**Segment**	2006-014T13:38	Jan14	Sat	19	1	2	TOST_020TI_T10 (2d); TOST Segment 020TI (T10)
S17	20	**20TI (t) [T10]**	**TITAN**	2006-015T11:41	Jan15	Sun	20	43	3	Inbound 2042.8 km flyby, v = 5.8 km/s, phase = 120 deg
S17	20	**Earth OCC**	TITAN	2006-015T11:49	Jan15	Sun	0	43	3	Egress = 2006-015T12:03 (14 min)
S17	20	**Sun OCC**	TITAN	2006-015T11:50	Jan15	Sun	0	43	3	Egress = 2006-015T12:03 (13 min)
S17	20	**SOST**	**Segment**	2006-016T12:05	Jan16	Mon	1	42	4	SOST_020 (3d); SOST Rev 20
S17	20	Ring CRX	Ascending	2006-017T06:24	Jan17	Tue	2	41	5	r = 5.590 Rs (337258 km)
S17	20	Periapse		2006-017T06:58	Jan17	Tue	2	41	5	R = 5.580 Rs, lat = 0 deg, phase = 73 deg
S17	20	**OTM-050**	20TI+3	2006-018T02:37	Jan18	Wed	3	40	6	D/L start, burn ~6h later. Backup D/L 019T02:45
S17	20	**XD**	**Segment**	2006-019T11:59	Jan19	Thu	4	39	1	XD_020_021 (36d); Cross Discipline Segment Outbound 20 - Inbound 21
S18	20	**S18 Begins**		2006-027T04:03	Jan27	Fri	12	31	9	S18 Sequence. Duration = 43 d
S18	20	**OTM-051**	~APO	2006-033T01:53	Feb02	Thu	18	25	15	D/L start, burn ~6h later. Backup D/L 034T01:38
S18	20	Ring CRX	Descending	2006-033T09:30	Feb02	Thu	18	25	0	r = 66.721 Rs (4025274 km)
S18	21	**Apoapse**		2006-036T20:58	Feb05	Sun	21	21	4	Per = 39.2 d, inc = 0.4 deg, r = 68.151 Rs, phase = 107 deg
S18	21	**OTM-052**	21TI-2	2006-055T00:26	Feb24	Fri	40	3	22	D/L start, burn ~6h later. Backup D/L 055T16:05
S18	21	**SATURN**	**Segment**	2006-055T09:22	Feb24	Fri	40	3	0	Saturn_021 (2d); Saturn Rev 21 Periapse
S18	21	21HE (nt)	**HELENE**	2006-056T04:48	Feb25	Sat	41	2	1	Inbound 67546.3 km flyby, v = 7.1 km/s, phase = 92 deg
S18	21	Ring CRX	Ascending	2006-056T10:21	Feb25	Sat	41	2	1	r = 5.587 Rs (337070 km)
S18	21	Periapse		2006-056T10:56	Feb25	Sat	41	2	1	R = 5.577 Rs, lat = 0 deg, phase = 72 deg
S18	21	**TOST**	**Segment**	2006-057T09:06	Feb26	Sun	42	1	2	TOST_021TI_T11 (2d); TOST Segment 021TI (T11)
S18	21	**21TI (t) [T11]**	**TITAN**	2006-058T08:25	Feb27	Mon	43	20	3	Outbound 1812.0 km flyby, v = 5.9 km/s, phase = 92 deg
S18	21	**XD**	**Segment**	2006-059T08:51	Feb28	Tue	1	19	4	XD_021_022 (18d); Cross Discipline Segment Outbound 21 - Inbound 22
S18	21	**OTM-053**	21TI+3	2006-060T23:51	Mar01	Wed	3	17	6	D/L start, burn ~6h later. Backup D/L 061T16:21
S18	21	**OTM-054**	~APO	2006-064T23:36	Mar05	Sun	7	13	4	D/L start, burn ~6h later. Backup D/L 065T15:51
S18	22	**Apoapse**		2006-068T03:35	Mar09	Thu	10	10	3	Per = 23.3 d, inc = 0.4 deg, r = 48.363 Rs, phase = 122 deg
S19	22	**S19 Begins**		2006-070T00:35	Mar11	Sat	12	8	5	S19 Sequence. Duration = 42 d
S19	22	Ring CRX	Descending	2006-071T15:21	Mar12	Sun	13	6	7	r = 45.450 Rs (2741987 km)
S19	22	**OTM-055**	22TI-3	2006-074T22:50	Mar15	Wed	17	3	10	D/L start, burn ~6h later. Backup D/L 075T22:50
S19	22	**TOST**	**Segment**	2006-077T07:49	Mar18	Sat	19	1	2	TOST_022TI_T12 (2d); TOST Segment 022TI (T12)
S19	22	**Sun OCC**	TITAN	2006-078T00:04	Mar19	Sun	20	0	3	Egress = 2006-078T00:19 (15 min)
S19	22	**Earth OCC**	TITAN	2006-078T00:06	Mar19	Sun	20	0	3	Egress = 2006-078T00:19 (14 min)
S19	22	**22TI (t) [T12]**	**TITAN**	2006-078T00:06	Mar19	Sun	20	43	3	Inbound 1949.4 km flyby, v = 5.8 km/s, phase = 148 deg
S19	22	**SATURN**	**Segment**	2006-079T00:04	Mar20	Mon	1	42	4	Saturn_022 (2d); Saturn Rev 22 Periapse (and SOST)
S19	22	Periapse		2006-079T20:05	Mar20	Mon	2	41	5	R = 5.477 Rs, lat = -0 deg, phase = 46 deg
S19	22	Ring CRX	Ascending	2006-079T22:43	Mar20	Mon	2	41	5	r = 5.691 Rs (343363 km)
S19	22	22RH (nt)	**RHEA**	2006-080T07:16	Mar21	Tue	2	41	5	Outbound 82000.5 km flyby, v = 5.3 km/s, phase = 137 deg
S19	22	**OTM-056**	22TI+3	2006-080T22:19	Mar21	Tue	3	40	6	D/L start, burn ~6h later. Backup D/L 081T22:19
S19	22	**XD**	**Segment**	2006-081T07:19	Mar22	Wed	3	40	0	XD_022_023 (36d); Cross Discipline Segment Outbound 22 - Inbound 23
S19	22	**OTM-057**	~APO	2006-095T21:32	Apr05	Wed	18	25	15	D/L start, burn ~6h later. Backup D/L 096T21:17
S19	23	**Apoapse**		2006-099T10:05	Apr09	Sun	21	21	4	Per = 39.2 d, inc = 0.4 deg, r = 68.250 Rs, phase = 135 deg
S20	23	**S20 Begins**		2006-112T05:15	Apr22	Sat	34	9	16	S20 Sequence. Duration = 42 d
S20	23	Ring CRX	Descending	2006-112T15:50	Apr22	Sat	35	8	17	r = 45.166 Rs (2724852 km)
S20	23	**OTM-058**	23TI-3	2006-116T19:59	Apr26	Wed	39	4	21	D/L start, burn ~6h later. Backup D/L 117T19:59
S20	23	**SATURN**	**Segment**	2006-117T04:59	Apr27	Thu	39	4	0	Saturn_023 (3d); Saturn Rev 23 Periapse
S20	23	Periapse		2006-118T23:59	Apr28	Fri	41	2	2	R = 5.470 Rs, lat = -0 deg, phase = 44 deg
S20	23	Ring CRX	Ascending	2006-119T02:43	Apr29	Sat	41	2	2	r = 5.702 Rs (343979 km)
S20	23	**TOST**	**Segment**	2006-120T04:44	Apr30	Sun	42	1	3	TOST_023TI_T13 (2d); TOST Segment 023TI (T13)
S20	23	**23TI (t) [T13]**	**TITAN**	2006-120T20:58	Apr30	Sun	43	20	4	Outbound 1855.7 km flyby, v = 5.8 km/s, phase = 121 deg
S20	23	**Sun OCC**	TITAN	2006-120T21:05	Apr30	Sun	0	20	4	Egress = 2006-120T21:21 (16 min)
S20	23	**Earth OCC**	TITAN	2006-120T21:11	Apr30	Sun	0	20	4	Egress = 2006-120T21:17 (7 min)
S20	23	**XD**	**Segment**	2006-121T21:14	May01	Mon	1	19	5	XD_023_024 (18d); Cross Discipline Segment Outbound 23 - Inbound 24
S20	23	**OTM-059**	23TI+3	2006-123T19:28	May03	Wed	3	17	7	D/L start, burn ~6h later. Backup D/L 124T20:58
S20	23	**OTM-060**	~APO	2006-127T19:13	May07	Sun	7	13	4	D/L start, burn ~6h later. Backup D/L 128T11:43
S20	24	**Apoapse**		2006-130T16:31	May10	Wed	10	10	3	Per = 23.3 d, inc = 0.4 deg, r = 48.416 Rs, phase = 148 deg
S20	24	**OTM-061**	24TI-3	2006-137T18:41	May17	Wed	17	3	10	D/L start, burn ~6h later. Backup D/L 138T18:41
S20	24	**TOST**	**Segment**	2006-139T19:56	May19	Fri	19	1	2	TOST_024TI_T14 (2d); TOST Segment 024TI (T14)
S20	24	Ring CRX	Descending	2006-140T05:31	May20	Sat	19	0	2	r = 22.703 Rs (1369692 km)
S20	24	**Sun OCC**	TITAN	2006-140T12:11	May20	Sat	20	0	3	Egress = 2006-140T12:24 (14 min)
S20	24	**Earth OCC**	TITAN	2006-140T12:12	May20	Sat	20	0	3	Egress = 2006-140T12:25 (14 min)
S20	24	**24TI (t) [T14]**	**TITAN**	2006-140T12:18	May20	Sat	20	43	3	Inbound 1879.3 km flyby, v = 5.8 km/s, phase = 163 deg
S20	24	**SATURN**	**Segment**	2006-141T19:56	May21	Sun	1	42	4	Saturn_024 (2d); Saturn Rev 24 Periapse

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S20	24	Periapse		2006-142T09:01	May22	Mon	2	41	5	R = 5.452 Rs, lat = -0 deg, phase = 21 deg
S20	24	24PO (nt)	POLYDEUCES	2006-142T14:53	May22	Mon	2	41	5	Outbound 64089.9 km flyby, v = 6.9 km/s, phase = 19 deg
S20	24	Ring CRX	Ascending	2006-142T15:14	May22	Mon	2	41	5	r = 6.538 Rs (394422 km)
S20	24	OTM-062	24TI+3	2006-143T10:41	May23	Tue	3	40	6	D/L start, burn ~6h later. Backup D/L 144T18:11
S20	24	XD	Segment	2006-143T19:41	May23	Tue	3	40	0	XD_024_025 (35d); Cross-Discipline Segment Outbound 24 - Inbound 25
S21	24	S21 Begins		2006-154T02:39	Jun03	Sat	14	29	11	S21 Sequence. Duration = 44 d
S21	24	OTM-063	~APO	2006-158T17:24	Jun07	Wed	18	25	15	D/L start, burn ~6h later. Backup D/L 159T17:09
S21	25	Apoapse		2006-161T23:07	Jun10	Sat	21	21	3	Per = 39.2 d, inc = 0.4 deg, r = 68.286 Rs, phase = 159 deg
S21	25	SATURN	Segment	2006-179T01:07	Jun28	Wed	39	4	20	Saturn_025 (4d); Saturn Rev 25 Periapse
S21	25	OTM-064	25TI-3	2006-179T16:07	Jun28	Wed	39	4	21	D/L start, burn ~6h later. Backup D/L 180T08:37
S21	25	Ring CRX	Descending	2006-180T01:39	Jun29	Thu	40	3	0	r = 17.569 Rs (1059963 km)
S21	25	Periapse		2006-181T13:05	Jun30	Fri	41	2	2	R = 5.437 Rs, lat = -0 deg, phase = 20 deg
S21	25	Ring CRX	Ascending	2006-181T20:54	Jun30	Fri	41	2	2	r = 7.051 Rs (425416 km)
S21	25	TOST	Segment	2006-182T17:22	Jul01	Sat	42	1	3	TOST_025TI_T15 (2d); TOST Segment 025TI (T15)
S21	25	Sun OCC	TITAN	2006-183T09:19	Jul02	Sun	43	0	4	Egress = 2006-183T09:34 (15 min)
S21	25	Earth OCC	TITAN	2006-183T09:20	Jul02	Sun	43	0	4	Egress = 2006-183T09:35 (15 min)
S21	25	25TI (t) [T15]	TITAN	2006-183T09:21	Jul02	Sun	43	20	4	Outbound 1905.9 km flyby, v = 5.8 km/s, phase = 148 deg
S21	25	XD	Segment	2006-184T17:22	Jul03	Mon	1	18	5	XD_025_026 (17d); Cross Discipline Segment Outbound 25 - Inbound 26
S21	25	OTM-065	25TI+3	2006-186T15:36	Jul05	Wed	3	16	7	D/L start, burn ~6h later. Backup D/L 187T15:36
S21	25	OTM-066	~APO	2006-191T15:21	Jul10	Mon	8	11	5	D/L start, burn ~6h later. Backup D/L 192T15:21
S21	26	Apoapse		2006-193T05:18	Jul12	Wed	10	10	2	Per = 23.3 d, inc = 0.4 deg, r = 48.376 Rs, phase = 164 deg
S22	26	S22 Begins		2006-198T00:06	Jul17	Mon	15	5	6	S22 Sequence. Duration = 34 d
S22	26	OTM-067	26TI-3	2006-199T14:51	Jul18	Tue	16	3	8	D/L start, burn ~6h later. Backup D/L 200T14:51
S22	26	TOST	Segment	2006-201T23:51	Jul20	Thu	19	1	2	TOST_025TI_T16 (2d); TOST Segment 026TI (T16)
S22	26	26TI (t) [T16]	TITAN	2006-203T00:25	Jul22	Sat	20	48	3	Inbound 949.9 km flyby, v = 6.0 km/s, phase = 105 deg
S22	26	Ring CRX	Descending	2006-203T02:36	Jul22	Sat	0	48	3	r = 20.126 Rs (1214207 km)
S22	26	RINGS	Segment	2006-203T23:36	Jul22	Sat	1	47	4	RINGS_Rev_26 (4d); Rings Rev 26
S22	26	26TE (nt)	TETHYS	2006-204T17:18	Jul23	Sun	2	46	5	Inbound 119806.3 km flyby, v = 9.9 km/s, phase = 29 deg
S22	26	Periapse		2006-204T21:48	Jul23	Sun	2	46	5	R = 4.192 Rs, lat = -10 deg, phase = 8 deg
S22	26	26AE (nt)	AEGAEON	2006-204T22:46	Jul23	Sun	2	46	5	Outbound 93379.5 km flyby, v = 3.8 km/s, phase = 16 deg
S22	26	Ring CRX	Ascending	2006-205T01:15	Jul24	Mon	2	46	5	r = 4.773 Rs (287949 km)
S22	26	26TL (nt)	TELESTO	2006-205T01:58	Jul24	Mon	2	46	5	Outbound 44778.6 km flyby, v = 8.6 km/s, phase = 39 deg
S22	26	OTM-068	26TI+3	2006-205T14:30	Jul24	Mon	3	45	6	D/L start, burn ~6h later. Backup D/L 206T14:30
S22	26	XD	Segment	2006-207T23:21	Jul26	Wed	5	43	2	XD_026_027 (18d); Cross Discipline Segment Outbound 26 - Inbound 27
S22	26	OTM-069	~APO	2006-213T14:05	Aug01	Tue	11	37	8	D/L start, burn ~6h later. Backup D/L 214T14:05
S22	26	SEP = 3.0 deg	Conjunction	2006-216T00:00	Aug04	Thu	13	35	2	Commanding must not be required below 3.0 deg
S22	27	Apoapse		2006-216T21:22	Aug04	Fri	14	34	3	Per = 24.0 d, inc = 14.9 deg, r = 48.960 Rs, phase = 172 deg
S22	27	SEP = 2.0 deg	Conjunction	2006-217T06:47	Aug05	Sat	14	34	4	Plan no SSR playback below 2.0 deg
S22	27	SEP = 1.0 deg	Conjunction	2006-218T18:34	Aug06	Sun	16	32	5	Require no DSN passes below 1.0 deg
S22	27	SEP = 1.0 deg	Conjunction	2006-220T05:42	Aug08	Tue	17	31	7	Require no DSN passes below 1.0 deg
S22	27	SEP = 2.0 deg	Conjunction	2006-221T17:27	Aug09	Wed	19	29	8	Plan no SSR playback below 2.0 deg
S22	27	SEP = 3.0 deg	Conjunction	2006-223T00:09	Aug11	Fri	20	28	9	Commanding must not be required below 3.0 deg
S22	27	SATURN	Segment	2006-225T22:12	Aug13	Sun	23	25	12	Saturn_027 (6d); Saturn_027 Inclined
S22	27	Ring CRX	Descending	2006-227T02:13	Aug15	Tue	24	24	14	r = 19.960 Rs (1204166 km)
S22	27	Periapse		2006-228T20:53	Aug16	Wed	26	22	15	R = 4.180 Rs, lat = -10 deg, phase = 8 deg
S22	27	Ring CRX	Ascending	2006-229T00:21	Aug17	Thu	26	22	15	r = 4.766 Rs (287519 km)
S22	27	27HE (nt)	HELENE	2006-229T03:29	Aug17	Thu	26	22	16	Outbound 48728.2 km flyby, v = 7.7 km/s, phase = 122 deg
S22	27	27TI (nt)	TITAN	2006-230T17:50	Aug18	Fri	28	20	17	Outbound 339768.2 km flyby, v = 4.8 km/s, phase = 121 deg
S23	27	XD	Segment	2006-231T22:06	Aug19	Sat	29	19	18	XD_027_028 (18d); Cross Discipline Segement Outbound 27 - Inbound 28
S23	27	S23 Begins		2006-231T22:06	Aug19	Sat	29	19	18	S23 Sequence. Duration = 32 d
S23	28	Apoapse		2006-240T19:16	Aug28	Mon	38	10	27	Per = 23.9 d, inc = 14.9 deg, r = 48.821 Rs, phase = 172 deg
S23	28	OTM-070	28TI-3	2006-247T12:21	Sep04	Mon	44	3	34	D/L start, burn ~6h later. Backup D/L 248T12:06
S23	28	TOST	Segment	2006-249T21:06	Sep06	Wed	47	1	2	TOST_028TI_T17 (2d); TOST 028TI (T17) Flyby
S23	28	28TI (t) [T17]	TITAN	2006-250T20:17	Sep07	Thu	48	16	3	Inbound 999.5 km flyby, v = 6.0 km/s, phase = 45 deg
S23	28	Ring CRX	Descending	2006-250T21:22	Sep07	Thu	0	16	3	r = 20.463 Rs (1234514 km)
S23	28	RINGS	Segment	2006-251T21:06	Sep08	Fri	1	15	4	RINGS_Rev_28A (3d); Rings Rev 28 Peri
S23	28	28AT (nt)	ATLAS	2006-252T17:33	Sep09	Sat	2	14	5	Inbound 119822.6 km flyby, v = 15.0 km/s, phase = 47 deg
S23	28	Periapse		2006-252T17:42	Sep09	Sat	2	14	5	R = 2.957 Rs, lat = -12 deg, phase = 7 deg
S23	28	Methone	Dust Crossing	2006-252T19:06	Sep09	Sat	2	14	5	Egress at 2006-252T19:06 (0.0 min); protection = none.
S23	28	Ring CRX	Ascending	2006-252T19:06	Sep09	Sat	2	14	5	r = 3.162 Rs (190756 km)
S23	28	28ME (nt)	METHONE	2006-252T19:17	Sep09	Sat	2	14	5	Outbound 15113.6 km flyby, v = 10.2 km/s, phase = 62 deg
S23	28	28CA (nt)	CALYPSO	2006-252T19:24	Sep09	Sat	2	14	5	Outbound 107224.5 km flyby, v = 9.7 km/s, phase = 162 deg
S23	28	28EN (nt)	ENCELADUS	2006-252T20:06	Sep09	Sat	2	14	5	Outbound 39935.4 km flyby, v = 10.4 km/s, phase = 111 deg
S23	28	OTM-071	28TI+3	2006-253T12:00	Sep10	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 254T12:00
S23	28	SATURN	Segment	2006-254T21:00	Sep11	Mon	4	12	1	Saturn_028 (1d); Saturn 028 Inclined
S23	28	MAG	Segment	2006-255T21:00	Sep12	Tue	5	11	2	MAG_028 (2d); MAGTWT Rev 28
S23	28	OTM-072	~APO	2006-257T04:07	Sep14	Thu	6	10	4	D/L start, burn ~6h later. Backup D/L 257T10:22
S23	28	Sun OCC	RING	2006-257T17:54	Sep14	Thu	7	9	1	Egress = 2006-258T17:42 (1429 min)
S23	28	Sun OCC	RING	2006-257T17:54	Sep14	Thu	7	9	1	Egress = 2006-258T17:42 (1429 min)
S23	28	RINGS	Segment	2006-257T19:22	Sep14	Thu	7	9	1	RINGS_Rev_28B (3d); RINGS Rev 28 APO
S23	28	Earth OCC	ENCELADUS	2006-258T07:20	Sep15	Fri	7	8	1	Egress = 2006-258T07:21 (1 min)
S23	28	Sun OCC	SATURN	2006-258T08:29	Sep15	Fri	8	8	1	Egress = 2006-258T23:28 (900 min)
S23	28	Earth OCC	RING	2006-258T18:26	Sep15	Fri	8	8	2	Egress = 2006-259T21:19 (1613 min)
S23	28	Earth OCC	RING	2006-258T18:26	Sep15	Fri	8	8	2	Egress = 2006-259T21:19 (1613 min)
S23	28	Earth OCC	SATURN	2006-259T06:13	Sep16	Sat	8	8	2	Egress = 2006-260T01:25 (1153 min)
S23	28	XD	Segment	2006-260T14:07	Sep17	Sun	10	6	3	XD_028_029 (5d); Cross Discipline Segment Outbound 28 - Inbound 29
S23	29	Apoapse		2006-260T17:10	Sep17	Sun	10	6	4	Per = 16.0 d, inc = 24.5 deg, r = 37.571 Rs, phase = 173 deg
S23	29	OTM-073	29TI-3	2006-263T10:32	Sep20	Wed	13	3	6	D/L start, burn ~6h later. Backup D/L 264T11:22
S24	29	S24 Begins		2006-263T20:22	Sep20	Wed	13	3	0	S24 Sequence. Duration = 32 d

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S24	29	**TOST**	**Segment**	2006-265T20:07	Sep22	Fri	15	1	2	TOST_029TI_T18 (2d); TOST Segment 029TI (T18)
S24	29	**29TI (t) [T18]**	**TITAN**	2006-266T18:59	Sep23	Sat	16	16	3	Inbound 959.8 km flyby, v = 6.0 km/s, phase = 90 deg
S24	29	Ring CRX	Descending	2006-266T19:55	Sep23	Sat	0	16	3	r = 20.534 Rs (1238795 km)
S24	29	**RINGS**	**Segment**	2006-267T20:00	Sep24	Sun	1	15	4	RINGS_Rev_29 (4d); RINGS Rev 29
S24	29	Periapse		2006-268T19:42	Sep25	Mon	2	14	5	R = 3.983 Rs, lat = -21 deg, phase = 7 deg
S24	29	29AE (nt)	**AEGAEON**	2006-268T20:19	Sep25	Mon	2	14	5	Outbound 107428.5 km flyby, v = 11.8 km/s, phase = 29 deg
S24	29	29ME (nt)	**METHONE**	2006-268T21:45	Sep25	Mon	2	14	5	Outbound 63839.0 km flyby, v = 10.1 km/s, phase = 39 deg
S24	29	E_ring_lg	Dust Crossing	2006-268T22:18	Sep25	Mon	2	14	5	Egress at 2006-268T22:17 (0.0 min); protection = none.
S24	29	Ring CRX	Ascending	2006-268T22:19	Sep25	Mon	2	14	5	r = 4.350 Rs (262433 km)
S24	29	**OTM-074**	29TI+3	2006-269T11:30	Sep26	Tue	3	13	6	D/L start, burn ~6h later. Backup D/L 270T11:30
S24	29	**XD**	**Segment**	2006-271T19:53	Sep28	Thu	5	11	2	XD_029_030 (10d); Cross Discipline Segment Outbound 29 - Inbound 30
S24	29	**OTM-075**	~APO	2006-274T03:08	Oct01	Sun	7	9	5	D/L start, burn ~6h later. Backup D/L 275T10:38
S24	30	**Apoapse**		2006-276T19:04	Oct03	Tue	10	6	3	Per = 15.9 d, inc = 37.6 deg, r = 36.521 Rs, phase = 173 deg
S24	30	**OTM-076**	30TI-3	2006-279T10:24	Oct06	Fri	13	3	5	D/L start, burn ~6h later. Backup D/L 280T10:24
S24	30	**TOST**	**Segment**	2006-281T19:24	Oct08	Sun	15	1	2	TOST_030TI_T19 (2d); TOST 030TI (T19) Flyby
S24	30	**30TI (t) [T19]**	**TITAN**	2006-282T17:30	Oct09	Mon	16	16	3	Inbound 979.7 km flyby, v = 6.0 km/s, phase = 81 deg
S24	30	Ring CRX	Descending	2006-282T18:09	Oct09	Mon	0	16	3	r = 20.615 Rs (1243731 km)
S24	30	**SATURN**	**Segment**	2006-283T19:00	Oct10	Tue	1	15	4	Saturn_030 (1d); Saturn 030 Inbound
S24	30	**RINGS**	**Segment**	2006-284T19:09	Oct11	Wed	2	14	5	RINGS_Rev_30 (4d); RINGS Rev 30
S24	30	Periapse		2006-284T22:59	Oct11	Wed	2	14	6	R = 5.507 Rs, lat = -29 deg, phase = 14 deg
S24	30	Ring CRX	Ascending	2006-285T04:14	Oct12	Thu	2	13	6	r = 6.184 Rs (373081 km)
S24	30	**OTM-077**	30TI+3	2006-285T10:10	Oct12	Thu	3	13	6	D/L start, burn ~6h later. Backup D/L 286T09:55
S24	30	**XD**	**Segment**	2006-288T18:55	Oct15	Sun	6	10	3	XD_030_031 (8d); Cross Discipline Segment Outbound 30 - Inbound 31
S24	30	**OTM-078**	~APO	2006-290T09:40	Oct17	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 291T02:10
S24	31	**Apoapse**		2006-292T22:17	Oct19	Thu	10	6	3	Per = 15.9 d, inc = 46.6 deg, r = 34.980 Rs, phase = 166 deg
S24	31	**OTM-079**	31TI-3	2006-295T09:26	Oct22	Sun	13	3	5	D/L start, burn ~6h later. Backup D/L 296T01:56
S25	31	**S25 Begins**		2006-295T18:26	Oct22	Sun	13	3	0	S25 Sequence. Duration = 33 d
S25	31	**TOST**	**Segment**	2006-296T10:56	Oct23	Mon	14	2	1	TOST_031TI_T20 (3d); TOST_031TI (T20) Flyby
S25	31	**31TI (t) [T20]**	**TITAN**	2006-298T15:58	Oct25	Wed	16	48	3	Inbound 1029.5 km flyby, v = 6.0 km/s, phase = 25 deg
S25	31	Ring CRX	Descending	2006-298T16:20	Oct25	Wed	0	48	3	r = 20.688 Rs (1248123 km)
S25	31	**RINGS**	**Segment**	2006-299T14:56	Oct26	Thu	1	47	4	RINGS_Rev_31 (3d); RINGS Rev 31
S25	31	Periapse		2006-301T00:10	Oct28	Sat	2	45	6	R = 4.656 Rs, lat = -26 deg, phase = 16 deg
S25	31	Ring CRX	Ascending	2006-301T03:09	Oct28	Sat	2	45	6	r = 4.972 Rs (299940 km)
S25	31	**SATURN**	**Segment**	2006-302T10:42	Oct29	Sun	4	44	7	Saturn_031 (2d); Saturn 031 Outbound
S25	31	**XD**	**Segment**	2006-304T17:57	Oct31	Tue	6	42	9	XD_031_032 (5d); Cross Discipline Segment Outbound 31 - Inbound 32
S25	32	**Apoapse**		2006-307T00:05	Nov03	Fri	8	39	12	Per = 12.0 d, inc = 55.0 deg, r = 28.836 Rs, phase = 164 deg
S25	32	**RINGS**	**Segment**	2006-309T17:42	Nov05	Sun	11	37	14	RINGS_Rev_32 (5d); RINGS Rev 32
S25	32	Ring CRX	Descending	2006-310T15:56	Nov06	Mon	12	36	15	r = 20.721 Rs (1250077 km)
S25	32	Periapse		2006-313T00:00	Nov09	Thu	14	33	18	R = 4.662 Rs, lat = -26 deg, phase = 16 deg
S25	32	32EN (nt)	**ENCELADUS**	2006-313T01:48	Nov09	Thu	14	33	18	Outbound 90219.9 km flyby, v = 14.0 km/s, phase = 26 deg
S25	32	Ring CRX	Ascending	2006-313T02:59	Nov09	Thu	14	33	18	r = 4.977 Rs (300289 km)
S25	32	32CA (nt)	**CALYPSO**	2006-313T03:45	Nov09	Thu	14	33	18	Outbound 116095.7 km flyby, v = 11.7 km/s, phase = 111 deg
S25	32	**OTM-080**	~PERI	2006-313T08:28	Nov09	Thu	15	33	18	D/L start, burn ~6h later. Backup D/L 314T00:58
S25	32	**SATURN**	**Segment**	2006-314T09:58	Nov10	Fri	16	32	1	Saturn_032 (2d); Saturn 032 Outbound
S25	32	**XD**	**Segment**	2006-316T17:13	Nov12	Sun	18	30	3	XD_032_033 (5d); Cross Discipline Segment Outbound 32 - Inbound 33
S25	33	**Apoapse**		2006-318T23:31	Nov14	Tue	20	28	6	Per = 12.0 d, inc = 55.0 deg, r = 28.779 Rs, phase = 164 deg
S25	33	**RINGS**	**Segment**	2006-321T16:59	Nov17	Fri	23	25	8	RINGS_Rev_33 (7d); RINGS Rev 33
S25	33	Ring CRX	Descending	2006-322T15:10	Nov18	Sat	24	24	9	r = 20.669 Rs (1246974 km)
S25	33	Periapse		2006-324T23:02	Nov20	Mon	26	22	12	R = 4.654 Rs, lat = -26 deg, phase = 16 deg
S25	33	Ring CRX	Ascending	2006-325T02:01	Nov21	Tue	26	21	12	r = 4.969 Rs (299769 km)
S25	33	33DI (nt)	**DIONE**	2006-325T02:28	Nov21	Tue	26	21	12	Outbound 74995.5 km flyby, v = 12.2 km/s, phase = 147 deg
S25	33	33TI (nt)	**TITAN**	2006-325T13:36	Nov21	Tue	27	21	12	Outbound 973181.6 km flyby, v = 8.0 km/s, phase = 78 deg
S26	33	**XD**	**Segment**	2006-328T16:30	Nov24	Fri	30	18	15	XD_033_034 (5d); Cross Discipline Segment Outbound 33 - Inbound 34
S26	33	**S26 Begins**		2006-328T16:30	Nov24	Fri	30	18	15	S26 Sequence. Duration = 42 d
S26	33	33TI (nt)	**TITAN**	2006-329T13:29	Nov25	Sat	31	17	16	Outbound 925037.1 km flyby, v = 4.5 km/s, phase = 115 deg
S26	33	**OTM-081**	~APO	2006-330T07:15	Nov26	Sun	32	16	17	D/L start, burn ~6h later. Backup D/L 331T07:15
S26	34	**Apoapse**		2006-330T22:20	Nov26	Sun	32	16	1	Per = 11.9 d, inc = 55.0 deg, r = 28.754 Rs, phase = 164 deg
S26	34	**RINGS**	**Segment**	2006-333T16:16	Nov29	Wed	35	13	3	RINGS_Rev_34in (2d); RINGS Rev 34-inbound
S26	34	Ring CRX	Descending	2006-334T13:45	Nov30	Thu	36	12	4	r = 20.673 Rs (1247198 km)
S26	34	**SATURN**	**Segment**	2006-335T08:31	Dec01	Fri	37	11	5	Saturn_034 (1d); Saturn 034 Inbound
S26	34	**RINGS**	**Segment**	2006-336T08:31	Dec02	Sat	38	10	6	RINGS_Rev_34out (4d); RINGS Rev 34-periapsis and outbound
S26	34	Periapse		2006-336T21:39	Dec02	Sat	38	10	7	R = 4.655 Rs, lat = -26 deg, phase = 15 deg
S26	34	34TI (nt)	**TITAN**	2006-336T23:17	Dec02	Sat	38	10	7	Outbound 945244.5 km flyby, v = 12.2 km/s, phase = 159 deg
S26	34	Ring CRX	Ascending	2006-337T00:37	Dec03	Sun	38	9	7	r = 4.970 Rs (299813 km)
S26	34	34TL (nt)	**TELESTO**	2006-337T01:20	Dec03	Sun	38	9	7	Outbound 102514.2 km flyby, v = 12.1 km/s, phase = 111 deg
S26	34	**XD**	**Segment**	2006-340T15:47	Dec06	Wed	42	6	10	XD_034_035 (5d); Cross Discipline Segment Outbound 34 - Inbound 35
S26	35	**Apoapse**		2006-342T21:01	Dec08	Fri	44	4	13	Per = 12.0 d, inc = 55.0 deg, r = 28.757 Rs, phase = 165 deg
S26	35	**OTM-082**	35TI-3	2006-343T06:32	Dec09	Sat	45	3	13	D/L start, burn ~6h later. Backup D/L 344T06:32
S26	35	**TOST**	**Segment**	2006-345T15:17	Dec11	Mon	47	1	2	TOST_035TI_T21 (2d); TOST Segment 035TI (T21)
S26	35	**35TI (t) [T21]**	**TITAN**	2006-346T11:42	Dec12	Tue	48	16	3	Inbound 1000.0 km flyby, v = 5.9 km/s, phase = 125 deg
S26	35	Ring CRX	Descending	2006-346T12:10	Dec12	Tue	0	16	3	r = 20.713 Rs (1249588 km)
S26	35	**SATURN**	**Segment**	2006-347T12:18	Dec13	Wed	1	15	4	Saturn_035 (1d); Saturn 035 Inbound
S26	35	**RINGS**	**Segment**	2006-348T06:28	Dec14	Thu	2	14	5	RINGS_Rev_35 (4d); RINGS Rev 35
S26	35	Periapse		2006-349T00:01	Dec15	Fri	3	13	6	R = 7.606 Rs, lat = -37 deg, phase = 23 deg
S26	35	**OTM-083**	35TI+3	2006-349T06:03	Dec15	Fri	3	13	6	D/L start, burn ~6h later. Backup D/L 350T05:45
S26	35	Ring CRX	Ascending	2006-349T10:35	Dec15	Fri	3	13	0	r = 8.802 Rs (531024 km)
S26	35	**XD**	**Segment**	2006-352T15:03	Dec18	Mon	6	10	3	XD_035_036 (9d); Cross Discipline Segment Outbound 35 - Inbound 36
S26	35	**OTM-084**	~APO	2006-354T05:48	Dec20	Wed	8	8	5	D/L start, burn ~6h later. Backup D/L 355T05:49

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S26	36	**Apoapse**		2006-356T23:09	Dec22	Fri	10	5	3	Per = 15.9 d, inc = 52.8 deg, r = 32.888 Rs, phase = 157 deg
S26	36	**OTM-085**	36TI-3	2006-359T05:34	Dec25	Mon	13	3	5	D/L start, burn ~6h later. Backup D/L 360T05:19
S26	36	**TOST**	**Segment**	2006-361T14:19	Dec27	Wed	15	1	2	TOST_036TI_T22 (2d); TOST Segment 036TI (T22)
S26	36	**36TI (t) [T22]**	**TITAN**	2006-362T10:05	Dec28	Thu	16	16	3	Inbound 1296.8 km flyby, v = 5.9 km/s, phase = 62 deg
S26	36	Ring CRX	Descending	2006-362T10:30	Dec28	Thu	0	16	3	r = 20.686 Rs (1247984 km)
S26	36	**RINGS**	**Segment**	2006-363T06:49	Dec29	Fri	1	15	4	RINGS_Rev_36 (6d); RINGS Rev 36
S26	36	**OTM-086**	36TI+3	2006-365T05:05	Dec31	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 001T05:05
S26	36	Periapse		2006-365T05:18	Dec31	Sun	3	13	0	R = 9.811 Rs, lat = -44 deg, phase = 31 deg
S26	36	Ring CRX	Ascending	2006-365T23:56	Dec31	Sun	4	12	1	r = 11.605 Rs (700137 km)
S26	36	**XD**	**Segment**	2007-004T13:50	Jan04	Thu	7	9	4	XD_036_037 (8d); Cross-Discipline Outbound 36 - Inbound 37
S26	36	**OTM-087**	~APO	2007-005T04:50	Jan05	Fri	8	8	5	D/L start, burn ~6h later. Backup D/L 006T04:35
S27	36	**S27 Begins**		2007-005T13:50	Jan05	Fri	8	8	0	S27 Sequence. Duration = 43 d
S27	37	**Apoapse**		2007-008T04:38	Jan08	Mon	11	5	3	Per = 15.9 d, inc = 56.6 deg, r = 30.692 Rs, phase = 149 deg
S27	37	**OTM-088**	37TI-3	2007-010T04:20	Jan10	Wed	13	3	5	D/L start, burn ~6h later. Backup D/L 011T04:21
S27	37	**TOST**	**Segment**	2007-012T13:34	Jan12	Fri	15	1	2	TOST_037TI_T23 (2d); TOST Segment 037TI (T23)
S27	37	**37TI (t) [T23]**	**TITAN**	2007-013T08:39	Jan13	Sat	16	16	3	Inbound 1000.3 km flyby, v = 6.0 km/s, phase = 53 deg
S27	37	Ring CRX	Descending	2007-013T08:58	Jan13	Sat	0	16	3	r = 20.714 Rs (1249678 km)
S27	37	**SATURN**	**Segment**	2007-014T13:34	Jan14	Sun	1	15	4	Saturn_037 (1d); Saturn 037 Inbound
S27	37	**RINGS**	**Segment**	2007-015T05:36	Jan15	Mon	2	14	5	RINGS_Rev_37 (6d); RINGS Rev 37
S27	37	**OTM-089**	37TI+3	2007-015T20:36	Jan15	Mon	2	13	6	D/L start, burn ~6h later. Backup D/L 017T03:51
S27	37	Periapse		2007-016T12:59	Jan16	Tue	3	13	1	R = 12.527 Rs, lat = -51 deg, phase = 39 deg
S27	37	Ring CRX	Ascending	2007-017T21:01	Jan17	Wed	5	11	2	r = 14.867 Rs (896900 km)
S27	37	**OTM-090**	~APO	2007-021T03:36	Jan21	Sun	8	8	5	D/L start, burn ~6h later. Backup D/L 022T03:36
S27	37	**XD**	**Segment**	2007-021T12:36	Jan21	Sun	8	8	0	XD_037_038 (7d); Cross Discipline Segment Outbound 37 - Inbound 38
S27	38	**Apoapse**		2007-024T12:19	Jan24	Wed	11	5	3	Per = 16.0 d, inc = 59.2 deg, r = 27.987 Rs, phase = 140 deg
S27	38	**OTM-091**	38TI-3	2007-026T03:21	Jan26	Fri	13	3	5	D/L start, burn ~6h later. Backup D/L 027T03:21
S27	38	**TOST**	**Segment**	2007-028T12:12	Jan28	Sun	15	1	2	TOST_038TI_T24 (2d); TOST Segment 038TI (T24)
S27	38	**38TI (t) [T24]**	**TITAN**	2007-029T07:16	Jan29	Mon	16	24	3	Inbound 2631.2 km flyby, v = 5.8 km/s, phase = 72 deg
S27	38	Ring CRX	Descending	2007-029T07:38	Jan29	Mon	0	24	3	r = 20.708 Rs (1249322 km)
S27	38	**SATURN**	**Segment**	2007-030T12:12	Jan30	Tue	1	23	4	Saturn_038 (7d); Saturn_038
S27	38	**OTM-092**	38TI+3	2007-032T02:52	Feb01	Thu	3	21	6	D/L start, burn ~6h later. Backup D/L 033T19:22
S27	38	Periapse		2007-032T09:57	Feb01	Thu	3	21	0	R = 15.572 Rs, lat = -58 deg, phase = 58 deg
S27	38	Ring CRX	Ascending	2007-035T01:08	Feb04	Sun	6	18	3	r = 19.612 Rs (1183214 km)
S27	38	38TI (nt)	**TITAN**	2007-036T07:42	Feb05	Mon	7	17	4	Outbound 910072.3 km flyby, v = 5.9 km/s, phase = 114 deg
S27	38	**XD**	**Segment**	2007-037T11:37	Feb06	Tue	8	16	5	XD_038_039 (8d); Cross Discipline Outbound 38 - Inbound 39
S27	38	**OTM-093**	~APO	2007-038T02:37	Feb07	Wed	9	15	6	D/L start, burn ~6h later. Backup D/L 039T02:22
S27	39	**Apoapse**		2007-041T10:53	Feb10	Sat	12	12	3	Per = 18.1 d, inc = 58.8 deg, r = 28.492 Rs, phase = 121 deg
S27	39	**RINGS**	**Segment**	2007-045T09:45	Feb14	Wed	16	8	7	RINGS_Rev_39 (7d); RINGS Rev 39
S27	39	39TI (nt)	**TITAN**	2007-046T03:10	Feb15	Thu	17	7	8	Inbound 867510.3 km flyby, v = 5.5 km/s, phase = 128 deg
S27	39	Ring CRX	Descending	2007-047T09:46	Feb16	Fri	18	6	9	r = 20.677 Rs (1247446 km)
S28	39	**S28 Begins**		2007-048T10:52	Feb17	Sat	19	5	10	S28 Sequence. Duration = 39 d
S28	39	**OTM-094**	39TI-3	2007-050T01:37	Feb19	Mon	21	3	12	D/L start, burn ~6h later. Backup D/L 051T01:37
S28	39	Periapse		2007-050T11:50	Feb19	Mon	21	3	0	R = 15.577 Rs, lat = -58 deg, phase = 59 deg
S28	39	**TOST**	**Segment**	2007-052T10:37	Feb21	Wed	23	1	2	TOST_039TI_T25 (2d); TOST Segment 039TI (T25)
S28	39	**Earth OCC**	TITAN	2007-053T03:01	Feb22	Thu	24	0	3	Egress = 2007-053T03:17 (16 min)
S28	39	**Sun OCC**	TITAN	2007-053T03:01	Feb22	Thu	24	0	3	Egress = 2007-053T03:17 (16 min)
S28	39	**39TI (t) [T25]**	**TITAN**	2007-053T03:12	Feb22	Thu	24	16	3	Outbound 1000.4 km flyby, v = 6.2 km/s, phase = 161 deg
S28	39	Ring CRX	Ascending	2007-053T03:18	Feb22	Thu	0	16	3	r = 19.654 Rs (1185738 km)
S28	39	**XD**	**Segment**	2007-054T10:11	Feb23	Fri	1	15	4	XD_039_040 (7d); Cross-Discipline Outbound 39 - Inbound 40
S28	39	**OTM-095**	39TI+3	2007-056T01:22	Feb25	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 057T01:06
S28	40	**Apoapse**		2007-058T07:46	Feb27	Tue	5	11	2	Per = 16.0 d, inc = 58.6 deg, r = 28.363 Rs, phase = 97 deg
S28	40	**OTM-096**	~APO	2007-061T00:51	Mar02	Fri	8	8	5	D/L start, burn ~6h later. Backup D/L 062T00:51
S28	40	**SATURN**	**Segment**	2007-061T09:51	Mar02	Fri	8	8	0	Saturn_040 (7d); Saturn_040 Early and Mid Inclined
S28	40	Ring CRX	Descending	2007-064T21:02	Mar05	Mon	12	4	4	r = 14.993 Rs (904502 km)
S28	40	**OTM-097**	40TI-3	2007-065T17:06	Mar06	Tue	13	3	5	D/L start, burn ~6h later. Backup D/L 067T00:21
S28	40	Periapse		2007-066T07:09	Mar07	Wed	13	3	1	R = 12.145 Rs, lat = -54 deg, phase = 83 deg
S28	40	**TOST**	**Segment**	2007-068T09:21	Mar09	Fri	15	1	3	TOST_040TI_T26 (2d); TOST Segment 040TI (T26)
S28	40	**Earth OCC**	TITAN	2007-069T01:37	Mar10	Sat	16	0	3	Egress = 2007-069T01:52 (15 min)
S28	40	**Sun OCC**	TITAN	2007-069T01:37	Mar10	Sat	16	0	3	Egress = 2007-069T01:52 (16 min)
S28	40	**40TI (t) [T26]**	**TITAN**	2007-069T01:49	Mar10	Sat	16	16	3	Outbound 980.6 km flyby, v = 6.2 km/s, phase = 150 deg
S28	40	Ring CRX	Ascending	2007-069T01:55	Mar10	Sat	0	16	3	r = 19.657 Rs (1185892 km)
S28	40	**XD**	**Segment**	2007-070T02:47	Mar11	Sun	1	15	4	XD_040_041 (8d); Cross Discipline Segment Outbound 40 - Inbound 41
S28	40	**OTM-098**	40TI+3	2007-072T00:06	Mar13	Tue	3	13	6	D/L start, burn ~6h later. Backup D/L 073T00:06
S28	41	**Apoapse**		2007-074T12:37	Mar15	Thu	5	10	3	Per = 15.9 d, inc = 56.0 deg, r = 30.793 Rs, phase = 88 deg
S28	41	**OTM-099**	~APO	2007-076T23:50	Mar17	Sat	8	8	5	D/L start, burn ~6h later. Backup D/L 077T23:35
S28	41	**RINGS**	**Segment**	2007-078T08:35	Mar19	Mon	9	7	1	RINGS_Rev_41 (6d); RINGS Rev 41
S28	41	**OTM-100**	41TI-3	2007-081T14:30	Mar22	Thu	13	3	5	D/L start, burn ~6h later. Backup D/L 082T23:20
S28	41	Ring CRX	Descending	2007-081T15:45	Mar22	Thu	13	3	0	r = 11.823 Rs (713289 km)
S28	41	Periapse		2007-082T11:57	Mar23	Fri	13	3	1	R = 9.710 Rs, lat = -47 deg, phase = 92 deg
S28	41	**TOST**	**Segment**	2007-084T08:20	Mar25	Sun	15	1	3	TOST_041TI_T27 (2d); TOST Segment 041TI_T27
S28	41	**Sun OCC**	TITAN	2007-085T00:08	Mar26	Mon	16	0	3	Egress = 2007-085T00:25 (17 min)
S28	41	**Earth OCC**	TITAN	2007-085T00:09	Mar26	Mon	16	0	3	Egress = 2007-085T00:25 (16 min)
S28	41	**41TI (t) [T27]**	**TITAN**	2007-085T00:23	Mar26	Mon	16	16	3	Outbound 1009.9 km flyby, v = 6.2 km/s, phase = 144 deg
S28	41	Ring CRX	Ascending	2007-085T00:28	Mar26	Mon	0	16	3	r = 19.661 Rs (1186145 km)
S28	41	**XD**	**Segment**	2007-086T07:22	Mar27	Tue	1	15	5	XD_041_042 (9d); Cross Discipline Outbound 41 - Inbound 42
S29	41	**S29 Begins**		2007-087T08:04	Mar28	Wed	2	14	6	S29 Sequence. Duration = 38 d
S29	41	**OTM-101**	41TI+3	2007-087T14:49	Mar28	Wed	3	13	6	D/L start, burn ~6h later. Backup D/L 088T22:49

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S29	42	**Apoapse**		2007-090T17:01	Mar31	Sat	6	10	3	Per = 15.9 d, inc = 52.3 deg, r = 32.939 Rs, phase = 80 deg
S29	42	**OTM-102**	~APO	2007-092T22:34	Apr02	Mon	8	8	5	D/L start, burn ~6h later. Backup D/L 093T22:33
S29	42	**SATURN**	**Segment**	2007-095T07:33	Apr05	Thu	10	6	2	Saturn_042 (3d); Saturn_042 Early and Mid Inclined
S29	42	**OTM-103**	42TI-3	2007-097T14:48	Apr07	Sat	13	3	5	D/L start, burn ~6h later. Backup D/L 098T22:18
S29	42	**RINGS**	**Segment**	2007-097T23:48	Apr07	Sat	13	3	0	RINGS_Rev_42 (2d); RINGS Rev 42
S29	42	Ring CRX	Descending	2007-098T04:57	Apr08	Sun	13	3	1	r = 8.954 Rs (540177 km)
S29	42	Periapse		2007-098T16:21	Apr08	Sun	14	2	1	R = 7.565 Rs, lat = -39 deg, phase = 100 deg
S29	42	**TOST**	**Segment**	2007-100T07:03	Apr10	Tue	15	1	3	TOST_042TI_T28 (2d); TOST Segment 042TI (T28)
S29	42	**Sun OCC**	TITAN	2007-100T22:40	Apr10	Tue	16	0	3	Egress = 2007-100T22:59 (20 min)
S29	42	**Earth OCC**	TITAN	2007-100T22:41	Apr10	Tue	16	0	3	Egress = 2007-100T22:59 (18 min)
S29	42	**42TI (t) [T28]**	**TITAN**	2007-100T22:58	Apr10	Tue	16	16	3	Outbound 990.9 km flyby, v = 6.2 km/s, phase = 137 deg
S29	42	Ring CRX	Ascending	2007-100T23:01	Apr10	Tue	0	16	3	r = 19.666 Rs (1186423 km)
S29	42	**XD**	**Segment**	2007-102T05:57	Apr12	Thu	1	15	5	XD_042_043 (9d); Cross Discipline Segment Outbound 42 - Inbound 43
S29	42	**OTM-104**	42TI+3	2007-103T21:47	Apr13	Fri	3	13	6	D/L start, burn ~6h later. Backup D/L 104T21:47
S29	43	**Apoapse**		2007-106T20:41	Apr16	Mon	6	10	3	Per = 15.9 d, inc = 46.9 deg, r = 34.766 Rs, phase = 73 deg
S29	43	**OTM-105**	~APO	2007-108T21:32	Apr18	Wed	8	8	5	D/L start, burn ~6h later. Backup D/L 109T21:32
S29	43	**RINGS**	**Segment**	2007-111T06:16	Apr21	Sat	10	6	2	RINGS_Rev_43 (5d); RINGS Rev 43
S29	43	**OTM-106**	43TI-3	2007-113T21:16	Apr23	Mon	13	3	5	D/L start, burn ~6h later. Backup D/L 114T21:01
S29	43	**Earth OCC**	RING	2007-114T09:05	Apr24	Tue	13	3	0	Egress = 2007-114T10:26 (82 min)
S29	43	**Sun OCC**	RING	2007-114T09:47	Apr24	Tue	13	2	1	Egress = 2007-114T11:26 (99 min)
S29	43	Ring CRX	Descending	2007-114T13:53	Apr24	Tue	14	2	1	r = 6.570 Rs (396380 km)
S29	43	43DI (nt)	**DIONE**	2007-114T14:52	Apr24	Tue	14	2	1	Inbound 114225.1 km flyby, v = 11.4 km/s, phase = 69 deg
S29	43	43TL (nt)	**TELESTO**	2007-114T16:00	Apr24	Tue	14	2	1	Inbound 107375.8 km flyby, v = 11.1 km/s, phase = 99 deg
S29	43	Periapse		2007-114T20:02	Apr24	Tue	14	2	1	R = 5.738 Rs, lat = -32 deg, phase = 107 deg
S29	43	**TOST**	**Segment**	2007-116T06:01	Apr26	Thu	15	1	2	TOST_043TI_T29 (2d); TOST Segment 043TI (T29)
S29	43	**Sun OCC**	TITAN	2007-116T21:11	Apr26	Thu	16	0	3	Egress = 2007-116T21:33 (22 min)
S29	43	**Earth OCC**	TITAN	2007-116T21:12	Apr26	Thu	16	0	3	Egress = 2007-116T21:33 (21 min)
S29	43	**43TI (t) [T29]**	**TITAN**	2007-116T21:33	Apr26	Thu	16	16	3	Outbound 980.8 km flyby, v = 6.2 km/s, phase = 130 deg
S29	43	Ring CRX	Ascending	2007-116T21:35	Apr26	Thu	0	16	3	r = 19.670 Rs (1186671 km)
S29	43	**XD**	**Segment**	2007-117T22:33	Apr27	Fri	1	15	4	XD_043_044 (9d); Cross Discipline Segment Outbound 43 - Inbound 44
S29	43	**OTM-107**	43TI+3	2007-119T20:45	Apr29	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 120T20:45
S29	44	**Apoapse**		2007-122T23:23	May02	Wed	6	10	3	Per = 15.9 d, inc = 39.1 deg, r = 36.222 Rs, phase = 67 deg
S29	44	**OTM-108**	~APO	2007-124T13:00	May04	Fri	8	8	5	D/L start, burn ~6h later. Backup D/L 125T20:29
S30	44	**S30 Begins**		2007-124T22:00	May04	Fri	8	8	0	S30 Sequence. Duration = 37 d
S30	44	**RINGS**	**Segment**	2007-127T05:14	May07	Mon	10	6	3	RINGS_Rev_44 (3d); RINGS Rev 44
S30	44	**OTM-109**	44TI-3	2007-128T20:14	May08	Tue	12	4	4	D/L start, burn ~6h later. Backup D/L 129T20:14
S30	44	**SATURN**	**Segment**	2007-130T05:14	May10	Thu	13	3	1	Saturn_044 (2d); Saturn_044
S30	44	**Earth OCC**	SATURN	2007-130T14:34	May10	Thu	14	2	2	Egress = 2007-130T16:37 (124 min)
S30	44	**Earth OCC**	RING	2007-130T15:19	May10	Thu	14	2	2	Egress = 2007-130T17:52 (154 min)
S30	44	**Sun OCC**	SATURN	2007-130T15:22	May10	Thu	14	2	2	Egress = 2007-130T17:36 (135 min)
S30	44	**Sun OCC**	RING	2007-130T16:13	May10	Thu	14	2	2	Egress = 2007-130T18:17 (124 min)
S30	44	Ring CRX	Descending	2007-130T19:31	May10	Thu	14	2	2	r = 4.757 Rs (287006 km)
S30	44	Periapse		2007-130T22:46	May10	Thu	14	2	2	R = 4.285 Rs, lat = -24 deg, phase = 114 deg
S30	44	**TOST**	**Segment**	2007-132T04:59	May12	Sat	15	1	3	TOST_044TI_T30 (2d); TOST Segment 044TI_T30
S30	44	**Sun OCC**	TITAN	2007-132T19:42	May12	Sat	16	0	4	Egress = 2007-132T20:08 (27 min)
S30	44	**Earth OCC**	TITAN	2007-132T19:44	May12	Sat	16	0	4	Egress = 2007-132T20:09 (25 min)
S30	44	**44TI (t) [T30]**	**TITAN**	2007-132T20:10	May12	Sat	16	16	4	Outbound 959.2 km flyby, v = 6.2 km/s, phase = 122 deg
S30	44	Ring CRX	Ascending	2007-132T20:11	May12	Sat	0	16	4	r = 19.674 Rs (1186904 km)
S30	44	**XD**	**Segment**	2007-133T21:13	May13	Sun	1	15	5	XD_044_045 (10d); Cross Discipline Segment Outbound 44 - Inbound 45
S30	44	**OTM-110**	44TI+3	2007-135T19:43	May15	Tue	3	13	7	D/L start, burn ~6h later. Backup D/L 136T19:43
S30	45	**Apoapse**		2007-139T01:03	May19	Sat	6	10	3	Per = 15.9 d, inc = 28.2 deg, r = 37.277 Rs, phase = 61 deg
S30	45	**OTM-111**	~APO	2007-140T19:27	May20	Sun	8	8	5	D/L start, burn ~6h later. Backup D/L 141T11:42
S30	45	**SATURN**	**Segment**	2007-144T04:12	May24	Thu	11	5	3	Saturn_045 (2d); Saturn_045
S30	45	**OTM-112**	45TI-3	2007-145T19:12	May25	Fri	13	3	5	D/L start, burn ~6h later. Backup D/L 146T09:57
S30	45	**SOST**	**Segment**	2007-146T04:12	May26	Sat	13	3	0	SOST_045 (2d); SOST segment in rev 45 (non-targeteds)
S30	45	**Earth OCC**	RING	2007-146T18:43	May26	Sat	14	2	1	Egress = 2007-146T21:49 (187 min)
S30	45	**Earth OCC**	SATURN	2007-146T19:02	May26	Sat	14	2	1	Egress = 2007-146T21:27 (146 min)
S30	45	**Sun OCC**	RING	2007-146T19:36	May26	Sat	14	2	1	Egress = 2007-146T22:02 (146 min)
S30	45	**Sun OCC**	SATURN	2007-146T19:39	May26	Sat	14	2	1	Egress = 2007-146T21:56 (137 min)
S30	45	45TE (nt)	**TETHYS**	2007-146T20:47	May26	Sat	14	2	1	Inbound 102788.1 km flyby, v = 12.0 km/s, phase = 78 deg
S30	45	Pallene	Dust Crossing	2007-146T22:39	May26	Sat	14	2	1	Egress at 2007-146T22:42 (0.0 min); protection = none.
S30	45	Ring CRX	Descending	2007-146T22:41	May26	Sat	14	2	1	r = 3.508 Rs (211648 km)
S30	45	45EP (nt)	**EPIMETHEUS**	2007-146T23:13	May26	Sat	14	2	1	Inbound 58279.1 km flyby, v = 8.7 km/s, phase = 148 deg
S30	45	Periapse		2007-147T00:28	May27	Sun	14	2	1	R = 3.236 Rs, lat = -15 deg, phase = 119 deg
S30	45	**TOST**	**Segment**	2007-148T03:56	May28	Mon	15	1	2	TOST_045TI_T31 (2d); TOST Segment 045TI (T31)
S30	45	**Sun OCC**	TITAN	2007-148T18:09	May28	Mon	16	0	3	Egress = 2007-148T18:41 (32 min)
S30	45	**Earth OCC**	TITAN	2007-148T18:10	May28	Mon	16	0	3	Egress = 2007-148T18:42 (32 min)
S30	45	Ring CRX	Ascending	2007-148T18:36	May28	Mon	16	0	3	r = 19.590 Rs (1181873 km)
S30	45	**45TI (t) [T31]**	**TITAN**	2007-148T18:52	May28	Mon	16	16	3	Outbound 2298.6 km flyby, v = 6.1 km/s, phase = 114 deg
S30	45	**XD**	**Segment**	2007-149T21:21	May29	Tue	1	15	4	XD_045_046 (10d); Cross Discipline Segment Outbound 45 - Inbound 46
S30	45	**OTM-113**	45TI+3	2007-151T18:41	May31	Thu	3	13	6	D/L start, burn ~6h later. Backup D/L 152T18:41
S30	46	**Apoapse**		2007-155T01:17	Jun04	Mon	6	10	3	Per = 16.0 d, inc = 18.0 deg, r = 37.779 Rs, phase = 58 deg
S30	46	**OTM-114**	~APO	2007-156T10:55	Jun05	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 157T18:25
S30	46	**SATURN**	**Segment**	2007-160T03:10	Jun09	Sat	11	5	4	Saturn_046 (2d); Saturn_046
S30	46	**OTM-115**	46TI-3	2007-161T18:10	Jun10	Sun	13	3	5	D/L start, burn ~6h later. Backup D/L 162T09:10
S31	46	**RINGS**	**Segment**	2007-162T03:10	Jun11	Mon	13	3	0	RINGS_Rev_46 (2d); RINGS Rev 46
S31	46	**S31 Begins**		2007-162T03:10	Jun11	Mon	13	3	0	S31 Sequence. Duration = 33 d
S31	46	**Earth OCC**	RING	2007-162T19:24	Jun11	Mon	14	2	1	Egress = 2007-162T22:52 (209 min)
S31	46	**Sun OCC**	RING	2007-162T20:05	Jun11	Mon	14	2	1	Egress = 2007-162T23:01 (177 min)
S31	46	**Earth OCC**	SATURN	2007-162T20:45	Jun11	Mon	14	2	1	Egress = 2007-162T22:47 (122 min)
S31	46	**Sun OCC**	SATURN	2007-162T21:08	Jun11	Mon	14	2	1	Egress = 2007-162T23:06 (119 min)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S31	46	46MI (nt)	**MIMAS**	2007-162T23:03	Jun11	Mon	14	2	1	Inbound 111975.3 km flyby, v = 16.7 km/s, phase = 95 deg
S31	46	G_ring	**DUST HAZARD**	2007-162T23:29	Jun11	Mon	14	2	1	Egress at 2007-162T23:30 (0.0 min); protection = HGA to ram and MEA cover closure.
S31	46	Ring CRX	Descending	2007-162T23:30	Jun11	Mon	14	2	1	r = 2.938 Rs (177239 km)
S31	46	46AT (nt)	**ATLAS**	2007-163T00:12	Jun12	Tue	14	2	1	Inbound 34016.6 km flyby, v = 7.5 km/s, phase = 117 deg
S31	46	46PM (nt)	**PROMETHEUS**	2007-163T00:19	Jun12	Tue	14	2	1	Inbound 69717.3 km flyby, v = 11.7 km/s, phase = 79 deg
S31	46	Periapse		2007-163T00:45	Jun12	Tue	14	2	1	R = 2.745 Rs, lat = -9 deg, phase = 123 deg
S31	46	46EP (nt)	**EPIMETHEUS**	2007-163T00:58	Jun12	Tue	14	2	1	Outbound 80596.3 km flyby, v = 10.6 km/s, phase = 144 deg
S31	46	46ME (nt)	**METHONE**	2007-163T02:26	Jun12	Tue	14	2	1	Outbound 61695.6 km flyby, v = 9.0 km/s, phase = 100 deg
S31	46	**TOST**	**Segment**	2007-164T02:54	Jun13	Wed	15	1	2	TOST_046TI_T32 (2d); TOST Segment 046TI (T32)
S31	46	**Earth OCC**	TITAN	2007-164T17:02	Jun13	Wed	16	0	3	Egress = 2007-164T17:41 (40 min)
S31	46	**Sun OCC**	TITAN	2007-164T17:02	Jun13	Wed	16	0	3	Egress = 2007-164T17:40 (39 min)
S31	46	Ring CRX	Ascending	2007-164T17:43	Jun13	Wed	16	0	3	r = 19.665 Rs (1186412 km)
S31	46	**46TI (t) [T32]**	**TITAN**	2007-164T17:46	Jun13	Wed	16	16	3	Outbound 964.9 km flyby, v = 6.2 km/s, phase = 107 deg
S31	46	**XD**	**Segment**	2007-165T20:47	Jun14	Thu	1	15	4	XD_046_047 (12d); Cross Discipline Segment Outbound 46 - Inbound 47
S31	46	**OTM-116**	46TI+3	2007-167T17:39	Jun16	Sat	3	13	6	D/L start, burn ~6h later. Backup D/L 168T17:39
S31	47	**Apoapse**		2007-171T01:10	Jun20	Wed	6	10	3	Per = 16.0 d, inc = 2.1 deg, r = 38.086 Rs, phase = 54 deg
S31	47	**OTM-117**	~APO	2007-172T17:23	Jun21	Thu	8	8	5	D/L start, burn ~6h later. Backup D/L 173T17:23
S31	47	**OTM-118**	47TI-3	2007-177T17:08	Jun26	Tue	13	3	5	D/L start, burn ~6h later. Backup D/L 178T08:10
S31	47	**SOST**	**Segment**	2007-178T02:08	Jun27	Wed	13	3	0	SOST_047 (2d); SOST segment in rev 47 (untargeted Tethys)
S31	47	47TE (nt)	**TETHYS**	2007-178T19:43	Jun27	Wed	14	2	1	Inbound 18417.8 km flyby, v = 10.6 km/s, phase = 116 deg
S31	47	**Earth OCC**	SATURN	2007-178T22:04	Jun27	Wed	14	2	1	Egress = 2007-178T23:23 (80 min)
S31	47	**Sun OCC**	SATURN	2007-178T22:10	Jun27	Wed	14	2	1	Egress = 2007-178T23:40 (90 min)
S31	47	47ME (nt)	**METHONE**	2007-178T22:23	Jun27	Wed	14	2	1	Inbound 22643.0 km flyby, v = 10.9 km/s, phase = 114 deg
S31	47	47MI (nt)	**MIMAS**	2007-178T22:48	Jun27	Wed	14	2	1	Inbound 102915.8 km flyby, v = 17.0 km/s, phase = 110 deg
S31	47	47AE (nt)	**AEGAEON**	2007-178T23:13	Jun27	Wed	14	2	1	Inbound 15433.9 km flyby, v = 9.4 km/s, phase = 107 deg
S31	47	G_ring	**DUST HAZARD**	2007-178T23:56	Jun27	Wed	14	2	1	Egress at 2007-179T00:00 (0.0 min); protection = HGA to ram and MEA cover closure.
S31	47	JE_ring	**DUST HAZARD**	2007-179T00:00	Jun28	Thu	14	2	1	Egress at 2007-179T00:21 (0.0 min); protection = HGA to ram and MEA cover closure.
S31	47	Ring CRX	Descending	2007-179T00:01	Jun28	Thu	14	2	1	r = 2.584 Rs (155889 km)
S31	47	47EN (nt)	**ENCELADUS**	2007-179T00:52	Jun28	Thu	14	2	1	Inbound 88670.3 km flyby, v = 9.3 km/s, phase = 54 deg
S31	47	Periapse		2007-179T00:54	Jun28	Thu	14	2	1	R = 2.462 Rs, lat = -1 deg, phase = 126 deg
S31	47	47PM (nt)	**PROMETHEUS**	2007-179T01:43	Jun28	Thu	14	2	1	Outbound 35159.7 km flyby, v = 9.1 km/s, phase = 153 deg
S31	47	47EP (nt)	**EPIMETHEUS**	2007-179T02:08	Jun28	Thu	14	2	1	Outbound 86621.8 km flyby, v = 15.7 km/s, phase = 153 deg
S31	47	47RH (nt)	**RHEA**	2007-179T12:00	Jun28	Thu	15	1	2	Outbound 104382.5 km flyby, v = 10.7 km/s, phase = 105 deg
S31	47	**TOST**	**Segment**	2007-180T02:05	Jun29	Fri	15	1	2	TOST_047TI_T33 (2d); TOST Segment 047TI (T33)
S31	47	**47TI (t) [T33]**	**TITAN**	2007-180T17:00	Jun29	Fri	16	19	3	Outbound 1932.6 km flyby, v = 6.2 km/s, phase = 96 deg
S31	47	**XD**	**Segment**	2007-182T02:35	Jul01	Sun	1	18	4	XD_047_048 (17d); Cross Discipline Segment Outbound 47 - Inbound 48
S31	47	Ring CRX	Ascending	2007-183T14:32	Jul02	Mon	3	16	6	r = 35.583 Rs (2146696 km)
S31	47	**OTM-119**	47TI+3	2007-184T16:37	Jul03	Tue	4	15	7	D/L start, burn ~6h later. Backup D/L 185T09:07
S31	47	**OTM-120**	~APO	2007-189T16:22	Jul08	Sun	9	10	5	D/L start, burn ~6h later. Backup D/L 190T08:52
S31	48	**Apoapse**		2007-190T10:08	Jul09	Mon	10	10	1	Per = 22.8 d, inc = 0.4 deg, r = 47.803 Rs, phase = 64 deg
S32	48	**S32 Begins**		2007-195T01:06	Jul14	Sat	14	5	5	S32 Sequence. Duration = 29 d
S32	48	**OTM-121**	48TI-3	2007-196T16:06	Jul15	Sun	16	3	7	D/L start, burn ~6h later. Backup D/L 197T15:51
S32	48	**TOST**	**Segment**	2007-199T00:51	Jul18	Wed	18	1	2	TOST_048TI_T34 (2d); TOST_048TI (T34) Flyby
S32	48	**48TI (t) [T34]**	**TITAN**	2007-200T01:11	Jul19	Thu	19	43	3	Inbound 1331.8 km flyby, v = 6.2 km/s, phase = 34 deg
S32	48	**SATURN**	**Segment**	2007-201T00:36	Jul20	Fri	1	42	4	Saturn_048 (2d); Saturn_048
S32	48	Ring CRX	Descending	2007-201T15:01	Jul20	Fri	2	42	5	r = 6.257 Rs (377507 km)
S32	48	48HE (nt)	**HELENE**	2007-201T17:18	Jul20	Fri	2	42	5	Inbound 38261.9 km flyby, v = 4.7 km/s, phase = 55 deg
S32	48	48TE (nt)	**TETHYS**	2007-201T18:16	Jul20	Fri	2	42	5	Inbound 95187.8 km flyby, v = 6.5 km/s, phase = 94 deg
S32	48	Periapse		2007-201T20:31	Jul20	Fri	2	41	5	R = 5.357 Rs, lat = -0 deg, phase = 130 deg
S32	48	**OTM-122**	48TI+3	2007-202T15:36	Jul21	Sat	3	41	6	D/L start, burn ~6h later. Backup D/L 203T15:36
S32	48	**XD**	**Segment**	2007-203T00:36	Jul22	Sun	3	40	0	XD_048_049 (37d); Cross Discipline Segment Outbound 48 - Inbound 49
S32	48	Ring CRX	Ascending	2007-204T04:09	Jul23	Mon	4	39	2	r = 24.108 Rs (1454449 km)
S32	48	**OTM-123**	~APO	2007-217T14:35	Aug05	Sun	18	26	15	D/L start, burn ~6h later. Backup D/L 218T14:35
S32	49	**Apoapse**		2007-221T16:46	Aug09	Thu	22	22	4	Per = 39.7 d, inc = 0.3 deg, r = 69.031 Rs, phase = 49 deg
S33	49	**S33 Begins**		2007-223T23:20	Aug11	Sat	24	19	6	S33 Sequence. Duration = 42 d
S33	49	SEP = 3.0 deg	Conjunction	2007-230T13:46	Aug18	Sat	31	13	13	Commanding must not be required below 3.0 deg
S33	49	SEP = 2.0 deg	Conjunction	2007-231T23:19	Aug19	Sun	32	11	14	Plan no SSR playback below 2.0 deg
S33	49	SEP = 2.0 deg	Conjunction	2007-235T17:59	Aug23	Thu	36	8	18	Plan no SSR playback below 2.0 deg
S33	49	SEP = 3.0 deg	Conjunction	2007-237T03:40	Aug25	Sat	37	6	20	Commanding must not be required below 3.0 deg
S33	49	**OTM-124**	49TI-3	2007-239T13:20	Aug27	Mon	40	4	22	D/L start, burn ~6h later. Backup D/L 240T13:20
S33	49	**SOST**	**Segment**	2007-239T22:20	Aug27	Mon	40	3	0	SOST_049RH (3d); SOST Rhea segment in Rev 49
S33	49	Ring CRX	Descending	2007-241T07:23	Aug29	Wed	41	2	2	r = 6.301 Rs (380129 km)
S33	49	49TE (nt)	**TETHYS**	2007-241T11:14	Aug29	Wed	41	2	2	Inbound 55392.9 km flyby, v = 4.8 km/s, phase = 102 deg
S33	49	Periapse		2007-241T13:03	Aug29	Wed	41	2	2	R = 5.351 Rs, lat = -0 deg, phase = 132 deg
S33	49	49RH (nt)	**RHEA**	2007-242T01:19	Aug30	Thu	42	1	2	Outbound 5727.2 km flyby, v = 6.7 km/s, phase = 46 deg
S33	49	**TOST**	**Segment**	2007-242T17:34	Aug30	Thu	43	1	3	TOST_049TI_T35 (2d); TOST_049TI (T35) Flyby
S33	49	**49TI (t) [T35]**	**TITAN**	2007-243T06:33	Aug31	Fri	43	10	4	Outbound 3324.2 km flyby, v = 6.1 km/s, phase = 87 deg
S33	49	**XD**	**Segment**	2007-244T16:20	Sep01	Sat	1	9	5	XD_049 (7d); Cross Discipline Segment 49
S33	49	**OTM-125**	49TI+2	2007-245T05:35	Sep02	Sun	2	8	6	D/L start, burn ~6h later. Backup D/L 246T13:05
S33	49	**OTM-126**	49IA-5	2007-248T12:50	Sep05	Wed	5	5	3	D/L start, burn ~6h later. Backup D/L 249T05:20
S33	49	**OTM-127**	49IA-2	2007-251T12:50	Sep08	Sat	8	2	3	D/L start, burn ~6h later. Backup D/L 252T12:45
S33	49	**SOST**	**Segment**	2007-251T21:50	Sep08	Sat	9	2	0	SOST_049IA (4d); SOST rev 49 targeted Iapetus
S33	49	**49IA (t) [I1]**	**IAPETUS**	2007-253T14:16	Sep10	Mon	10	22	2	Outbound 1621.6 km flyby, v = 2.4 km/s, phase = 59 deg
S33	49	**XD**	**Segment**	2007-255T21:35	Sep12	Wed	2	19	4	XD_049_050 (16d); Cross Discipline Segment Outbound 49 - Inbound 50
S33	49	**OTM-128**	49IA+3	2007-256T12:20	Sep13	Thu	3	19	5	D/L start, burn ~6h later. Backup D/L 257T12:21
S33	50	**Apoapse**		2007-257T11:38	Sep14	Fri	4	18	1	Per = 32.0 d, inc = 6.1 deg, r = 59.736 Rs, phase = 42 deg
S33	50	**OTM-129**	49IA+7	2007-260T12:21	Sep17	Mon	7	15	4	D/L start, burn ~6h later. Backup D/L 261T04:51
S34	50	**S34 Begins**		2007-265T20:51	Sep22	Sat	12	9	5	S34 Sequence. Duration = 39 d
S34	50	**OTM-130**	50TI-3	2007-271T11:36	Sep28	Fri	18	4	11	D/L start, burn ~6h later. Backup D/L 272T04:06
S34	50	**SOST**	**Segment**	2007-271T20:36	Sep28	Fri	18	3	0	SOST_050 (3d); SOST segment in rev 50 (non-targeteds)
S34	50	50DI (nt)	**DIONE**	2007-273T05:48	Sep30	Sun	20	2	2	Inbound 43433.5 km flyby, v = 5.7 km/s, phase = 49 deg

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S34	50	Ring CRX	Ascending	2007-273T06:06	Sep30	Sun	20	2	2	r = 5.776 Rs (348456 km)
S34	50	**Earth OCC**	SATURN	2007-273T07:10	Sep30	Sun	20	2	2	Egress = 2007-273T08:49 (99 min)
S34	50	**Sun OCC**	SATURN	2007-273T07:15	Sep30	Sun	20	2	2	Egress = 2007-273T08:10 (56 min)
S34	50	50EN (nt)	**ENCELADUS**	2007-273T10:51	Sep30	Sun	20	2	2	Inbound 101494.2 km flyby, v = 6.5 km/s, phase = 95 deg
S34	50	Periapse		2007-273T11:29	Sep30	Sun	20	2	2	R = 4.701 Rs, lat = 5 deg, phase = 138 deg
S34	50	50TL (nt)	**TELESTO**	2007-273T15:08	Sep30	Sun	20	2	2	Outbound 75929.3 km flyby, v = 7.7 km/s, phase = 163 deg
S34	50	**TOST**	**Segment**	2007-274T12:51	Oct01	Mon	21	1	3	TOST_050TI_T36 (2d); TOST_050TI (T36) Flyby
S34	50	Ring CRX	Descending	2007-274T22:50	Oct01	Mon	21	0	3	r = 17.690 Rs (1067231 km)
S34	**50**	**50TI (t) [T36]**	**TITAN**	2007-275T04:43	Oct02	Tue	22	48	4	Outbound 973.0 km flyby, v = 6.3 km/s, phase = 67 deg
S34	50	Ring CRX	Ascending	2007-275T15:08	Oct02	Tue	0	47	4	r = 22.856 Rs (1378899 km)
S34	50	**XD**	**Segment**	2007-276T14:19	Oct03	Wed	1	46	5	XD_50 (3d); XD rev 50 before CDS V10
S34	50	**OTM-131**	50TI+3	2007-278T11:22	Oct05	Fri	3	45	7	D/L start, burn ~6h later. Backup D/L 279T11:07
S34	50	**V10**	**Segment**	2007-279T20:07	Oct06	Sat	5	43	1	V10_Transition (12d); CDS V10 upload

S34	51	**Apoapse**		2007-285T09:52	Oct12	Fri	10	38	7	Per = 23.8 d, inc = 5.0 deg, r = 49.165 Rs, phase = 36 deg
S34	51	**XD**	**Segment**	2007-291T19:38	Oct18	Thu	17	31	13	XD_51 (4d); XD rev 51 post CDS V10
S34	51	51HY (nt)	**HYPERION**	2007-294T15:01	Oct21	Sun	19	28	16	Inbound 121741.8 km flyby, v = 4.9 km/s, phase = 116 deg
S34	51	51TI (nt)	**TITAN**	2007-295T01:12	Oct22	Mon	20	28	17	Inbound 438394.5 km flyby, v = 4.2 km/s, phase = 30 deg
S34	51	**RINGS**	**Segment**	2007-295T19:23	Oct22	Mon	21	27	17	RINGS_Rev_51 (3d); RINGS Rev 51
S34	51	51HE (nt)	**HELENE**	2007-297T00:33	Oct24	Wed	22	26	19	Inbound 23943.1 km flyby, v = 7.7 km/s, phase = 76 deg
S34	51	**Sun OCC**	SATURN	2007-297T05:05	Oct24	Wed	22	26	19	Egress = 2007-297T06:18 (73 min)
S34	51	**Earth OCC**	SATURN	2007-297T05:14	Oct24	Wed	22	26	19	Egress = 2007-297T06:48 (94 min)
S34	51	E_ring_lg	Dust Crossing	2007-297T05:16	Oct24	Wed	22	26	19	Egress at 2007-297T05:39 (0.0 min); protection = none.
S34	51	Ring CRX	Descending	2007-297T05:28	Oct24	Wed	22	26	19	r = 4.184 Rs (252413 km)
S34	51	51PM (nt)	**PROMETHEUS**	2007-297T06:30	Oct24	Wed	22	26	19	Inbound 104597.4 km flyby, v = 1.5 km/s, phase = 166 deg
S34	51	Periapse		2007-297T07:56	Oct24	Wed	22	26	19	R = 3.807 Rs, lat = -3 deg, phase = 145 deg
S34	51	**XD**	**Segment**	2007-298T19:09	Oct25	Thu	24	24	20	XD_051_052 (20d); Cross Discipline Segment Outbound 51 - Inbound 52
S34	51	Ring CRX	Ascending	2007-299T10:28	Oct26	Fri	24	24	21	r = 22.606 Rs (1363833 km)
S35	51	**S35 Begins**		2007-304T18:40	Oct31	Wed	30	18	26	S35 Sequence. Duration = 44 d
S35	51	**OTM-132**	~APO	2007-305T09:40	Nov01	Thu	30	18	27	D/L start, burn ~6h later. Backup D/L 306T09:40

S35	52	**Apoapse**		2007-309T07:54	Nov05	Mon	34	14	4	Per = 24.0 d, inc = 5.0 deg, r = 49.390 Rs, phase = 35 deg
S35	52	**SOST**	**Segment**	2007-318T17:56	Nov14	Wed	44	4	13	SOST_052 (2d); SOST segment in rev 52 (non-targeteds)
S35	52	**OTM-133**	52TI-3	2007-319T08:56	Nov15	Thu	44	4	14	D/L start, burn ~6h later. Backup D/L 321T08:42
S35	52	**RINGS**	**Segment**	2007-320T17:41	Nov16	Fri	46	2	1	RINGS_Rev_52 (2d); RINGS Rev 52
S35	52	52RH (nt)	**RHEA**	2007-320T19:43	Nov16	Fri	46	2	1	Inbound 91621.1 km flyby, v = 9.3 km/s, phase = 150 deg
S35	52	**Sun OCC**	SATURN	2007-321T05:02	Nov17	Sat	46	2	2	Egress = 2007-321T06:18 (76 min)
S35	52	E_ring_lg	Dust Crossing	2007-321T05:09	Nov17	Sat	46	2	2	Egress at 2007-321T05:32 (0.0 min); protection = none.
S35	52	**Earth OCC**	SATURN	2007-321T05:14	Nov17	Sat	46	2	2	Egress = 2007-321T06:52 (98 min)
S35	52	Ring CRX	Descending	2007-321T05:21	Nov17	Sat	46	2	2	r = 4.191 Rs (252864 km)
S35	52	52AE (nt)	**AEGAEON**	2007-321T06:15	Nov17	Sat	46	2	2	Inbound 99721.6 km flyby, v = 3.3 km/s, phase = 150 deg
S35	52	52PA (nt)	**PANDORA**	2007-321T06:59	Nov17	Sat	46	2	2	Inbound 93408.3 km flyby, v = 1.6 km/s, phase = 163 deg
S35	52	Periapse		2007-321T07:52	Nov17	Sat	46	2	2	R = 3.801 Rs, lat = -3 deg, phase = 145 deg
S35	52	52PN (nt)	**PAN**	2007-321T08:05	Nov17	Sat	46	2	2	Outbound 96661.9 km flyby, v = 1.3 km/s, phase = 138 deg
S35	52	52EP (nt)	**EPIMETHEUS**	2007-321T09:20	Nov17	Sat	46	2	2	Outbound 99466.0 km flyby, v = 1.9 km/s, phase = 102 deg
S35	52	52CA (nt)	**CALYPSO**	2007-321T12:35	Nov17	Sat	46	2	2	Outbound 87970.0 km flyby, v = 10.5 km/s, phase = 147 deg
S35	52	**TOST**	**Segment**	2007-322T10:12	Nov18	Sun	47	1	3	TOST_052TI_T37 (2d); TOST_052TI (T37) Flyby
S35	**52**	**52TI (t) [T37]**	**TITAN**	2007-323T00:47	Nov19	Mon	48	16	4	Outbound 999.4 km flyby, v = 6.3 km/s, phase = 51 deg
S35	52	Ring CRX	Ascending	2007-323T04:36	Nov19	Mon	0	16	4	r = 20.832 Rs (1256788 km)
S35	52	**XD**	**Segment**	2007-324T09:57	Nov20	Tue	1	15	5	XD_052_053 (5d); Cross Discipline Segment Outbound 52 - Inbound 53
S35	52	**OTM-134**	52TI+3	2007-326T00:57	Nov22	Thu	3	13	7	D/L start, burn ~6h later. Backup D/L 327T08:28

S35	53	**Apoapse**		2007-329T07:54	Nov25	Sun	6	10	3	Per = 16.0 d, inc = 12.0 deg, r = 38.004 Rs, phase = 25 deg
S35	53	**MAG**	**Segment**	2007-329T17:13	Nov25	Sun	7	9	4	MAG_053_A (3d); MAG rev 53 doy 329
S35	53	**OTM-135**	~APO	2007-331T00:43	Nov27	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 332T07:58
S35	53	**RINGS**	**Segment**	2007-332T16:58	Nov28	Wed	10	6	2	RINGS_Rev_53in (4d); RINGS Rev 53in
S35	53	**OTM-136**	53TI-2	2007-336T07:44	Dec02	Sun	13	3	5	D/L start, burn ~6h later. Backup D/L 337T09:10
S35	53	**MAG**	**Segment**	2007-336T16:44	Dec02	Sun	14	2	0	MAG_053_B (2d); MAG rev 53 doy 336
S35	53	**Sun OCC**	RING	2007-337T04:56	Dec03	Mon	14	2	1	Egress = 2007-337T06:24 (89 min)
S35	53	**Earth OCC**	RING	2007-337T05:16	Dec03	Mon	14	2	1	Egress = 2007-337T06:30 (75 min)
S35	53	53TL (nt)	**TELESTO**	2007-337T05:23	Dec03	Mon	14	2	1	Inbound 119570.5 km flyby, v = 8.9 km/s, phase = 20 deg
S35	53	53MI (nt)	**MIMAS**	2007-337T05:26	Dec03	Mon	14	2	1	Inbound 84254.0 km flyby, v = 15.0 km/s, phase = 139 deg
S35	53	53AE (nt)	**AEGAEON**	2007-337T06:03	Dec03	Mon	14	2	1	Inbound 81097.7 km flyby, v = 15.0 km/s, phase = 131 deg
S35	53	**Sun OCC**	SATURN	2007-337T06:10	Dec03	Mon	14	2	1	Egress = 2007-337T07:33 (83 min)
S35	53	**Earth OCC**	SATURN	2007-337T06:21	Dec03	Mon	14	2	1	Egress = 2007-337T07:47 (86 min)
S35	53	G_ring	Dust Crossing	2007-337T06:41	Dec03	Mon	14	2	1	Egress at 2007-337T06:42 (0.0 min); protection = none.
S35	53	Ring CRX	Descending	2007-337T06:42	Dec03	Mon	14	2	1	r = 2.672 Rs (161217 km)
S35	53	53EP (nt)	**EPIMETHEUS**	2007-337T06:52	Dec03	Mon	14	2	1	Inbound 9176.0 km flyby, v = 7.6 km/s, phase = 141 deg
S35	53	Periapse		2007-337T07:40	Dec03	Mon	14	2	1	R = 2.535 Rs, lat = -6 deg, phase = 155 deg
S35	53	**TOST**	**Segment**	2007-338T09:14	Dec04	Tue	15	1	2	TOST_053TI_T38 (2d); TOST_053TI (T38) Flyby
S35	**53**	**53TI (t) [T38]**	**TITAN**	2007-339T00:07	Dec05	Wed	16	16	3	Outbound 1298.3 km flyby, v = 6.3 km/s, phase = 70 deg
S35	53	Ring CRX	Ascending	2007-339T02:17	Dec05	Wed	0	16	3	r = 20.329 Rs (1226457 km)
S35	53	**RINGS**	**Segment**	2007-340T09:29	Dec06	Thu	1	15	4	RINGS_Rev_53out (2d); RINGS Rev 53out
S35	53	**OTM-137**	53TI+3	2007-342T00:00	Dec08	Sat	3	13	6	D/L start, burn ~6h later. Backup D/L 343T07:30
S35	53	**MAG**	**Segment**	2007-342T09:00	Dec08	Sat	3	13	0	MAG_53_54 (6d); MAG rev 53/54

S35	54	**Apoapse**		2007-345T05:34	Dec11	Tue	6	10	3	Per = 16.0 d, inc = 26.0 deg, r = 37.492 Rs, phase = 30 deg
S35	54	**OTM-138**	~APO	2007-347T01:10	Dec13	Thu	8	8	5	D/L start, burn ~6h later. Backup D/L 348T07:00
S36	54	**SATURN**	**Segment**	2007-348T16:00	Dec14	Fri	10	6	2	Saturn_054 (6d); Saturn_054
S36	54	**S36 Begins**		2007-348T16:00	Dec14	Fri	10	6	2	S36 Sequence. Duration = 39 d
S36	54	**OTM-139**	54TI-3	2007-351T23:16	Dec17	Mon	13	3	5	D/L start, burn ~6h later. Backup D/L 353T06:46
S36	54	**Sun OCC**	RING	2007-353T02:04	Dec19	Wed	14	2	1	Egress = 2007-353T03:13 (70 min)
S36	54	**Earth OCC**	RING	2007-353T02:30	Dec19	Wed	14	2	1	Egress = 2007-353T03:20 (50 min)
S36	54	**Sun OCC**	SATURN	2007-353T03:07	Dec19	Wed	14	2	1	Egress = 2007-353T04:22 (75 min)
S36	54	**Earth OCC**	SATURN	2007-353T03:24	Dec19	Wed	14	2	1	Egress = 2007-353T04:39 (76 min)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S36	54	Ring CRX	Descending	2007-353T03:35	Dec19	Wed	14	2	1	r = 3.241 Rs (195514 km)
S36	54	Periapse		2007-353T05:04	Dec19	Wed	14	2	1	R = 3.023 Rs, lat = -13 deg, phase = 150 deg
S36	54	**TOST**	**Segment**	2007-354T08:16	Dec20	Thu	15	1	2	TOST_054TI_T39 (2d); TOST_054TI (T39) Flyby
S36	54	**54TI (t) [T39]**	**TITAN**	2007-354T22:58	Dec20	Thu	16	16	3	Outbound 969.5 km flyby, v = 6.3 km/s, phase = 61 deg
S36	54	Ring CRX	Ascending	2007-355T00:15	Dec21	Fri	0	16	3	r = 20.062 Rs (1210337 km)
S36	54	**MAG**	**Segment**	2007-356T08:57	Dec22	Sat	1	15	4	MAG_54_55 (10d); MAG rev 54/55
S36	54	**OTM-140**	54TI+3	2007-357T23:02	Dec23	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 359T06:17
S36	55	**Apoapse**		2007-361T01:29	Dec27	Thu	6	10	3	Per = 15.9 d, inc = 37.7 deg, r = 36.521 Rs, phase = 37 deg
S36	55	**OTM-141**	~APO	2007-363T06:02	Dec29	Sat	8	8	5	D/L start, burn ~6h later. Backup D/L 364T06:02
S36	55	**SATURN**	**Segment**	2008-001T15:03	Jan01	Tue	12	4	3	Saturn_055 (4d); Saturn_055 Periapse
S36	55	**OTM-142**	55TI-3	2008-002T22:18	Jan02	Wed	13	3	5	D/L start, burn ~6h later. Backup D/L 004T07:18
S36	55	55DI (nt)	**DIONE**	2008-003T20:13	Jan03	Thu	14	2	1	Inbound 121807.0 km flyby, v = 11.1 km/s, phase = 76 deg
S36	55	**Sun OCC**	RING	2008-003T20:28	Jan03	Thu	14	2	1	Egress = 2008-003T21:24 (57 min)
S36	55	**Earth OCC**	RING	2008-003T20:54	Jan03	Thu	14	2	1	Egress = 2008-003T21:34 (40 min)
S36	55	55PL (nt)	**PALLENE**	2008-003T21:52	Jan03	Thu	14	2	1	Inbound 105665.4 km flyby, v = 9.4 km/s, phase = 120 deg
S36	55	Ring CRX	Descending	2008-003T22:05	Jan03	Thu	14	2	1	r = 4.361 Rs (263075 km)
S36	55	55JA (nt)	**JANUS**	2008-003T22:11	Jan03	Thu	14	2	1	Inbound 118143.2 km flyby, v = 10.1 km/s, phase = 154 deg
S36	55	55DA (nt)	**DAPHNIS**	2008-003T22:50	Jan03	Thu	14	2	1	Inbound 119011.2 km flyby, v = 10.4 km/s, phase = 157 deg
S36	55	55PM (nt)	**PROMETHEUS**	2008-003T23:11	Jan03	Thu	14	2	1	Inbound 112567.1 km flyby, v = 10.5 km/s, phase = 152 deg
S36	55	55PA (nt)	**PANDORA**	2008-004T00:34	Jan04	Fri	14	2	1	Inbound 123126.8 km flyby, v = 11.3 km/s, phase = 119 deg
S36	55	Periapse		2008-004T00:48	Jan04	Fri	14	2	1	R = 3.970 Rs, lat = -22 deg, phase = 143 deg
S36	55	**TOST**	**Segment**	2008-005T05:48	Jan05	Sat	15	1	2	TOST_055TI_T40 (2d); TOST_055TI (T40) Flyby
S36	55	**55TI (t) [T40]**	**TITAN**	2008-005T21:30	Jan05	Sat	16	48	3	Outbound 1014.0 km flyby, v = 6.3 km/s, phase = 38 deg
S36	55	Ring CRX	Ascending	2008-005T22:21	Jan05	Sat	0	48	3	r = 19.934 Rs (1202592 km)
S36	55	**MAG**	**Segment**	2008-007T07:26	Jan07	Mon	1	46	4	MAG_55_56 (6d); MAG rev 55/56
S36	56	**Apoapse**		2008-009T22:55	Jan09	Wed	4	44	7	Per = 12.0 d, inc = 46.6 deg, r = 30.201 Rs, phase = 31 deg
S36	56	**RINGS**	**Segment**	2008-013T14:04	Jan13	Sun	8	40	11	RINGS_Rev_56 (6d); RINGS Rev 56
S36	56	**Sun OCC**	RING	2008-015T19:59	Jan15	Tue	10	38	13	Egress = 2008-015T20:49 (50 min)
S36	56	**Earth OCC**	RING	2008-015T20:14	Jan15	Tue	10	38	13	Egress = 2008-015T20:52 (38 min)
S36	56	**Sun OCC**	SATURN	2008-015T20:34	Jan15	Tue	10	38	13	Egress = 2008-015T21:33 (60 min)
S36	56	**Earth OCC**	SATURN	2008-015T20:55	Jan15	Tue	10	38	13	Egress = 2008-015T21:38 (43 min)
S36	56	Ring CRX	Descending	2008-015T21:04	Jan15	Tue	10	38	13	r = 3.451 Rs (208171 km)
S36	56	56ME (nt)	**METHONE**	2008-015T21:11	Jan15	Tue	10	38	13	Inbound 15618.6 km flyby, v = 13.7 km/s, phase = 142 deg
S36	56	**OTM-143**	55TI+P	2008-015T22:15	Jan15	Tue	10	38	13	D/L start, burn ~6h later. Backup D/L 016T21:19
S36	56	56PA (nt)	**PANDORA**	2008-015T22:20	Jan15	Tue	10	38	0	Inbound 95184.3 km flyby, v = 16.2 km/s, phase = 116 deg
S36	56	Periapse		2008-015T22:37	Jan15	Tue	10	38	0	R = 3.257 Rs, lat = -21 deg, phase = 149 deg
S36	56	Ring CRX	Ascending	2008-017T21:15	Jan17	Thu	12	36	2	r = 19.825 Rs (1196063 km)
S36	56	**MAG**	**Segment**	2008-019T13:49	Jan19	Sat	14	34	4	MAG_56_57 (5d); MAG rev 56/57
S36	57	**Apoapse**		2008-021T21:43	Jan21	Mon	16	32	6	Per = 11.9 d, inc = 46.6 deg, r = 30.120 Rs, phase = 31 deg
S37	57	**S37 Begins**		2008-022T13:35	Jan22	Tue	17	31	7	S37 Sequence. Duration = 25 d
S37	57	57TI (nt)	**TITAN**	2008-022T21:22	Jan22	Tue	17	31	7	Inbound 855499.6 km flyby, v = 4.5 km/s, phase = 69 deg
S37	57	**RINGS**	**Segment**	2008-024T13:20	Jan24	Thu	19	29	9	RINGS_Rev_57 (7d); RINGS Rev 57
S37	57	57TI (nt)	**TITAN**	2008-027T12:32	Jan27	Sun	22	26	12	Inbound 985542.3 km flyby, v = 9.4 km/s, phase = 60 deg
S37	57	**Sun OCC**	RING	2008-027T18:11	Jan27	Sun	22	26	12	Egress = 2008-027T18:59 (48 min)
S37	57	**Earth OCC**	RING	2008-027T18:21	Jan27	Sun	22	26	12	Egress = 2008-027T19:01 (40 min)
S37	57	**Sun OCC**	SATURN	2008-027T18:47	Jan27	Sun	22	26	12	Egress = 2008-027T19:44 (57 min)
S37	57	**Earth OCC**	SATURN	2008-027T19:03	Jan27	Sun	22	26	12	Egress = 2008-027T19:46 (43 min)
S37	57	Ring CRX	Descending	2008-027T19:14	Jan27	Sun	22	26	12	r = 3.452 Rs (208244 km)
S37	57	57AT (nt)	**ATLAS**	2008-027T19:15	Jan27	Sun	22	26	12	Inbound 75195.9 km flyby, v = 13.6 km/s, phase = 151 deg
S37	57	57EP (nt)	**EPIMETHEUS**	2008-027T20:17	Jan27	Sun	22	26	12	Inbound 89730.5 km flyby, v = 16.2 km/s, phase = 116 deg
S37	57	57PM (nt)	**PROMETHEUS**	2008-027T20:18	Jan27	Sun	22	26	12	Inbound 82364.0 km flyby, v = 15.4 km/s, phase = 125 deg
S37	57	57PA (nt)	**PANDORA**	2008-027T20:45	Jan27	Sun	22	26	12	Inbound 114391.8 km flyby, v = 17.3 km/s, phase = 109 deg
S37	57	Periapse		2008-027T20:47	Jan27	Sun	22	26	12	R = 3.257 Rs, lat = -21 deg, phase = 149 deg
S37	57	Ring CRX	Ascending	2008-029T19:10	Jan29	Tue	24	24	14	r = 19.764 Rs (1192387 km)
S37	57	**MAG**	**Segment**	2008-031T12:50	Jan31	Thu	26	22	16	MAG_57_58 (6d); MAG rev 57/58
S37	58	**Apoapse**		2008-033T20:08	Feb02	Sat	28	20	18	Per = 11.9 d, inc = 46.5 deg, r = 30.150 Rs, phase = 31 deg
S37	58	**OTM-144**	~APO	2008-036T20:06	Feb05	Tue	31	17	21	D/L start, burn ~6h later. Backup D/L 038T03:36
S37	58	**SATURN**	**Segment**	2008-037T05:06	Feb06	Wed	31	17	0	Saturn_058 (6d); Saturn_058
S37	58	**Sun OCC**	RING	2008-039T16:51	Feb08	Fri	34	14	3	Egress = 2008-039T17:37 (47 min)
S37	58	**Earth OCC**	RING	2008-039T16:56	Feb08	Fri	34	14	3	Egress = 2008-039T17:39 (43 min)
S37	58	58EP (nt)	**EPIMETHEUS**	2008-039T17:00	Feb08	Fri	34	14	3	Inbound 116981.6 km flyby, v = 13.4 km/s, phase = 121 deg
S37	58	**Sun OCC**	SATURN	2008-039T17:24	Feb08	Fri	34	14	3	Egress = 2008-039T18:23 (60 min)
S37	58	**Earth OCC**	SATURN	2008-039T17:33	Feb08	Fri	34	14	3	Egress = 2008-039T18:24 (51 min)
S37	58	Ring CRX	Descending	2008-039T17:52	Feb08	Fri	34	14	3	r = 3.461 Rs (208811 km)
S37	58	58PA (nt)	**PANDORA**	2008-039T19:23	Feb08	Fri	34	14	3	Inbound 114438.7 km flyby, v = 17.3 km/s, phase = 110 deg
S37	58	Periapse		2008-039T19:24	Feb08	Fri	34	14	3	R = 3.273 Rs, lat = -21 deg, phase = 149 deg
S37	58	58AT (nt)	**ATLAS**	2008-039T19:26	Feb08	Fri	34	14	3	Outbound 114718.7 km flyby, v = 17.2 km/s, phase = 110 deg
S37	58	Ring CRX	Ascending	2008-041T19:03	Feb10	Sun	36	12	5	r = 20.073 Rs (1211033 km)
S37	58	**MAG**	**Segment**	2008-043T12:06	Feb12	Tue	38	10	7	MAG_58_59 (6d); MAG rev 58/59
S37	59	**Apoapse**		2008-045T19:24	Feb14	Thu	40	8	9	Per = 12.0 d, inc = 47.0 deg, r = 30.236 Rs, phase = 30 deg
S38	59	**S38 Begins**		2008-047T11:51	Feb16	Sat	42	6	11	S38 Sequence. Duration = 36 d
S38	59	**SATURN**	**Segment**	2008-049T11:36	Feb18	Mon	44	4	13	Saturn_059 (4d); Saturn_059
S38	59	**OTM-145**	59TI-3	2008-050T02:36	Feb19	Tue	44	4	13	D/L start, burn ~6h later. Backup D/L 052T02:36
S38	59	**Earth OCC**	RING	2008-051T16:51	Feb20	Wed	46	2	2	Egress = 2008-051T17:36 (46 min)
S38	59	**Sun OCC**	RING	2008-051T16:51	Feb20	Wed	46	2	2	Egress = 2008-051T17:36 (46 min)
S38	59	**Sun OCC**	SATURN	2008-051T17:25	Feb20	Wed	46	2	2	Egress = 2008-051T18:21 (57 min)
S38	59	**Earth OCC**	SATURN	2008-051T17:28	Feb20	Wed	46	2	2	Egress = 2008-051T18:21 (53 min)
S38	59	59PN (nt)	**PAN**	2008-051T17:51	Feb20	Wed	46	2	2	Inbound 79784.7 km flyby, v = 13.8 km/s, phase = 152 deg
S38	59	Ring CRX	Descending	2008-051T17:51	Feb20	Wed	46	2	2	r = 3.459 Rs (208703 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S38	59	59PM (nt)	**PROMETHEUS**	2008-051T18:19	Feb20	Wed	46	2	2	Inbound 65853.1 km flyby, v = 14.2 km/s, phase = 152 deg
S38	59	59PA (nt)	**PANDORA**	2008-051T18:29	Feb20	Wed	46	2	2	Inbound 66345.2 km flyby, v = 14.5 km/s, phase = 145 deg
S38	59	59JA (nt)	**JANUS**	2008-051T19:08	Feb20	Wed	46	2	2	Inbound 110348.0 km flyby, v = 17.3 km/s, phase = 111 deg
S38	59	Periapse		2008-051T19:23	Feb20	Wed	46	2	2	R = 3.269 Rs, lat = -21 deg, phase = 150 deg
S38	59	**TOST**	**Segment**	2008-053T03:51	Feb22	Fri	47	1	3	TOST_059TI_T41 (2d); TOST_059TI (T41) Flyby
S38	59	**59TI (t) [T41]**	**TITAN**	2008-053T17:32	Feb22	Fri	48	19	4	Outbound 999.7 km flyby, v = 6.3 km/s, phase = 30 deg
S38	59	Ring CRX	Ascending	2008-053T18:22	Feb22	Fri	0	19	4	r = 19.908 Rs (1201040 km)
S38	59	**SATURN**	**Segment**	2008-055T11:21	Feb24	Sun	2	17	5	Saturn_059_060 (4d); Saturn Rev 59 Apoapsis
S38	60	**Apoapse**		2008-056T22:03	Feb25	Mon	3	16	7	Per = 10.6 d, inc = 56.1 deg, r = 27.074 Rs, phase = 31 deg
S38	60	**RINGS**	**Segment**	2008-059T10:51	Feb28	Thu	6	13	9	RINGS_Rev_60 (2d); RINGS_Rev_60
S38	60	**MAG**	**Segment**	2008-061T10:51	Mar01	Sat	8	11	11	MAG_60 (5d); MAG rev 60
S38	60	**OTM-146**	59TI+P	2008-061T16:56	Mar01	Sat	8	11	12	D/L start, burn ~6h later. Backup D/L 063T01:36
S38	60	**Earth OCC**	RING	2008-062T02:08	Mar02	Sun	8	11	0	Egress = 2008-062T02:53 (46 min)
S38	60	**Sun OCC**	RING	2008-062T02:12	Mar02	Sun	8	11	0	Egress = 2008-062T02:54 (42 min)
S38	60	**Earth OCC**	SATURN	2008-062T03:00	Mar02	Sun	8	11	0	Egress = 2008-062T03:13 (14 min)
S38	60	E_ring_lg	**DUST HAZARD**	2008-062T03:12	Mar02	Sun	8	11	0	Egress at 2008-062T03:14 (0.0 min); protection = MEA cover closure.
S38	60	Ring CRX	Descending	2008-062T03:13	Mar02	Sun	8	11	0	r = 3.943 Rs (237911 km)
S38	60	60DA (nt)	**DAPHNIS**	2008-062T03:45	Mar02	Sun	8	11	0	Inbound 97745.5 km flyby, v = 15.8 km/s, phase = 155 deg
S38	60	60PM (nt)	**PROMETHEUS**	2008-062T04:16	Mar02	Sun	8	11	0	Inbound 99680.5 km flyby, v = 16.3 km/s, phase = 141 deg
S38	60	Periapse		2008-062T05:04	Mar02	Sun	8	11	1	R = 3.747 Rs, lat = -23 deg, phase = 149 deg
S38	60	Ring CRX	Ascending	2008-064T08:33	Mar04	Tue	11	8	3	r = 19.987 Rs (1205797 km)
S38	60	**RINGS**	**Segment**	2008-066T10:21	Mar06	Thu	13	6	5	RINGS_Rev_60_61 (6d); RINGS Rev 60/61
S38	60	**OTM-147**	~APO	2008-067T01:21	Mar07	Fri	13	6	5	D/L start, burn ~6h later. Backup D/L 068T01:21
S38	61	**Apoapse**		2008-067T13:01	Mar07	Fri	14	5	0	Per = 10.7 d, inc = 56.1 deg, r = 27.222 Rs, phase = 31 deg
S38	61	**OTM-148**	61EN-3	2008-070T01:06	Mar10	Mon	16	3	3	D/L start, burn ~6h later. Backup D/L 071T01:06
S38	61	61TI (nt)	**TITAN**	2008-070T19:50	Mar10	Mon	17	2	1	Inbound 918311.7 km flyby, v = 6.3 km/s, phase = 124 deg
S38	61	**SOST**	**Segment**	2008-072T02:36	Mar12	Wed	18	1	2	SOST_061EN (1d); SOST segment in Rev 61 (targeted Enceladus)
S38	61	**Earth OCC**	RING	2008-072T17:57	Mar12	Wed	19	0	3	Egress = 2008-072T18:45 (48 min)
S38	61	**Sun OCC**	RING	2008-072T18:06	Mar12	Wed	19	0	3	Egress = 2008-072T18:47 (41 min)
S38	61	**Earth OCC**	SATURN	2008-072T18:43	Mar12	Wed	19	0	3	Egress = 2008-072T19:09 (26 min)
S38	61	E_ring_lg	**DUST HAZARD**	2008-072T19:05	Mar12	Wed	19	0	3	Egress at 2008-072T19:07 (0.0 min); protection = MEA cover closure.
S38	61	**Earth OCC**	ENCELADUS	2008-072T19:06	Mar12	Wed	19	0	3	Egress = 2008-072T19:06 (0 min)
S38	61	**Sun OCC**	ENCELADUS	2008-072T19:06	Mar12	Wed	19	0	3	Egress = 2008-072T19:06 (0 min)
S38	61	Ring CRX	Descending	2008-072T19:06	Mar12	Wed	19	0	3	r = 3.948 Rs (238198 km)
S38	61	**61EN (t) [E3]**	**ENCELADUS**	2008-072T19:06	Mar12	Wed	19	13	3	Inbound 47.9 km flyby, v = 14.4 km/s, phase = 137 deg
S38	61	61DA (nt)	**DAPHNIS**	2008-072T20:15	Mar12	Wed	0	13	3	Inbound 104055.9 km flyby, v = 16.4 km/s, phase = 139 deg
S38	61	Periapse		2008-072T20:57	Mar12	Wed	0	13	3	R = 3.751 Rs, lat = -23 deg, phase = 149 deg
S38	61	**RINGS**	**Segment**	2008-073T09:51	Mar13	Thu	1	12	3	RINGS_Rev_61_62 (12d); RINGS Rev 61/62
S38	61	**OTM-149**	61EN+1	2008-073T17:21	Mar13	Thu	1	12	4	D/L start, burn ~6h later. Backup D/L 075T00:51
S38	61	Ring CRX	Ascending	2008-075T00:16	Mar15	Sat	2	11	1	r = 19.937 Rs (1202819 km)
S38	61	**OTM-150**	~APO	2008-078T00:35	Mar18	Tue	5	8	4	D/L start, burn ~6h later. Backup D/L 079T00:35
S38	62	**Apoapse**		2008-078T04:34	Mar18	Tue	5	7	0	Per = 10.6 d, inc = 56.2 deg, r = 27.168 Rs, phase = 31 deg
S38	62	**OTM-151**	62TI-3	2008-082T16:50	Mar22	Sat	10	3	5	D/L start, burn ~6h later. Backup D/L 084T00:20
S39	62	**S39 Begins**		2008-083T01:50	Mar23	Sun	10	3	0	S39 Sequence. Duration = 27 d
S39	62	**Earth OCC**	RING	2008-083T09:08	Mar23	Sun	11	2	1	Egress = 2008-083T09:58 (50 min)
S39	62	**Sun OCC**	RING	2008-083T09:22	Mar23	Sun	11	2	1	Egress = 2008-083T10:01 (40 min)
S39	62	**Earth OCC**	SATURN	2008-083T09:51	Mar23	Sun	11	2	1	Egress = 2008-083T10:24 (33 min)
S39	62	E_ring_lg	**DUST HAZARD**	2008-083T10:19	Mar23	Sun	11	2	1	Egress at 2008-083T10:21 (0.0 min); protection = MEA cover closure.
S39	62	Ring CRX	Descending	2008-083T10:20	Mar23	Sun	11	2	1	r = 3.945 Rs (238020 km)
S39	62	62PL (nt)	**PALLENE**	2008-083T10:37	Mar23	Sun	11	2	1	Inbound 32685.5 km flyby, v = 15.0 km/s, phase = 135 deg
S39	62	62EP (nt)	**EPIMETHEUS**	2008-083T10:55	Mar23	Sun	11	2	1	Inbound 84178.8 km flyby, v = 15.6 km/s, phase = 152 deg
S39	62	62ME (nt)	**METHONE**	2008-083T11:24	Mar23	Sun	11	2	1	Inbound 102765.3 km flyby, v = 17.0 km/s, phase = 114 deg
S39	62	Periapse		2008-083T12:11	Mar23	Sun	11	2	1	R = 3.747 Rs, lat = -23 deg, phase = 149 deg
S39	62	**TOST**	**Segment**	2008-085T01:35	Mar25	Tue	12	1	2	TOST_062TI_T42 (2d); TOST_062TI (T42) Flyby
S39	62	**62TI (t) [T42]**	**TITAN**	2008-085T14:28	Mar25	Tue	13	48	3	Outbound 999.4 km flyby, v = 6.3 km/s, phase = 21 deg
S39	62	Ring CRX	Ascending	2008-085T15:10	Mar25	Tue	0	48	3	r = 19.859 Rs (1198064 km)
S39	62	**SATURN**	**Segment**	2008-087T01:20	Mar27	Thu	1	46	4	Saturn_062 (4d); Saturn_062/063 Apo
S39	63	**Apoapse**		2008-088T01:23	Mar28	Fri	2	45	5	Per = 9.6 d, inc = 63.1 deg, r = 24.525 Rs, phase = 31 deg
S39	63	**RINGS**	**Segment**	2008-091T08:34	Mar31	Mon	6	42	9	RINGS_Rev_63_64 (13d); RINGS Rev 63/64
S39	63	63TI (nt)	**TITAN**	2008-092T15:53	Apr01	Tue	7	41	10	Inbound 973782.7 km flyby, v = 12.9 km/s, phase = 36 deg
S39	63	**Earth OCC**	RING	2008-092T16:48	Apr01	Tue	7	41	10	Egress = 2008-092T17:39 (51 min)
S39	63	**Sun OCC**	RING	2008-092T17:07	Apr01	Tue	7	41	10	Egress = 2008-092T17:45 (38 min)
S39	63	63PL (nt)	**PALLENE**	2008-092T17:20	Apr01	Tue	7	41	10	Inbound 104465.7 km flyby, v = 14.8 km/s, phase = 123 deg
S39	63	Ring CRX	Descending	2008-092T18:09	Apr01	Tue	7	41	10	r = 4.490 Rs (270898 km)
S39	63	63AE (nt)	**AEGAEON**	2008-092T18:17	Apr01	Tue	7	41	10	Inbound 103559.2 km flyby, v = 15.9 km/s, phase = 157 deg
S39	63	63JA (nt)	**JANUS**	2008-092T19:09	Apr01	Tue	7	41	10	Inbound 117131.4 km flyby, v = 16.6 km/s, phase = 148 deg
S39	63	Periapse		2008-092T20:17	Apr01	Tue	7	41	10	R = 4.305 Rs, lat = -23 deg, phase = 149 deg
S39	63	Ring CRX	Ascending	2008-095T04:56	Apr04	Fri	10	38	13	r = 19.866 Rs (1198509 km)
S39	64	**Apoapse**		2008-097T15:15	Apr06	Sun	12	36	15	Per = 9.6 d, inc = 63.1 deg, r = 24.533 Rs, phase = 30 deg
S39	64	**OTM-152**	62TI+3d	2008-101T19:04	Apr10	Thu	16	32	19	D/L start, burn ~6h later. Backup D/L 102T22:48
S39	64	**Earth OCC**	RING	2008-102T06:42	Apr11	Fri	17	31	0	Egress = 2008-102T07:34 (52 min)
S39	64	**Sun OCC**	RING	2008-102T07:04	Apr11	Fri	17	31	1	Egress = 2008-102T07:41 (37 min)
S39	64	Ring CRX	Descending	2008-102T08:05	Apr11	Fri	17	31	1	r = 4.493 Rs (271035 km)
S39	64	64MI (nt)	**MIMAS**	2008-102T09:29	Apr11	Fri	17	31	1	Inbound 106553.0 km flyby, v = 17.1 km/s, phase = 129 deg
S39	64	Periapse		2008-102T10:13	Apr11	Fri	17	31	1	R = 4.307 Rs, lat = -23 deg, phase = 150 deg
S39	64	**SATURN**	**Segment**	2008-104T07:48	Apr13	Sun	19	29	3	Saturn_064 (6d); Saturn_064/065 Apo
S39	64	Ring CRX	Ascending	2008-104T18:44	Apr13	Sun	19	29	3	r = 19.830 Rs (1196317 km)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S39	65	**Apoapse**		2008-107T04:56	Apr16	Wed	22	26	5	Per = 9.6 d, inc = 63.1 deg, r = 24.493 Rs, phase = 30 deg
S40	65	**RINGS**	**Segment**	2008-110T07:18	Apr19	Sat	25	23	9	RINGS_Rev_65_66 (14d); RINGS Rev 65/66
S40	65	**S40 Begins**		2008-110T07:18	Apr19	Sat	25	23	9	S40 sequence. Duration = 42 d
S40	65	**Earth OCC**	RING	2008-111T20:08	Apr20	Sun	26	22	10	Egress = 2008-111T21:00 (53 min)
S40	65	**Sun OCC**	RING	2008-111T20:32	Apr20	Sun	26	22	10	Egress = 2008-111T21:08 (36 min)
S40	65	65MI (nt)	**MIMAS**	2008-111T20:44	Apr20	Sun	26	22	10	Inbound 116382.2 km flyby, v = 15.6 km/s, phase = 135 deg
S40	65	Ring CRX	Descending	2008-111T21:31	Apr20	Sun	26	22	10	r = 4.487 Rs (270725 km)
S40	65	65TL (nt)	**TELESTO**	2008-111T21:47	Apr20	Sun	26	22	10	Inbound 77619.0 km flyby, v = 15.5 km/s, phase = 88 deg
S40	65	65EP (nt)	**EPIMETHEUS**	2008-111T22:34	Apr20	Sun	26	21	10	Inbound 119100.9 km flyby, v = 16.8 km/s, phase = 148 deg
S40	65	Periapse		2008-111T23:39	Apr20	Sun	26	21	10	R = 4.301 Rs, lat = -23 deg, phase = 150 deg
S40	65	Ring CRX	Ascending	2008-114T08:05	Apr23	Wed	29	19	13	r = 19.816 Rs (1195507 km)
S40	66	**Apoapse**		2008-116T18:16	Apr25	Fri	31	17	15	Per = 9.6 d, inc = 63.1 deg, r = 24.485 Rs, phase = 30 deg
S40	66	**OTM-153**	~apo	2008-116T21:47	Apr25	Fri	31	17	15	D/L start, burn ~6h later. Backup D/L 117T21:47
S40	66	66TI (nt)	**TITAN**	2008-117T18:57	Apr26	Sat	32	16	1	Inbound 780171.4 km flyby, v = 5.5 km/s, phase = 94 deg
S40	66	**Earth OCC**	RING	2008-121T09:20	Apr30	Wed	36	12	4	Egress = 2008-121T10:13 (54 min)
S40	66	**Sun OCC**	RING	2008-121T09:46	Apr30	Wed	36	12	4	Egress = 2008-121T10:21 (35 min)
S40	66	66TL (nt)	**TELESTO**	2008-121T10:26	Apr30	Wed	36	12	5	Inbound 35745.3 km flyby, v = 14.5 km/s, phase = 45 deg
S40	66	Ring CRX	Descending	2008-121T10:44	Apr30	Wed	36	12	5	r = 4.480 Rs (270265 km)
S40	66	66PL (nt)	**PALLENE**	2008-121T11:51	Apr30	Wed	36	12	5	Inbound 93069.4 km flyby, v = 16.4 km/s, phase = 123 deg
S40	66	Periapse		2008-121T12:51	Apr30	Wed	36	12	5	R = 4.294 Rs, lat = -23 deg, phase = 150 deg
S40	66	Ring CRX	Ascending	2008-123T21:15	May02	Fri	38	10	7	r = 19.816 Rs (1195508 km)
S40	66	**SATURN**	**Segment**	2008-124T06:16	May03	Sat	39	9	7	Saturn_066 (6d); Saturn_066/067 Apo
S40	67	**Apoapse**		2008-126T07:32	May05	Mon	41	7	9	Per = 9.6 d, inc = 63.1 deg, r = 24.496 Rs, phase = 30 deg
S40	67	**OTM-154**	67TI-3d	2008-129T21:00	May08	Thu	44	4	13	D/L start, burn ~6h later. Backup D/L 130T12:00
S40	67	**RINGS**	**Segment**	2008-130T06:00	May09	Fri	45	3	0	RINGS_REV_67 (3d); RINGS_Rev_67
S40	67	**Earth OCC**	RING	2008-130T22:41	May09	Fri	45	2	1	Egress = 2008-130T23:34 (53 min)
S40	67	**Sun OCC**	RING	2008-130T23:09	May09	Fri	45	2	1	Egress = 2008-130T23:42 (34 min)
S40	67	67ME (nt)	**METHONE**	2008-130T23:36	May09	Fri	45	2	1	Inbound 95843.1 km flyby, v = 15.2 km/s, phase = 138 deg
S40	67	Ring CRX	Descending	2008-131T00:05	May10	Sat	45	2	1	r = 4.478 Rs (270164 km)
S40	67	67JA (nt)	**JANUS**	2008-131T00:06	May10	Sat	45	2	1	Inbound 120787.7 km flyby, v = 16.3 km/s, phase = 156 deg
S40	67	Periapse		2008-131T02:12	May10	Sat	45	2	1	R = 4.292 Rs, lat = -23 deg, phase = 150 deg
S40	67	**TOST**	**Segment**	2008-132T22:30	May11	Sun	47	0	3	TOST_067TI_T43 (2d); TOST_067TI (T43) Flyby
S40	67	**67TI (t) [T43]**	**TITAN**	2008-133T10:02	May12	Mon	48	16	4	Outbound 1001.4 km flyby, v = 6.3 km/s, phase = 36 deg
S40	67	Ring CRX	Ascending	2008-133T10:31	May12	Mon	0	16	4	r = 19.798 Rs (1194415 km)
S40	67	**RINGS**	**Segment**	2008-134T22:14	May13	Tue	2	14	5	RINGS_REV_67_68 (3d); RINGS_Rev_67_68
S40	68	**Apoapse**		2008-135T00:20	May14	Wed	2	14	5	Per = 8.0 d, inc = 69.3 deg, r = 22.293 Rs, phase = 24 deg
S40	68	**OTM-155**	67TI+5d	2008-137T19:20	May16	Fri	4	12	8	D/L start, burn ~6h later. Backup D/L 139T13:44
S40	68	**SATURN**	**Segment**	2008-138T04:20	May17	Sat	5	11	0	Saturn_068 (10d); Saturn_068_069
S40	68	**Earth OCC**	SATURN	2008-138T22:08	May17	Sat	6	10	1	Egress = 2008-138T23:19 (72 min)
S40	68	**Earth OCC**	RING	2008-138T22:09	May17	Sat	6	10	1	Egress = 2008-138T22:53 (45 min)
S40	68	**Sun OCC**	RING	2008-138T22:26	May17	Sat	6	10	1	Egress = 2008-138T22:56 (30 min)
S40	68	**Sun OCC**	SATURN	2008-138T22:44	May17	Sat	6	10	1	Egress = 2008-138T23:13 (30 min)
S40	68	68CA (nt)	**CALYPSO**	2008-138T23:03	May17	Sat	6	10	1	Inbound 111500.9 km flyby, v = 18.4 km/s, phase = 52 deg
S40	68	Ring CRX	Descending	2008-138T23:04	May17	Sat	6	10	1	r = 3.270 Rs (197266 km)
S40	68	68ME (nt)	**METHONE**	2008-138T23:36	May17	Sat	6	10	1	Inbound 78909.1 km flyby, v = 20.1 km/s, phase = 105 deg
S40	68	Periapse		2008-138T23:55	May17	Sat	6	10	1	R = 3.210 Rs, lat = -16 deg, phase = 156 deg
S40	68	68EP (nt)	**EPIMETHEUS**	2008-139T00:05	May18	Sun	6	10	1	Outbound 98913.1 km flyby, v = 21.5 km/s, phase = 118 deg
S40	68	Ring CRX	Ascending	2008-141T09:44	May20	Tue	8	8	4	r = 19.808 Rs (1195035 km)
S40	68	**OTM-156**	~apo	2008-142T20:13	May21	Wed	9	7	5	D/L start, burn ~6h later. Backup D/L 143T19:58
S40	69	**Apoapse**		2008-142T23:31	May21	Wed	10	6	0	Per = 8.0 d, inc = 69.3 deg, r = 22.291 Rs, phase = 24 deg
S40	69	**OTM-157**	69TI-3d	2008-145T19:58	May24	Sat	12	4	3	D/L start, burn ~6h later. Backup D/L 147T19:43
S40	69	**Earth OCC**	RING	2008-146T21:21	May25	Sun	13	2	1	Egress = 2008-146T22:05 (44 min)
S40	69	**Earth OCC**	SATURN	2008-146T21:21	May25	Sun	13	2	1	Egress = 2008-146T22:30 (70 min)
S40	69	**Sun OCC**	RING	2008-146T21:39	May25	Sun	13	2	1	Egress = 2008-146T22:08 (29 min)
S40	69	69AE (nt)	**AEGAEON**	2008-146T21:56	May25	Sun	13	2	1	Inbound 52211.8 km flyby, v = 19.0 km/s, phase = 119 deg
S40	69	**Sun OCC**	SATURN	2008-146T21:59	May25	Sun	13	2	1	Egress = 2008-146T22:22 (23 min)
S40	69	69PL (nt)	**PALLENE**	2008-146T22:06	May25	Sun	14	2	1	Inbound 28256.8 km flyby, v = 18.3 km/s, phase = 51 deg
S40	69	69AT (nt)	**ATLAS**	2008-146T22:14	May25	Sun	14	2	1	Inbound 60519.4 km flyby, v = 19.9 km/s, phase = 155 deg
S40	69	Ring CRX	Descending	2008-146T22:15	May25	Sun	14	2	1	r = 3.270 Rs (197262 km)
S40	69	69JA (nt)	**JANUS**	2008-146T22:50	May25	Sun	14	2	1	Inbound 63865.7 km flyby, v = 20.4 km/s, phase = 134 deg
S40	69	Periapse		2008-146T23:07	May25	Sun	14	2	1	R = 3.211 Rs, lat = -16 deg, phase = 156 deg
S40	69	**TOST**	**Segment**	2008-148T04:43	May27	Tue	15	1	2	TOST_069TI_T44 (3d); TOST_069TI (T44) Flyby
S40	69	**69TI (t) [T44]**	**TITAN**	2008-149T08:25	May28	Wed	16	64	4	Outbound 1400.0 km flyby, v = 6.3 km/s, phase = 21 deg
S40	69	Ring CRX	Ascending	2008-149T08:58	May28	Wed	0	64	4	r = 19.802 Rs (1194684 km)
S40	70	**Apoapse**		2008-150T09:16	May29	Thu	1	63	5	Per = 7.1 d, inc = 74.7 deg, r = 20.985 Rs, phase = 22 deg
S40	70	**RINGS**	**Segment**	2008-150T21:12	May29	Thu	2	62	5	RINGS_REV_70 (2d); RINGS_Rev_70
S41	70	**S41 Begins**		2008-152T04:27	May31	Sat	3	61	6	S41 sequence. Duration = 32 d
S41	70	**OTM-158**	69TI+4d	2008-152T18:27	May31	Sat	3	60	7	D/L start, burn ~6h later. Backup D/L 154T11:57
S41	70	**MAG**	**Segment**	2008-153T03:27	Jun01	Sun	4	60	0	MAG_70 (3d); MAG rev 70
S41	70	**Earth OCC**	SATURN	2008-153T21:33	Jun01	Sun	5	59	1	Egress = 2008-153T22:47 (74 min)
S41	70	**Earth OCC**	RING	2008-153T21:44	Jun01	Sun	5	59	1	Egress = 2008-153T22:22 (39 min)
S41	70	**Sun OCC**	SATURN	2008-153T21:53	Jun01	Sun	5	59	1	Egress = 2008-153T22:46 (53 min)
S41	70	**Sun OCC**	RING	2008-153T21:57	Jun01	Sun	5	59	1	Egress = 2008-153T22:23 (26 min)
S41	70	70PL (nt)	**PALLENE**	2008-153T22:01	Jun01	Sun	5	59	1	Inbound 115590.8 km flyby, v = 21.4 km/s, phase = 66 deg
S41	70	70JA (nt)	**JANUS**	2008-153T22:14	Jun01	Sun	5	59	1	Inbound 32923.3 km flyby, v = 22.1 km/s, phase = 109 deg
S41	70	G_ring	**DUST HAZARD**	2008-153T22:26	Jun01	Sun	5	59	1	Egress at 2008-153T22:26 (0.0 min); protection = MEA cover closure.
S41	70	Ring CRX	Descending	2008-153T22:26	Jun01	Sun	5	59	1	r = 2.731 Rs (164772 km)
S41	70	Periapse		2008-153T22:52	Jun01	Sun	5	59	1	R = 2.710 Rs, lat = -10 deg, phase = 158 deg
S41	70	**RINGS**	**Segment**	2008-156T04:11	Jun04	Wed	7	57	3	RINGS_REV_70_71 (4d); RINGS_Rev_70_71

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S41	70	Ring CRX	Ascending	2008-156T12:25	Jun04	Wed	7	57	4	r = 19.836 Rs (1196682 km)
S41	71	**Apoapse**		2008-157T12:31	Jun05	Thu	8	56	5	Per = 7.1 d, inc = 74.7 deg, r = 20.991 Rs, phase = 22 deg
S41	71	**MAG**	**Segment**	2008-160T02:45	Jun08	Sun	11	53	7	MAG_71 (2d); MAG rev 71
S41	71	71PN (nt)	**PAN**	2008-161T00:43	Jun09	Mon	12	52	8	Inbound 116078.3 km flyby, v = 23.6 km/s, phase = 103 deg
S41	71	**Earth OCC**	SATURN	2008-161T00:52	Jun09	Mon	12	52	8	Egress = 2008-161T02:05 (73 min)
S41	71	**Earth OCC**	RING	2008-161T01:03	Jun09	Mon	12	52	8	Egress = 2008-161T01:40 (38 min)
S41	71	**Sun OCC**	SATURN	2008-161T01:13	Jun09	Mon	12	52	8	Egress = 2008-161T02:03 (51 min)
S41	71	**Sun OCC**	RING	2008-161T01:16	Jun09	Mon	12	52	8	Egress = 2008-161T01:41 (26 min)
S41	71	G_ring	**DUST HAZARD**	2008-161T01:44	Jun09	Mon	12	52	8	Egress at 2008-161T01:44 (0.0 min); protection = MEA cover closure.
S41	71	Ring CRX	Descending	2008-161T01:44	Jun09	Mon	12	52	8	r = 2.730 Rs (164708 km)
S41	71	Periapse		2008-161T02:09	Jun09	Mon	12	52	8	R = 2.710 Rs, lat = -10 deg, phase = 158 deg
S41	71	71PA (nt)	**PANDORA**	2008-161T02:16	Jun09	Mon	12	52	8	Outbound 64064.0 km flyby, v = 23.5 km/s, phase = 119 deg
S41	71	**RINGS**	**Segment**	2008-162T02:50	Jun10	Tue	13	51	9	RINGS_REV_71_72 (5d); RINGS_Rev_71_72
S41	71	Ring CRX	Ascending	2008-163T15:56	Jun11	Wed	14	49	11	r = 19.855 Rs (1197859 km)
S41	72	**Apoapse**		2008-164T15:44	Jun12	Thu	15	48	12	Per = 7.1 d, inc = 74.7 deg, r = 20.985 Rs, phase = 22 deg
S41	72	72TI (nt)	**TITAN**	2008-165T04:28	Jun13	Fri	16	48	12	Inbound 364575.1 km flyby, v = 5.8 km/s, phase = 88 deg
S41	72	**SATURN**	**Segment**	2008-167T03:40	Jun15	Sun	18	46	14	Saturn_072 (7d); Saturn_072_073
S41	72	**Earth OCC**	SATURN	2008-168T04:00	Jun16	Mon	19	45	15	Egress = 2008-168T05:11 (71 min)
S41	72	**Earth OCC**	RING	2008-168T04:10	Jun16	Mon	19	45	15	Egress = 2008-168T04:47 (37 min)
S41	72	**Sun OCC**	SATURN	2008-168T04:20	Jun16	Mon	19	45	15	Egress = 2008-168T05:09 (50 min)
S41	72	**Sun OCC**	RING	2008-168T04:23	Jun16	Mon	19	45	15	Egress = 2008-168T04:48 (25 min)
S41	72	G_ring	**DUST HAZARD**	2008-168T04:50	Jun16	Mon	19	45	15	Egress at 2008-168T04:50 (0.0 min); protection = MEA cover closure.
S41	72	Ring CRX	Descending	2008-168T04:50	Jun16	Mon	19	45	15	r = 2.713 Rs (163704 km)
S41	72	72EP (nt)	**EPIMETHEUS**	2008-168T05:00	Jun16	Mon	19	45	15	Inbound 23095.8 km flyby, v = 22.5 km/s, phase = 127 deg
S41	72	72PM (nt)	**PROMETHEUS**	2008-168T05:03	Jun16	Mon	19	45	15	Inbound 31846.0 km flyby, v = 22.8 km/s, phase = 142 deg
S41	72	Periapse		2008-168T05:15	Jun16	Mon	19	45	15	R = 2.694 Rs, lat = -10 deg, phase = 158 deg
S41	72	72ME (nt)	**METHONE**	2008-168T05:19	Jun16	Mon	19	45	15	Outbound 113224.3 km flyby, v = 23.3 km/s, phase = 100 deg
S41	72	Ring CRX	Ascending	2008-170T19:09	Jun18	Wed	21	42	18	r = 19.875 Rs (1199066 km)
S41	73	**Apoapse**		2008-171T18:47	Jun19	Thu	22	41	19	Per = 7.1 d, inc = 74.7 deg, r = 20.989 Rs, phase = 22 deg
S41	73	**MAG**	**Segment**	2008-174T03:09	Jun22	Sun	25	39	21	MAG_73 (2d); MAG rev 73
S41	73	**OTM-159**	~apo	2008-175T00:24	Jun23	Mon	26	38	22	D/L start, burn ~6h later. Backup D/L 175T09:24
S41	73	**Earth OCC**	SATURN	2008-175T07:07	Jun23	Mon	26	38	0	Egress = 2008-175T08:16 (69 min)
S41	73	**Earth OCC**	RING	2008-175T07:17	Jun23	Mon	26	38	0	Egress = 2008-175T07:52 (36 min)
S41	73	**Sun OCC**	SATURN	2008-175T07:26	Jun23	Mon	26	38	0	Egress = 2008-175T08:14 (48 min)
S41	73	**Sun OCC**	RING	2008-175T07:28	Jun23	Mon	26	38	0	Egress = 2008-175T07:53 (25 min)
S41	73	G_ring	**DUST HAZARD**	2008-175T07:55	Jun23	Mon	26	38	0	Egress at 2008-175T07:55 (0.0 min); protection = MEA cover closure.
S41	73	Ring CRX	Descending	2008-175T07:55	Jun23	Mon	26	38	0	r = 2.711 Rs (163573 km)
S41	73	73ME (nt)	**METHONE**	2008-175T08:06	Jun23	Mon	26	38	0	Inbound 55473.8 km flyby, v = 21.9 km/s, phase = 76 deg
S41	73	73PN (nt)	**PAN**	2008-175T08:18	Jun23	Mon	26	38	0	Inbound 48788.9 km flyby, v = 23.3 km/s, phase = 132 deg
S41	73	Periapse		2008-175T08:20	Jun23	Mon	26	38	0	R = 2.692 Rs, lat = -10 deg, phase = 158 deg
S41	73	**SATURN**	**Segment**	2008-175T18:24	Jun23	Mon	26	37	1	Saturn_073 (6d); Saturn_073_074
S41	73	Ring CRX	Ascending	2008-177T21:40	Jun25	Wed	29	35	3	r = 19.705 Rs (1188785 km)
S41	74	**Apoapse**		2008-178T20:45	Jun26	Thu	30	34	4	Per = 7.0 d, inc = 74.7 deg, r = 20.785 Rs, phase = 22 deg
S41	74	74TI (nt)	**TITAN**	2008-180T15:58	Jun28	Sat	31	32	6	Inbound 729791.6 km flyby, v = 7.9 km/s, phase = 107 deg
S41	74	**MAG**	**Segment**	2008-181T19:08	Jun29	Sun	32	31	7	MAG_74 (2d); MAG rev 74
S41	74	**Earth OCC**	SATURN	2008-182T07:59	Jun30	Mon	33	31	7	Egress = 2008-182T09:06 (67 min)
S41	74	**Earth OCC**	RING	2008-182T08:08	Jun30	Mon	33	31	7	Egress = 2008-182T08:42 (34 min)
S41	74	74PM (nt)	**PROMETHEUS**	2008-182T08:16	Jun30	Mon	33	31	7	Inbound 65577.1 km flyby, v = 22.6 km/s, phase = 106 deg
S41	74	**Sun OCC**	SATURN	2008-182T08:18	Jun30	Mon	33	31	7	Egress = 2008-182T09:03 (46 min)
S41	74	**Sun OCC**	RING	2008-182T08:19	Jun30	Mon	33	31	7	Egress = 2008-182T08:43 (24 min)
S41	74	G_ring	**DUST HAZARD**	2008-182T08:45	Jun30	Mon	33	31	7	Egress at 2008-182T08:45 (0.0 min); protection = MEA cover closure.
S41	74	Ring CRX	Descending	2008-182T08:45	Jun30	Mon	33	31	7	r = 2.708 Rs (163395 km)
S41	74	74EN (nt)	**ENCELADUS**	2008-182T08:47	Jun30	Mon	33	31	7	Inbound 83739.2 km flyby, v = 21.1 km/s, phase = 49 deg
S41	74	74JA (nt)	**JANUS**	2008-182T08:58	Jun30	Mon	33	31	7	Inbound 29622.3 km flyby, v = 22.6 km/s, phase = 124 deg
S41	74	74ME (nt)	**METHONE**	2008-182T09:05	Jun30	Mon	33	31	7	Inbound 84849.9 km flyby, v = 22.5 km/s, phase = 92 deg
S41	74	Periapse		2008-182T09:09	Jun30	Mon	33	31	7	R = 2.689 Rs, lat = -10 deg, phase = 158 deg
S41	74	74AE (nt)	**AEGAEON**	2008-182T09:21	Jun30	Mon	33	31	7	Outbound 95502.8 km flyby, v = 23.7 km/s, phase = 107 deg
S41	74	74AT (nt)	**ATLAS**	2008-182T09:22	Jun30	Mon	33	31	7	Outbound 73450.8 km flyby, v = 24.0 km/s, phase = 119 deg
S42	74	**END OF PRIME MISSION**		2008-183T19:08	Jul01	Tue	34	29	9	End of prime mission / start of Equinox mission (2008-2010)
S42	74	**RINGS**	**Segment**	2008-183T19:08	Jul01	Tue	34	29	9	RINGS_REV_74_75 (8d); Rings Rev 74/75
S42	74	**S42 Begins**		2008-183T19:08	Jul01	Tue	34	29	9	S42 sequence. Duration = 40 d
S42	74	Ring CRX	Ascending	2008-184T22:39	Jul02	Wed	36	28	10	r = 19.728 Rs (1190172 km)
S42	75	**Apoapse**		2008-185T21:30	Jul03	Thu	37	27	11	Per = 7.0 d, inc = 74.7 deg, r = 20.785 Rs, phase = 22 deg
S42	75	**Earth OCC**	SATURN	2008-189T08:47	Jul07	Mon	40	24	14	Egress = 2008-189T09:51 (65 min)
S42	75	75JA (nt)	**JANUS**	2008-189T08:55	Jul07	Mon	40	24	14	Inbound 79292.7 km flyby, v = 22.6 km/s, phase = 94 deg
S42	75	**Earth OCC**	RING	2008-189T08:55	Jul07	Mon	40	24	14	Egress = 2008-189T09:27 (33 min)
S42	75	**Sun OCC**	SATURN	2008-189T09:04	Jul07	Mon	40	24	14	Egress = 2008-189T09:48 (44 min)
S42	75	**Sun OCC**	RING	2008-189T09:05	Jul07	Mon	40	24	14	Egress = 2008-189T09:28 (24 min)
S42	75	G_ring	**DUST HAZARD**	2008-189T09:30	Jul07	Mon	40	24	14	Egress at 2008-189T09:30 (0.0 min); protection = MEA cover closure.
S42	75	Ring CRX	Descending	2008-189T09:31	Jul07	Mon	40	24	14	r = 2.707 Rs (163306 km)
S42	75	75PA (nt)	**PANDORA**	2008-189T09:37	Jul07	Mon	40	24	14	Inbound 23846.6 km flyby, v = 22.6 km/s, phase = 156 deg
S42	75	Periapse		2008-189T09:54	Jul07	Mon	40	24	14	R = 2.688 Rs, lat = -10 deg, phase = 158 deg
S42	75	75ME (nt)	**METHONE**	2008-189T10:01	Jul07	Mon	40	24	14	Outbound 118376.1 km flyby, v = 23.4 km/s, phase = 102 deg
S42	75	Ring CRX	Ascending	2008-191T23:43	Jul09	Wed	43	21	17	r = 19.751 Rs (1191557 km)
S42	75	**SATURN**	**Segment**	2008-192T02:07	Jul10	Thu	43	21	17	Saturn_076 (10d); Saturn_075_076_077
S42	76	**Apoapse**		2008-192T22:19	Jul10	Thu	44	20	18	Per = 7.0 d, inc = 74.7 deg, r = 20.786 Rs, phase = 22 deg
S42	76	**Earth OCC**	SATURN	2008-196T09:39	Jul14	Mon	47	17	21	Egress = 2008-196T10:41 (62 min)
S42	76	**Earth OCC**	RING	2008-196T09:46	Jul14	Mon	47	17	21	Egress = 2008-196T10:17 (31 min)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S42	76	Sun OCC	RING	2008-196T09:55	Jul14	Mon	47	17	21	Egress = 2008-196T10:18 (23 min)
S42	76	Sun OCC	SATURN	2008-196T09:56	Jul14	Mon	47	17	21	Egress = 2008-196T10:37 (42 min)
S42	76	76DA (nt)	DAPHNIS	2008-196T10:05	Jul14	Mon	47	17	21	Inbound 42589.1 km flyby, v = 22.6 km/s, phase = 121 deg
S42	76	76PM (nt)	PROMETHEUS	2008-196T10:12	Jul14	Mon	47	17	21	Inbound 31440.5 km flyby, v = 22.5 km/s, phase = 130 deg
S42	76	G_ring	DUST HAZARD	2008-196T10:20	Jul14	Mon	47	17	21	Egress at 2008-196T10:20 (0.0 min); protection = MEA cover closure.
S42	76	Ring CRX	Descending	2008-196T10:21	Jul14	Mon	47	17	21	r = 2.707 Rs (163332 km)
S42	76	76PL (nt)	PALLENE	2008-196T10:35	Jul14	Mon	47	17	21	Inbound 93050.4 km flyby, v = 22.1 km/s, phase = 85 deg
S42	76	Periapse		2008-196T10:44	Jul14	Mon	47	17	21	R = 2.689 Rs, lat = -10 deg, phase = 158 deg
S42	76	76TI (nt)	TITAN	2008-197T21:25	Jul15	Tue	49	15	23	Outbound 542450.1 km flyby, v = 8.4 km/s, phase = 127 deg
S42	76	Ring CRX	Ascending	2008-199T00:49	Jul17	Thu	50	14	24	r = 19.779 Rs (1193266 km)
S42	77	Apoapse		2008-199T23:11	Jul17	Thu	51	13	25	Per = 7.0 d, inc = 74.7 deg, r = 20.792 Rs, phase = 22 deg
S42	77	RINGS	Segment	2008-202T01:21	Jul20	Sun	53	11	27	RINGS_REV_77 (4d); Rings Rev 77
S42	77	Earth OCC	SATURN	2008-203T10:37	Jul21	Mon	54	10	28	Egress = 2008-203T11:35 (59 min)
S42	77	Earth OCC	RING	2008-203T10:43	Jul21	Mon	54	10	28	Egress = 2008-203T11:12 (30 min)
S42	77	Sun OCC	RING	2008-203T10:50	Jul21	Mon	54	10	28	Egress = 2008-203T11:13 (23 min)
S42	77	Sun OCC	SATURN	2008-203T10:52	Jul21	Mon	54	10	28	Egress = 2008-203T11:31 (40 min)
S42	77	77PL (nt)	PALLENE	2008-203T11:06	Jul21	Mon	54	10	28	Inbound 59024.9 km flyby, v = 21.0 km/s, phase = 26 deg
S42	77	77AE (nt)	AEGAEON	2008-203T11:11	Jul21	Mon	54	10	28	Inbound 11577.8 km flyby, v = 21.8 km/s, phase = 59 deg
S42	77	G_ring	DUST HAZARD	2008-203T11:15	Jul21	Mon	54	10	28	Egress at 2008-203T11:15 (0.0 min); protection = MEA cover closure.
S42	77	Ring CRX	Descending	2008-203T11:15	Jul21	Mon	54	10	28	r = 2.710 Rs (163480 km)
S42	77	77AT (nt)	ATLAS	2008-203T11:24	Jul21	Mon	54	10	28	Inbound 28464.8 km flyby, v = 22.7 km/s, phase = 155 deg
S42	77	Periapse		2008-203T11:39	Jul21	Mon	54	10	28	R = 2.692 Rs, lat = -10 deg, phase = 158 deg
S42	77	77DA (nt)	DAPHNIS	2008-203T12:00	Jul21	Mon	54	10	28	Outbound 86141.4 km flyby, v = 24.4 km/s, phase = 118 deg
S42	77	77PN (nt)	PAN	2008-203T12:04	Jul21	Mon	54	10	28	Outbound 91246.9 km flyby, v = 24.6 km/s, phase = 118 deg
S42	77	SATURN	Segment	2008-206T01:21	Jul24	Thu	57	7	31	Saturn_078 (3d); Saturn_077_078
S42	77	Ring CRX	Ascending	2008-206T02:01	Jul24	Thu	57	7	31	r = 19.806 Rs (1194901 km)
S42	78	Apoapse		2008-207T00:08	Jul25	Fri	58	6	32	Per = 7.0 d, inc = 74.7 deg, r = 20.796 Rs, phase = 22 deg
S42	78	RINGS	Segment	2008-209T01:06	Jul27	Sun	60	4	34	RINGS_REV_78 (3d); Rings Rev 78
S42	78	OTM-160	78TI-3d	2008-209T08:36	Jul27	Sun	60	4	34	D/L start, burn ~6h later. Backup D/L 210T17:00
S42	78	78PN (nt)	PAN	2008-210T11:26	Jul28	Mon	61	3	1	Inbound 97453.9 km flyby, v = 23.3 km/s, phase = 102 deg
S42	78	Earth OCC	SATURN	2008-210T11:39	Jul28	Mon	61	3	1	Egress = 2008-210T12:34 (55 min)
S42	78	Earth OCC	RING	2008-210T11:44	Jul28	Mon	61	3	1	Egress = 2008-210T12:12 (28 min)
S42	78	Sun OCC	RING	2008-210T11:50	Jul28	Mon	61	3	1	Egress = 2008-210T12:12 (22 min)
S42	78	Sun OCC	SATURN	2008-210T11:53	Jul28	Mon	61	3	1	Egress = 2008-210T12:30 (37 min)
S42	78	G_ring	DUST HAZARD	2008-210T12:14	Jul28	Mon	61	3	1	Egress at 2008-210T12:14 (0.0 min); protection = MEA cover closure.
S42	78	Ring CRX	Descending	2008-210T12:15	Jul28	Mon	61	3	1	r = 2.710 Rs (163488 km)
S42	78	78PM (nt)	PROMETHEUS	2008-210T12:19	Jul28	Mon	61	3	1	Inbound 24662.9 km flyby, v = 22.6 km/s, phase = 160 deg
S42	78	Periapse		2008-210T12:38	Jul28	Mon	61	3	1	R = 2.693 Rs, lat = -10 deg, phase = 158 deg
S42	78	78EP (nt)	EPIMETHEUS	2008-210T13:02	Jul28	Mon	61	3	1	Outbound 106965.4 km flyby, v = 24.6 km/s, phase = 113 deg
S42	78	TOST	Segment	2008-212T07:20	Jul30	Wed	63	1	3	TOST_78_79_T45 (2d); Titan Flyby T45
S42	78	78TI (t) [T45]	TITAN	2008-213T02:13	Jul31	Thu	64	12	4	Outbound 1613.4 km flyby, v = 6.2 km/s, phase = 49 deg
S42	78	Ring CRX	Ascending	2008-213T03:01	Jul31	Thu	0	12	4	r = 19.792 Rs (1194063 km)
S42	79	Apoapse		2008-213T19:52	Jul31	Thu	1	11	4	Per = 7.4 d, inc = 74.5 deg, r = 20.317 Rs, phase = 22 deg
S42	79	RINGS	Segment	2008-214T00:50	Aug01	Fri	1	11	5	RINGS_79_80 (10d); RINGS_79_80
S42	79	OTM-162	78TI+4d	2008-216T16:15	Aug03	Sun	4	8	7	D/L start, burn ~6h later. Backup D/L 218T08:05
S42	79	79MI (nt)	MIMAS	2008-217T10:28	Aug04	Mon	4	7	1	Inbound 122382.3 km flyby, v = 19.0 km/s, phase = 110 deg
S42	79	Earth OCC	RING	2008-217T11:10	Aug04	Mon	4	7	1	Egress = 2008-217T11:39 (30 min)
S42	79	Sun OCC	RING	2008-217T11:18	Aug04	Mon	4	7	1	Egress = 2008-217T11:41 (24 min)
S42	79	E_ring_lg	DUST HAZARD	2008-217T11:53	Aug04	Mon	4	7	1	Egress at 2008-217T11:54 (0.0 min); protection = MEA cover closure.
S42	79	Ring CRX	Descending	2008-217T11:54	Aug04	Mon	4	7	1	r = 3.962 Rs (239025 km)
S42	79	Periapse		2008-217T12:33	Aug04	Mon	4	7	1	R = 3.942 Rs, lat = -9 deg, phase = 158 deg
S42	79	79EP (nt)	EPIMETHEUS	2008-217T12:49	Aug04	Mon	4	7	1	Outbound 101808.7 km flyby, v = 19.8 km/s, phase = 143 deg
S42	79	Ring CRX	Ascending	2008-220T12:21	Aug07	Thu	7	4	4	r = 19.781 Rs (1193365 km)
S42	80	Apoapse		2008-221T05:09	Aug08	Fri	8	4	5	Per = 7.4 d, inc = 74.4 deg, r = 20.301 Rs, phase = 22 deg
S42	80	OTM-163	E4-3d	2008-221T15:20	Aug08	Fri	9	3	5	D/L start, burn ~6h later. Backup D/L 222T15:20
S43	80	SOST	Segment	2008-224T00:20	Aug11	Mon	11	1	2	SOST_80 (2d); E4 Flyby
S43	80	S43 Begins		2008-224T00:20	Aug11	Mon	11	1	2	S43 sequence. Duration = 34 d
S43	80	Earth OCC	RING	2008-224T20:25	Aug11	Mon	12	0	3	Egress = 2008-224T20:53 (28 min)
S43	80	Sun OCC	RING	2008-224T20:31	Aug11	Mon	12	0	3	Egress = 2008-224T20:54 (23 min)
S43	80	80ME (nt)	METHONE	2008-224T20:41	Aug11	Mon	12	0	3	Inbound 60464.2 km flyby, v = 18.4 km/s, phase = 127 deg
S43	80	E_ring_lg	DUST HAZARD	2008-224T21:05	Aug11	Mon	12	0	3	Egress at 2008-224T21:07 (0.0 min); protection = MEA cover closure.
S43	80	Ring CRX	Descending	2008-224T21:06	Aug11	Mon	12	0	3	r = 3.961 Rs (238973 km)
S43	80	80EN (t) [E4]	ENCELADUS	2008-224T21:06	Aug11	Mon	12	59	3	Inbound 45.8 km flyby, v = 17.7 km/s, phase = 110 deg
S43	80	Periapse		2008-224T21:45	Aug11	Mon	0	59	3	R = 3.941 Rs, lat = -9 deg, phase = 158 deg
S43	80	80MI (nt)	MIMAS	2008-224T21:49	Aug11	Mon	0	59	3	Outbound 81523.0 km flyby, v = 19.1 km/s, phase = 137 deg
S43	80	80DA (nt)	DAPHNIS	2008-224T21:56	Aug11	Mon	0	59	3	Outbound 110003.9 km flyby, v = 20.3 km/s, phase = 149 deg
S43	80	RINGS	Segment	2008-226T00:04	Aug13	Wed	1	58	4	RINGS_80_82 (10d); RINGS_80_82
S43	80	Ring CRX	Ascending	2008-227T21:30	Aug14	Thu	3	56	6	r = 19.755 Rs (1191849 km)
S43	81	Apoapse		2008-228T14:08	Aug15	Fri	4	55	7	Per = 7.4 d, inc = 74.5 deg, r = 20.267 Rs, phase = 22 deg
S43	81	81TI (nt)	TITAN	2008-228T20:47	Aug15	Fri	4	55	7	Inbound 302605.5 km flyby, v = 5.9 km/s, phase = 85 deg
S43	81	Earth OCC	RING	2008-232T05:08	Aug19	Tue	7	52	11	Egress = 2008-232T05:33 (25 min)
S43	81	Sun OCC	RING	2008-232T05:12	Aug19	Tue	7	52	11	Egress = 2008-232T05:34 (22 min)
S43	81	E_ring_lg	DUST HAZARD	2008-232T05:45	Aug19	Tue	7	52	11	Egress at 2008-232T05:46 (0.0 min); protection = MEA cover closure.
S43	81	Ring CRX	Descending	2008-232T05:46	Aug19	Tue	7	52	11	r = 3.935 Rs (237405 km)
S43	81	81PL (nt)	PALLENE	2008-232T06:10	Aug19	Tue	7	52	11	Inbound 44152.3 km flyby, v = 18.4 km/s, phase = 134 deg
S43	81	Periapse		2008-232T06:24	Aug19	Tue	7	52	11	R = 3.915 Rs, lat = -9 deg, phase = 158 deg
S43	81	81PN (nt)	PAN	2008-232T06:44	Aug19	Tue	7	52	11	Outbound 114866.0 km flyby, v = 20.5 km/s, phase = 146 deg
S43	81	Ring CRX	Ascending	2008-235T06:06	Aug22	Fri	10	49	14	r = 19.761 Rs (1192168 km)
S43	81	OTM-164	E4+11d	2008-235T20:49	Aug22	Fri	11	48	14	D/L start, burn ~6h later. Backup D/L 236T20:49

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S43	82	**Apoapse**		2008-235T22:41	Aug22	Fri	11	48	0	Per = 7.4 d, inc = 74.5 deg, r = 20.271 Rs, phase = 23 deg
S43	82	**MAG**	**Segment**	2008-236T05:49	Aug23	Sat	11	48	0	MAG_82_84 (22d); MAG_82_84
S43	82	**Earth OCC**	RING	2008-239T13:57	Aug26	Tue	15	44	4	Egress = 2008-239T14:20 (23 min)
S43	82	**Sun OCC**	RING	2008-239T13:59	Aug26	Tue	15	44	4	Egress = 2008-239T14:20 (22 min)
S43	82	82AT (nt)	**ATLAS**	2008-239T14:08	Aug26	Tue	15	44	4	Inbound 107394.5 km flyby, v = 19.9 km/s, phase = 150 deg
S43	82	E_ring_lg	**DUST HAZARD**	2008-239T14:31	Aug26	Tue	15	44	4	Egress at 2008-239T14:33 (0.0 min); protection = MEA cover closure.
S43	82	Ring CRX	Descending	2008-239T14:32	Aug26	Tue	15	44	4	r = 3.976 Rs (239850 km)
S43	82	Periapse		2008-239T15:11	Aug26	Tue	15	44	4	R = 3.956 Rs, lat = -9 deg, phase = 157 deg
S43	82	Ring CRX	Ascending	2008-242T15:18	Aug29	Fri	18	41	7	r = 19.773 Rs (1192882 km)
S43	83	**Apoapse**		2008-243T07:39	Aug30	Sat	18	40	7	Per = 7.4 d, inc = 74.5 deg, r = 20.267 Rs, phase = 23 deg
S43	83	83TI (nt)	**TITAN**	2008-244T13:12	Aug31	Sun	20	39	9	Inbound 603610.8 km flyby, v = 6.9 km/s, phase = 97 deg
S43	83	SEP = 3.0 deg	Conjunction	2008-245T03:20	Sep01	Mon	20	39	9	Commanding must not be required below 3.0 deg
S43	83	SEP = 2.0 deg	Conjunction	2008-246T18:49	Sep02	Tue	22	37	11	Plan no SSR playback below 2.0 deg
S43	83	**Sun OCC**	RING	2008-246T22:56	Sep02	Tue	22	37	11	Egress = 2008-246T23:17 (21 min)
S43	83	**Earth OCC**	RING	2008-246T22:57	Sep02	Tue	22	37	11	Egress = 2008-246T23:17 (21 min)
S43	83	E_ring_lg	**DUST HAZARD**	2008-246T23:27	Sep02	Tue	22	37	11	Egress at 2008-246T23:29 (0.0 min); protection = MEA cover closure.
S43	83	83JA (nt)	**JANUS**	2008-246T23:28	Sep02	Tue	22	37	11	Inbound 87030.4 km flyby, v = 19.4 km/s, phase = 156 deg
S43	83	Ring CRX	Descending	2008-246T23:28	Sep02	Tue	22	37	11	r = 3.968 Rs (239389 km)
S43	83	Periapse		2008-247T00:06	Sep03	Wed	22	37	11	R = 3.949 Rs, lat = -9 deg, phase = 157 deg
S43	83	83ME (nt)	**METHONE**	2008-247T00:26	Sep03	Wed	22	37	11	Outbound 90915.2 km flyby, v = 19.2 km/s, phase = 128 deg
S43	83	SEP = 2.0 deg	Conjunction	2008-249T09:22	Sep05	Fri	25	34	14	Plan no SSR playback below 2.0 deg
S43	83	Ring CRX	Ascending	2008-250T00:17	Sep06	Sat	25	34	14	r = 19.777 Rs (1193150 km)
S43	84	**Apoapse**		2008-250T16:33	Sep06	Sat	26	33	15	Per = 7.4 d, inc = 74.5 deg, r = 20.266 Rs, phase = 23 deg
S43	84	SEP = 3.0 deg	Conjunction	2008-251T00:37	Sep07	Sun	26	33	15	Commanding must not be required below 3.0 deg
S43	84	**Sun OCC**	RING	2008-254T07:50	Sep10	Wed	29	29	18	Egress = 2008-254T08:10 (21 min)
S43	84	**Earth OCC**	RING	2008-254T07:53	Sep10	Wed	29	29	18	Egress = 2008-254T08:11 (19 min)
S43	84	E_ring_lg	**DUST HAZARD**	2008-254T08:21	Sep10	Wed	29	29	18	Egress at 2008-254T08:22 (0.0 min); protection = MEA cover closure.
S43	84	Ring CRX	Descending	2008-254T08:22	Sep10	Wed	29	29	18	r = 3.966 Rs (239249 km)
S43	84	84AE (nt)	**AEGAEON**	2008-254T08:45	Sep10	Wed	29	29	18	Inbound 74493.7 km flyby, v = 19.1 km/s, phase = 156 deg
S43	84	84EP (nt)	**EPIMETHEUS**	2008-254T08:51	Sep10	Wed	29	29	19	Inbound 91395.5 km flyby, v = 19.6 km/s, phase = 155 deg
S43	84	Periapse		2008-254T08:59	Sep10	Wed	29	29	19	R = 3.947 Rs, lat = -9 deg, phase = 157 deg
S43	84	Ring CRX	Ascending	2008-257T09:14	Sep13	Sat	33	26	22	r = 19.780 Rs (1193356 km)
S44	84	**RINGS**	**Segment**	2008-257T22:19	Sep13	Sat	33	26	22	RINGS_84_87 (15d); RINGS_84_87
S44	84	**S44 Begins**		2008-257T22:19	Sep13	Sat	33	26	22	S44 sequence. Duration = 35 d
S44	85	**Apoapse**		2008-258T01:24	Sep14	Sun	33	26	22	Per = 7.4 d, inc = 74.5 deg, r = 20.264 Rs, phase = 23 deg
S44	85	85TI (nt)	**TITAN**	2008-259T23:39	Sep15	Mon	35	24	24	Inbound 902767.5 km flyby, v = 8.7 km/s, phase = 99 deg
S44	85	85AE (nt)	**AEGAEON**	2008-261T16:08	Sep17	Wed	37	22	26	Inbound 109583.9 km flyby, v = 19.2 km/s, phase = 121 deg
S44	85	85PA (nt)	**PANDORA**	2008-261T16:37	Sep17	Wed	37	22	26	Inbound 106669.4 km flyby, v = 19.7 km/s, phase = 143 deg
S44	85	**Sun OCC**	RING	2008-261T16:41	Sep17	Wed	37	22	26	Egress = 2008-261T17:00 (20 min)
S44	85	**Earth OCC**	RING	2008-261T16:46	Sep17	Wed	37	22	26	Egress = 2008-261T17:02 (17 min)
S44	85	85MI (nt)	**MIMAS**	2008-261T16:50	Sep17	Wed	37	22	26	Inbound 61673.0 km flyby, v = 18.3 km/s, phase = 139 deg
S44	85	E_ring_lg	**DUST HAZARD**	2008-261T17:11	Sep17	Wed	37	22	26	Egress at 2008-261T17:12 (0.0 min); protection = MEA cover closure.
S44	85	Ring CRX	Descending	2008-261T17:12	Sep17	Wed	37	22	26	r = 3.965 Rs (239185 km)
S44	85	85CA (nt)	**CALYPSO**	2008-261T17:20	Sep17	Wed	37	22	26	Inbound 60484.0 km flyby, v = 17.5 km/s, phase = 45 deg
S44	85	Periapse		2008-261T17:49	Sep17	Wed	37	22	26	R = 3.946 Rs, lat = -8 deg, phase = 157 deg
S44	85	85DA (nt)	**DAPHNIS**	2008-261T17:56	Sep17	Wed	37	22	26	Outbound 108431.4 km flyby, v = 20.2 km/s, phase = 151 deg
S44	85	**OTM-164A**	E4+40	2008-264T12:49	Sep20	Sat	40	19	29	D/L start, burn ~6h later. Backup D/L 265T12:49
S44	85	Ring CRX	Ascending	2008-264T18:11	Sep20	Sat	40	19	0	r = 19.788 Rs (1193826 km)
S44	86	**Apoapse**		2008-265T10:14	Sep21	Sun	41	18	1	Per = 7.4 d, inc = 74.5 deg, r = 20.264 Rs, phase = 23 deg
S44	86	86TE (nt)	**TETHYS**	2008-269T01:25	Sep25	Thu	44	15	5	Inbound 110411.6 km flyby, v = 17.5 km/s, phase = 50 deg
S44	86	**Sun OCC**	RING	2008-269T01:32	Sep25	Thu	44	15	5	Egress = 2008-269T01:50 (19 min)
S44	86	**Earth OCC**	RING	2008-269T01:38	Sep25	Thu	44	15	5	Egress = 2008-269T01:52 (15 min)
S44	86	86AT (nt)	**ATLAS**	2008-269T01:41	Sep25	Thu	44	15	5	Inbound 105294.8 km flyby, v = 19.9 km/s, phase = 150 deg
S44	86	E_ring_lg	**DUST HAZARD**	2008-269T02:00	Sep25	Thu	44	15	5	Egress at 2008-269T02:02 (0.0 min); protection = MEA cover closure.
S44	86	Ring CRX	Descending	2008-269T02:01	Sep25	Thu	44	15	5	r = 3.963 Rs (239070 km)
S44	86	86PN (nt)	**PAN**	2008-269T02:14	Sep25	Thu	44	15	5	Inbound 105237.0 km flyby, v = 20.1 km/s, phase = 157 deg
S44	86	Periapse		2008-269T02:38	Sep25	Thu	44	15	5	R = 3.944 Rs, lat = -8 deg, phase = 157 deg
S44	86	86PL (nt)	**PALLENE**	2008-269T03:12	Sep25	Thu	44	15	5	Outbound 114653.0 km flyby, v = 19.3 km/s, phase = 118 deg
S44	86	Ring CRX	Ascending	2008-272T03:06	Sep28	Sun	47	12	8	r = 19.794 Rs (1194165 km)
S44	87	**Apoapse**		2008-272T19:03	Sep28	Sun	48	11	8	Per = 7.4 d, inc = 74.5 deg, r = 20.264 Rs, phase = 23 deg
S44	87	**MAG**	**Segment**	2008-272T21:20	Sep28	Sun	48	11	8	MAG_87 (7d); MAG_87
S44	87	**OTM-165**	E5-7d	2008-276T04:19	Oct02	Thu	51	8	12	D/L start, burn ~6h later. Backup D/L 277T04:35
S44	87	**Sun OCC**	RING	2008-276T10:22	Oct02	Thu	52	7	0	Egress = 2008-276T10:40 (19 min)
S44	87	**Earth OCC**	RING	2008-276T10:30	Oct02	Thu	52	7	0	Egress = 2008-276T10:43 (13 min)
S44	87	E_ring_lg	**DUST HAZARD**	2008-276T10:50	Oct02	Thu	52	7	0	Egress at 2008-276T10:51 (0.0 min); protection = MEA cover closure.
S44	87	Ring CRX	Descending	2008-276T10:51	Oct02	Thu	52	7	0	r = 3.964 Rs (239130 km)
S44	87	87TE (nt)	**TETHYS**	2008-276T11:04	Oct02	Thu	52	7	0	Inbound 78633.8 km flyby, v = 17.6 km/s, phase = 70 deg
S44	87	Periapse		2008-276T11:28	Oct02	Thu	52	7	0	R = 3.945 Rs, lat = -8 deg, phase = 157 deg
S44	87	87TI (nt)	**TITAN**	2008-277T23:03	Oct03	Fri	53	6	2	Outbound 780488.0 km flyby, v = 9.3 km/s, phase = 125 deg
S44	87	**RINGS**	**Segment**	2008-279T11:05	Oct05	Sun	55	4	3	RINGS_87_88 (3d); RINGS_87_88
S44	87	Ring CRX	Ascending	2008-279T11:50	Oct05	Sun	55	4	3	r = 19.751 Rs (1191577 km)
S44	88	**Apoapse**		2008-280T03:35	Oct06	Mon	55	4	4	Per = 7.3 d, inc = 74.5 deg, r = 20.212 Rs, phase = 23 deg
S44	88	**OTM-166**	E5-3d	2008-280T12:05	Oct06	Mon	56	3	4	D/L start, burn ~6h later. Backup D/L 281T11:50
S44	88	**SOST**	**Segment**	2008-282T20:50	Oct08	Wed	58	1	2	SOST_88 (2d); E5 Flyby
S44	88	**Sun OCC**	RING	2008-283T18:38	Oct09	Thu	59	0	3	Egress = 2008-283T18:56 (18 min)
S44	88	88TL (nt)	**TELESTO**	2008-283T18:46	Oct09	Thu	59	0	3	Inbound 67249.8 km flyby, v = 17.2 km/s, phase = 22 deg
S44	88	**Earth OCC**	RING	2008-283T18:49	Oct09	Thu	59	0	3	Egress = 2008-283T18:59 (11 min)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S44	88	88JA (nt)	**JANUS**	2008-283T18:56	Oct09	Thu	59	0	3	Inbound 89531.3 km flyby, v = 19.4 km/s, phase = 151 deg
S44	88	E_ring_lg	**DUST HAZARD**	2008-283T19:05	Oct09	Thu	59	0	3	Egress at 2008-283T19:07 (0.0 min); protection = MEA cover closure.
S44	88	**Earth OCC**	ENCELADUS	2008-283T19:07	Oct09	Thu	59	0	3	Egress = 2008-283T19:06 (0 min)
S44	88	**Sun OCC**	ENCELADUS	2008-283T19:07	Oct09	Thu	59	0	3	Egress = 2008-283T19:06 (0 min)
S44	88	Ring CRX	Descending	2008-283T19:07	Oct09	Thu	59	0	3	r = 3.964 Rs (239162 km)
S44	88	**88EN (t) [E5]**	**ENCELADUS**	2008-283T19:07	Oct09	Thu	59	22	3	Inbound 19.2 km flyby, v = 17.7 km/s, phase = 113 deg
S44	88	Periapse		2008-283T19:43	Oct09	Thu	0	22	3	R = 3.946 Rs, lat = -8 deg, phase = 157 deg
S44	88	**RINGS**	**Segment**	2008-284T20:50	Oct10	Fri	1	21	4	RINGS_88_89 (8d); RINGS_88_89
S44	88	**OTM-167**	E5+3d	2008-286T17:51	Oct12	Sun	3	19	6	D/L start, burn ~6h later. Backup D/L 287T11:36
S44	88	Ring CRX	Ascending	2008-286T20:04	Oct12	Sun	3	19	0	r = 19.730 Rs (1190281 km)
S44	89	**Apoapse**		2008-287T11:42	Oct13	Mon	4	18	1	Per = 7.3 d, inc = 74.5 deg, r = 20.184 Rs, phase = 23 deg
S44	89	**Sun OCC**	RING	2008-291T02:34	Oct17	Fri	7	15	4	Egress = 2008-291T02:51 (18 min)
S44	89	**Earth OCC**	RING	2008-291T02:46	Oct17	Fri	7	15	4	Egress = 2008-291T02:55 (10 min)
S44	89	89PL (nt)	**PALLENE**	2008-291T02:58	Oct17	Fri	7	15	4	Inbound 29037.1 km flyby, v = 18.1 km/s, phase = 141 deg
S44	89	E_ring_lg	**DUST HAZARD**	2008-291T03:00	Oct17	Fri	7	15	4	Egress at 2008-291T03:02 (0.0 min); protection = MEA cover closure.
S44	89	Ring CRX	Descending	2008-291T03:01	Oct17	Fri	7	15	4	r = 3.956 Rs (238685 km)
S44	89	**OTM-168**	~apo	2008-291T03:10	Oct17	Fri	7	15	4	D/L start, burn ~6h later. Backup D/L 291T08:25
S44	89	Periapse		2008-291T03:38	Oct17	Fri	7	15	0	R = 3.939 Rs, lat = -8 deg, phase = 156 deg
S44	89	89EP (nt)	**EPIMETHEUS**	2008-291T04:22	Oct17	Fri	7	15	0	Outbound 118902.9 km flyby, v = 20.3 km/s, phase = 136 deg
S45	89	**RINGS**	**Segment**	2008-292T20:21	Oct18	Sat	9	13	2	RINGS_89_91 (12d); RINGS_89_91
S45	89	**S45 Begins**		2008-292T20:21	Oct18	Sat	9	13	2	S45 sequence. Duration = 39 d
S45	89	89TI (nt)	**TITAN**	2008-293T05:25	Oct19	Sun	9	12	2	Outbound 397750.4 km flyby, v = 7.1 km/s, phase = 125 deg
S45	89	Ring CRX	Ascending	2008-294T03:50	Oct20	Mon	10	12	3	r = 19.665 Rs (1186370 km)
S45	90	**Apoapse**		2008-294T19:08	Oct20	Mon	11	11	4	Per = 7.3 d, inc = 74.5 deg, r = 20.103 Rs, phase = 24 deg
S45	90	90JA (nt)	**JANUS**	2008-298T09:14	Oct24	Fri	15	7	7	Inbound 108155.1 km flyby, v = 19.5 km/s, phase = 132 deg
S45	90	**Sun OCC**	RING	2008-298T09:37	Oct24	Fri	15	7	7	Egress = 2008-298T09:54 (17 min)
S45	90	**Earth OCC**	RING	2008-298T09:51	Oct24	Fri	15	7	7	Egress = 2008-298T09:58 (8 min)
S45	90	E_ring_lg	**DUST HAZARD**	2008-298T10:03	Oct24	Fri	15	7	7	Egress at 2008-298T10:05 (0.0 min); protection = MEA cover closure.
S45	90	Ring CRX	Descending	2008-298T10:04	Oct24	Fri	15	7	7	r = 3.965 Rs (239201 km)
S45	90	90PN (nt)	**PAN**	2008-298T10:08	Oct24	Fri	15	7	7	Inbound 105477.8 km flyby, v = 20.1 km/s, phase = 156 deg
S45	90	90MI (nt)	**MIMAS**	2008-298T10:25	Oct24	Fri	15	7	7	Inbound 57294.8 km flyby, v = 18.3 km/s, phase = 155 deg
S45	90	Periapse		2008-298T10:40	Oct24	Fri	15	7	7	R = 3.948 Rs, lat = -8 deg, phase = 156 deg
S45	90	Ring CRX	Ascending	2008-301T11:03	Oct27	Mon	18	4	10	r = 19.677 Rs (1187090 km)
S45	91	**Apoapse**		2008-302T02:15	Oct28	Tue	18	4	11	Per = 7.3 d, inc = 74.5 deg, r = 20.109 Rs, phase = 24 deg
S45	91	**OTM-169**	91TI-4d or TBD	2008-303T10:37	Oct29	Wed	20	2	12	D/L start, burn ~6h later. Backup D/L 304T10:37
S45	91	**SOST**	**Segment**	2008-304T19:37	Oct30	Thu	21	1	1	SOST_91 (3d); E6 Flyby
S45	91	91ME (nt)	**METHONE**	2008-305T16:02	Oct31	Fri	22	0	2	Inbound 116142.2 km flyby, v = 18.8 km/s, phase = 102 deg
S45	91	91PA (nt)	**PANDORA**	2008-305T16:31	Oct31	Fri	22	0	2	Inbound 111487.2 km flyby, v = 19.8 km/s, phase = 138 deg
S45	91	**Sun OCC**	RING	2008-305T16:49	Oct31	Fri	22	0	2	Egress = 2008-305T17:05 (16 min)
S45	91	**Earth OCC**	RING	2008-305T17:03	Oct31	Fri	22	0	2	Egress = 2008-305T17:09 (7 min)
S45	91	E_ring_lg	**DUST HAZARD**	2008-305T17:14	Oct31	Fri	22	0	2	Egress at 2008-305T17:15 (0.0 min); protection = MEA cover closure.
S45	91	91DA (nt)	**DAPHNIS**	2008-305T17:14	Oct31	Fri	22	0	2	Inbound 102902.6 km flyby, v = 19.9 km/s, phase = 155 deg
S45	91	**91EN (dt) [E6]**	**ENCELADUS**	2008-305T17:15	Oct31	Fri	22	3	2	Inbound 190.4 km flyby, v = 17.7 km/s, phase = 113 deg
S45	91	Ring CRX	Descending	2008-305T17:15	Oct31	Fri	22	0	2	r = 3.965 Rs (239220 km)
S45	91	Periapse		2008-305T17:50	Oct31	Fri	0	3	2	R = 3.948 Rs, lat = -8 deg, phase = 156 deg
S45	91	91AT (nt)	**ATLAS**	2008-305T18:11	Oct31	Fri	0	3	2	Outbound 112345.2 km flyby, v = 20.3 km/s, phase = 147 deg
S45	91	91EP (nt)	**EPIMETHEUS**	2008-305T18:35	Oct31	Fri	0	3	2	Outbound 118944.6 km flyby, v = 20.3 km/s, phase = 136 deg
S45	91	**TOST**	**Segment**	2008-307T19:22	Nov02	Sun	2	1	4	TOST_91_T46 (2d); Titan Flyby T46
S45	91	**Sun OCC**	TITAN	2008-308T17:29	Nov03	Mon	3	0	5	Egress = 2008-308T17:44 (15 min)
S45	91	**Earth OCC**	TITAN	2008-308T17:30	Nov03	Mon	3	0	5	Egress = 2008-308T17:44 (15 min)
S45	91	**91TI (dt) [T46]**	**TITAN**	2008-308T17:35	Nov03	Mon	3	16	5	Outbound 1100.0 km flyby, v = 6.3 km/s, phase = 171 deg
S45	91	Ring CRX	Ascending	2008-308T18:08	Nov03	Mon	0	16	5	r = 19.688 Rs (1187760 km)
S45	91	**RINGS**	**Segment**	2008-309T19:08	Nov04	Tue	1	15	6	RINGS_91_93 (8d); RINGS_91_93
S45	92	**Apoapse**		2008-309T23:30	Nov04	Tue	1	15	7	Per = 7.9 d, inc = 70.6 deg, r = 21.149 Rs, phase = 26 deg
S45	92	**OTM-170**	91TI+5d	2008-313T16:23	Nov08	Sat	5	11	10	D/L start, burn ~6h later. Backup D/L 314T09:53
S45	92	**Sun OCC**	RING	2008-313T20:53	Nov08	Sat	5	11	0	Egress = 2008-313T21:07 (15 min)
S45	92	92PO (nt)	**POLYDEUCES**	2008-313T21:02	Nov08	Sat	5	11	0	Inbound 121407.5 km flyby, v = 15.3 km/s, phase = 12 deg
S45	92	**Earth OCC**	RING	2008-313T21:10	Nov08	Sat	5	11	0	Egress = 2008-313T21:14 (5 min)
S45	92	E_ring_lg	Dust Crossing	2008-313T21:19	Nov08	Sat	5	11	0	Egress at 2008-313T21:21 (0.0 min); protection = none.
S45	92	Ring CRX	Descending	2008-313T21:20	Nov08	Sat	5	11	0	r = 4.334 Rs (261455 km)
S45	92	92TL (nt)	**TELESTO**	2008-313T21:31	Nov08	Sat	5	11	0	Inbound 64756.4 km flyby, v = 16.4 km/s, phase = 84 deg
S45	92	92EN (nt)	**ENCELADUS**	2008-313T21:51	Nov08	Sat	5	11	0	Inbound 52795.2 km flyby, v = 16.8 km/s, phase = 128 deg
S45	92	Periapse		2008-313T22:36	Nov08	Sat	5	11	0	R = 4.269 Rs, lat = -15 deg, phase = 154 deg
S45	92	Ring CRX	Ascending	2008-316T16:52	Nov11	Tue	8	8	3	r = 19.815 Rs (1195433 km)
S45	92	**OTM-171**	~apo	2008-317T16:09	Nov12	Wed	9	7	4	D/L start, burn ~6h later. Backup D/L 318T16:09
S45	93	**Apoapse**		2008-317T22:29	Nov12	Wed	9	7	0	Per = 8.0 d, inc = 70.5 deg, r = 21.284 Rs, phase = 26 deg
S45	93	**MAG**	**Segment**	2008-318T01:09	Nov13	Thu	9	7	0	MAG_93 (6d); MAG_93
S45	93	**OTM-172**	93TI-3d	2008-321T02:09	Nov16	Sun	12	4	3	D/L start, burn ~6h later. Backup D/L 322T09:24
S45	93	**Sun OCC**	RING	2008-321T20:40	Nov16	Sun	13	3	1	Egress = 2008-321T20:54 (14 min)
S45	93	93TE (nt)	**TETHYS**	2008-321T20:41	Nov16	Sun	13	3	1	Inbound 57056.6 km flyby, v = 15.9 km/s, phase = 41 deg
S45	93	**Earth OCC**	RING	2008-321T20:58	Nov16	Sun	13	3	1	Egress = 2008-321T21:01 (4 min)
S45	93	E_ring_lg	Dust Crossing	2008-321T21:05	Nov16	Sun	13	3	1	Egress at 2008-321T21:07 (0.0 min); protection = none.
S45	93	Ring CRX	Descending	2008-321T21:07	Nov16	Sun	13	3	1	r = 4.333 Rs (261440 km)
S45	93	Periapse		2008-321T22:22	Nov16	Sun	13	3	1	R = 4.269 Rs, lat = -15 deg, phase = 154 deg
S45	93	**TOST**	**Segment**	2008-323T18:24	Nov18	Tue	15	1	3	TOST_93_94_T47 (2d); Titan Flyby T47
S45	93	**93TI (t) [T47]**	**TITAN**	2008-324T15:56	Nov19	Wed	16	16	4	Outbound 1022.6 km flyby, v = 6.3 km/s, phase = 28 deg
S45	93	Ring CRX	Ascending	2008-324T16:32	Nov19	Wed	0	16	4	r = 19.787 Rs (1193736 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S45	94	**Apoapse**		2008-325T10:14	Nov20	Thu	1	15	4	Per = 8.0 d, inc = 72.3 deg, r = 20.296 Rs, phase = 25 deg
S45	94	**RINGS**	**Segment**	2008-325T23:25	Nov20	Thu	1	15	5	RINGS_94A (6d); RINGS_94A
S45	94	**OTM-173**	93TI+5d	2008-328T15:25	Nov23	Sun	4	12	8	D/L start, burn ~6h later. Backup D/L 329T15:25
S45	94	**Sun OCC**	RING	2008-329T08:11	Nov24	Mon	5	11	1	Egress = 2008-329T08:17 (7 min)
S45	94	94HE (nt)	**HELENE**	2008-329T08:25	Nov24	Mon	5	11	1	Inbound 65066.7 km flyby, v = 14.3 km/s, phase = 14 deg
S45	94	Ring CRX	Descending	2008-329T08:40	Nov24	Mon	5	11	1	r = 5.248 Rs (316584 km)
S45	94	94TE (nt)	**TETHYS**	2008-329T08:45	Nov24	Mon	5	11	1	Inbound 24236.9 km flyby, v = 15.2 km/s, phase = 159 deg
S45	94	Periapse		2008-329T09:51	Nov24	Mon	5	11	1	R = 5.213 Rs, lat = -10 deg, phase = 155 deg
S46	94	**RINGS**	**Segment**	2008-331T17:55	Nov26	Wed	7	9	3	RINGS_94B (1d); RINGS_94B
S46	94	**S46 Begins**		2008-331T17:55	Nov26	Wed	7	9	3	S46 sequence. Duration = 44 d
S46	94	**OTM-174**	~apo	2008-332T15:10	Nov27	Thu	8	8	4	D/L start, burn ~6h later. Backup D/L 333T08:56
S46	94	Ring CRX	Ascending	2008-332T15:47	Nov27	Thu	8	8	0	r = 19.788 Rs (1193800 km)
S46	94	**SATURN**	**Segment**	2008-333T00:10	Nov28	Fri	8	8	0	SATURN_94_95 (7d); SATURN_094_095
S46	95	**Apoapse**		2008-333T09:29	Nov28	Fri	9	7	1	Per = 8.0 d, inc = 72.3 deg, r = 20.294 Rs, phase = 25 deg
S46	95	**OTM-175**	95TI-3d	2008-336T14:56	Dec01	Mon	12	4	4	D/L start, burn ~6h later. Backup D/L 337T14:56
S46	95	95EN (nt)	**ENCELADUS**	2008-337T06:35	Dec02	Tue	13	3	1	Inbound 122555.1 km flyby, v = 15.7 km/s, phase = 115 deg
S46	95	95CA (nt)	**CALYPSO**	2008-337T07:16	Dec02	Tue	13	3	1	Inbound 68178.3 km flyby, v = 15.2 km/s, phase = 93 deg
S46	95	**Sun OCC**	RING	2008-337T07:28	Dec02	Tue	13	3	1	Egress = 2008-337T07:33 (6 min)
S46	95	Ring CRX	Descending	2008-337T07:55	Dec02	Tue	13	3	1	r = 5.247 Rs (316573 km)
S46	95	Periapse		2008-337T09:07	Dec02	Tue	13	3	1	R = 5.213 Rs, lat = -10 deg, phase = 155 deg
S46	95	**TOST**	**Segment**	2008-339T17:26	Dec04	Thu	15	1	3	TOST_95_96_T48 (2d); Titan Flyby T48
S46	95	**95TI (t) [T48]**	**TITAN**	2008-340T14:26	Dec05	Fri	16	16	4	Outbound 960.0 km flyby, v = 6.3 km/s, phase = 25 deg
S46	95	Ring CRX	Ascending	2008-340T14:57	Dec05	Fri	0	16	4	r = 19.759 Rs (1192081 km)
S46	96	**Apoapse**		2008-340T18:20	Dec05	Fri	0	16	4	Per = 8.0 d, inc = 73.1 deg, r = 19.778 Rs, phase = 24 deg
S46	96	**RINGS**	**Segment**	2008-341T21:27	Dec06	Sat	1	15	5	Rings_96_97 (7d); Rings_96_97
S46	96	**OTM-176**	95TI+4d	2008-344T14:27	Dec09	Tue	4	12	8	D/L start, burn ~6h later. Backup D/L 345T08:12
S46	96	Ring CRX	Descending	2008-344T17:50	Dec09	Tue	4	12	0	r = 5.766 Rs (347886 km)
S46	96	Periapse		2008-344T18:07	Dec09	Tue	4	12	0	R = 5.765 Rs, lat = -2 deg, phase = 155 deg
S46	96	96CA (nt)	**CALYPSO**	2008-344T18:39	Dec09	Tue	4	12	0	Outbound 74907.0 km flyby, v = 14.9 km/s, phase = 147 deg
S46	96	**OTM-177**	~apo	2008-348T14:13	Dec13	Sat	8	8	4	D/L start, burn ~6h later. Backup D/L 349T07:58
S46	96	Ring CRX	Ascending	2008-348T14:27	Dec13	Sat	8	8	0	r = 19.728 Rs (1190219 km)
S46	97	**Apoapse**		2008-348T17:46	Dec13	Sat	8	8	0	Per = 8.0 d, inc = 72.9 deg, r = 19.746 Rs, phase = 25 deg
S46	97	**SATURN**	**Segment**	2008-348T23:13	Dec13	Sat	8	8	0	Saturn_097 (7d); Saturn_097
S46	97	**OTM-178**	97TI-3d	2008-352T13:58	Dec17	Wed	12	4	4	D/L start, burn ~6h later. Backup D/L 353T23:58
S46	97	Ring CRX	Descending	2008-352T17:08	Dec17	Wed	12	4	0	r = 5.766 Rs (347879 km)
S46	97	Periapse		2008-352T17:25	Dec17	Wed	12	4	0	R = 5.765 Rs, lat = -2 deg, phase = 155 deg
S46	97	**TOST**	**Segment**	2008-355T16:29	Dec20	Sat	15	1	3	TOST_97_98_T49 (2d); Titan Flyby T49
S46	97	**97TI (t) [T49]**	**TITAN**	2008-356T13:00	Dec21	Sun	16	48	4	Outbound 970.0 km flyby, v = 6.3 km/s, phase = 83 deg
S46	98	**Apoapse**		2008-356T13:04	Dec21	Sun	0	48	4	Per = 9.4 d, inc = 74.7 deg, r = 19.715 Rs, phase = 25 deg
S46	98	Ring CRX	Ascending	2008-356T13:34	Dec21	Sun	0	48	4	r = 19.713 Rs (1189267 km)
S46	98	**RINGS**	**Segment**	2008-357T22:14	Dec22	Mon	1	46	5	RINGS_98 (6d); RINGS_98
S46	98	**OTM-179**	97TI+3d	2008-359T13:44	Dec24	Wed	3	45	7	D/L start, burn ~6h later. Backup D/L 361T23:29
S46	98	Periapse		2008-361T05:11	Dec26	Fri	5	43	2	R = 9.095 Rs, lat = 2 deg, phase = 154 deg
S46	98	Ring CRX	Descending	2008-361T05:52	Dec26	Fri	5	43	2	r = 9.097 Rs (548850 km)
S46	98	**SATURN**	**Segment**	2008-363T16:00	Dec28	Sun	7	41	4	SATURN_98_100 (12d); SATURN_098_100
S46	99	**Apoapse**		2008-366T00:06	Dec31	Wed	9	38	6	Per = 9.6 d, inc = 69.0 deg, r = 19.732 Rs, phase = 26 deg
S46	99	Ring CRX	Ascending	2008-366T03:15	Dec31	Wed	10	38	7	r = 19.721 Rs (1189783 km)
S46	99	Periapse		2009-004T19:01	Jan04	Sun	14	34	11	R = 9.100 Rs, lat = 2 deg, phase = 154 deg
S46	99	Ring CRX	Descending	2009-004T19:42	Jan04	Sun	14	34	11	r = 9.102 Rs (549129 km)
S46	100	**Apoapse**		2009-009T13:53	Jan09	Fri	19	29	16	Per = 9.6 d, inc = 69.0 deg, r = 19.727 Rs, phase = 27 deg
S47	100	**RINGS**	**Segment**	2009-009T15:16	Jan09	Fri	19	29	16	RINGS_100 (9d); RINGS_100
S47	100	**S47 Begins**		2009-009T15:16	Jan09	Fri	19	29	16	S47 sequence. Duration = 39 d
S47	100	Ring CRX	Ascending	2009-009T17:06	Jan09	Fri	19	29	16	r = 19.716 Rs (1189442 km)
S47	100	Periapse		2009-014T08:45	Jan14	Wed	24	24	21	R = 9.098 Rs, lat = 2 deg, phase = 153 deg
S47	100	100TI (nt)	**TITAN**	2009-014T09:26	Jan14	Wed	24	24	21	Outbound 704012.6 km flyby, v = 9.3 km/s, phase = 25 deg
S47	100	Ring CRX	Descending	2009-014T09:26	Jan14	Wed	24	24	21	r = 9.101 Rs (549034 km)
S47	100	**SATURN**	**Segment**	2009-018T14:32	Jan18	Sun	28	20	25	SATURN_100_102 (12d); SATURN_100_102
S47	101	**Apoapse**		2009-019T03:38	Jan19	Mon	29	19	26	Per = 9.6 d, inc = 69.0 deg, r = 19.727 Rs, phase = 27 deg
S47	101	Ring CRX	Ascending	2009-019T06:53	Jan19	Mon	29	19	26	r = 19.715 Rs (1189431 km)
S47	101	**OTM-180**	~apo	2009-023T21:48	Jan23	Fri	33	14	30	D/L start, burn ~6h later. Backup D/L 024T21:48
S47	101	Periapse		2009-023T22:29	Jan23	Fri	33	14	0	R = 9.100 Rs, lat = 2 deg, phase = 153 deg
S47	101	Ring CRX	Descending	2009-023T23:11	Jan23	Fri	33	14	0	r = 9.102 Rs (549127 km)
S47	102	**Apoapse**		2009-028T17:02	Jan28	Wed	38	10	5	Per = 9.5 d, inc = 69.0 deg, r = 19.674 Rs, phase = 27 deg
S47	102	Ring CRX	Ascending	2009-028T20:25	Jan28	Wed	38	10	5	r = 19.662 Rs (1186222 km)
S47	102	**RINGS**	**Segment**	2009-030T06:18	Jan30	Fri	40	8	6	RINGS_102 (7d); RINGS_102
S47	102	102RH (nt)	**RHEA**	2009-033T10:43	Feb02	Mon	43	5	10	Inbound 97433.5 km flyby, v = 10.4 km/s, phase = 82 deg
S47	102	Periapse		2009-033T11:36	Feb02	Mon	43	5	10	R = 9.095 Rs, lat = 2 deg, phase = 153 deg
S47	102	Ring CRX	Descending	2009-033T12:19	Feb02	Mon	43	5	10	r = 9.097 Rs (548843 km)
S47	102	**OTM-181**	102TI-3d	2009-035T04:34	Feb04	Wed	45	3	11	D/L start, burn ~6h later. Backup D/L 036T04:19
S47	102	**TOST**	**Segment**	2009-037T13:19	Feb06	Fri	47	1	2	TOST_102_103_T50 (2d); Titan Flyby T50
S47	102	**Earth OCC**	TITAN	2009-038T08:48	Feb07	Sat	48	0	3	Egress = 2009-038T08:58 (10 min)
S47	102	**Sun OCC**	TITAN	2009-038T08:49	Feb07	Sat	48	0	3	Egress = 2009-038T08:57 (9 min)
S47	102	**102TI (t) [T50]**	**TITAN**	2009-038T08:51	Feb07	Sat	48	48	3	Outbound 960.0 km flyby, v = 6.3 km/s, phase = 136 deg
S47	102	Ring CRX	Ascending	2009-038T09:20	Feb07	Sat	0	48	3	r = 19.671 Rs (1186760 km)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S47	103	**Apoapse**		2009-039T02:29	Feb08	Sun	1	47	4	Per = 11.9 d, inc = 65.3 deg, r = 19.828 Rs, phase = 26 deg
S47	103	**MAG**	**Segment**	2009-039T16:19	Feb08	Sun	1	47	4	MAG_103 (9d); MAG_103
S47	103	**OTM-182**	102TI+3d	2009-041T04:04	Feb10	Tue	3	45	6	D/L start, burn ~6h later. Backup D/L 042T04:04
S47	103	Ring CRX	Descending	2009-044T17:37	Feb13	Fri	6	41	4	r = 13.647 Rs (823315 km)
S47	103	103TI (nt)	**TITAN**	2009-044T18:27	Feb13	Fri	6	41	4	Inbound 706923.8 km flyby, v = 7.7 km/s, phase = 98 deg
S47	103	Periapse		2009-045T01:43	Feb14	Sat	7	41	4	R = 13.573 Rs, lat = -14 deg, phase = 154 deg
S48	103	**MAG**	**Segment**	2009-048T12:35	Feb17	Tue	10	38	7	MAG_103_104 (5d); MAG_103_104
S48	103	**S48 Begins**		2009-048T12:35	Feb17	Tue	10	38	7	S48 sequence. Duration = 37 d
S48	103	Ring CRX	Ascending	2009-050T07:51	Feb19	Thu	12	36	9	r = 19.668 Rs (1186578 km)
S48	104	**Apoapse**		2009-051T01:02	Feb20	Fri	13	35	10	Per = 11.9 d, inc = 65.3 deg, r = 19.825 Rs, phase = 26 deg
S48	104	**RINGS**	**Segment**	2009-053T04:50	Feb22	Sun	15	33	12	RINGS_104_105 (24d); RINGS_104_105
S48	104	Ring CRX	Descending	2009-056T16:12	Feb25	Wed	18	30	16	r = 13.657 Rs (823956 km)
S48	104	Periapse		2009-057T00:16	Feb26	Thu	19	29	16	R = 13.584 Rs, lat = -14 deg, phase = 154 deg
S48	104	Ring CRX	Ascending	2009-062T06:39	Mar03	Tue	24	24	21	r = 19.685 Rs (1187592 km)
S48	105	**Apoapse**		2009-062T23:47	Mar03	Tue	25	23	22	Per = 12.0 d, inc = 65.3 deg, r = 19.840 Rs, phase = 26 deg
S48	105	**OTM-183**	~apo	2009-068T02:20	Mar09	Mon	30	18	27	D/L start, burn ~6h later. Backup D/L 069T08:35
S48	105	Ring CRX	Descending	2009-068T15:08	Mar09	Mon	30	18	1	r = 13.659 Rs (824045 km)
S48	105	Periapse		2009-068T23:08	Mar09	Mon	31	17	1	R = 13.586 Rs, lat = -13 deg, phase = 153 deg
S48	105	Ring CRX	Ascending	2009-074T05:57	Mar15	Sun	36	12	6	r = 19.737 Rs (1190739 km)
S48	106	**Apoapse**		2009-074T22:48	Mar15	Sun	37	11	7	Per = 12.0 d, inc = 65.3 deg, r = 19.888 Rs, phase = 27 deg
S48	106	**MAG**	**Segment**	2009-077T03:05	Mar18	Wed	39	9	9	MAG_106 (8d); MAG_106
S48	106	106TI (nt)	**TITAN**	2009-080T11:50	Mar21	Sat	42	6	12	Inbound 991403.6 km flyby, v = 8.0 km/s, phase = 63 deg
S48	106	Ring CRX	Descending	2009-080T14:36	Mar21	Sat	42	6	13	r = 13.646 Rs (823275 km)
S48	106	Periapse		2009-080T22:34	Mar21	Sat	43	5	13	R = 13.574 Rs, lat = -13 deg, phase = 153 deg
S48	106	**OTM-184**	106TI-3d	2009-083T01:20	Mar24	Tue	45	3	15	D/L start, burn ~6h later. Backup D/L 084T01:05
S49	106	**TOST**	**Segment**	2009-085T10:05	Mar26	Thu	47	1	2	TOST_106_107_T51 (2d); Titan Flyby T51
S49	106	**S49 Begins**		2009-085T10:05	Mar26	Thu	47	1	2	S49 sequence. Duration = 40 d
S49	106	**106TI (t) [T51]**	**TITAN**	2009-086T04:44	Mar27	Fri	48	8	3	Outbound 960.0 km flyby, v = 6.3 km/s, phase = 84 deg
S49	107	**Apoapse**		2009-086T04:52	Mar27	Fri	0	8	3	Per = 16.5 d, inc = 65.2 deg, r = 19.723 Rs, phase = 29 deg
S49	107	Ring CRX	Ascending	2009-086T05:10	Mar27	Fri	0	8	3	r = 19.722 Rs (1189834 km)
S49	107	Periapse		2009-087T05:33	Mar28	Sat	1	7	4	R = 19.667 Rs, lat = 21 deg, phase = 28 deg
S49	107	**MAG**	**Segment**	2009-087T16:50	Mar28	Sat	2	6	5	MAG_107 (6d); MAG_107
S49	107	**OTM-186**	106TI+2d	2009-088T07:05	Mar29	Sun	2	6	5	D/L start, burn ~6h later. Backup D/L 089T00:50
S49	107	**OTM-186A**	T52-3d	2009-091T00:35	Apr01	Wed	5	3	3	D/L start, burn ~6h later. Backup D/L 091T17:05
S49	107	**TOST**	**Segment**	2009-093T09:34	Apr03	Fri	7	1	2	TOST_107_108_T52 (2d); Titan Flyby T52
S49	108	**Apoapse**		2009-094T01:30	Apr04	Sat	8	0	3	Per = 16.8 d, inc = 63.3 deg, r = 20.921 Rs, phase = 151 deg
S49	108	**Earth OCC**	TITAN	2009-094T01:37	Apr04	Sat	8	0	3	Egress = 2009-094T01:46 (10 min)
S49	108	**Sun OCC**	TITAN	2009-094T01:39	Apr04	Sat	8	0	3	Egress = 2009-094T01:45 (6 min)
S49	108	**108TI (t) [T52]**	**TITAN**	2009-094T01:48	Apr04	Sat	8	16	3	Inbound 4150.0 km flyby, v = 5.8 km/s, phase = 151 deg
S49	108	Ring CRX	Descending	2009-094T02:10	Apr04	Sat	0	16	3	r = 20.914 Rs (1261741 km)
S49	108	**MAG**	**Segment**	2009-095T09:19	Apr05	Sun	1	15	4	MAG_108_109 (14d); MAG_108_109
S49	108	**OTM-188**	108TI+3d	2009-097T00:19	Apr07	Tue	3	13	6	D/L start, burn ~6h later. Backup D/L 098T00:04
S49	108	Periapse		2009-098T23:42	Apr08	Wed	5	11	2	R = 18.686 Rs, lat = -51 deg, phase = 76 deg
S49	108	Ring CRX	Ascending	2009-101T08:56	Apr11	Sat	7	9	4	r = 19.415 Rs (1171336 km)
S49	108	108TI (nt)	**TITAN**	2009-101T17:34	Apr11	Sat	8	8	5	Outbound 326591.0 km flyby, v = 6.0 km/s, phase = 67 deg
S49	108	**OTM-189**	~apo	2009-102T06:04	Apr12	Sun	8	8	5	D/L start, burn ~6h later. Backup D/L 102T23:49
S49	109	**Apoapse**		2009-106T23:06	Apr16	Thu	13	3	5	Per = 16.0 d, inc = 61.7 deg, r = 21.814 Rs, phase = 104 deg
S49	109	**OTM-190**	109TI-3d	2009-106T23:33	Apr16	Thu	13	3	5	D/L start, burn ~6h later. Backup D/L 107T23:33
S49	109	**TOST**	**Segment**	2009-109T08:18	Apr19	Sun	15	1	2	TOST_109_T53 (2d); Titan Flyby T53
S49	109	**Earth OCC**	TITAN	2009-110T00:07	Apr20	Mon	16	0	3	Egress = 2009-110T00:19 (13 min)
S49	109	**Sun OCC**	TITAN	2009-110T00:09	Apr20	Mon	16	0	3	Egress = 2009-110T00:18 (9 min)
S49	109	**109TI (t) [T53]**	**TITAN**	2009-110T00:21	Apr20	Mon	16	16	3	Inbound 3600.0 km flyby, v = 5.8 km/s, phase = 149 deg
S49	109	Ring CRX	Descending	2009-110T00:41	Apr20	Mon	0	16	3	r = 20.894 Rs (1260529 km)
S49	109	**RINGS**	**Segment**	2009-111T10:18	Apr21	Tue	1	15	4	RINGS_109 (5d); RINGS_109
S49	109	**OTM-191**	109TI+4d	2009-112T23:03	Apr22	Wed	3	13	6	D/L start, burn ~6h later. Backup D/L 113T23:03
S49	109	Periapse		2009-114T02:19	Apr24	Fri	4	12	1	R = 16.939 Rs, lat = -55 deg, phase = 84 deg
S49	109	**TOST**	**Segment**	2009-116T00:33	Apr26	Sun	6	10	3	TOST_109 (3d); Titan Non-Targeted Flyby
S49	109	Ring CRX	Ascending	2009-116T10:40	Apr26	Sun	6	10	3	r = 18.659 Rs (1125686 km)
S49	109	109TI (nt)	**TITAN**	2009-117T04:04	Apr27	Mon	7	9	4	Outbound 695883.9 km flyby, v = 6.3 km/s, phase = 64 deg
S49	109	**OTM-192**	~apo	2009-118T05:02	Apr28	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 119T05:02
S49	109	**MAG**	**Segment**	2009-118T14:02	Apr28	Tue	9	7	0	MAG_109_110 (7d); MAG_109_110
S49	110	**Apoapse**		2009-122T01:38	May02	Sat	12	4	4	Per = 16.0 d, inc = 61.3 deg, r = 23.562 Rs, phase = 96 deg
S49	110	**OTM-193**	110TI-3d	2009-122T15:02	May02	Sat	13	3	4	D/L start, burn ~6h later. Backup D/L 123T14:46
S50	110	**TOST**	**Segment**	2009-125T07:16	May05	Tue	15	1	3	TOST_110_T54 (2d); Titan Flyby T54
S50	110	**S50 Begins**		2009-125T07:16	May05	Tue	15	1	3	S50 sequence. Duration = 39 d
S50	110	**Earth OCC**	TITAN	2009-125T22:38	May05	Tue	16	0	3	Egress = 2009-125T22:53 (15 min)
S50	110	**Sun OCC**	TITAN	2009-125T22:40	May05	Tue	16	0	3	Egress = 2009-125T22:52 (12 min)
S50	110	**110TI (t) [T54]**	**TITAN**	2009-125T22:54	May05	Tue	16	16	3	Inbound 3244.4 km flyby, v = 5.8 km/s, phase = 146 deg
S50	110	Ring CRX	Descending	2009-125T23:13	May05	Tue	0	16	3	r = 20.877 Rs (1259504 km)
S50	110	**MAG**	**Segment**	2009-127T07:16	May07	Thu	1	15	5	MAG_110_111 (14d); MAG_110_111
S50	110	**OTM-194**	110TI+4d	2009-128T22:01	May08	Fri	3	13	6	D/L start, burn ~6h later. Backup D/L 129T22:01
S50	110	Periapse		2009-129T14:39	May09	Sat	4	12	1	R = 15.068 Rs, lat = -54 deg, phase = 84 deg
S50	110	Ring CRX	Ascending	2009-131T12:45	May11	Mon	6	10	3	r = 17.315 Rs (1044637 km)
S50	110	**OTM-195**	~apo	2009-134T04:00	May14	Thu	8	8	5	D/L start, burn ~6h later. Backup D/L 135T03:45
S50	111	**Apoapse**		2009-137T13:45	May17	Sun	12	4	3	Per = 15.9 d, inc = 60.6 deg, r = 25.434 Rs, phase = 96 deg
S50	111	**OTM-196**	111TI-3d	2009-138T13:45	May18	Mon	13	3	4	D/L start, burn ~6h later. Backup D/L 139T13:44

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S50	111	**TOST**	**Segment**	2009-141T06:14	May21	Thu	15	1	3	TOST_111_T55 (2d); Titan Flyby T55
S50	111	**Earth OCC**	TITAN	2009-141T21:11	May21	Thu	16	0	3	Egress = 2009-141T21:29 (18 min)
S50	111	**Sun OCC**	TITAN	2009-141T21:12	May21	Thu	16	0	3	Egress = 2009-141T21:28 (17 min)
S50	111	**111TI (t) [T55]**	**TITAN**	2009-141T21:27	May21	Thu	16	16	3	Inbound 965.0 km flyby, v = 6.0 km/s, phase = 141 deg
S50	111	Ring CRX	Descending	2009-141T21:47	May21	Thu	0	16	3	r = 20.819 Rs (1256014 km)
S50	111	**MAG**	**Segment**	2009-143T07:59	May23	Sat	1	15	5	MAG_111_112 (14d); MAG_111_112
S50	111	**OTM-197**	111TI+4d	2009-144T20:59	May24	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 145T20:59
S50	111	Periapse		2009-145T01:25	May25	Mon	3	13	0	R = 12.164 Rs, lat = -50 deg, phase = 80 deg
S50	111	Ring CRX	Ascending	2009-146T07:00	May26	Tue	4	12	1	r = 14.396 Rs (868503 km)
S50	111	**OTM-198**	~apo	2009-150T02:58	May30	Sat	8	8	5	D/L start, burn ~6h later. Backup D/L 151T12:58
S50	112	**Apoapse**		2009-153T00:45	Jun02	Tue	11	5	3	Per = 16.0 d, inc = 58.7 deg, r = 28.339 Rs, phase = 100 deg
S50	112	**OTM-199**	112TI-3d	2009-154T12:42	Jun03	Wed	13	3	4	D/L start, burn ~6h later. Backup D/L 155T12:42
S50	112	**TOST**	**Segment**	2009-157T05:12	Jun06	Sat	15	1	3	TOST_112_T56 (2d); Titan Flyby T56
S50	112	**Earth OCC**	TITAN	2009-157T19:41	Jun06	Sat	16	0	3	Egress = 2009-157T20:01 (20 min)
S50	112	**Sun OCC**	TITAN	2009-157T19:42	Jun06	Sat	16	0	3	Egress = 2009-157T20:01 (19 min)
S50	112	**112TI (t) [T56]**	**TITAN**	2009-157T20:00	Jun06	Sat	16	16	3	Inbound 965.0 km flyby, v = 6.0 km/s, phase = 135 deg
S50	112	Ring CRX	Descending	2009-157T20:20	Jun06	Sat	0	16	3	r = 20.799 Rs (1254784 km)
S50	112	**RINGS**	**Segment**	2009-159T06:57	Jun08	Mon	1	14	5	RINGS_112 (5d); RINGS EQUINOX
S50	112	Periapse		2009-160T14:48	Jun09	Tue	3	13	6	R = 9.523 Rs, lat = -43 deg, phase = 75 deg
S50	112	**OTM-200**	112TI+3d	2009-161T02:12	Jun10	Wed	3	13	7	D/L start, burn ~6h later. Backup D/L 162T02:11
S50	112	Ring CRX	Ascending	2009-161T08:04	Jun10	Wed	4	12	0	r = 11.213 Rs (676489 km)
S51	112	**MAG**	**Segment**	2009-164T04:41	Jun13	Sat	6	10	3	MAG_112_113 (9d); MAG_112_113
S51	112	**S51 Begins**		2009-164T04:41	Jun13	Sat	6	10	3	S51 sequence. Duration = 41 d
S51	112	**OTM-201**	~apo	2009-165T19:26	Jun14	Sun	8	8	5	D/L start, burn ~6h later. Backup D/L 166T19:26
S51	112	**End of reconstructed traj.**		2009-168T12:00	Jun17	Wed	11	5	3	End use of reconstructed s/c trajectory / Start use of predicted reference trajectory
S51	113	**Apoapse**		2009-168T14:09	Jun17	Wed	11	5	3	Per = 16.0 d, inc = 55.7 deg, r = 30.977 Rs, phase = 104 deg
S51	113	**OTM-202**	113TI-4d	2009-170T11:40	Jun19	Fri	13	3	5	D/L start, burn ~6h later. Backup D/L 171T11:40
S51	113	**TOST**	**Segment**	2009-173T04:10	Jun22	Mon	15	1	3	TOST_113_T57 (2d); Titan Flyby T57
S51	113	**Earth OCC**	TITAN	2009-173T18:09	Jun22	Mon	16	0	3	Egress = 2009-173T18:32 (24 min)
S51	113	**Sun OCC**	TITAN	2009-173T18:10	Jun22	Mon	16	0	3	Egress = 2009-173T18:32 (22 min)
S51	113	**113TI (t) [T57]**	**TITAN**	2009-173T18:33	Jun22	Mon	16	16	3	Inbound 955.0 km flyby, v = 6.0 km/s, phase = 128 deg
S51	113	Ring CRX	Descending	2009-173T18:54	Jun22	Mon	0	16	3	r = 20.775 Rs (1253345 km)
S51	113	**RINGS**	**Segment**	2009-175T05:40	Jun24	Wed	1	14	5	RINGS_113 (5d); RINGS EQUINOX
S51	113	Periapse		2009-176T05:45	Jun25	Thu	2	13	6	R = 7.212 Rs, lat = -36 deg, phase = 71 deg
S51	113	Ring CRX	Ascending	2009-176T15:06	Jun25	Thu	3	13	6	r = 8.296 Rs (500498 km)
S51	113	**OTM-203**	113TI+4d	2009-177T01:09	Jun26	Fri	3	13	7	D/L start, burn ~6h later. Backup D/L 178T01:09
S51	113	**MAG**	**Segment**	2009-180T03:39	Jun29	Mon	6	10	3	MAG_113_114 (9d); MAG_113_114
S51	113	**OTM-204**	~apo	2009-181T18:24	Jun30	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 182T18:24
S51	114	**Apoapse**		2009-184T05:05	Jul03	Fri	10	5	2	Per = 15.9 d, inc = 51.2 deg, r = 33.287 Rs, phase = 109 deg
S51	114	**OTM-205**	114TI-3d	2009-186T10:38	Jul05	Sun	13	3	5	D/L start, burn ~6h later. Backup D/L 187T10:38
S51	114	**TOST**	**Segment**	2009-189T04:00	Jul08	Wed	15	1	3	TOST_114_T58 (2d); Titan Flyby T58
S51	114	**Earth OCC**	TITAN	2009-189T16:33	Jul08	Wed	16	0	3	Egress = 2009-189T17:02 (29 min)
S51	114	**Sun OCC**	TITAN	2009-189T16:36	Jul08	Wed	16	0	3	Egress = 2009-189T17:02 (27 min)
S51	114	**114TI (t) [T58]**	**TITAN**	2009-189T17:04	Jul08	Wed	16	16	3	Inbound 965.0 km flyby, v = 6.0 km/s, phase = 120 deg
S51	114	Ring CRX	Descending	2009-189T17:28	Jul08	Wed	0	16	3	r = 20.744 Rs (1251475 km)
S51	114	**RINGS**	**Segment**	2009-191T04:53	Jul10	Fri	1	14	5	RINGS_114_115 (14d); RINGS EQUINOX
S51	114	Periapse		2009-191T22:10	Jul10	Fri	2	14	5	R = 5.309 Rs, lat = -28 deg, phase = 67 deg
S51	114	Ring CRX	Ascending	2009-192T02:58	Jul11	Sat	2	14	6	r = 5.931 Rs (357823 km)
S51	114	114DI (nt)	**DIONE**	2009-192T03:13	Jul11	Sat	2	14	6	Outbound 86641.8 km flyby, v = 9.2 km/s, phase = 62 deg
S51	114	**OTM-206**	114TI+4d	2009-193T10:22	Jul12	Sun	4	12	7	D/L start, burn ~6h later. Backup D/L 194T10:07
S51	114	**OTM-207**	~apo	2009-198T09:52	Jul17	Fri	9	7	5	D/L start, burn ~6h later. Backup D/L 199T09:52
S51	115	**Apoapse**		2009-199T21:29	Jul18	Sat	10	6	1	Per = 15.9 d, inc = 44.6 deg, r = 35.191 Rs, phase = 112 deg
S51	115	**OTM-208**	115TI-3d	2009-202T09:36	Jul21	Tue	13	3	4	D/L start, burn ~6h later. Backup D/L 203T09:36
S52	115	**TOST**	**Segment**	2009-204T21:51	Jul23	Thu	15	1	3	TOST_115_T59 (1d); Titan Flyby T59
S52	115	**S52 Begins**		2009-204T21:51	Jul23	Thu	15	1	3	S52 sequence. Duration = 32 d
S52	115	**Earth OCC**	TITAN	2009-205T14:53	Jul24	Fri	16	0	3	Egress = 2009-205T15:30 (38 min)
S52	115	**Sun OCC**	TITAN	2009-205T14:56	Jul24	Fri	16	0	3	Egress = 2009-205T15:30 (34 min)
S52	115	**115TI (t) [T59]**	**TITAN**	2009-205T15:34	Jul24	Fri	16	16	3	Inbound 955.0 km flyby, v = 6.0 km/s, phase = 112 deg
S52	115	Ring CRX	Descending	2009-205T16:05	Jul24	Fri	0	16	3	r = 20.694 Rs (1248440 km)
S52	115	**RINGS**	**Segment**	2009-206T08:21	Jul25	Sat	1	15	4	RINGS_115_116 (15d); RINGS EQUINOX
S52	115	Periapse		2009-207T16:04	Jul26	Sun	2	14	5	R = 3.865 Rs, lat = -19 deg, phase = 65 deg
S52	115	115PM (nt)	**PROMETHEUS**	2009-207T16:19	Jul26	Sun	2	14	5	Outbound 113276.5 km flyby, v = 10.4 km/s, phase = 79 deg
S52	115	115PA (nt)	**PANDORA**	2009-207T16:49	Jul26	Sun	2	14	5	Outbound 103474.7 km flyby, v = 10.1 km/s, phase = 65 deg
S52	115	115JA (nt)	**JANUS**	2009-207T17:06	Jul26	Sun	2	14	5	Outbound 93992.6 km flyby, v = 9.9 km/s, phase = 59 deg
S52	115	115TE (nt)	**TETHYS**	2009-207T18:15	Jul26	Sun	2	14	5	Outbound 68411.4 km flyby, v = 9.5 km/s, phase = 89 deg
S52	115	E_ring_lg	Dust Crossing	2009-207T18:29	Jul26	Sun	2	14	5	Egress at 2009-207T18:32 (0.0 min); protection = none.
S52	115	Ring CRX	Ascending	2009-207T18:31	Jul26	Sun	2	14	5	r = 4.205 Rs (253670 km)
S52	115	**OTM-209**	115TI+4d	2009-209T09:21	Jul28	Tue	4	12	7	D/L start, burn ~6h later. Backup D/L 210T09:06
S52	115	**OTM-210**	~apo	2009-213T16:35	Aug01	Sat	8	8	4	D/L start, burn ~6h later. Backup D/L 214T16:20
S52	116	**Apoapse**		2009-215T15:22	Aug03	Mon	10	6	2	Per = 15.9 d, inc = 34.7 deg, r = 36.641 Rs, phase = 115 deg
S52	116	**OTM-211**	116TI-3d	2009-218T16:05	Aug06	Thu	13	3	5	D/L start, burn ~6h later. Backup D/L 219T16:05
S52	116	**TOST**	**Segment**	2009-221T01:05	Aug09	Sun	15	1	2	TOST_116_T60 (1d); Titan Flyby T60
S52	116	**Earth OCC**	TITAN	2009-221T13:02	Aug09	Sun	16	0	3	Egress = 2009-221T13:56 (54 min)
S52	116	**Sun OCC**	TITAN	2009-221T13:09	Aug09	Sun	16	0	3	Egress = 2009-221T13:56 (48 min)
S52	116	**116TI (t) [T60]**	**TITAN**	2009-221T14:04	Aug09	Sun	16	16	3	Inbound 970.0 km flyby, v = 6.0 km/s, phase = 104 deg
S52	116	Ring CRX	Descending	2009-221T14:55	Aug09	Sun	0	16	3	r = 20.578 Rs (1241468 km)
S52	116	**RINGS**	**Segment**	2009-222T02:05	Aug10	Mon	1	15	3	RINGS_116_117 (15d); RINGS EQUINOX
S52	116	**NORTHERN VERNAL EQUINOX**		2009-223T00:25	Aug11	Tue	1	15	4	Northern vernal equinox (Sun passes northward through ring plane)
S52	116	116AT (nt)	**ATLAS**	2009-223T11:26	Aug11	Tue	2	14	5	Inbound 118686.9 km flyby, v = 14.4 km/s, phase = 117 deg

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S52	116	Periapse		2009-223T11:37	Aug11	Tue	2	14	5	R = 2.932 Rs, lat = -10 deg, phase = 63 deg
S52	116	Ring CRX	Ascending	2009-223T12:59	Aug11	Tue	2	14	5	r = 3.130 Rs (188831 km)
S52	116	OTM-212	116TI+4d	2009-225T15:49	Aug13	Thu	4	12	7	D/L start, burn ~6h later. Backup D/L 226T15:34
S52	116	OTM-213	~apo	2009-228T08:04	Aug16	Sun	7	9	3	D/L start, burn ~6h later. Backup D/L 229T08:04
S52	117	Apoapse		2009-231T11:07	Aug19	Wed	10	6	3	Per = 16.0 d, inc = 20.8 deg, r = 37.603 Rs, phase = 117 deg
S52	117	OTM-214	117TI-3d	2009-233T21:34	Aug21	Fri	12	4	6	D/L start, burn ~6h later. Backup D/L 234T21:34
S53	117	TOST	Segment	2009-237T00:04	Aug25	Tue	15	1	3	TOST_117_T61 (2d); Titan Flyby T61
S53	117	S53 Begins		2009-237T00:04	Aug25	Tue	15	1	3	S53 sequence. Duration = 41 d
S53	117	117TI (t) [T61]	TITAN	2009-237T12:52	Aug25	Tue	16	48	4	Inbound 970.0 km flyby, v = 6.0 km/s, phase = 86 deg
S53	117	Ring CRX	Descending	2009-237T14:56	Aug25	Tue	0	48	4	r = 20.194 Rs (1218278 km)
S53	117	RINGS	Segment	2009-239T02:04	Aug27	Thu	2	46	5	RINGS_117_118 (18d); RINGS EQUINOX
S53	117	117AE (nt)	AEGAEON	2009-239T09:37	Aug27	Thu	2	46	6	Inbound 107227.7 km flyby, v = 3.0 km/s, phase = 105 deg
S53	117	Periapse		2009-239T10:33	Aug27	Thu	2	46	6	R = 4.278 Rs, lat = -8 deg, phase = 76 deg
S53	117	117PL (nt)	PALLENE	2009-239T11:03	Aug27	Thu	2	46	6	Outbound 101630.2 km flyby, v = 7.5 km/s, phase = 120 deg
S53	117	117TL (nt)	TELESTO	2009-239T13:12	Aug27	Thu	2	46	6	Outbound 24898.5 km flyby, v = 5.8 km/s, phase = 86 deg
S53	117	Ring CRX	Ascending	2009-239T14:09	Aug27	Thu	2	46	6	r = 4.880 Rs (294416 km)
S53	117	OTM-215	117TI+4d	2009-241T07:19	Aug29	Sat	4	44	7	D/L start, burn ~6h later. Backup D/L 242T07:19
S53	117	OTM-216	~apo	2009-247T20:48	Sep04	Fri	10	37	7	D/L start, burn ~6h later. Backup D/L 248T20:48
S53	118	Apoapse		2009-251T10:14	Sep08	Tue	14	34	4	Per = 24.0 d, inc = 12.1 deg, r = 48.887 Rs, phase = 104 deg
S53	118	MAG	Segment	2009-256T23:03	Sep13	Sun	19	28	9	MAG_118 (6d); MAG_118
S53	118	SEP = 3.0 deg	Conjunction	2009-258T00:02	Sep15	Tue	20	27	10	Commanding must not be required below 3.0 deg
S53	118	SEP = 2.0 deg	Conjunction	2009-260T04:59	Sep17	Thu	23	25	12	Plan no SSR playback below 2.0 deg
S53	118	SEP = 2.0 deg	Conjunction	2009-261T05:05	Sep18	Fri	24	24	13	Plan no SSR playback below 2.0 deg
S53	118	Ring CRX	Descending	2009-261T14:50	Sep18	Fri	24	24	14	r = 20.010 Rs (1207227 km)
S53	118	RINGS	Segment	2009-262T22:33	Sep19	Sat	25	22	15	RINGS_118 (5d); RINGS EQUINOX
S53	118	Periapse		2009-263T09:52	Sep20	Sun	26	22	16	R = 4.269 Rs, lat = -8 deg, phase = 77 deg
S53	118	118PM (nt)	PROMETHEUS	2009-263T10:09	Sep20	Sun	26	22	16	Outbound 121105.8 km flyby, v = 2.7 km/s, phase = 74 deg
S53	118	SEP = 3.0 deg	Conjunction	2009-263T10:14	Sep20	Sun	26	22	16	Commanding must not be required below 3.0 deg
S53	118	Ring CRX	Ascending	2009-263T13:29	Sep20	Sun	26	22	16	r = 4.879 Rs (294350 km)
S53	118	118DI (nt)	DIONE	2009-263T16:52	Sep20	Sun	26	22	16	Outbound 97729.0 km flyby, v = 9.8 km/s, phase = 94 deg
S53	118	118TI (nt)	TITAN	2009-265T06:54	Sep22	Tue	28	20	17	Outbound 293850.2 km flyby, v = 4.8 km/s, phase = 113 deg
S53	118	MAG	Segment	2009-267T22:18	Sep24	Thu	30	17	20	MAG_118_119 (10d); MAG_118_119
S53	119	Apoapse		2009-275T08:03	Oct02	Fri	38	10	27	Per = 23.8 d, inc = 12.2 deg, r = 48.712 Rs, phase = 103 deg
S54	119	MAG	Segment	2009-278T04:03	Oct05	Mon	41	7	30	MAG_119 (6d); MAG_119
S54	119	S54 Begins		2009-278T04:03	Oct05	Mon	41	7	30	S54 sequence. Duration = 40 d
S54	119	OTM-217	119TI-3d	2009-282T05:04	Oct09	Fri	45	3	34	D/L start, burn ~6h later. Backup D/L 283T05:04
S54	119	TOST	Segment	2009-284T13:49	Oct11	Sun	47	1	2	TOST_119_T62 (2d); Titan Flyby T61
S54	119	Sun OCC	TITAN	2009-285T07:39	Oct12	Mon	48	0	3	Egress = 2009-285T08:19 (41 min)
S54	119	Earth OCC	TITAN	2009-285T07:46	Oct12	Mon	48	0	3	Egress = 2009-285T08:22 (36 min)
S54	119	119TI (t) [T62]	TITAN	2009-285T08:36	Oct12	Mon	48	21	3	Inbound 1300.0 km flyby, v = 6.0 km/s, phase = 99 deg
S54	119	Ring CRX	Descending	2009-286T05:35	Oct13	Tue	1	20	4	r = 13.625 Rs (821992 km)
S54	119	SATURN	Segment	2009-286T13:49	Oct13	Tue	1	20	4	SATURN_119_120 (19d); SATURN_119_120
S54	119	Sun OCC	SATURN	2009-286T14:58	Oct13	Tue	1	20	4	Egress = 2009-286T19:34 (277 min)
S54	119	Earth OCC	SATURN	2009-286T15:49	Oct13	Tue	1	20	4	Egress = 2009-286T20:03 (255 min)
S54	119	119RH (nt)	RHEA	2009-286T16:55	Oct13	Tue	1	20	4	Inbound 40380.3 km flyby, v = 9.0 km/s, phase = 82 deg
S54	119	119ME (nt)	METHONE	2009-287T03:18	Oct14	Wed	2	19	5	Inbound 86126.0 km flyby, v = 11.8 km/s, phase = 39 deg
S54	119	119AT (nt)	ATLAS	2009-287T04:20	Oct14	Wed	2	19	5	Inbound 70611.5 km flyby, v = 2.9 km/s, phase = 114 deg
S54	119	119CA (nt)	CALYPSO	2009-287T04:33	Oct14	Wed	2	19	5	Inbound 100835.5 km flyby, v = 7.7 km/s, phase = 107 deg
S54	119	Periapse		2009-287T05:10	Oct14	Wed	2	19	5	R = 3.194 Rs, lat = -0 deg, phase = 70 deg
S54	119	119MI (nt)	MIMAS	2009-287T06:33	Oct14	Wed	2	19	5	Outbound 43913.4 km flyby, v = 7.8 km/s, phase = 102 deg
S54	119	Pallene	Dust Crossing	2009-287T06:58	Oct14	Wed	2	19	5	Egress at 2009-287T07:13 (0.0 min); protection = none.
S54	119	E_ring_lg	DUST HAZARD	2009-287T07:11	Oct14	Wed	2	19	5	Egress at 2009-287T09:06 (0.1 min); protection = MEA cover closure.
S54	119	Ring CRX	Ascending	2009-287T07:51	Oct14	Wed	2	19	5	r = 3.788 Rs (228512 km)
S54	119	119TE (nt)	TETHYS	2009-287T10:11	Oct14	Wed	2	19	5	Outbound 84938.1 km flyby, v = 11.9 km/s, phase = 76 deg
S54	119	OTM-218	119TI+3d	2009-288T18:34	Oct15	Thu	3	18	7	D/L start, burn ~6h later. Backup D/L 289T18:19
S54	119	OTM-219	~apo	2009-293T18:04	Oct20	Tue	8	13	5	D/L start, burn ~6h later. Backup D/L 294T18:04
S54	120	Apoapse		2009-296T16:58	Oct23	Fri	11	10	3	Per = 19.0 d, inc = 0.5 deg, r = 42.295 Rs, phase = 110 deg
S54	120	OTM-220	E7-3d	2009-302T17:35	Oct29	Thu	17	4	9	D/L start, burn ~6h later. Backup D/L 303T17:35
S54	120	SOST	Segment	2009-305T12:35	Nov01	Sun	20	1	3	SOST_120 (2d); E7 Flyby
S54	120	Sun OCC	SATURN	2009-305T14:35	Nov01	Sun	20	1	3	Egress = 2009-305T19:06 (272 min)
S54	120	Ring CRX	Descending	2009-305T15:07	Nov01	Sun	20	1	3	r = 9.238 Rs (557309 km)
S54	120	Earth OCC	SATURN	2009-305T16:02	Nov01	Sun	20	1	3	Egress = 2009-305T19:51 (229 min)
S54	120	120TI (nt)	TITAN	2009-305T23:57	Nov01	Sun	21	0	3	Inbound 978723.8 km flyby, v = 9.9 km/s, phase = 73 deg
S54	120	120PL (nt)	PALLENE	2009-306T02:08	Nov02	Mon	21	0	3	Inbound 93565.0 km flyby, v = 12.6 km/s, phase = 46 deg
S54	120	Periapse		2009-306T04:39	Nov02	Mon	21	0	3	R = 3.201 Rs, lat = -0 deg, phase = 70 deg
S54	120	120EP (nt)	EPIMETHEUS	2009-306T05:27	Nov02	Mon	21	0	3	Outbound 105180.8 km flyby, v = 8.0 km/s, phase = 5 deg
S54	120	E_ring_lg	DUST HAZARD	2009-306T06:39	Nov02	Mon	21	0	4	Egress at 2009-306T08:34 (0.1 min); protection = MEA cover closure.
S54	120	120CA (nt)	CALYPSO	2009-306T06:56	Nov02	Mon	21	0	4	Outbound 86772.7 km flyby, v = 6.9 km/s, phase = 117 deg
S54	120	Earth OCC	ENCELADUS	2009-306T07:28	Nov02	Mon	21	0	4	Egress = 2009-306T07:35 (7 min)
S54	120	120EN (t) [E7]	ENCELADUS	2009-306T07:42	Nov02	Mon	21	19	4	Outbound 99.0 km flyby, v = 7.7 km/s, phase = 90 deg
S54	120	Ring CRX	Ascending	2009-306T08:36	Nov02	Mon	0	19	4	r = 4.356 Rs (262811 km)
S54	120	SATURN	Segment	2009-307T12:35	Nov03	Tue	1	18	5	SATURN_120_121 (10d); SATURN_120_121
S54	120	OTM-221	E7+3d	2009-309T03:20	Nov05	Thu	3	16	6	D/L start, burn ~6h later. Backup D/L 310T03:21
S54	121	Apoapse		2009-315T16:57	Nov11	Wed	9	9	7	Per = 19.0 d, inc = 0.5 deg, r = 42.367 Rs, phase = 110 deg
S54	121	OTM-222	~apo	2009-316T03:06	Nov12	Thu	10	9	7	D/L start, burn ~6h later. Backup D/L 317T10:21
S55	121	SATURN	Segment	2009-317T19:21	Nov13	Fri	11	7	2	SATURN_121 (7d); SATURN_121
S55	121	S55 Begins		2009-317T19:21	Nov13	Fri	11	7	2	S55 sequence. Duration = 39 d
S55	121	OTM-223	E8-3d	2009-321T16:37	Nov17	Tue	15	3	6	D/L start, burn ~6h later. Backup D/L 322T16:37
S55	121	SOST	Segment	2009-324T11:37	Nov20	Fri	18	1	3	SOST_121 (2d); E8 Flyby

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S55	121	Ring CRX	Descending	2009-324T15:15	Nov20	Fri	18	0	3	r = 9.455 Rs (570449 km)
S55	121	Sun OCC	SATURN	2009-324T15:19	Nov20	Fri	18	0	3	Egress = 2009-324T19:46 (267 min)
S55	121	Earth OCC	SATURN	2009-324T17:14	Nov20	Fri	18	0	3	Egress = 2009-324T20:41 (207 min)
S55	121	121HE (nt)	HELENE	2009-324T22:04	Nov20	Fri	19	0	3	Inbound 82577.5 km flyby, v = 6.6 km/s, phase = 105 deg
S55	121	121EN (t) [E8]	ENCELADUS	2009-325T02:10	Nov21	Sat	19	21	3	Inbound 1602.9 km flyby, v = 7.8 km/s, phase = 87 deg
S55	121	121PA (nt)	PANDORA	2009-325T04:45	Nov21	Sat	0	21	4	Inbound 59395.7 km flyby, v = 3.0 km/s, phase = 103 deg
S55	121	Periapse		2009-325T05:15	Nov21	Sat	0	21	4	R = 3.202 Rs, lat = -0 deg, phase = 70 deg
S55	121	E_ring_lg	DUST HAZARD	2009-325T07:15	Nov21	Sat	0	21	4	Egress at 2009-325T09:10 (0.1 min); protection = MEA cover closure.
S55	121	121TI (nt)	TITAN	2009-325T08:30	Nov21	Sat	0	21	4	Outbound 965101.8 km flyby, v = 11.1 km/s, phase = 138 deg
S55	121	Ring CRX	Ascending	2009-325T09:07	Nov21	Sat	0	21	4	r = 4.315 Rs (260323 km)
S55	121	121CA (nt)	CALYPSO	2009-325T10:20	Nov21	Sat	0	21	4	Outbound 36955.9 km flyby, v = 10.0 km/s, phase = 78 deg
S55	121	121RH (nt)	RHEA	2009-325T17:36	Nov21	Sat	1	20	4	Outbound 24359.6 km flyby, v = 8.7 km/s, phase = 58 deg
S55	121	SATURN	Segment	2009-326T01:22	Nov22	Sun	1	20	4	SATURN_121_122 (19d); SATURN_121_122
S55	121	OTM-224	E8+2d	2009-326T16:22	Nov22	Sun	2	19	5	D/L start, burn ~6h later. Backup D/L 327T16:07
S55	122	Apoapse		2009-334T17:37	Nov30	Mon	10	11	8	Per = 19.0 d, inc = 0.5 deg, r = 42.380 Rs, phase = 109 deg
S55	122	OTM-225	~apo	2009-338T01:39	Dec04	Fri	13	8	11	D/L start, burn ~6h later. Backup D/L 339T01:39
S55	122	OTM-226	122TI-3d	2009-342T15:24	Dec08	Tue	18	3	5	D/L start, burn ~6h later. Backup D/L 343T01:24
S55	122	Ring CRX	Descending	2009-343T15:26	Dec09	Wed	19	2	1	r = 9.759 Rs (588741 km)
S55	122	Sun OCC	SATURN	2009-343T16:19	Dec09	Wed	19	2	1	Egress = 2009-343T20:41 (262 min)
S55	122	Earth OCC	SATURN	2009-343T18:32	Dec09	Wed	19	2	1	Egress = 2009-343T21:42 (190 min)
S55	122	122HE (nt)	HELENE	2009-343T22:28	Dec09	Wed	19	2	1	Inbound 31798.4 km flyby, v = 9.6 km/s, phase = 73 deg
S55	122	122PL (nt)	PALLENE	2009-344T03:34	Dec10	Thu	19	2	2	Inbound 111237.7 km flyby, v = 13.7 km/s, phase = 47 deg
S55	122	Periapse		2009-344T06:04	Dec10	Thu	19	2	2	R = 3.196 Rs, lat = -0 deg, phase = 71 deg
S55	122	E_ring_lg	DUST HAZARD	2009-344T08:04	Dec10	Thu	19	2	2	Egress at 2009-344T09:59 (0.1 min); protection = MEA cover closure.
S55	122	Ring CRX	Ascending	2009-344T09:49	Dec10	Thu	19	2	2	r = 4.259 Rs (256972 km)
S55	122	TOST	Segment	2009-345T10:10	Dec11	Fri	20	1	3	TOST_122_T63 (3d); Titan Flyby T63
S55	122	122TI (t) [T63]	TITAN	2009-346T01:03	Dec12	Sat	21	16	3	Outbound 4850.0 km flyby, v = 5.7 km/s, phase = 124 deg
S55	122	Ring CRX	Descending	2009-346T16:22	Dec12	Sat	1	15	4	r = 24.334 Rs (1468089 km)
S55	122	MAG	Segment	2009-348T10:10	Dec14	Mon	2	14	6	MAG_122_123 (9d); MAG_122_123
S55	122	OTM-227	122TI+3d	2009-349T00:55	Dec15	Tue	3	13	6	D/L start, burn ~6h later. Backup D/L 350T00:55
S55	123	Apoapse		2009-352T05:55	Dec18	Fri	6	10	3	Per = 16.0 d, inc = 4.9 deg, r = 37.964 Rs, phase = 104 deg
S55	123	OTM-228	~apo	2009-354T00:41	Dec20	Sun	8	8	5	D/L start, burn ~6h later. Backup D/L 355T00:41
S56	123	RINGS	Segment	2009-356T23:26	Dec22	Tue	11	5	3	RINGS_123 (4d); RINGS_123
S56	123	S56 Begins		2009-356T23:26	Dec22	Tue	11	5	3	S56 sequence. Duration = 32 d
S56	123	OTM-229	123TI-3d	2009-358T14:26	Dec24	Thu	13	3	5	D/L start, burn ~6h later. Backup D/L 359T07:57
S56	123	Earth OCC	RING	2009-359T19:27	Dec25	Fri	14	2	1	Egress = 2009-360T03:27 (480 min)
S56	123	Sun OCC	SATURN	2009-359T20:41	Dec25	Fri	14	2	1	Egress = 2009-360T00:12 (211 min)
S56	123	Earth OCC	SATURN	2009-359T21:39	Dec25	Fri	14	2	1	Egress = 2009-360T01:02 (203 min)
S56	123	123TE (nt)	TETHYS	2009-360T00:36	Dec26	Sat	14	2	1	Inbound 52988.1 km flyby, v = 11.8 km/s, phase = 77 deg
S56	123	G_ring	DUST HAZARD	2009-360T04:48	Dec26	Sat	14	2	2	Egress at 2009-360T04:52 (0.0 min); protection = MEA cover closure.
S56	123	Ring CRX	Ascending	2009-360T04:50	Dec26	Sat	14	2	2	r = 2.692 Rs (162400 km)
S56	123	123PM (nt)	PROMETHEUS	2009-360T05:01	Dec26	Sat	14	2	2	Inbound 56487.7 km flyby, v = 10.5 km/s, phase = 36 deg
S56	123	Periapse		2009-360T05:41	Dec26	Sat	14	2	2	R = 2.590 Rs, lat = 2 deg, phase = 76 deg
S56	123	TOST	Segment	2009-361T09:12	Dec27	Sun	15	1	3	TOST_123_T64 (3d); Titan Flyby T64
S56	123	123TI (t) [T64]	TITAN	2009-362T00:17	Dec28	Mon	16	16	3	Outbound 955.0 km flyby, v = 5.9 km/s, phase = 86 deg
S56	123	Ring CRX	Descending	2009-362T02:57	Dec28	Mon	0	16	4	r = 21.038 Rs (1269242 km)
S56	123	MAG	Segment	2009-364T09:12	Dec30	Wed	2	14	6	MAG_123_124 (7d); MAG_123_124
S56	123	OTM-230	123TI+3d	2009-364T23:57	Dec30	Wed	3	13	6	D/L start, burn ~6h later. Backup D/L 001T23:58
S56	124	Apoapse		2010-003T03:51	Jan03	Sun	6	10	3	Per = 16.0 d, inc = 21.6 deg, r = 37.515 Rs, phase = 104 deg
S56	124	OTM-231	~apo	2010-004T23:43	Jan04	Mon	8	8	5	D/L start, burn ~6h later. Backup D/L 005T23:43
S56	124	RINGS	Segment	2010-006T08:43	Jan06	Wed	9	7	1	RINGS_124 (6d); RINGS_124
S56	124	OTM-232	124TI-3d	2010-009T13:29	Jan09	Sat	13	3	5	D/L start, burn ~6h later. Backup D/L 010T06:59
S56	124	Ring CRX	Ascending	2010-011T01:57	Jan11	Mon	14	2	2	r = 3.209 Rs (193604 km)
S56	124	124PA (nt)	PANDORA	2010-011T02:09	Jan11	Mon	14	2	2	Inbound 73247.9 km flyby, v = 7.1 km/s, phase = 141 deg
S56	124	124ME (nt)	METHONE	2010-011T02:26	Jan11	Mon	14	2	2	Inbound 27153.9 km flyby, v = 8.0 km/s, phase = 145 deg
S56	124	Periapse		2010-011T03:23	Jan11	Mon	14	2	2	R = 2.999 Rs, lat = 11 deg, phase = 76 deg
S56	124	124EN (nt)	ENCELADUS	2010-011T06:48	Jan11	Mon	14	2	2	Outbound 105785.9 km flyby, v = 11.4 km/s, phase = 98 deg
S56	124	124CA (nt)	CALYPSO	2010-011T08:17	Jan11	Mon	14	2	2	Outbound 112932.7 km flyby, v = 8.0 km/s, phase = 102 deg
S56	124	TOST	Segment	2010-012T08:14	Jan12	Tue	15	1	3	TOST_124_T65 (3d); Titan Flyby T65
S56	124	124TI (t) [T65]	TITAN	2010-012T23:11	Jan12	Tue	16	16	3	Outbound 1072.8 km flyby, v = 5.9 km/s, phase = 95 deg
S56	124	Ring CRX	Descending	2010-013T02:03	Jan13	Wed	0	16	4	r = 21.101 Rs (1273003 km)
S56	124	MAG	Segment	2010-015T08:14	Jan15	Fri	2	14	6	MAG_124_125 (8d); MAG_124_125
S56	124	OTM-233	124TI+3d	2010-015T22:59	Jan15	Fri	3	13	6	D/L start, burn ~6h later. Backup D/L 016T23:00
S56	125	Apoapse		2010-019T04:18	Jan19	Tue	6	10	3	Per = 16.0 d, inc = 5.2 deg, r = 37.978 Rs, phase = 103 deg
S56	125	OTM-234	~apo	2010-020T22:45	Jan20	Wed	8	8	5	D/L start, burn ~6h later. Backup D/L 021T22:45
S57	125	RINGS	Segment	2010-023T15:00	Jan23	Sat	11	5	3	RINGS_125 (5d); RINGS_125
S57	125	S57 Begins		2010-023T15:00	Jan23	Sat	11	5	3	S57 sequence. Duration = 37 d
S57	125	OTM-235	125TI-3d	2010-025T12:16	Jan25	Mon	13	3	5	D/L start, burn ~6h later. Backup D/L 025T22:31
S57	125	Earth OCC	RING	2010-026T18:00	Jan26	Tue	14	2	1	Egress = 2010-027T01:43 (464 min)
S57	125	Sun OCC	SATURN	2010-026T19:15	Jan26	Tue	14	2	1	Egress = 2010-026T22:47 (213 min)
S57	125	Earth OCC	SATURN	2010-026T20:06	Jan26	Tue	14	2	1	Egress = 2010-026T23:29 (204 min)
S57	125	125TL (nt)	TELESTO	2010-027T00:30	Jan27	Wed	14	2	2	Inbound 97062.5 km flyby, v = 7.2 km/s, phase = 91 deg
S57	125	125ME (nt)	METHONE	2010-027T01:40	Jan27	Wed	14	2	2	Inbound 72072.5 km flyby, v = 14.0 km/s, phase = 60 deg
S57	125	G_ring	DUST HAZARD	2010-027T02:59	Jan27	Wed	14	2	2	Egress at 2010-027T03:03 (0.0 min); protection = HGA to ram and MEA cover closure.
S57	125	Ring CRX	Ascending	2010-027T03:01	Jan27	Wed	14	2	2	r = 2.740 Rs (165299 km)
S57	125	Periapse		2010-027T04:07	Jan27	Wed	14	2	2	R = 2.572 Rs, lat = 3 deg, phase = 77 deg
S57	125	125PN (nt)	PAN	2010-027T04:31	Jan27	Wed	14	2	2	Outbound 97945.4 km flyby, v = 14.8 km/s, phase = 123 deg
S57	125	125PM (nt)	PROMETHEUS	2010-027T04:36	Jan27	Wed	14	2	2	Outbound 32517.2 km flyby, v = 7.2 km/s, phase = 112 deg
S57	125	125AE (nt)	AEGAEON	2010-027T05:19	Jan27	Wed	14	2	2	Outbound 13308.1 km flyby, v = 7.4 km/s, phase = 97 deg
S57	125	125DI (nt)	DIONE	2010-027T11:47	Jan27	Wed	15	1	2	Outbound 45082.5 km flyby, v = 8.7 km/s, phase = 106 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S57	125	**TOST**	**Segment**	2010-028T07:16	Jan28	Thu	15	1	3	TOST_125_T66 (3d); Titan Flyby T66
S57	125	Ring CRX	Descending	2010-028T20:30	Jan28	Thu	16	0	3	r = 19.635 Rs (1184552 km)
S57	125	**125TI (t) [T66]**	**TITAN**	2010-028T22:29	Jan28	Thu	16	16	3	Outbound 7490.4 km flyby, v = 5.7 km/s, phase = 69 deg
S57	125	**SATURN**	**Segment**	2010-031T07:01	Jan31	Sun	2	13	6	SATURN_125_127 (30d); SATURN_125_127
S57	125	**OTM-236**	125TI+3d	2010-031T22:01	Jan31	Sun	3	13	6	D/L start, burn ~6h later. Backup D/L 032T22:01
S57	125	Ring CRX	Ascending	2010-032T05:48	Feb01	Mon	3	12	0	r = 35.396 Rs (2135437 km)
S57	126	**Apoapse**		2010-035T22:19	Feb04	Thu	7	9	4	Per = 17.5 d, inc = 0.3 deg, r = 40.258 Rs, phase = 106 deg
S57	126	**Sun OCC**	SATURN	2010-044T05:58	Feb13	Sat	15	0	12	Egress = 2010-044T09:39 (222 min)
S57	126	**Earth OCC**	SATURN	2010-044T07:08	Feb13	Sat	15	0	12	Egress = 2010-044T10:12 (184 min)
S57	126	126CA (nt)	**CALYPSO**	2010-044T11:45	Feb13	Sat	16	0	13	Inbound 21305.3 km flyby, v = 10.1 km/s, phase = 72 deg
S57	126	E_ring_lg	**DUST HAZARD**	2010-044T13:04	Feb13	Sat	16	0	13	Egress at 2010-044T14:22 (0.1 min); protection = MEA cover closure.
S57	126	Pallene	**DUST HAZARD**	2010-044T14:20	Feb13	Sat	16	0	13	Egress at 2010-044T14:31 (0.0 min); protection = MEA cover closure.
S57	126	G_ring	**DUST HAZARD**	2010-044T15:51	Feb13	Sat	16	0	13	Egress at 2010-044T17:08 (0.1 min); protection = HGA to ram and MEA cover closure.
S57	126	126EP (nt)	**EPIMETHEUS**	2010-044T16:16	Feb13	Sat	16	0	13	Inbound 93226.3 km flyby, v = 12.4 km/s, phase = 30 deg
S57	126	126JA (nt)	**JANUS**	2010-044T16:22	Feb13	Sat	16	0	13	Inbound 115080.2 km flyby, v = 14.4 km/s, phase = 29 deg
S57	126	Ring CRX	Descending	2010-044T16:24	Feb13	Sat	16	0	13	r = 2.939 Rs (177298 km)
S57	126	Periapse		2010-044T16:55	Feb13	Sat	16	0	13	R = 2.908 Rs, lat = -0 deg, phase = 74 deg
S57	126	**126MI (t) [M1]**	**MIMAS**	2010-044T17:22	Feb13	Sat	16	17	13	Outbound 9545.9 km flyby, v = 5.8 km/s, phase = 99 deg
S57	126	126TE (nt)	**TETHYS**	2010-044T18:16	Feb13	Sat	0	17	13	Outbound 111979.3 km flyby, v = 8.2 km/s, phase = 114 deg
S57	126	E_ring_lg	**DUST HAZARD**	2010-044T19:27	Feb13	Sat	0	17	13	Egress at 2010-044T19:52 (0.0 min); protection = MEA cover closure.
S57	126	126TI (nt)	**TITAN**	2010-045T22:47	Feb14	Sun	1	16	14	Outbound 650561.8 km flyby, v = 8.8 km/s, phase = 32 deg
S57	126	Ring CRX	Ascending	2010-049T14:56	Feb18	Thu	5	12	18	r = 34.923 Rs (2106903 km)
S57	127	**Apoapse**		2010-053T11:27	Feb22	Mon	9	8	22	Per = 17.5 d, inc = 0.3 deg, r = 40.256 Rs, phase = 106 deg
S57	127	**OTM-237**	~apo	2010-054T10:33	Feb23	Tue	10	7	23	D/L start, burn ~6h later. Backup D/L 055T10:18
S57	127	**OTM-238**	R2-3d	2010-058T10:19	Feb27	Sat	14	3	4	D/L start, burn ~6h later. Backup D/L 059T10:04
S58	127	**SOST**	**Segment**	2010-060T19:04	Mar01	Mon	16	1	2	SOST_127 (2d); Rhea Flyby
S58	127	**S58 Begins**		2010-060T19:04	Mar01	Mon	16	1	2	S58 sequence. Duration = 34 d
S58	127	**127RH (t) [R2]**	**RHEA**	2010-061T17:41	Mar02	Tue	17	34	3	Inbound 100.9 km flyby, v = 8.6 km/s, phase = 87 deg
S58	127	**Sun OCC**	SATURN	2010-061T19:11	Mar02	Tue	0	34	3	Egress = 2010-061T22:49 (218 min)
S58	127	**Earth OCC**	SATURN	2010-061T19:51	Mar02	Tue	0	34	3	Egress = 2010-061T23:09 (199 min)
S58	127	Ring CRX	Descending	2010-062T01:51	Mar03	Wed	0	34	4	r = 4.362 Rs (263178 km)
S58	127	E_ring_lg	**DUST HAZARD**	2010-062T01:54	Mar03	Wed	0	34	4	Egress at 2010-062T03:30 (0.1 min); protection = MEA cover closure.
S58	127	Pallene	**DUST HAZARD**	2010-062T03:29	Mar03	Wed	0	34	4	Egress at 2010-062T03:40 (0.0 min); protection = MEA cover closure.
S58	127	127PN (nt)	**PAN**	2010-062T04:38	Mar03	Wed	0	33	4	Inbound 103812.7 km flyby, v = 6.0 km/s, phase = 149 deg
S58	127	Periapse		2010-062T06:02	Mar03	Wed	1	33	4	R = 2.915 Rs, lat = -0 deg, phase = 75 deg
S58	127	E_ring_lg	**DUST HAZARD**	2010-062T10:02	Mar03	Wed	1	33	4	Egress at 2010-062T10:11 (0.0 min); protection = MEA cover closure.
S58	127	127CA (nt)	**CALYPSO**	2010-062T11:03	Mar03	Wed	1	33	4	Outbound 91794.7 km flyby, v = 12.8 km/s, phase = 77 deg
S58	127	127HE (nt)	**HELENE**	2010-062T13:41	Mar03	Wed	1	33	4	Outbound 1817.5 km flyby, v = 9.1 km/s, phase = 106 deg
S58	127	Ring CRX	Ascending	2010-062T15:20	Mar03	Wed	1	33	4	r = 7.158 Rs (431844 km)
S58	127	**MAG**	**Segment**	2010-063T05:04	Mar04	Thu	1	32	5	MAG_127_128 (15d); MAG_127_128
S58	127	**OTM-239**	R2+8d	2010-069T19:34	Mar10	Wed	8	26	11	D/L start, burn ~6h later. Backup D/L 070T19:19
S58	128	**Apoapse**		2010-071T01:44	Mar12	Fri	9	25	1	Per = 17.6 d, inc = 0.4 deg, r = 40.401 Rs, phase = 105 deg
S58	128	**RINGS**	**Segment**	2010-078T03:49	Mar19	Fri	16	18	8	RINGS_128 (6d); RINGS_128
S58	128	128TI (nt)	**TITAN**	2010-078T14:41	Mar19	Fri	17	17	9	Inbound 648209.1 km flyby, v = 8.9 km/s, phase = 119 deg
S58	128	**Sun OCC**	SATURN	2010-079T10:39	Mar20	Sat	18	16	10	Egress = 2010-079T14:11 (213 min)
S58	128	**Earth OCC**	SATURN	2010-079T10:41	Mar20	Sat	18	16	10	Egress = 2010-079T14:16 (215 min)
S58	128	128TL (nt)	**TELESTO**	2010-079T16:22	Mar20	Sat	18	16	10	Inbound 89168.5 km flyby, v = 12.7 km/s, phase = 72 deg
S58	128	Ring CRX	Descending	2010-079T16:31	Mar20	Sat	18	16	10	r = 4.732 Rs (285472 km)
S58	128	E_ring_lg	**DUST HAZARD**	2010-079T17:14	Mar20	Sat	18	16	10	Egress at 2010-079T18:50 (0.1 min); protection = MEA cover closure.
S58	128	Pallene	**DUST HAZARD**	2010-079T18:49	Mar20	Sat	18	16	10	Egress at 2010-079T18:58 (0.0 min); protection = MEA cover closure.
S58	128	128AT (nt)	**ATLAS**	2010-079T20:53	Mar20	Sat	18	16	10	Inbound 106900.7 km flyby, v = 10.5 km/s, phase = 137 deg
S58	128	128PM (nt)	**PROMETHEUS**	2010-079T21:15	Mar20	Sat	18	16	10	Inbound 44031.3 km flyby, v = 4.3 km/s, phase = 107 deg
S58	128	Periapse		2010-079T21:23	Mar20	Sat	18	16	10	R = 2.913 Rs, lat = -0 deg, phase = 75 deg
S58	128	128DA (nt)	**DAPHNIS**	2010-079T21:45	Mar20	Sat	18	16	10	Outbound 60529.2 km flyby, v = 5.2 km/s, phase = 26 deg
S58	128	128ME (nt)	**METHONE**	2010-079T23:35	Mar20	Sat	18	16	10	Outbound 56778.8 km flyby, v = 11.3 km/s, phase = 99 deg
S58	128	128PL (nt)	**PALLENE**	2010-080T00:08	Mar21	Sun	18	16	10	Outbound 75132.7 km flyby, v = 12.8 km/s, phase = 94 deg
S58	128	E_ring_lg	**DUST HAZARD**	2010-080T00:13	Mar21	Sun	18	16	10	Egress at 2010-080T01:31 (0.1 min); protection = MEA cover closure.
S58	128	128PO (nt)	**POLYDEUCES**	2010-080T04:52	Mar21	Sun	18	15	10	Outbound 33330.7 km flyby, v = 10.3 km/s, phase = 70 deg
S58	128	Ring CRX	Ascending	2010-080T05:09	Mar21	Sun	18	15	10	r = 6.331 Rs (381952 km)
S58	128	**SATURN**	**Segment**	2010-084T03:34	Mar25	Thu	22	12	14	SATURN_128_129 (11d); SATURN_128_129
S58	128	**OTM-240**	~apo	2010-085T08:19	Mar26	Fri	24	10	16	D/L start, burn ~6h later. Backup D/L 086T18:19
S58	129	**Apoapse**		2010-088T17:02	Mar29	Mon	27	7	3	Per = 17.6 d, inc = 0.4 deg, r = 40.401 Rs, phase = 105 deg
S58	129	**OTM-241**	129TI-3d	2010-092T07:49	Apr02	Fri	31	3	7	D/L start, burn ~6h later. Backup D/L 093T07:34
S59	129	**TOST**	**Segment**	2010-095T02:49	Apr05	Mon	33	1	3	TOST_129_T67 (2d); Titan Flyby T67
S59	129	**S59 Begins**		2010-095T02:49	Apr05	Mon	33	1	3	S59 sequence. Duration = 42 d
S59	129	**129TI (dt) [T67]**	**TITAN**	2010-095T15:51	Apr05	Mon	34	2	3	Inbound 7461.9 km flyby, v = 5.7 km/s, phase = 73 deg
S59	129	**SOST**	**Segment**	2010-096T16:34	Apr06	Tue	1	1	4	SOST_129 (2d); Dione Flyby
S59	129	**Earth OCC**	SATURN	2010-097T00:17	Apr07	Wed	1	0	5	Egress = 2010-097T04:01 (224 min)
S59	129	**Sun OCC**	SATURN	2010-097T00:55	Apr07	Wed	1	0	5	Egress = 2010-097T04:15 (200 min)
S59	129	**129DI (dt) [D2]**	**DIONE**	2010-097T05:16	Apr07	Wed	2	21	5	Inbound 503.0 km flyby, v = 8.4 km/s, phase = 79 deg
S59	129	Ring CRX	Descending	2010-097T06:01	Apr07	Wed	0	21	5	r = 5.887 Rs (355175 km)
S59	129	129CA (nt)	**CALYPSO**	2010-097T08:00	Apr07	Wed	0	21	5	Inbound 97948.3 km flyby, v = 11.7 km/s, phase = 69 deg
S59	129	E_ring_lg	Dust Crossing	2010-097T09:21	Apr07	Wed	0	21	5	Egress at 2010-097T10:14 (0.0 min); protection = none.
S59	129	129EP (nt)	**EPIMETHEUS**	2010-097T12:46	Apr07	Wed	0	20	5	Inbound 61716.2 km flyby, v = 2.4 km/s, phase = 89 deg
S59	129	Periapse		2010-097T12:53	Apr07	Wed	0	20	5	R = 3.525 Rs, lat = -0 deg, phase = 80 deg
S59	129	129JA (nt)	**JANUS**	2010-097T13:44	Apr07	Wed	0	20	5	Outbound 74603.7 km flyby, v = 2.8 km/s, phase = 40 deg
S59	129	129MI (nt)	**MIMAS**	2010-097T13:47	Apr07	Wed	0	20	5	Outbound 97424.4 km flyby, v = 10.3 km/s, phase = 124 deg
S59	129	129TE (nt)	**TETHYS**	2010-097T14:58	Apr07	Wed	0	20	5	Outbound 70784.2 km flyby, v = 6.4 km/s, phase = 103 deg
S59	129	Ring CRX	Ascending	2010-097T22:34	Apr07	Wed	1	20	6	r = 7.289 Rs (439731 km)
S59	129	**MAG**	**Segment**	2010-098T10:04	Apr08	Thu	1	20	6	MAG_129_130 (19d); MAG_129_130

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S59	129	**OTM-242**	D2+4d	2010-100T17:19	Apr10	Sat	4	17	8	D/L start, burn ~6h later. Backup D/L 101T17:19
S59	130	**Apoapse**		2010-107T17:21	Apr17	Sat	11	10	7	Per = 20.4 d, inc = 0.3 deg, r = 44.158 Rs, phase = 100 deg
S59	130	**OTM-243**	~apo	2010-108T06:33	Apr18	Sun	11	10	8	D/L start, burn ~6h later. Backup D/L 109T06:33
S59	130	**OTM-244**	E9-3d	2010-114T16:18	Apr24	Sat	17	3	6	D/L start, burn ~6h later. Backup D/L 115T16:18
S59	130	Ring CRX	Descending	2010-117T01:11	Apr27	Tue	20	1	2	r = 12.236 Rs (738185 km)
S59	130	**SOST**	**Segment**	2010-117T07:47	Apr27	Tue	20	1	3	SOST_130 (2d); E9 Flyby
S59	130	**Earth OCC**	SATURN	2010-117T08:30	Apr27	Tue	20	1	3	Egress = 2010-117T12:31 (242 min)
S59	130	**Sun OCC**	SATURN	2010-117T09:49	Apr27	Tue	20	1	3	Egress = 2010-117T13:07 (198 min)
S59	130	130PM (nt)	**PROMETHEUS**	2010-117T20:39	Apr27	Tue	21	0	3	Inbound 84529.2 km flyby, v = 1.6 km/s, phase = 116 deg
S59	130	130TL (nt)	**TELESTO**	2010-117T20:45	Apr27	Tue	21	0	3	Inbound 79587.6 km flyby, v = 6.7 km/s, phase = 97 deg
S59	130	130TI (nt)	**TITAN**	2010-117T21:09	Apr27	Tue	21	0	3	Inbound 989104.4 km flyby, v = 12.5 km/s, phase = 94 deg
S59	130	Periapse		2010-117T21:41	Apr27	Tue	21	0	3	R = 3.529 Rs, lat = -0 deg, phase = 81 deg
S59	130	E_ring_lg	Dust Crossing	2010-117T22:11	Apr27	Tue	21	0	3	Egress at 2010-118T01:12 (0.1 min); protection = none.
S59	130	130DA (nt)	**DAPHNIS**	2010-117T22:42	Apr27	Tue	21	0	3	Outbound 86357.7 km flyby, v = 1.4 km/s, phase = 47 deg
S59	130	130PA (nt)	**PANDORA**	2010-117T23:05	Apr27	Tue	21	0	3	Outbound 91560.1 km flyby, v = 1.8 km/s, phase = 34 deg
S59	130	130PL (nt)	**PALLENE**	2010-117T23:46	Apr27	Tue	21	0	3	Outbound 71194.8 km flyby, v = 9.7 km/s, phase = 112 deg
S59	130	130PN (nt)	**PAN**	2010-117T23:53	Apr27	Tue	21	0	3	Outbound 113814.9 km flyby, v = 0.5 km/s, phase = 24 deg
S59	130	**130EN (t) [E9]**	**ENCELADUS**	2010-118T00:10	Apr28	Wed	21	20	3	Outbound 99.0 km flyby, v = 6.5 km/s, phase = 93 deg
S59	130	Ring CRX	Ascending	2010-118T01:22	Apr28	Wed	0	20	3	r = 4.409 Rs (266008 km)
S59	130	**SATURN**	**Segment**	2010-119T01:03	Apr29	Thu	1	19	4	SATURN_130_131 (19d); SATURN_130_131
S59	130	**OTM-245**	E9+2d	2010-119T05:47	Apr29	Thu	1	19	5	D/L start, burn ~6h later. Backup D/L 120T05:47
S59	131	**Apoapse**		2010-128T03:04	May08	Sat	10	10	9	Per = 20.5 d, inc = 0.3 deg, r = 44.288 Rs, phase = 99 deg
S59	131	**OTM-246**	~apo	2010-131T05:01	May11	Tue	13	7	12	D/L start, burn ~6h later. Backup D/L 132T05:01
S59	131	**OTM-247**	131TI-4d or TBD	2010-135T22:31	May15	Sat	18	2	5	D/L start, burn ~6h later. Backup D/L 137T04:31
S60	131	**SOST**	**Segment**	2010-137T13:31	May17	Mon	20	1	2	SOST_131 (2d); E10 Flyby
S60	131	**S60 Begins**		2010-137T13:31	May17	Mon	20	1	2	S60 sequence. Duration = 39 d
S60	131	**Earth OCC**	SATURN	2010-137T19:03	May17	Mon	20	0	2	Egress = 2010-137T23:08 (246 min)
S60	131	**Sun OCC**	SATURN	2010-137T20:56	May17	Mon	20	0	2	Egress = 2010-137T23:59 (183 min)
S60	131	Ring CRX	Descending	2010-138T04:01	May18	Tue	20	0	2	r = 4.759 Rs (287088 km)
S60	131	E_ring_lg	Dust Crossing	2010-138T04:59	May18	Tue	20	0	2	Egress at 2010-138T07:54 (0.1 min); protection = none.
S60	131	**131EN (t) [E10]**	**ENCELADUS**	2010-138T06:05	May18	Tue	20	2	2	Inbound 434.1 km flyby, v = 6.5 km/s, phase = 114 deg
S60	131	131ME (nt)	**METHONE**	2010-138T08:21	May18	Tue	0	2	2	Inbound 18325.9 km flyby, v = 4.3 km/s, phase = 75 deg
S60	131	Periapse		2010-138T08:32	May18	Tue	0	2	2	R = 3.518 Rs, lat = -0 deg, phase = 81 deg
S60	131	E_ring_lg	Dust Crossing	2010-138T09:09	May18	Tue	0	2	2	Egress at 2010-138T12:03 (0.1 min); protection = none.
S60	131	Ring CRX	Ascending	2010-139T00:13	May19	Wed	1	1	3	r = 10.153 Rs (612513 km)
S60	131	**TOST**	**Segment**	2010-139T07:15	May19	Wed	1	1	3	TOST_131_T68 (3d); Titan Flyby T68
S60	131	**Sun OCC**	TITAN	2010-140T03:10	May20	Thu	2	0	4	Egress = 2010-140T03:13 (4 min)
S60	131	**131TI (t) [T68]**	**TITAN**	2010-140T03:24	May20	Thu	2	16	4	Outbound 1400.0 km flyby, v = 5.9 km/s, phase = 112 deg
S60	131	**MAG**	**Segment**	2010-142T07:00	May22	Sat	2	14	6	MAG_131_132 (7d); MAG_131_132
S60	131	**OTM-248**	131TI+3d	2010-143T04:15	May23	Sun	3	13	7	D/L start, burn ~6h later. Backup D/L 144T04:00
S60	132	**Apoapse**		2010-146T08:05	May26	Wed	6	10	3	Per = 16.0 d, inc = 12.1 deg, r = 37.792 Rs, phase = 91 deg
S60	132	**OTM-249**	~apo	2010-148T03:44	May28	Fri	8	8	5	D/L start, burn ~6h later. Backup D/L 149T03:44
S60	132	**RINGS**	**Segment**	2010-149T12:44	May29	Sat	9	7	1	RINGS_132 (6d); RINGS_132
S60	132	**OTM-250**	132TI-3d	2010-152T13:44	Jun01	Tue	12	4	4	D/L start, burn ~6h later. Backup D/L 153T13:44
S60	132	132HE (nt)	**HELENE**	2010-154T00:39	Jun03	Thu	14	2	1	Inbound 114165.7 km flyby, v = 11.7 km/s, phase = 85 deg
S60	132	132CA (nt)	**CALYPSO**	2010-154T06:11	Jun03	Thu	14	2	2	Inbound 119927.2 km flyby, v = 8.9 km/s, phase = 80 deg
S60	132	G_ring	**DUST HAZARD**	2010-154T06:23	Jun03	Thu	14	2	2	Egress at 2010-154T06:25 (0.0 min); protection = MEA cover closure.
S60	132	Ring CRX	Descending	2010-154T06:24	Jun03	Thu	14	2	2	r = 2.950 Rs (177993 km)
S60	132	132PA (nt)	**PANDORA**	2010-154T07:16	Jun03	Thu	14	2	2	Inbound 97168.0 km flyby, v = 14.1 km/s, phase = 44 deg
S60	132	Periapse		2010-154T07:43	Jun03	Thu	14	2	2	R = 2.739 Rs, lat = -6 deg, phase = 89 deg
S60	132	132TE (nt)	**TETHYS**	2010-154T12:57	Jun03	Thu	14	2	2	Outbound 52578.3 km flyby, v = 9.6 km/s, phase = 99 deg
S60	132	132RH (nt)	**RHEA**	2010-154T20:32	Jun03	Thu	15	1	2	Outbound 69355.8 km flyby, v = 8.0 km/s, phase = 102 deg
S60	132	**TOST**	**Segment**	2010-155T12:13	Jun04	Fri	15	1	3	TOST_132_T69 (4d); Titan Flyby T69
S60	132	Ring CRX	Ascending	2010-155T21:44	Jun04	Fri	16	0	3	r = 18.763 Rs (1131982 km)
S60	132	**132TI (t) [T69]**	**TITAN**	2010-156T02:26	Jun05	Sat	16	16	4	Outbound 2044.1 km flyby, v = 5.8 km/s, phase = 87 deg
S60	132	Ring CRX	Descending	2010-158T04:55	Jun07	Mon	2	14	6	r = 30.915 Rs (1865086 km)
S60	132	**OTM-251**	132TI+3d	2010-158T20:43	Jun07	Mon	3	13	6	D/L start, burn ~6h later. Backup D/L 159T20:43
S60	132	**MAG**	**Segment**	2010-159T05:43	Jun08	Tue	3	13	0	MAG_132_133 (12d); MAG_132_133
S60	133	**Apoapse**		2010-162T07:28	Jun11	Fri	6	10	3	Per = 16.0 d, inc = 2.0 deg, r = 37.903 Rs, phase = 91 deg
S60	133	**OTM-252**	~apo	2010-164T02:42	Jun13	Sun	8	8	5	D/L start, burn ~6h later. Backup D/L 165T02:42
S60	133	**OTM-253**	T70-3d	2010-168T20:11	Jun17	Thu	13	3	5	D/L start, burn ~6h later. Backup D/L 169T12:41; complexes G/M
S60	133	133RH (nt)	**RHEA**	2010-169T19:50	Jun18	Fri	14	2	1	Inbound 120515.3 km flyby, v = 10.7 km/s, phase = 109 deg
S60	133	**Earth OCC**	RING	2010-169T21:08	Jun18	Fri	14	2	1	Egress = 2010-170T05:03 (475 min)
S60	133	**Earth OCC**	SATURN	2010-169T23:18	Jun18	Fri	14	2	1	Egress = 2010-170T02:35 (197 min)
S60	133	**Sun OCC**	SATURN	2010-170T00:17	Jun19	Sat	14	2	1	Egress = 2010-170T03:10 (174 min)
S60	133	G_ring	**DUST HAZARD**	2010-170T06:22	Jun19	Sat	14	2	1	Egress at 2010-170T06:33 (0.0 min); protection = HGA to ram and MEA cover closure.
S60	133	Ring CRX	Ascending	2010-170T06:28	Jun19	Sat	14	2	1	r = 2.670 Rs (161065 km)
S60	133	Periapse		2010-170T07:03	Jun19	Sat	14	2	1	R = 2.623 Rs, lat = 1 deg, phase = 89 deg
S60	133	133PN (nt)	**PAN**	2010-170T07:10	Jun19	Sat	14	2	1	Outbound 29344.4 km flyby, v = 5.0 km/s, phase = 115 deg
S60	133	133PO (nt)	**POLYDEUCES**	2010-170T14:09	Jun19	Sat	14	1	2	Outbound 97489.3 km flyby, v = 12.4 km/s, phase = 79 deg
S60	133	**TOST**	**Segment**	2010-171T11:11	Jun20	Sun	15	1	3	TOST_133_T70 (5d); Titan Flyby T70
S60	133	**133TI (t) [T70]**	**TITAN**	2010-172T01:28	Jun21	Mon	16	16	3	Outbound 880.0 km flyby, v = 5.9 km/s, phase = 82 deg
S60	133	Ring CRX	Descending	2010-172T04:16	Jun21	Mon	0	16	3	r = 20.995 Rs (1266604 km)
S60	133	**OTM-254**	T70+3d	2010-175T01:56	Jun24	Thu	3	13	6	D/L start, burn ~6h later. Backup D/L 176T12:10; complexes C/M
S61	133	**MAG**	**Segment**	2010-176T21:10	Jun25	Fri	5	11	2	MAG_133_134 (8d); MAG_133_134
S61	133	**S61 Begins**		2010-176T21:10	Jun25	Fri	5	11	2	S61 sequence. Duration = 35 d
S61	134	**Apoapse**		2010-178T05:26	Jun27	Sun	6	10	3	Per = 16.0 d, inc = 19.1 deg, r = 37.556 Rs, phase = 92 deg
S61	134	**OTM-255**	Apo	2010-181T01:40	Jun30	Wed	9	7	6	D/L start, burn ~6h later. Backup D/L 182T01:40; complexes C/C
S61	134	**OTM-256**	T71-3d	2010-184T19:09	Jul03	Sat	13	3	4	D/L start, burn ~6h later. Backup D/L 185T11:24; complexes G/M

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S61	134	**RINGS**	**Segment**	2010-185T04:09	Jul04	Sun	13	3	0	RINGS_134 (2d); Rings 134 periapse
S61	134	134CA (nt)	**CALYPSO**	2010-186T00:35	Jul05	Mon	14	2	1	Inbound 114636.1 km flyby, v = 12.7 km/s, phase = 70 deg
S61	134	134EN (nt)	**ENCELADUS**	2010-186T02:02	Jul05	Mon	14	2	1	Inbound 58130.5 km flyby, v = 11.4 km/s, phase = 58 deg
S61	134	Ring CRX	Ascending	2010-186T03:37	Jul05	Mon	14	2	1	r = 3.174 Rs (191481 km)
S61	134	134DA (nt)	**DAPHNIS**	2010-186T05:01	Jul05	Mon	14	2	1	Inbound 72576.7 km flyby, v = 9.2 km/s, phase = 48 deg
S61	134	Periapse		2010-186T05:02	Jul05	Mon	14	2	1	R = 2.967 Rs, lat = 10 deg, phase = 88 deg
S61	134	**TOST**	**Segment**	2010-187T10:09	Jul06	Tue	15	1	3	TOST_134_T71 (3d); Titan Flyby T71
S61	134	**134TI (t) [T71]**	**TITAN**	2010-188T00:23	Jul07	Wed	16	38	3	Outbound 1005.0 km flyby, v = 5.9 km/s, phase = 82 deg
S61	134	Ring CRX	Descending	2010-188T03:18	Jul07	Wed	0	38	3	r = 21.054 Rs (1270208 km)
S61	134	**SATURN**	**Segment**	2010-190T10:09	Jul09	Fri	2	36	6	SATURN_134_135 (17d); Saturn 134_135 apo and 135 peri
S61	134	**OTM-257**	T71+3d	2010-191T00:53	Jul10	Sat	3	35	6	D/L start, burn ~6h later. Backup D/L 192T00:53; complexes C/C
S61	135	**Apoapse**		2010-196T04:19	Jul15	Thu	8	30	5	Per = 19.9 d, inc = 4.5 deg, r = 43.515 Rs, phase = 97 deg
S61	135	**OTM-258**	Apo	2010-199T00:37	Jul18	Sun	11	27	8	D/L start, burn ~6h later. Backup D/L 200T00:22; complexes C/C
S61	135	**Sun OCC**	RING	2010-205T13:50	Jul24	Sat	18	20	7	Egress = 2010-205T20:52 (423 min)
S61	135	**Earth OCC**	SATURN	2010-205T14:42	Jul24	Sat	18	20	7	Egress = 2010-205T18:42 (241 min)
S61	135	**Sun OCC**	SATURN	2010-205T15:59	Jul24	Sat	18	20	7	Egress = 2010-205T19:44 (225 min)
S61	135	E_ring_lg	Dust Crossing	2010-206T01:07	Jul25	Sun	18	20	7	Egress at 2010-206T01:30 (0.0 min); protection = none.
S61	135	Ring CRX	Ascending	2010-206T01:19	Jul25	Sun	18	20	7	r = 3.719 Rs (224348 km)
S61	135	Periapse		2010-206T03:14	Jul25	Sun	18	20	7	R = 3.438 Rs, lat = 3 deg, phase = 83 deg
S61	135	135PM (nt)	**PROMETHEUS**	2010-206T03:30	Jul25	Sun	18	20	7	Outbound 70085.7 km flyby, v = 2.2 km/s, phase = 70 deg
S61	135	135AT (nt)	**ATLAS**	2010-206T04:32	Jul25	Sun	18	20	7	Outbound 89552.9 km flyby, v = 2.1 km/s, phase = 37 deg
S61	135	**MAG**	**Segment**	2010-207T09:06	Jul26	Mon	19	19	8	MAG_135 (4d); MAG apo 135_136 split at S61/S62
S61	135	Ring CRX	Descending	2010-208T04:42	Jul27	Tue	20	18	9	r = 22.106 Rs (1333626 km)
S62	135	**MAG**	**Segment**	2010-211T18:51	Jul30	Fri	24	14	13	MAG_135_136 (14d); MAG apo 135_136 split at S61/S62
S62	135	**S62 Begins**		2010-211T18:51	Jul30	Fri	24	14	13	S62 sequence. Duration = 37 d
S62	136	**Apoapse**		2010-216T02:12	Aug04	Wed	28	10	17	Per = 19.9 d, inc = 4.7 deg, r = 43.524 Rs, phase = 97 deg
S62	136	**OTM-259**	E11-3d	2010-222T16:35	Aug10	Tue	35	3	24	D/L start, burn ~6h later. Backup D/L 223T16:35; complexes G/G
S62	136	136TI (nt)	**TITAN**	2010-223T13:35	Aug11	Wed	36	2	1	Inbound 417278.3 km flyby, v = 4.0 km/s, phase = 80 deg
S62	136	**Sun OCC**	DIONE	2010-225T03:14	Aug13	Fri	37	1	2	Egress = 2010-225T03:16 (3 min)
S62	136	**SOST**	**Segment**	2010-225T07:49	Aug13	Fri	37	1	3	SOST_136 (2d); E11 Flyby
S62	136	**Earth OCC**	RING	2010-225T11:38	Aug13	Fri	37	0	3	Egress = 2010-225T19:28 (471 min)
S62	136	**Sun OCC**	RING	2010-225T12:27	Aug13	Fri	38	0	3	Egress = 2010-225T18:05 (338 min)
S62	136	**Earth OCC**	SATURN	2010-225T12:52	Aug13	Fri	38	0	3	Egress = 2010-225T16:51 (240 min)
S62	136	**Sun OCC**	SATURN	2010-225T13:55	Aug13	Fri	38	0	3	Egress = 2010-225T17:36 (222 min)
S62	136	136DI (nt)	**DIONE**	2010-225T17:44	Aug13	Fri	38	0	3	Inbound 108729.7 km flyby, v = 11.3 km/s, phase = 84 deg
S62	136	136CA (nt)	**CALYPSO**	2010-225T20:31	Aug13	Fri	38	0	3	Inbound 22711.7 km flyby, v = 7.4 km/s, phase = 99 deg
S62	136	**136EN (t) [E11]**	**ENCELADUS**	2010-225T22:31	Aug13	Fri	38	42	3	Inbound 2550.4 km flyby, v = 6.8 km/s, phase = 85 deg
S62	136	E_ring_lg	Dust Crossing	2010-225T22:53	Aug13	Fri	0	42	3	Egress at 2010-225T23:17 (0.0 min); protection = none.
S62	136	Ring CRX	Ascending	2010-225T23:05	Aug13	Fri	0	42	3	r = 3.753 Rs (226418 km)
S62	136	136EP (nt)	**EPIMETHEUS**	2010-225T23:21	Aug13	Fri	0	42	3	Inbound 101927.5 km flyby, v = 3.7 km/s, phase = 149 deg
S62	136	136AT (nt)	**ATLAS**	2010-226T00:33	Aug14	Sat	0	42	3	Inbound 74217.3 km flyby, v = 2.1 km/s, phase = 102 deg
S62	136	Periapse		2010-226T01:04	Aug14	Sat	0	42	3	R = 3.457 Rs, lat = 3 deg, phase = 83 deg
S62	136	136PA (nt)	**PANDORA**	2010-226T01:08	Aug14	Sat	0	42	3	Outbound 67749.3 km flyby, v = 2.3 km/s, phase = 78 deg
S62	136	136DA (nt)	**DAPHNIS**	2010-226T01:25	Aug14	Sat	0	42	3	Outbound 74370.0 km flyby, v = 2.0 km/s, phase = 68 deg
S62	136	136JA (nt)	**JANUS**	2010-226T02:01	Aug14	Sat	0	42	3	Outbound 79685.7 km flyby, v = 3.3 km/s, phase = 35 deg
S62	136	136PN (nt)	**PAN**	2010-226T02:05	Aug14	Sat	0	42	3	Outbound 86676.9 km flyby, v = 1.8 km/s, phase = 48 deg
S62	136	136TE (nt)	**TETHYS**	2010-226T06:03	Aug14	Sat	0	42	4	Outbound 38087.2 km flyby, v = 9.3 km/s, phase = 91 deg
S62	136	**XD**	**Segment**	2010-227T07:49	Aug15	Sun	1	40	5	XD_136_137 (18d); XD 136_137 apo
S62	136	Ring CRX	Descending	2010-228T01:07	Aug16	Mon	2	40	5	r = 21.702 Rs (1309269 km)
S62	136	**OTM-260**	E11+3d	2010-228T22:34	Aug16	Mon	3	39	6	D/L start, burn ~6h later. Backup D/L 229T22:34; complexes C/C
S62	137	**Apoapse**		2010-236T01:24	Aug24	Tue	10	32	7	Per = 20.0 d, inc = 4.6 deg, r = 43.691 Rs, phase = 97 deg
S62	137	**RINGS**	**Segment**	2010-245T00:18	Sep02	Thu	19	23	16	RINGS_137 (2d); Rings 137 periapse
S62	137	**Earth OCC**	RING	2010-245T11:22	Sep02	Thu	20	22	17	Egress = 2010-245T19:19 (478 min)
S62	137	**Earth OCC**	SATURN	2010-245T13:56	Sep02	Thu	20	22	17	Egress = 2010-245T17:48 (232 min)
S62	137	**Sun OCC**	RING	2010-245T14:09	Sep02	Thu	20	22	17	Egress = 2010-245T17:53 (224 min)
S62	137	**Sun OCC**	SATURN	2010-245T14:36	Sep02	Thu	20	22	17	Egress = 2010-245T18:14 (218 min)
S62	137	137PO (nt)	**POLYDEUCES**	2010-245T18:31	Sep02	Thu	20	22	17	Inbound 66879.2 km flyby, v = 7.0 km/s, phase = 94 deg
S62	137	137TL (nt)	**TELESTO**	2010-245T20:44	Sep02	Thu	20	22	17	Inbound 70828.4 km flyby, v = 10.8 km/s, phase = 73 deg
S62	137	**OTM-261**	Peri	2010-245T21:33	Sep02	Thu	20	22	17	D/L start, burn ~6h later. Backup D/L 246T21:33; complexes C/C
S62	137	E_ring_lg	Dust Crossing	2010-245T23:27	Sep02	Thu	20	22	0	Egress at 2010-245T23:51 (0.0 min); protection = none.
S62	137	Ring CRX	Ascending	2010-245T23:39	Sep02	Thu	20	22	0	r = 3.767 Rs (227268 km)
S62	137	Periapse		2010-246T01:40	Sep03	Fri	20	22	0	R = 3.462 Rs, lat = 3 deg, phase = 84 deg
S62	137	137AE (nt)	**AEGAEON**	2010-246T01:51	Sep03	Fri	20	22	0	Outbound 110165.6 km flyby, v = 9.3 km/s, phase = 27 deg
S62	137	137TI (nt)	**TITAN**	2010-246T02:30	Sep03	Fri	20	22	0	Outbound 985285.8 km flyby, v = 12.5 km/s, phase = 103 deg
S62	137	137EP (nt)	**EPIMETHEUS**	2010-246T03:00	Sep03	Fri	20	22	0	Outbound 92914.4 km flyby, v = 3.9 km/s, phase = 24 deg
S62	137	137DI (nt)	**DIONE**	2010-246T09:16	Sep03	Fri	20	21	0	Outbound 39311.7 km flyby, v = 9.2 km/s, phase = 83 deg
S62	137	**SATURN**	**Segment**	2010-247T06:33	Sep04	Sat	21	21	1	SATURN_137 (2d); Saturn 137_138 apo split at S62/S63
S62	137	Ring CRX	Descending	2010-248T00:48	Sep05	Sun	22	20	2	r = 21.427 Rs (1292672 km)
S63	137	**SATURN**	**Segment**	2010-249T06:33	Sep06	Mon	23	19	3	SATURN_137_138 (18d); Saturn 137_138 apo, 138 peri
S63	137	**S63 Begins**		2010-249T06:33	Sep06	Mon	23	19	3	S63 sequence. Duration = 35 d
S63	138	**Apoapse**		2010-256T00:46	Sep13	Mon	30	12	10	Per = 19.9 d, inc = 4.6 deg, r = 43.542 Rs, phase = 96 deg
S63	138	**OTM-261a**	Apo	2010-258T20:47	Sep15	Wed	33	9	13	D/L start, burn ~6h later. Backup D/L 259T20:47; complexes C/C
S63	138	**OTM-262**	T72-3d	2010-264T06:47	Sep21	Tue	38	3	5	D/L start, burn ~6h later. Backup D/L 264T20:32; complexes M/C
S63	138	**Earth OCC**	SATURN	2010-265T12:51	Sep22	Wed	40	2	1	Egress = 2010-265T16:29 (218 min)
S63	138	**Sun OCC**	SATURN	2010-265T13:04	Sep22	Wed	40	2	1	Egress = 2010-265T16:38 (214 min)
S63	138	**Earth OCC**	RING	2010-265T13:17	Sep22	Wed	40	2	1	Egress = 2010-265T15:24 (127 min)
S63	138	E_ring_lg	Dust Crossing	2010-265T21:44	Sep22	Wed	40	2	2	Egress at 2010-265T22:08 (0.0 min); protection = none.
S63	138	Ring CRX	Ascending	2010-265T21:57	Sep22	Wed	40	2	2	r = 3.768 Rs (227308 km)
S63	138	Periapse		2010-265T23:59	Sep22	Wed	40	2	2	R = 3.456 Rs, lat = 3 deg, phase = 84 deg
S63	138	138EN (nt)	**ENCELADUS**	2010-266T03:07	Sep23	Thu	40	2	2	Outbound 111991.6 km flyby, v = 12.6 km/s, phase = 104 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
S63	138	138CA (nt)	**CALYPSO**	2010-266T04:54	Sep23	Thu	40	2	2	Outbound 28921.8 km flyby, v = 8.4 km/s, phase = 86 deg
S63	138	138PO (nt)	**POLYDEUCES**	2010-266T07:25	Sep23	Thu	40	1	2	Outbound 100315.3 km flyby, v = 11.2 km/s, phase = 82 deg
S63	138	**TOST**	**Segment**	2010-267T05:17	Sep24	Fri	41	1	3	TOST_138_T72 (2d); Titan Flyby T72
S63	138	**138TI (t) [T72]**	**TITAN**	2010-267T18:39	Sep24	Fri	42	48	3	Outbound 8174.9 km flyby, v = 5.7 km/s, phase = 60 deg
S63	138	Ring CRX	Descending	2010-267T23:33	Sep24	Fri	0	48	4	r = 21.621 Rs (1304380 km)
S63	138	**XD**	**Segment**	2010-269T06:17	Sep26	Sun	1	46	5	XD_138_139 (15d); XD 138_139 apo split at S63/S64
S63	138	**OTM-263**	T72+3d	2010-270T20:02	Sep27	Mon	3	45	7	D/L start, burn ~6h later. Backup D/L 271T06:17; complexes C/M
S63	138	SEP = 3.0 deg	Conjunction	2010-271T11:25	Sep28	Tue	4	44	1	Commanding must not be required below 3.0 deg
S63	138	SEP = 3.0 deg	Conjunction	2010-276T09:29	Oct03	Sun	9	39	6	Commanding must not be required below 3.0 deg
S63	139	**Apoapse**		2010-277T20:48	Oct04	Mon	10	38	7	Per = 23.8 d, inc = 3.0 deg, r = 48.780 Rs, phase = 101 deg
---	139	**END OF EQUINOX MISSION**		2010-284T04:17	Oct11	Mon	16	31	13	End of Equinox mission / start of Solstice mission (2010-2017)
---	139	139TI (nt)	**TITAN**	2010-287T17:07	Oct14	Thu	20	28	17	Inbound 172365.7 km flyby, v = 4.8 km/s, phase = 80 deg
---	139	**OTM-264**	Peri	2010-287T19:02	Oct14	Thu	20	28	17	D/L start, burn ~6h later. Backup D/L 288T18:20; complexes C/C
---	139	**Sun OCC**	SATURN	2010-289T03:19	Oct16	Sat	21	26	1	Egress = 2010-289T06:30 (192 min)
---	139	**Earth OCC**	SATURN	2010-289T04:03	Oct16	Sat	21	26	1	Egress = 2010-289T06:35 (152 min)
---	139	139PO (nt)	**POLYDEUCES**	2010-289T10:46	Oct16	Sat	22	26	2	Inbound 116530.3 km flyby, v = 10.8 km/s, phase = 73 deg
---	139	Ring CRX	Ascending	2010-289T14:41	Oct16	Sat	22	26	2	r = 4.648 Rs (280435 km)
---	139	139MI (nt)	**MIMAS**	2010-289T17:32	Oct16	Sat	22	26	2	Inbound 69946.3 km flyby, v = 2.4 km/s, phase = 101 deg
---	139	139PL (nt)	**PALLENE**	2010-289T17:56	Oct16	Sat	22	26	2	Inbound 36118.0 km flyby, v = 3.5 km/s, phase = 82 deg
---	139	Periapse		2010-289T17:57	Oct16	Sat	22	26	2	R = 4.101 Rs, lat = 2 deg, phase = 80 deg
---	139	139TL (nt)	**TELESTO**	2010-289T18:03	Oct16	Sat	22	26	2	Outbound 48455.1 km flyby, v = 5.5 km/s, phase = 99 deg
---	139	139ME (nt)	**METHONE**	2010-289T18:39	Oct16	Sat	22	26	2	Outbound 105871.1 km flyby, v = 5.8 km/s, phase = 23 deg
---	139	139AE (nt)	**AEGAEON**	2010-289T19:18	Oct16	Sat	22	26	2	Outbound 96756.6 km flyby, v = 1.9 km/s, phase = 40 deg
---	139	139DI (nt)	**DIONE**	2010-290T01:03	Oct17	Sun	22	26	2	Outbound 31716.9 km flyby, v = 6.8 km/s, phase = 91 deg
---	139	139RH (nt)	**RHEA**	2010-290T06:47	Oct17	Sun	23	25	2	Outbound 38754.2 km flyby, v = 7.1 km/s, phase = 100 deg
---	139	Ring CRX	Descending	2010-291T13:15	Oct18	Mon	24	24	4	r = 20.223 Rs (1220041 km)
---	140	**Apoapse**		2010-301T18:06	Oct28	Thu	34	14	14	Per = 24.0 d, inc = 3.0 deg, r = 49.120 Rs, phase = 100 deg
---	140	**OTM-265**	T73-3d	2010-312T11:34	Nov08	Mon	45	3	25	D/L start, burn ~6h later. Backup D/L 313T11:19; complexes G/G
---	140	**Sun OCC**	SATURN	2010-313T03:59	Nov09	Tue	45	2	1	Egress = 2010-313T06:54 (175 min)
---	140	**Earth OCC**	SATURN	2010-313T05:59	Nov09	Tue	45	2	1	Egress = 2010-313T06:53 (54 min)
---	140	140DI (nt)	**DIONE**	2010-313T10:56	Nov09	Tue	46	2	1	Inbound 100031.7 km flyby, v = 10.2 km/s, phase = 74 deg
---	140	Ring CRX	Ascending	2010-313T14:58	Nov09	Tue	46	2	1	r = 4.660 Rs (281113 km)
---	140	140EN (nt)	**ENCELADUS**	2010-313T16:22	Nov09	Tue	46	2	1	Inbound 45119.0 km flyby, v = 6.5 km/s, phase = 50 deg
---	140	Periapse		2010-313T18:16	Nov09	Tue	46	2	1	R = 4.100 Rs, lat = 2 deg, phase = 80 deg
---	140	Ring CRX	Descending	2010-315T13:00	Nov11	Thu	48	0	3	r = 20.053 Rs (1209809 km)
---	140	**140TI (t) [T73]**	**TITAN**	2010-315T13:37	Nov11	Thu	48	19	3	Outbound 7920.7 km flyby, v = 5.6 km/s, phase = 121 deg
---	140	**OTM-266**	T73+3d	2010-318T17:19	Nov14	Sun	3	16	6	D/L start, burn ~6h later. Backup D/L 319T17:19; complexes C/C
---	141	**Apoapse**		2010-324T01:57	Nov20	Sat	9	10	5	Per = 20.6 d, inc = 0.0 deg, r = 44.521 Rs, phase = 95 deg
---	141	**OTM-267**	Apo	2010-325T17:05	Nov21	Sun	10	9	7	D/L start, burn ~6h later. Backup D/L 326T16:50; complexes C/C
---	141	**OTM-268**	E12-3d	2010-331T10:20	Nov27	Sat	16	3	6	D/L start, burn ~6h later. Backup D/L 332T10:21; complexes G/G
---	141	141HY (nt)	**HYPERION**	2010-332T03:29	Nov28	Sun	17	2	1	Inbound 71756.6 km flyby, v = 4.9 km/s, phase = 73 deg
---	141	141TI (nt)	**TITAN**	2010-333T06:26	Nov29	Mon	18	1	2	Inbound 926201.0 km flyby, v = 10.5 km/s, phase = 130 deg
---	141	**Sun OCC**	SATURN	2010-333T23:58	Nov29	Mon	18	0	3	Egress = 2010-334T01:07 (70 min)
---	141	E_ring_lg	**DUST HAZARD**	2010-334T06:05	Nov30	Tue	19	0	3	Egress at 2010-334T13:00 (0.3 min); protection = MEA cover closure.
---	141	141AT (nt)	**ATLAS**	2010-334T07:18	Nov30	Tue	19	0	3	Inbound 114446.0 km flyby, v = 0.7 km/s, phase = 144 deg
---	141	Ring CRX	Ascending	2010-334T08:48	Nov30	Tue	19	0	3	r = 3.609 Rs (217722 km)
---	141	Periapse		2010-334T09:33	Nov30	Tue	19	0	3	R = 3.567 Rs, lat = 0 deg, phase = 86 deg
---	141	141JA (nt)	**JANUS**	2010-334T11:13	Nov30	Tue	19	0	3	Outbound 95624.7 km flyby, v = 2.4 km/s, phase = 29 deg
---	141	**141EN (t) [E12]**	**ENCELADUS**	2010-334T11:54	Nov30	Tue	19	21	3	Outbound 47.9 km flyby, v = 6.3 km/s, phase = 78 deg
---	141	**OTM-269**	E12+3d	2010-335T02:36	Dec01	Wed	1	20	4	D/L start, burn ~6h later. Backup D/L 335T16:21; complexes M/C
---	141	Ring CRX	Descending	2010-337T08:10	Dec03	Fri	3	18	2	r = 27.398 Rs (1652899 km)
---	141	**OTM-270**	Apo	2010-342T16:07	Dec08	Wed	8	12	8	D/L start, burn ~6h later. Backup D/L 343T15:52; complexes C/C
---	142	**Apoapse**		2010-344T16:15	Dec10	Fri	10	10	2	Per = 20.6 d, inc = 0.1 deg, r = 44.403 Rs, phase = 94 deg
---	142	142TI (nt)	**TITAN**	2010-350T19:58	Dec16	Thu	16	4	8	Inbound 848540.0 km flyby, v = 3.6 km/s, phase = 91 deg
---	142	**OTM-271**	E13-3d	2010-351T15:23	Dec17	Fri	17	3	9	D/L start, burn ~6h later. Backup D/L 352T15:23; complexes C/C
---	142	142TI (nt)	**TITAN**	2010-354T12:17	Dec20	Mon	20	1	3	Inbound 893250.3 km flyby, v = 6.3 km/s, phase = 46 deg
---	142	**Sun OCC**	SATURN	2010-354T13:35	Dec20	Mon	20	0	3	Egress = 2010-354T14:12 (38 min)
---	142	142DI (nt)	**DIONE**	2010-354T16:51	Dec20	Mon	20	0	3	Inbound 100117.0 km flyby, v = 5.9 km/s, phase = 87 deg
---	142	E_ring_lg	**DUST HAZARD**	2010-354T19:22	Dec20	Mon	20	0	3	Egress at 2010-355T02:15 (0.3 min); protection = MEA cover closure.
---	142	142PN (nt)	**PAN**	2010-354T20:36	Dec20	Mon	20	0	3	Inbound 115568.9 km flyby, v = 0.4 km/s, phase = 141 deg
---	142	142PA (nt)	**PANDORA**	2010-354T20:42	Dec20	Mon	20	0	3	Inbound 109071.3 km flyby, v = 1.1 km/s, phase = 143 deg
---	142	Ring CRX	Ascending	2010-354T21:12	Dec20	Mon	20	0	3	r = 3.765 Rs (227136 km)
---	142	Periapse		2010-354T22:49	Dec20	Mon	20	0	3	R = 3.578 Rs, lat = 0 deg, phase = 86 deg
---	142	142DA (nt)	**DAPHNIS**	2010-354T23:36	Dec20	Mon	20	0	3	Outbound 85033.3 km flyby, v = 1.3 km/s, phase = 61 deg
---	142	**142EN (t) [E13]**	**ENCELADUS**	2010-355T01:08	Dec21	Tue	21	21	3	Outbound 47.8 km flyby, v = 6.2 km/s, phase = 89 deg
---	142	Ring CRX	Descending	2010-356T07:18	Dec22	Wed	1	20	5	r = 16.782 Rs (1012478 km)
---	142	**OTM-272**	E13+3d	2010-358T01:09	Dec24	Fri	3	18	6	D/L start, burn ~6h later. Backup D/L 359T01:09; complexes M/M
---	143	**Apoapse**		2010-365T07:31	Dec31	Fri	10	11	7	Per = 20.7 d, inc = 0.1 deg, r = 44.665 Rs, phase = 94 deg
---	143	**OTM-273**	Apo	2011-001T00:40	Jan01	Sat	11	10	8	D/L start, burn ~6h later. Backup D/L 002T00:40; complexes M/M
---	143	**OTM-274**	R3-3d	2011-008T00:26	Jan08	Sat	18	3	7	D/L start, burn ~6h later. Backup D/L 009T00:11; complexes M/M
---	143	E_ring_lg	**DUST HAZARD**	2011-010T12:47	Jan10	Mon	20	1	3	Egress at 2011-010T19:40 (0.3 min); protection = MEA cover closure.
---	143	Ring CRX	Ascending	2011-010T13:18	Jan10	Mon	21	1	3	r = 4.152 Rs (250469 km)
---	143	143PA (nt)	**PANDORA**	2011-010T14:43	Jan10	Mon	21	1	3	Inbound 94681.2 km flyby, v = 1.5 km/s, phase = 133 deg
---	143	Periapse		2011-010T16:14	Jan10	Mon	21	1	3	R = 3.578 Rs, lat = 0 deg, phase = 86 deg
---	143	143ME (nt)	**METHONE**	2011-010T17:22	Jan10	Mon	21	0	3	Outbound 85732.8 km flyby, v = 9.4 km/s, phase = 127 deg
---	143	143TI (nt)	**TITAN**	2011-011T03:19	Jan11	Tue	21	0	3	Outbound 838659.9 km flyby, v = 5.9 km/s, phase = 140 deg
---	143	**Earth OCC**	RHEA	2011-011T04:39	Jan11	Tue	21	0	3	Egress = 2011-011T04:52 (14 min)
---	143	**Sun OCC**	RHEA	2011-011T04:44	Jan11	Tue	21	0	3	Egress = 2011-011T04:52 (8 min)
---	143	**143RH (t) [R3]**	**RHEA**	2011-011T04:53	Jan11	Tue	21	38	3	Outbound 75.9 km flyby, v = 8.0 km/s, phase = 101 deg

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	143	**OTM-275**	R3+3d	2011-014T07:27	Jan14	Fri	3	35	6	D/L start, burn ~6h later. Backup D/L 015T07:27; complexes G/G
---	143	143TI (nt)	**TITAN**	2011-014T09:20	Jan14	Fri	3	35	0	Outbound 772833.4 km flyby, v = 3.6 km/s, phase = 96 deg
---	144	**Apoapse**		2011-020T21:28	Jan20	Thu	10	29	7	Per = 20.4 d, inc = 0.3 deg, r = 44.239 Rs, phase = 93 deg
---	144	Ring CRX	Descending	2011-030T21:45	Jan30	Sun	20	19	17	r = 5.036 Rs (303845 km)
---	144	E_ring_lg	Dust Crossing	2011-030T23:22	Jan30	Sun	20	19	17	Egress at 2011-031T01:53 (0.1 min); protection = none.
---	144	144EP (nt)	**EPIMETHEUS**	2011-031T02:15	Jan31	Mon	20	19	17	Inbound 68382.4 km flyby, v = 2.5 km/s, phase = 115 deg
---	144	Periapse		2011-031T02:50	Jan31	Mon	20	19	17	R = 3.559 Rs, lat = -0 deg, phase = 87 deg
---	144	144CA (nt)	**CALYPSO**	2011-031T03:48	Jan31	Mon	20	19	17	Outbound 77752.9 km flyby, v = 6.6 km/s, phase = 94 deg
---	144	144PM (nt)	**PROMETHEUS**	2011-031T04:47	Jan31	Mon	20	18	17	Outbound 105611.4 km flyby, v = 1.1 km/s, phase = 33 deg
---	144	144EN (nt)	**ENCELADUS**	2011-031T05:52	Jan31	Mon	20	18	17	Outbound 60036.8 km flyby, v = 9.6 km/s, phase = 111 deg
---	144	144HE (nt)	**HELENE**	2011-031T10:17	Jan31	Mon	20	18	17	Outbound 27762.4 km flyby, v = 7.5 km/s, phase = 92 deg
---	144	Ring CRX	Ascending	2011-031T17:04	Jan31	Mon	21	18	17	r = 9.463 Rs (570926 km)
---	144	**OTM-276**	Peri	2011-031T22:44	Jan31	Mon	21	18	18	D/L start, burn ~6h later. Backup D/L 032T22:44; complexes M/M
---	145	**Apoapse**		2011-041T08:06	Feb10	Thu	30	8	9	Per = 20.4 d, inc = 0.3 deg, r = 44.226 Rs, phase = 93 deg
---	145	**OTM-277**	T74-3d	2011-046T11:46	Feb15	Tue	35	3	15	D/L start, burn ~6h later. Backup D/L 047T11:46; complexes C/C
---	145	**145TI (t) [T74]**	**TITAN**	2011-049T16:04	Feb18	Fri	38	60	3	Inbound 3650.7 km flyby, v = 5.8 km/s, phase = 62 deg
---	145	Ring CRX	Descending	2011-051T04:39	Feb20	Sun	2	58	5	r = 7.192 Rs (433864 km)
---	145	Periapse		2011-051T13:44	Feb20	Sun	2	58	5	R = 4.712 Rs, lat = -0 deg, phase = 97 deg
---	145	145EN (nt)	**ENCELADUS**	2011-051T14:03	Feb20	Sun	2	58	5	Outbound 67915.2 km flyby, v = 4.2 km/s, phase = 134 deg
---	145	145PL (nt)	**PALLENE**	2011-051T14:49	Feb20	Sun	2	58	5	Outbound 92220.0 km flyby, v = 2.7 km/s, phase = 56 deg
---	145	145TE (nt)	**TETHYS**	2011-051T17:13	Feb20	Sun	2	57	5	Outbound 47662.1 km flyby, v = 6.7 km/s, phase = 124 deg
---	145	Ring CRX	Ascending	2011-052T07:32	Feb21	Mon	3	57	6	r = 10.873 Rs (655953 km)
---	145	**OTM-278**	T74+3d	2011-052T21:31	Feb21	Mon	3	56	6	D/L start, burn ~6h later. Backup D/L 053T21:16; complexes M/M
---	145	**OTM-279**	Apo	2011-061T04:17	Mar02	Wed	12	48	8	D/L start, burn ~6h later. Backup D/L 062T04:17; complexes G/G
---	146	**Apoapse**		2011-065T12:39	Mar06	Sun	16	44	4	Per = 27.9 d, inc = 0.4 deg, r = 54.127 Rs, phase = 83 deg
---	146	Ring CRX	Descending	2011-079T02:37	Mar20	Sun	29	30	18	r = 7.215 Rs (435270 km)
---	146	Periapse		2011-079T11:45	Mar20	Sun	30	30	18	R = 4.716 Rs, lat = -0 deg, phase = 98 deg
---	146	146TL (nt)	**TELESTO**	2011-079T12:15	Mar20	Sun	30	30	18	Outbound 10036.9 km flyby, v = 4.3 km/s, phase = 83 deg
---	146	Ring CRX	Ascending	2011-080T05:30	Mar21	Mon	31	29	19	r = 10.849 Rs (654533 km)
---	147	**Apoapse**		2011-093T10:51	Apr03	Sun	44	16	32	Per = 27.9 d, inc = 0.4 deg, r = 54.127 Rs, phase = 82 deg
---	147	**OTM-280**	T75-3d	2011-105T17:48	Apr15	Fri	56	3	45	D/L start, burn ~6h later. Backup D/L 106T17:48; complexes M/M
---	147	Ring CRX	Descending	2011-107T00:37	Apr17	Sun	57	2	1	r = 7.220 Rs (435582 km)
---	147	147AE (nt)	**AEGAEON**	2011-107T09:12	Apr17	Sun	58	2	2	Inbound 118148.9 km flyby, v = 0.6 km/s, phase = 109 deg
---	147	Periapse		2011-107T09:46	Apr17	Sun	58	2	2	R = 4.711 Rs, lat = -0 deg, phase = 98 deg
---	147	147HE (nt)	**HELENE**	2011-107T13:39	Apr17	Sun	58	2	2	Outbound 69503.6 km flyby, v = 5.2 km/s, phase = 81 deg
---	147	Ring CRX	Ascending	2011-108T03:24	Apr18	Mon	58	1	2	r = 10.805 Rs (651860 km)
---	147	**147TI (t) [T75]**	**TITAN**	2011-109T05:01	Apr19	Tue	60	20	3	Outbound 10052.8 km flyby, v = 5.6 km/s, phase = 101 deg
---	147	**OTM-281**	T75+3d	2011-112T00:48	Apr22	Fri	3	17	6	D/L start, burn ~6h later. Backup D/L 113T00:48; complexes G/G
---	147	**OTM-282**	Apo	2011-119T00:18	Apr29	Fri	10	10	7	D/L start, burn ~6h later. Backup D/L 120T00:18; complexes G/G
---	148	**Apoapse**		2011-119T03:06	Apr29	Fri	10	10	0	Per = 23.4 d, inc = 0.4 deg, r = 48.220 Rs, phase = 76 deg
---	148	**OTM-283**	T76-3d	2011-125T16:17	May05	Thu	16	3	7	D/L start, burn ~6h later. Backup D/L 126T16:17; complexes M/M
---	148	**148TI (t) [T76]**	**TITAN**	2011-128T22:54	May08	Sun	20	43	3	Inbound 1873.2 km flyby, v = 5.9 km/s, phase = 47 deg
---	148	Ring CRX	Descending	2011-130T12:12	May10	Tue	2	41	5	r = 7.528 Rs (454154 km)
---	148	Periapse		2011-130T20:44	May10	Tue	2	41	5	R = 5.888 Rs, lat = -0 deg, phase = 118 deg
---	148	**OTM-284**	T76+3d	2011-131T23:32	May11	Wed	3	40	6	D/L start, burn ~6h later. Backup D/L 132T23:32; complexes G/G
---	148	Ring CRX	Ascending	2011-132T13:55	May12	Thu	4	39	1	r = 19.276 Rs (1162943 km)
---	148	**OTM-285**	Apo	2011-143T22:46	May23	Mon	15	28	12	D/L start, burn ~6h later. Backup D/L 144T22:30; complexes G/G
---	149	**Apoapse**		2011-150T10:16	May30	Mon	21	21	6	Per = 39.1 d, inc = 0.4 deg, r = 67.794 Rs, phase = 62 deg
---	149	**OTM-286**	T77-3d	2011-168T13:12	Jun17	Fri	40	3	25	D/L start, burn ~6h later. Backup D/L 169T03:12; complexes M/C
---	149	Ring CRX	Descending	2011-169T15:24	Jun18	Sat	41	2	1	r = 7.469 Rs (450614 km)
---	149	149HE (nt)	**HELENE**	2011-169T19:32	Jun18	Sat	41	2	1	Inbound 6972.8 km flyby, v = 4.6 km/s, phase = 102 deg
---	149	149TL (nt)	**TELESTO**	2011-169T23:40	Jun18	Sat	41	2	1	Inbound 61995.9 km flyby, v = 2.8 km/s, phase = 108 deg
---	149	Periapse		2011-169T23:46	Jun18	Sat	41	2	1	R = 5.884 Rs, lat = -0 deg, phase = 119 deg
---	149	Ring CRX	Ascending	2011-171T18:03	Jun20	Mon	43	0	3	r = 19.647 Rs (1185280 km)
---	149	**149TI (t) [T77]**	**TITAN**	2011-171T18:32	Jun20	Mon	43	83	3	Outbound 1358.7 km flyby, v = 5.9 km/s, phase = 73 deg
---	149	**OTM-287**	T77+3d	2011-175T02:42	Jun24	Fri	3	80	7	D/L start, burn ~6h later. Backup D/L 176T02:42; complexes C/C
---	150	**Apoapse**		2011-180T19:56	Jun29	Wed	9	74	6	Per = 21.7 d, inc = 0.4 deg, r = 45.646 Rs, phase = 45 deg
---	150	150TI (nt)	**TITAN**	2011-190T07:00	Jul09	Sat	19	65	15	Inbound 627604.2 km flyby, v = 8.8 km/s, phase = 168 deg
---	150	150HE (nt)	**HELENE**	2011-191T10:47	Jul10	Sun	20	64	16	Inbound 93013.2 km flyby, v = 5.6 km/s, phase = 40 deg
---	150	**Earth OCC**	SATURN	2011-191T11:17	Jul10	Sun	20	64	16	Egress = 2011-191T12:57 (100 min)
---	150	**Sun OCC**	SATURN	2011-191T12:12	Jul10	Sun	20	64	16	Egress = 2011-191T13:08 (57 min)
---	150	Ring CRX	Descending	2011-191T12:48	Jul10	Sun	20	64	16	r = 4.573 Rs (275905 km)
---	150	E_ring_lg	Dust Crossing	2011-191T13:38	Jul10	Sun	20	64	16	Egress at 2011-191T15:52 (0.1 min); protection = none.
---	150	150JA (nt)	**JANUS**	2011-191T15:44	Jul10	Sun	20	63	17	Inbound 92987.2 km flyby, v = 1.0 km/s, phase = 141 deg
---	150	150PN (nt)	**PAN**	2011-191T15:51	Jul10	Sun	20	63	17	Inbound 110201.4 km flyby, v = 0.1 km/s, phase = 138 deg
---	150	Periapse		2011-191T16:00	Jul10	Sun	20	63	17	R = 4.039 Rs, lat = -0 deg, phase = 135 deg
---	150	150PM (nt)	**PROMETHEUS**	2011-191T17:15	Jul10	Sun	20	63	17	Outbound 113052.9 km flyby, v = 0.2 km/s, phase = 108 deg
---	150	Ring CRX	Ascending	2011-193T09:45	Jul12	Tue	22	62	18	r = 19.569 Rs (1180578 km)
---	151	**Apoapse**		2011-202T12:10	Jul21	Thu	31	53	27	Per = 21.7 d, inc = 0.4 deg, r = 45.664 Rs, phase = 45 deg
---	151	151HE (nt)	**HELENE**	2011-213T00:39	Aug01	Mon	41	42	38	Inbound 76554.1 km flyby, v = 9.6 km/s, phase = 129 deg
---	151	**Earth OCC**	SATURN	2011-213T03:40	Aug01	Mon	41	42	38	Egress = 2011-213T05:11 (91 min)
---	151	**Sun OCC**	SATURN	2011-213T04:31	Aug01	Mon	41	42	38	Egress = 2011-213T05:19 (49 min)
---	151	Ring CRX	Descending	2011-213T05:00	Aug01	Mon	41	42	38	r = 4.583 Rs (276496 km)
---	151	E_ring_lg	Dust Crossing	2011-213T05:52	Aug01	Mon	41	42	38	Egress at 2011-213T08:03 (0.1 min); protection = none.
---	151	151TI (nt)	**TITAN**	2011-213T07:08	Aug01	Mon	42	42	38	Inbound 983667.8 km flyby, v = 11.2 km/s, phase = 37 deg
---	151	151TL (nt)	**TELESTO**	2011-213T08:11	Aug01	Mon	42	42	38	Inbound 50589.4 km flyby, v = 5.6 km/s, phase = 45 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	151	Periapse		2011-213T08:12	Aug01	Mon	42	42	38	R = 4.045 Rs, lat = -0 deg, phase = 135 deg
---	151	151ME (nt)	**METHONE**	2011-213T08:41	Aug01	Mon	42	42	38	Outbound 97384.6 km flyby, v = 5.8 km/s, phase = 81 deg
---	151	151RH (nt)	**RHEA**	2011-213T21:02	Aug01	Mon	42	41	39	Outbound 5843.6 km flyby, v = 7.7 km/s, phase = 131 deg
---	151	Ring CRX	Ascending	2011-215T01:09	Aug03	Wed	43	40	40	r = 19.327 Rs (1166008 km)
---	152	**Apoapse**		2011-224T06:02	Aug12	Fri	52	31	49	Per = 21.8 d, inc = 0.4 deg, r = 45.884 Rs, phase = 45 deg
---	152	**OTM-288**	Peri	2011-234T09:04	Aug22	Mon	63	21	59	D/L start, burn ~6h later. Backup D/L 235T09:04; complexes M/M
---	152	**Earth OCC**	SATURN	2011-234T23:42	Aug22	Mon	63	20	1	Egress = 2011-235T00:58 (77 min)
---	152	**Sun OCC**	SATURN	2011-235T00:24	Aug23	Tue	63	20	1	Egress = 2011-235T01:03 (40 min)
---	152	Ring CRX	Descending	2011-235T00:42	Aug23	Tue	63	20	1	r = 4.600 Rs (277527 km)
---	152	E_ring_lg	Dust Crossing	2011-235T01:38	Aug23	Tue	63	20	1	Egress at 2011-235T03:43 (0.1 min); protection = none.
---	152	152EP (nt)	**EPIMETHEUS**	2011-235T03:17	Aug23	Tue	63	20	1	Inbound 96015.1 km flyby, v = 1.1 km/s, phase = 153 deg
---	152	Periapse		2011-235T03:58	Aug23	Tue	63	20	1	R = 4.044 Rs, lat = -0 deg, phase = 136 deg
---	152	152AT (nt)	**ATLAS**	2011-235T05:29	Aug23	Tue	63	20	1	Outbound 118246.3 km flyby, v = 0.1 km/s, phase = 104 deg
---	152	152TI (nt)	**TITAN**	2011-236T12:12	Aug24	Wed	65	19	2	Outbound 585590.2 km flyby, v = 8.5 km/s, phase = 101 deg
---	152	Ring CRX	Ascending	2011-236T20:38	Aug24	Wed	65	18	2	r = 19.242 Rs (1160891 km)
---	152	152HY (nt)	**HYPERION**	2011-237T16:48	Aug25	Thu	66	17	3	Outbound 24985.6 km flyby, v = 5.1 km/s, phase = 107 deg
---	152	**OTM-288a**	Apo	2011-244T16:03	Sep01	Thu	73	10	10	D/L start, burn ~6h later. Backup D/L 245T16:03; complexes G/G
---	153	**Apoapse**		2011-246T01:48	Sep03	Sat	74	9	1	Per = 21.8 d, inc = 0.4 deg, r = 45.867 Rs, phase = 44 deg
---	153	**OTM-289**	T78-3d	2011-251T21:48	Sep08	Thu	80	3	7	D/L start, burn ~6h later. Backup D/L 252T21:47; complexes C/C
---	153	**Earth OCC**	TITAN	2011-255T02:34	Sep12	Mon	83	0	3	Egress = 2011-255T02:48 (14 min)
---	153	**Sun OCC**	TITAN	2011-255T02:36	Sep12	Mon	83	0	3	Egress = 2011-255T02:49 (13 min)
---	153	**153TI (t) [T78]**	**TITAN**	2011-255T02:50	Sep12	Mon	83	19	3	Inbound 5821.5 km flyby, v = 5.8 km/s, phase = 159 deg
---	153	153RH (nt)	**RHEA**	2011-256T10:34	Sep13	Tue	1	18	5	Inbound 25404.8 km flyby, v = 8.7 km/s, phase = 141 deg
---	153	153TE (nt)	**TETHYS**	2011-256T18:19	Sep13	Tue	2	18	5	Inbound 32824.6 km flyby, v = 7.3 km/s, phase = 53 deg
---	153	E_ring_lg	**DUST HAZARD**	2011-256T19:06	Sep13	Tue	2	18	5	Egress at 2011-256T21:05 (0.1 min); protection = MEA cover closure.
---	153	**Earth OCC**	SATURN	2011-256T19:26	Sep13	Tue	2	18	5	Egress = 2011-256T20:48 (83 min)
---	153	**Sun OCC**	SATURN	2011-256T19:42	Sep13	Tue	2	18	5	Egress = 2011-256T20:55 (73 min)
---	153	Ring CRX	Descending	2011-256T20:28	Sep13	Tue	2	18	5	r = 3.764 Rs (227094 km)
---	153	Pallene	**DUST HAZARD**	2011-256T21:03	Sep13	Tue	2	18	5	Egress at 2011-256T21:20 (0.0 min); protection = MEA cover closure.
---	153	153PA (nt)	**PANDORA**	2011-256T22:37	Sep13	Tue	2	18	5	Inbound 58073.4 km flyby, v = 2.7 km/s, phase = 154 deg
---	153	153EN (nt)	**ENCELADUS**	2011-256T22:49	Sep13	Tue	2	18	5	Inbound 42224.8 km flyby, v = 6.3 km/s, phase = 51 deg
---	153	Periapse		2011-256T23:01	Sep13	Tue	2	18	5	R = 3.231 Rs, lat = -0 deg, phase = 129 deg
---	153	153JA (nt)	**JANUS**	2011-256T23:09	Sep13	Tue	2	18	5	Outbound 45574.5 km flyby, v = 3.5 km/s, phase = 109 deg
---	153	153ME (nt)	**METHONE**	2011-257T00:48	Sep14	Wed	2	18	5	Outbound 119919.5 km flyby, v = 14.0 km/s, phase = 159 deg
---	153	153PL (nt)	**PALLENE**	2011-257T01:21	Sep14	Wed	2	18	5	Outbound 25972.4 km flyby, v = 8.2 km/s, phase = 152 deg
---	153	153TL (nt)	**TELESTO**	2011-257T03:53	Sep14	Wed	2	17	5	Outbound 23337.7 km flyby, v = 7.7 km/s, phase = 53 deg
---	153	**OTM-290**	T78+3d	2011-257T21:32	Sep14	Wed	3	17	6	D/L start, burn ~6h later. Backup D/L 258T21:32; complexes C/C
---	153	Ring CRX	Ascending	2011-258T00:57	Sep15	Thu	3	17	0	r = 14.431 Rs (870643 km)
---	153	153HY (nt)	**HYPERION**	2011-259T13:23	Sep16	Fri	4	15	2	Outbound 58016.7 km flyby, v = 4.8 km/s, phase = 84 deg
---	153	**OTM-291**	Apo	2011-262T21:17	Sep19	Mon	8	12	5	D/L start, burn ~6h later. Backup D/L 263T21:17; complexes C/C
---	154	**Apoapse**		2011-265T19:59	Sep22	Thu	11	9	3	Per = 17.7 d, inc = 0.3 deg, r = 40.259 Rs, phase = 50 deg
---	154	154TI (nt)	**TITAN**	2011-271T03:23	Sep28	Wed	16	3	8	Inbound 679802.4 km flyby, v = 3.7 km/s, phase = 43 deg
---	154	**OTM-292**	E14-3d	2011-271T07:02	Sep28	Wed	16	3	8	D/L start, burn ~6h later. Backup D/L 272T07:02; complexes M/M
---	154	154TI (nt)	**TITAN**	2011-273T17:17	Sep30	Fri	19	1	2	Inbound 835923.8 km flyby, v = 3.1 km/s, phase = 12 deg
---	154	Ring CRX	Descending	2011-274T12:21	Oct01	Sat	19	0	3	r = 4.643 Rs (280096 km)
---	154	E_ring_lg	**DUST HAZARD**	2011-274T12:59	Oct01	Sat	19	0	3	Egress at 2011-274T15:00 (0.1 min); protection = MEA cover closure.
---	154	**Earth OCC**	SATURN	2011-274T13:29	Oct01	Sat	19	0	3	Egress = 2011-274T14:42 (74 min)
---	154	**Sun OCC**	SATURN	2011-274T13:35	Oct01	Sat	19	0	3	Egress = 2011-274T14:45 (71 min)
---	154	**154EN (t) [E14]**	**ENCELADUS**	2011-274T13:52	Oct01	Sat	19	18	3	Inbound 99.0 km flyby, v = 7.4 km/s, phase = 102 deg
---	154	Pallene	**DUST HAZARD**	2011-274T14:58	Oct01	Sat	0	18	3	Egress at 2011-274T15:15 (0.0 min); protection = MEA cover closure.
---	154	Periapse		2011-274T16:53	Oct01	Sat	0	18	3	R = 3.250 Rs, lat = -0 deg, phase = 130 deg
---	154	154EP (nt)	**EPIMETHEUS**	2011-274T17:31	Oct01	Sat	0	18	3	Outbound 67937.3 km flyby, v = 4.2 km/s, phase = 81 deg
---	154	E_ring_lg	**DUST HAZARD**	2011-274T18:45	Oct01	Sat	0	18	3	Egress at 2011-274T20:46 (0.1 min); protection = MEA cover closure.
---	154	154CA (nt)	**CALYPSO**	2011-274T21:46	Oct01	Sat	0	17	4	Outbound 124828.5 km flyby, v = 13.3 km/s, phase = 136 deg
---	154	Ring CRX	Ascending	2011-275T04:29	Oct02	Sun	1	17	4	r = 8.247 Rs (497560 km)
---	154	**OTM-293**	E14+3d	2011-277T20:17	Oct04	Tue	3	15	7	D/L start, burn ~6h later. Backup D/L 278T20:16; complexes C/C
---	154	**OTM-294**	Apo	2011-282T20:01	Oct09	Sun	8	10	5	D/L start, burn ~6h later. Backup D/L 283T20:01; complexes C/C
---	155	**Apoapse**		2011-283T14:41	Oct10	Mon	9	9	1	Per = 17.8 d, inc = 0.2 deg, r = 40.357 Rs, phase = 50 deg
---	155	SEP = 3.0 deg	Conjunction	2011-284T11:32	Oct11	Tue	10	8	2	Commanding must not be required below 3.0 deg
---	155	SEP = 3.0 deg	Conjunction	2011-289T03:32	Oct16	Sun	15	3	6	Commanding must not be required below 3.0 deg
---	155	**OTM-295**	E15-3d	2011-289T06:01	Oct16	Sun	15	3	6	D/L start, burn ~6h later. Backup D/L 290T06:01; complexes M/M
---	155	155TL (nt)	**TELESTO**	2011-292T07:36	Oct19	Wed	18	0	3	Inbound 25567.9 km flyby, v = 7.5 km/s, phase = 53 deg
---	155	Ring CRX	Descending	2011-292T07:41	Oct19	Wed	18	0	3	r = 4.736 Rs (285727 km)
---	155	E_ring_lg	**DUST HAZARD**	2011-292T08:30	Oct19	Wed	18	0	3	Egress at 2011-292T10:32 (0.1 min); protection = MEA cover closure.
---	155	**Sun OCC**	SATURN	2011-292T09:08	Oct19	Wed	18	0	3	Egress = 2011-292T10:15 (68 min)
---	155	**Earth OCC**	SATURN	2011-292T09:13	Oct19	Wed	18	0	3	Egress = 2011-292T10:14 (61 min)
---	155	**155EN (t) [E15]**	**ENCELADUS**	2011-292T09:22	Oct19	Wed	18	18	3	Inbound 1229.9 km flyby, v = 7.5 km/s, phase = 118 deg
---	155	Pallene	**DUST HAZARD**	2011-292T10:30	Oct19	Wed	0	18	3	Egress at 2011-292T10:47 (0.0 min); protection = MEA cover closure.
---	155	Periapse		2011-292T12:24	Oct19	Wed	0	18	3	R = 3.259 Rs, lat = -0 deg, phase = 130 deg
---	155	E_ring_lg	**DUST HAZARD**	2011-292T14:15	Oct19	Wed	0	18	3	Egress at 2011-292T16:16 (0.1 min); protection = MEA cover closure.
---	155	Ring CRX	Ascending	2011-292T23:58	Oct19	Wed	1	17	4	r = 8.228 Rs (496366 km)
---	155	**OTM-296**	E15+3d	2011-293T19:31	Oct20	Thu	1	16	5	D/L start, burn ~6h later. Backup D/L 294T19:16; complexes C/C
---	155	**OTM-297**	Apo	2011-301T05:17	Oct28	Fri	9	9	7	D/L start, burn ~6h later. Backup D/L 302T05:17; complexes M/M
---	156	**Apoapse**		2011-301T10:11	Oct28	Fri	9	9	0	Per = 17.8 d, inc = 0.2 deg, r = 40.358 Rs, phase = 50 deg
---	156	**OTM-298**	E16-3d	2011-306T18:47	Nov02	Wed	14	3	6	D/L start, burn ~6h later. Backup D/L 307T18:47; complexes C/C
---	156	Ring CRX	Descending	2011-310T03:14	Nov06	Sun	18	0	3	r = 4.753 Rs (286778 km)
---	156	E_ring_lg	**DUST HAZARD**	2011-310T04:05	Nov06	Sun	18	0	3	Egress at 2011-310T06:07 (0.1 min); protection = MEA cover closure.
---	156	**Sun OCC**	SATURN	2011-310T04:45	Nov06	Sun	18	0	3	Egress = 2011-310T05:49 (64 min)
---	156	**156EN (t) [E16]**	**ENCELADUS**	2011-310T04:59	Nov06	Sun	18	36	3	Inbound 495.9 km flyby, v = 7.4 km/s, phase = 74 deg

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	156	**Earth OCC**	SATURN	2011-310T05:02	Nov06	Sun	0	36	3	Egress = 2011-310T05:49 (48 min)
---	156	Pallene	**DUST HAZARD**	2011-310T06:05	Nov06	Sun	0	36	3	Egress at 2011-310T06:22 (0.0 min); protection = MEA cover closure.
---	156	156ME (nt)	**METHONE**	2011-310T06:37	Nov06	Sun	0	36	3	Inbound 19395.8 km flyby, v = 6.3 km/s, phase = 107 deg
---	156	Periapse		2011-310T07:58	Nov06	Sun	0	36	4	R = 3.260 Rs, lat = -0 deg, phase = 130 deg
---	156	156PA (nt)	**PANDORA**	2011-310T08:58	Nov06	Sun	0	36	4	Outbound 76384.3 km flyby, v = 3.2 km/s, phase = 81 deg
---	156	E_ring_lg	**DUST HAZARD**	2011-310T09:49	Nov06	Sun	0	36	4	Egress at 2011-310T11:51 (0.1 min); protection = MEA cover closure.
---	156	156CA (nt)	**CALYPSO**	2011-310T10:00	Nov06	Sun	0	36	4	Outbound 86238.0 km flyby, v = 6.9 km/s, phase = 55 deg
---	156	Ring CRX	Ascending	2011-310T19:22	Nov06	Sun	1	36	4	r = 8.142 Rs (491233 km)
---	156	**OTM-299**	E16+3d	2011-312T18:17	Nov08	Tue	3	34	6	D/L start, burn ~6h later. Backup D/L 313T18:17; complexes C/C
---	157	**Apoapse**		2011-319T06:13	Nov15	Tue	9	27	6	Per = 17.9 d, inc = 0.2 deg, r = 40.424 Rs, phase = 50 deg
---	157	157HE (nt)	**HELENE**	2011-327T21:05	Nov23	Wed	18	19	15	Inbound 103260.3 km flyby, v = 11.4 km/s, phase = 134 deg
---	157	Ring CRX	Descending	2011-327T22:47	Nov23	Wed	18	18	15	r = 5.252 Rs (316870 km)
---	157	**OTM-300**	Peri	2011-327T23:18	Nov23	Wed	18	18	15	D/L start, burn ~6h later. Backup D/L 329T03:33; complexes C+M/M
---	157	E_ring_lg	**DUST HAZARD**	2011-328T00:38	Nov24	Thu	18	18	0	Egress at 2011-328T02:39 (0.1 min); protection = MEA cover closure.
---	157	**Sun OCC**	SATURN	2011-328T01:22	Nov24	Thu	18	18	0	Egress = 2011-328T02:23 (61 min)
---	157	**Earth OCC**	SATURN	2011-328T01:50	Nov24	Thu	18	18	0	Egress = 2011-328T02:22 (32 min)
---	157	Pallene	**DUST HAZARD**	2011-328T02:37	Nov24	Thu	18	18	0	Egress at 2011-328T02:54 (0.0 min); protection = MEA cover closure.
---	157	157EN (nt)	**ENCELADUS**	2011-328T03:03	Nov24	Thu	18	18	0	Inbound 34524.3 km flyby, v = 6.2 km/s, phase = 52 deg
---	157	Periapse		2011-328T04:32	Nov24	Thu	18	18	0	R = 3.255 Rs, lat = -0 deg, phase = 130 deg
---	157	157EP (nt)	**EPIMETHEUS**	2011-328T04:44	Nov24	Thu	18	18	0	Outbound 51246.9 km flyby, v = 3.5 km/s, phase = 103 deg
---	157	E_ring_lg	**DUST HAZARD**	2011-328T06:23	Nov24	Thu	18	18	0	Egress at 2011-328T08:24 (0.1 min); protection = MEA cover closure.
---	157	Ring CRX	Ascending	2011-328T13:40	Nov24	Thu	18	18	1	r = 7.013 Rs (423111 km)
---	157	157TI (nt)	**TITAN**	2011-328T15:00	Nov24	Thu	18	18	1	Outbound 863147.4 km flyby, v = 6.1 km/s, phase = 97 deg
---	157	157TI (nt)	**TITAN**	2011-331T23:52	Nov27	Sun	22	14	4	Outbound 748422.4 km flyby, v = 3.8 km/s, phase = 52 deg
---	157	**OTM-300a**	Apo	2011-335T17:04	Dec01	Thu	26	11	8	D/L start, burn ~6h later. Backup D/L 336T16:49; complexes C/C
---	158	**Apoapse**		2011-337T03:13	Dec03	Sat	27	9	1	Per = 17.9 d, inc = 0.2 deg, r = 40.497 Rs, phase = 49 deg
---	158	**OTM-301**	T79-4d	2011-343T02:49	Dec09	Fri	33	3	7	D/L start, burn ~6h later. Backup D/L 344T02:50; complexes M/M
---	158	Ring CRX	Descending	2011-345T20:17	Dec11	Sun	36	1	3	r = 5.244 Rs (316399 km)
---	158	E_ring_lg	**DUST HAZARD**	2011-345T22:06	Dec11	Sun	36	0	3	Egress at 2011-346T00:06 (0.1 min); protection = MEA cover closure.
---	158	**Sun OCC**	SATURN	2011-345T22:55	Dec11	Sun	36	0	3	Egress = 2011-345T23:53 (58 min)
---	158	**Earth OCC**	SATURN	2011-345T23:37	Dec11	Sun	36	0	3	Egress = 2011-345T23:47 (11 min)
---	158	Pallene	**DUST HAZARD**	2011-346T00:04	Dec12	Mon	36	0	3	Egress at 2011-346T00:20 (0.0 min); protection = MEA cover closure.
---	158	158CA (nt)	**CALYPSO**	2011-346T00:26	Dec12	Mon	36	0	3	Inbound 91792.3 km flyby, v = 7.1 km/s, phase = 42 deg
---	158	Periapse		2011-346T02:00	Dec12	Mon	36	0	3	R = 3.241 Rs, lat = -0 deg, phase = 131 deg
---	158	E_ring_lg	**DUST HAZARD**	2011-346T03:54	Dec12	Mon	36	0	3	Egress at 2011-346T05:54 (0.1 min); protection = MEA cover closure.
---	158	158EN (nt)	**ENCELADUS**	2011-346T04:27	Dec12	Mon	36	0	3	Outbound 19898.6 km flyby, v = 6.5 km/s, phase = 41 deg
---	158	158TE (nt)	**TETHYS**	2011-346T07:05	Dec12	Mon	36	0	3	Outbound 60602.9 km flyby, v = 10.8 km/s, phase = 139 deg
---	158	**158DI (dt) [D3]**	**DIONE**	2011-346T09:39	Dec12	Mon	36	1	3	Outbound 98.6 km flyby, v = 8.7 km/s, phase = 49 deg
---	158	Ring CRX	Ascending	2011-346T11:10	Dec12	Mon	0	1	3	r = 7.018 Rs (423399 km)
---	158	**158TI (dt) [T79]**	**TITAN**	2011-347T20:11	Dec13	Tue	1	20	5	Outbound 3585.8 km flyby, v = 5.8 km/s, phase = 100 deg
---	158	Ring CRX	Descending	2011-348T14:20	Dec14	Wed	1	19	5	r = 24.966 Rs (1506212 km)
---	158	**OTM-302**	T79+4d	2011-351T02:20	Dec17	Sat	3	17	8	D/L start, burn ~6h later. Backup D/L 352T02:21; complexes M/M
---	158	**OTM-303**	Apo	2011-356T15:51	Dec22	Thu	9	11	6	D/L start, burn ~6h later. Backup D/L 357T15:51; complexes C/C
---	159	**Apoapse**		2011-357T19:14	Dec23	Fri	10	10	1	Per = 23.5 d, inc = 0.9 deg, r = 48.124 Rs, phase = 58 deg
---	159	**159TI (t) [T80]**	**TITAN**	2012-002T15:14	Jan02	Mon	20	28	11	Inbound 29415.1 km flyby, v = 5.5 km/s, phase = 76 deg
---	159	Ring CRX	Ascending	2012-003T22:03	Jan03	Tue	1	27	12	r = 9.665 Rs (583107 km)
---	159	159HE (nt)	**HELENE**	2012-004T05:38	Jan04	Wed	2	26	13	Inbound 42189.8 km flyby, v = 8.1 km/s, phase = 112 deg
---	159	159MI (nt)	**MIMAS**	2012-004T11:17	Jan04	Wed	2	26	13	Inbound 103411.4 km flyby, v = 2.0 km/s, phase = 161 deg
---	159	159JA (nt)	**JANUS**	2012-004T12:21	Jan04	Wed	2	26	13	Inbound 117846.1 km flyby, v = 0.2 km/s, phase = 135 deg
---	159	Periapse		2012-004T13:03	Jan04	Wed	2	26	13	R = 4.419 Rs, lat = 2 deg, phase = 123 deg
---	159	Ring CRX	Descending	2012-004T21:50	Jan04	Wed	2	26	13	r = 6.966 Rs (420250 km)
---	159	**OTM-306**	Apo	2012-016T00:39	Jan16	Mon	13	15	24	D/L start, burn ~6h later. Backup D/L 017T00:25; complexes M/M
---	160	**Apoapse**		2012-016T15:47	Jan16	Mon	14	14	1	Per = 24.2 d, inc = 1.6 deg, r = 49.116 Rs, phase = 57 deg
---	160	Ring CRX	Ascending	2012-028T03:21	Jan28	Sat	26	2	12	r = 9.738 Rs (587463 km)
---	160	Periapse		2012-028T18:30	Jan28	Sat	26	2	13	R = 4.418 Rs, lat = 2 deg, phase = 124 deg
---	160	Ring CRX	Descending	2012-029T03:12	Jan29	Sun	26	1	13	r = 6.926 Rs (417866 km)
---	160	**160TI (t) [T81]**	**TITAN**	2012-030T13:40	Jan30	Mon	28	20	15	Outbound 31130.7 km flyby, v = 5.4 km/s, phase = 96 deg
---	160	**OTM-308**	T81+4d	2012-033T23:27	Feb02	Thu	3	16	18	D/L start, burn ~6h later. Backup D/L 034T23:27; complexes M/M
---	161	**Apoapse**		2012-040T12:16	Feb09	Thu	10	10	7	Per = 23.5 d, inc = 1.4 deg, r = 48.111 Rs, phase = 55 deg
---	161	**OTM-309**	Apo	2012-041T06:28	Feb10	Fri	11	9	7	D/L start, burn ~6h later. Backup D/L 042T06:28; complexes G/G
---	161	**OTM-310**	T82-3d	2012-046T22:43	Feb15	Wed	16	3	6	D/L start, burn ~6h later. Backup D/L 047T22:29; complexes M/M
---	161	Ring CRX	Ascending	2012-048T06:25	Feb17	Fri	18	2	1	r = 32.588 Rs (1966018 km)
---	161	**Sun OCC**	TITAN	2012-050T08:31	Feb19	Sun	20	0	3	Egress = 2012-050T08:36 (5 min)
---	161	**161TI (t) [T82]**	**TITAN**	2012-050T08:43	Feb19	Sun	20	37	3	Inbound 3803.3 km flyby, v = 5.8 km/s, phase = 143 deg
---	161	161PO (nt)	**POLYDEUCES**	2012-051T22:33	Feb20	Mon	2	36	5	Inbound 90810.3 km flyby, v = 6.3 km/s, phase = 59 deg
---	161	161TL (nt)	**TELESTO**	2012-052T01:14	Feb21	Tue	2	36	5	Inbound 55042.1 km flyby, v = 6.6 km/s, phase = 64 deg
---	161	E_ring_lg	**DUST HAZARD**	2012-052T01:28	Feb21	Tue	2	36	5	Egress at 2012-052T03:26 (0.1 min); protection = MEA cover closure.
---	161	Ring CRX	Descending	2012-052T01:37	Feb21	Tue	2	36	5	r = 4.275 Rs (257913 km)
---	161	161EN (nt)	**ENCELADUS**	2012-052T02:01	Feb21	Tue	2	36	5	Inbound 46648.2 km flyby, v = 10.1 km/s, phase = 100 deg
---	161	161PL (nt)	**PALLENE**	2012-052T02:55	Feb21	Tue	2	36	5	Inbound 69218.7 km flyby, v = 11.0 km/s, phase = 92 deg
---	161	Pallene	**DUST HAZARD**	2012-052T03:25	Feb21	Tue	2	36	5	Egress at 2012-052T03:29 (0.0 min); protection = MEA cover closure.
---	161	Periapse		2012-052T05:23	Feb21	Tue	2	36	5	R = 3.231 Rs, lat = -0 deg, phase = 117 deg
---	161	161PA (nt)	**PANDORA**	2012-052T05:51	Feb21	Tue	2	36	5	Outbound 60396.5 km flyby, v = 2.9 km/s, phase = 87 deg
---	161	161DI (nt)	**DIONE**	2012-052T11:10	Feb21	Tue	2	35	6	Outbound 119721.9 km flyby, v = 6.0 km/s, phase = 74 deg
---	161	161RH (nt)	**RHEA**	2012-052T17:16	Feb21	Tue	2	35	6	Outbound 100507.5 km flyby, v = 9.9 km/s, phase = 104 deg
---	161	Ring CRX	Ascending	2012-052T20:25	Feb21	Tue	2	35	6	r = 9.891 Rs (596726 km)
---	161	**OTM-311**	T82+4d	2012-053T22:14	Feb22	Wed	4	34	7	D/L start, burn ~6h later. Backup D/L 054T21:59; complexes M/M

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	162	**Apoapse**		2012-061T03:51	Mar01	Thu	11	27	7	Per = 17.9 d, inc = 0.4 deg, r = 40.464 Rs, phase = 63 deg
---	162	162TI (nt)	**TITAN**	2012-066T07:50	Mar06	Tue	16	21	12	Inbound 717788.1 km flyby, v = 3.7 km/s, phase = 56 deg
---	162	162HY (nt)	**HYPERION**	2012-068T01:31	Mar08	Thu	18	20	14	Inbound 104043.9 km flyby, v = 5.5 km/s, phase = 148 deg
---	162	162TI (nt)	**TITAN**	2012-069T10:14	Mar09	Fri	19	18	16	Inbound 864013.4 km flyby, v = 4.5 km/s, phase = 16 deg
---	162	162HE (nt)	**HELENE**	2012-069T18:49	Mar09	Fri	19	18	16	Inbound 109821.9 km flyby, v = 11.6 km/s, phase = 121 deg
---	162	**OTM-312**	Peri	2012-069T21:01	Mar09	Fri	20	18	16	D/L start, burn ~6h later. Backup D/L 070T21:01; complexes M/M
---	162	E_ring_lg	**DUST HAZARD**	2012-069T22:20	Mar09	Fri	20	18	0	Egress at 2012-070T00:19 (0.1 min); protection = MEA cover closure.
---	162	Ring CRX	Descending	2012-069T22:24	Mar09	Fri	20	18	0	r = 4.307 Rs (259855 km)
---	162	162EN (nt)	**ENCELADUS**	2012-069T23:22	Mar09	Fri	20	18	0	Inbound 9176.7 km flyby, v = 7.0 km/s, phase = 75 deg
---	162	Periapse		2012-070T02:13	Mar10	Sat	20	18	0	R = 3.245 Rs, lat = -0 deg, phase = 117 deg
---	162	162PM (nt)	**PROMETHEUS**	2012-070T02:30	Mar10	Sat	20	18	0	Outbound 58008.8 km flyby, v = 2.5 km/s, phase = 102 deg
---	162	162EP (nt)	**EPIMETHEUS**	2012-070T03:29	Mar10	Sat	20	18	0	Outbound 111451.3 km flyby, v = 6.9 km/s, phase = 49 deg
---	162	162CA (nt)	**CALYPSO**	2012-070T06:36	Mar10	Sat	20	17	0	Outbound 47339.7 km flyby, v = 6.8 km/s, phase = 63 deg
---	162	162RH (nt)	**RHEA**	2012-070T15:03	Mar10	Sat	20	17	1	Outbound 41864.6 km flyby, v = 7.5 km/s, phase = 73 deg
---	162	Ring CRX	Ascending	2012-070T17:13	Mar10	Sat	20	17	1	r = 9.860 Rs (594871 km)
---	162	**OTM-312a**	Apo	2012-075T20:46	Mar15	Thu	26	12	6	D/L start, burn ~6h later. Backup D/L 076T20:31; complexes M/M

---	163	**Apoapse**		2012-078T23:55	Mar18	Sun	29	9	3	Per = 17.8 d, inc = 0.4 deg, r = 40.349 Rs, phase = 62 deg
---	163	**OTM-313**	E17-3d	2012-084T10:02	Mar24	Sat	34	3	9	D/L start, burn ~6h later. Backup D/L 085T09:47; complexes C/C
---	163	163PO (nt)	**POLYDEUCES**	2012-087T15:54	Mar27	Tue	37	0	3	Inbound 118110.1 km flyby, v = 6.0 km/s, phase = 53 deg
---	163	E_ring_lg	**DUST HAZARD**	2012-087T17:38	Mar27	Tue	37	0	3	Egress at 2012-087T19:39 (0.1 min); protection = MEA cover closure.
---	163	Ring CRX	Descending	2012-087T17:40	Mar27	Tue	37	0	3	r = 4.330 Rs (261210 km)
---	163	**Earth OCC**	ENCELADUS	2012-087T18:28	Mar27	Tue	37	0	3	Egress = 2012-087T18:29 (2 min)
---	163	**Sun OCC**	ENCELADUS	2012-087T18:28	Mar27	Tue	37	0	3	Egress = 2012-087T18:29 (2 min)
---	163	**163EN (t) [E17]**	**ENCELADUS**	2012-087T18:30	Mar27	Tue	37	18	3	Inbound 74.0 km flyby, v = 7.5 km/s, phase = 105 deg
---	163	Pallene	**DUST HAZARD**	2012-087T19:37	Mar27	Tue	0	18	3	Egress at 2012-087T19:37 (0.0 min); protection = MEA cover closure.
---	163	163PM (nt)	**PROMETHEUS**	2012-087T21:01	Mar27	Tue	0	18	3	Inbound 62926.6 km flyby, v = 2.5 km/s, phase = 145 deg
---	163	Periapse		2012-087T21:32	Mar27	Tue	0	18	3	R = 3.253 Rs, lat = -0 deg, phase = 118 deg
---	163	163JA (nt)	**JANUS**	2012-087T21:34	Mar27	Tue	0	18	3	Outbound 43862.5 km flyby, v = 3.2 km/s, phase = 116 deg
---	163	163DI (nt)	**DIONE**	2012-088T05:07	Mar28	Wed	0	17	4	Outbound 44007.1 km flyby, v = 7.5 km/s, phase = 64 deg
---	163	Ring CRX	Ascending	2012-088T12:09	Mar28	Wed	1	17	4	r = 9.688 Rs (584450 km)
---	163	**OTM-314**	E17+4d	2012-090T19:32	Mar30	Fri	3	15	6	D/L start, burn ~6h later. Backup D/L 091T19:32; complexes M/M
---	163	**OTM-315**	Apo	2012-096T02:47	Apr05	Thu	8	9	5	D/L start, burn ~6h later. Backup D/L 097T02:47; complexes G/G

---	164	**Apoapse**		2012-096T19:17	Apr05	Thu	9	9	1	Per = 17.8 d, inc = 0.4 deg, r = 40.361 Rs, phase = 62 deg
---	164	**OTM-316**	E18-3d	2012-102T08:48	Apr11	Wed	15	3	6	D/L start, burn ~6h later. Backup D/L 103T08:33; complexes C/C
---	164	164HE (nt)	**HELENE**	2012-105T09:37	Apr14	Sat	18	0	3	Inbound 104845.1 km flyby, v = 11.5 km/s, phase = 121 deg
---	164	164CA (nt)	**CALYPSO**	2012-105T12:44	Apr14	Sat	18	0	3	Inbound 48734.3 km flyby, v = 6.7 km/s, phase = 62 deg
---	164	Ring CRX	Descending	2012-105T13:08	Apr14	Sat	18	0	3	r = 4.356 Rs (262824 km)
---	164	E_ring_lg	**DUST HAZARD**	2012-105T13:10	Apr14	Sat	18	0	3	Egress at 2012-105T15:11 (0.1 min); protection = MEA cover closure.
---	164	**Earth OCC**	ENCELADUS	2012-105T14:00	Apr14	Sat	18	0	3	Egress = 2012-105T14:00 (1 min)
---	164	**Sun OCC**	ENCELADUS	2012-105T14:00	Apr14	Sat	18	0	3	Egress = 2012-105T14:00 (1 min)
---	164	**164EN (t) [E18]**	**ENCELADUS**	2012-105T14:02	Apr14	Sat	18	18	3	Inbound 74.0 km flyby, v = 7.5 km/s, phase = 104 deg
---	164	Pallene	**DUST HAZARD**	2012-105T15:09	Apr14	Sat	0	18	3	Egress at 2012-105T15:12 (0.0 min); protection = MEA cover closure.
---	164	164PM (nt)	**PROMETHEUS**	2012-105T15:15	Apr14	Sat	0	18	3	Inbound 97634.6 km flyby, v = 2.4 km/s, phase = 166 deg
---	164	164PA (nt)	**PANDORA**	2012-105T15:43	Apr14	Sat	0	18	3	Inbound 87204.6 km flyby, v = 3.3 km/s, phase = 167 deg
---	164	Periapse		2012-105T17:03	Apr14	Sat	0	18	3	R = 3.253 Rs, lat = -0 deg, phase = 118 deg
---	164	164TI (nt)	**TITAN**	2012-105T18:44	Apr14	Sat	0	18	3	Outbound 995036.5 km flyby, v = 12.5 km/s, phase = 78 deg
---	164	164TE (nt)	**TETHYS**	2012-105T22:06	Apr14	Sat	0	17	4	Outbound 9037.0 km flyby, v = 8.7 km/s, phase = 126 deg
---	164	Ring CRX	Ascending	2012-106T07:14	Apr15	Sun	1	17	4	r = 9.484 Rs (572178 km)
---	164	164HY (nt)	**HYPERION**	2012-107T21:30	Apr16	Mon	2	16	6	Outbound 123196.5 km flyby, v = 5.7 km/s, phase = 87 deg
---	164	**OTM-317**	E18+3d	2012-108T18:18	Apr17	Tue	3	15	6	D/L start, burn ~6h later. Backup D/L 109T18:18; complexes M/M

---	165	**Apoapse**		2012-114T14:46	Apr23	Mon	9	9	6	Per = 17.8 d, inc = 0.4 deg, r = 40.363 Rs, phase = 62 deg
---	165	**OTM-318**	Apo	2012-115T01:33	Apr24	Tue	9	8	6	D/L start, burn ~6h later. Backup D/L 116T01:32; complexes G/G
---	165	**OTM-319**	E19-3d	2012-120T01:17	Apr29	Sun	14	3	5	D/L start, burn ~6h later. Backup D/L 121T01:02; complexes G/G
---	165	165PO (nt)	**POLYDEUCES**	2012-123T06:55	May02	Wed	18	0	3	Inbound 117990.2 km flyby, v = 6.0 km/s, phase = 53 deg
---	165	Ring CRX	Descending	2012-123T08:38	May02	Wed	18	0	3	r = 4.358 Rs (262909 km)
---	165	E_ring_lg	**DUST HAZARD**	2012-123T08:40	May02	Wed	18	0	3	Egress at 2012-123T10:40 (0.1 min); protection = MEA cover closure.
---	165	**165EN (t) [E19]**	**ENCELADUS**	2012-123T09:31	May02	Wed	18	20	3	Inbound 73.6 km flyby, v = 7.5 km/s, phase = 100 deg
---	165	Pallene	**DUST HAZARD**	2012-123T10:38	May02	Wed	0	20	3	Egress at 2012-123T10:47 (0.0 min); protection = MEA cover closure.
---	165	Periapse		2012-123T12:34	May02	Wed	0	20	3	R = 3.245 Rs, lat = -0 deg, phase = 118 deg
---	165	165AT (nt)	**ATLAS**	2012-123T13:54	May02	Wed	0	19	4	Outbound 84769.4 km flyby, v = 2.5 km/s, phase = 65 deg
---	165	165DI (nt)	**DIONE**	2012-123T20:13	May02	Wed	0	19	4	Outbound 8063.4 km flyby, v = 8.5 km/s, phase = 65 deg
---	165	165TI (nt)	**TITAN**	2012-124T01:17	May03	Thu	1	19	4	Outbound 834425.1 km flyby, v = 5.2 km/s, phase = 111 deg
---	165	Ring CRX	Ascending	2012-124T02:41	May03	Thu	1	19	4	r = 9.457 Rs (570549 km)
---	165	**OTM-320**	E19+4d	2012-127T00:47	May06	Sun	4	16	7	D/L start, burn ~6h later. Backup D/L 128T00:32; complexes G/G
---	165	165TI (nt)	**TITAN**	2012-127T03:41	May06	Sun	4	16	0	Outbound 699945.2 km flyby, v = 3.7 km/s, phase = 65 deg

---	166	**Apoapse**		2012-132T09:28	May11	Fri	9	11	5	Per = 17.8 d, inc = 0.4 deg, r = 40.266 Rs, phase = 62 deg
---	166	**OTM-321**	Apo	2012-135T00:01	May14	Mon	12	8	8	D/L start, burn ~6h later. Backup D/L 136T00:01; complexes G/G
---	166	**OTM-322**	T83-3d	2012-139T23:46	May18	Fri	17	3	5	D/L start, burn ~6h later. Backup D/L 140T16:16; complexes G/M
---	166	Ring CRX	Descending	2012-140T19:21	May19	Sat	17	2	1	r = 7.995 Rs (482345 km)
---	166	166TE (nt)	**TETHYS**	2012-141T02:15	May20	Sun	18	2	1	Inbound 53799.0 km flyby, v = 6.6 km/s, phase = 62 deg
---	166	166EN (nt)	**ENCELADUS**	2012-141T03:03	May20	Sun	18	2	1	Inbound 70348.6 km flyby, v = 11.4 km/s, phase = 102 deg
---	166	166DA (nt)	**DAPHNIS**	2012-141T03:58	May20	Sun	18	2	1	Inbound 115191.6 km flyby, v = 1.4 km/s, phase = 161 deg
---	166	166PM (nt)	**PROMETHEUS**	2012-141T04:33	May20	Sun	18	2	1	Inbound 100899.8 km flyby, v = 2.6 km/s, phase = 164 deg
---	166	166PN (nt)	**PAN**	2012-141T06:03	May20	Sun	18	2	1	Inbound 63573.9 km flyby, v = 2.2 km/s, phase = 137 deg
---	166	Periapse		2012-141T06:26	May20	Sun	18	2	1	R = 3.222 Rs, lat = -0 deg, phase = 119 deg
---	166	166ME (nt)	**METHONE**	2012-141T06:57	May20	Sun	18	2	1	Outbound 1861.6 km flyby, v = 5.1 km/s, phase = 126 deg
---	166	166MI (nt)	**MIMAS**	2012-141T08:04	May20	Sun	18	2	1	Outbound 113316.1 km flyby, v = 13.7 km/s, phase = 148 deg
---	166	E_ring_lg	**DUST HAZARD**	2012-141T08:23	May20	Sun	18	2	1	Egress at 2012-141T10:21 (0.1 min); protection = MEA cover closure.
---	166	166PL (nt)	**PALLENE**	2012-141T08:57	May20	Sun	18	2	1	Outbound 75148.5 km flyby, v = 11.3 km/s, phase = 142 deg
---	166	Ring CRX	Ascending	2012-141T11:11	May20	Sun	18	2	1	r = 4.739 Rs (285932 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	166	166TL (nt)	TELESTO	2012-141T11:31	May20	Sun	18	2	1	Outbound 10989.0 km flyby, v = 8.9 km/s, phase = 128 deg
---	166	166TI (t) [T83]	TITAN	2012-143T01:10	May22	Tue	20	16	3	Outbound 955.0 km flyby, v = 5.9 km/s, phase = 71 deg
---	166	Ring CRX	Descending	2012-143T03:58	May22	Tue	0	16	3	r = 20.855 Rs (1258167 km)
---	166	OTM-323	T83+3d	2012-145T23:16	May24	Thu	3	13	6	D/L start, burn ~6h later. Backup D/L 146T23:16; complexes G/G
---	167	Apoapse		2012-149T05:16	May28	Mon	6	10	3	Per = 16.0 d, inc = 15.8 deg, r = 37.437 Rs, phase = 61 deg
---	167	OTM-324	Apo	2012-150T23:00	May29	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 151T23:00; complexes G/G
---	167	OTM-325	T84-3d	2012-155T15:15	Jun03	Sun	13	3	5	D/L start, burn ~6h later. Backup D/L 156T04:45; complexes M/C
---	167	Earth OCC	RING	2012-156T22:32	Jun04	Mon	14	2	1	Egress = 2012-157T02:17 (225 min)
---	167	Sun OCC	RING	2012-156T22:50	Jun04	Mon	14	2	1	Egress = 2012-157T02:04 (195 min)
---	167	Earth OCC	SATURN	2012-157T00:02	Jun05	Tue	14	2	1	Egress = 2012-157T02:15 (134 min)
---	167	Sun OCC	SATURN	2012-157T00:30	Jun05	Tue	14	2	1	Egress = 2012-157T02:25 (115 min)
---	167	167PL (nt)	PALLENE	2012-157T02:32	Jun05	Tue	14	2	1	Inbound 74739.6 km flyby, v = 12.1 km/s, phase = 93 deg
---	167	167PM (nt)	PROMETHEUS	2012-157T03:01	Jun05	Tue	14	2	1	Inbound 99965.3 km flyby, v = 5.7 km/s, phase = 161 deg
---	167	Ring CRX	Ascending	2012-157T03:22	Jun05	Tue	14	2	2	r = 3.315 Rs (199981 km)
---	167	167PA (nt)	PANDORA	2012-157T04:22	Jun05	Tue	14	2	2	Inbound 49933.1 km flyby, v = 5.5 km/s, phase = 126 deg
---	167	Periapse		2012-157T04:53	Jun05	Tue	14	2	2	R = 3.097 Rs, lat = 8 deg, phase = 119 deg
---	167	167MI (nt)	MIMAS	2012-157T05:08	Jun05	Tue	14	2	2	Outbound 43551.8 km flyby, v = 7.2 km/s, phase = 114 deg
---	167	167DA (nt)	DAPHNIS	2012-157T05:13	Jun05	Tue	14	2	2	Outbound 75879.1 km flyby, v = 7.1 km/s, phase = 74 deg
---	167	167AT (nt)	ATLAS	2012-157T05:35	Jun05	Tue	14	2	2	Outbound 100458.6 km flyby, v = 8.7 km/s, phase = 60 deg
---	167	167TI (t) [T84]	TITAN	2012-159T00:07	Jun07	Thu	16	48	3	Outbound 959.0 km flyby, v = 5.9 km/s, phase = 75 deg
---	167	Ring CRX	Descending	2012-159T01:30	Jun07	Thu	0	48	3	r = 20.453 Rs (1233941 km)
---	167	OTM-326	T84+3d	2012-162T04:29	Jun10	Sun	3	45	7	D/L start, burn ~6h later. Backup D/L 163T04:29; complexes C/C
---	168	Apoapse		2012-169T01:46	Jun17	Sun	10	38	7	Per = 23.9 d, inc = 21.1 deg, r = 47.978 Rs, phase = 76 deg
---	168	OTM-327	Apo	2012-172T21:28	Jun20	Wed	14	34	11	D/L start, burn ~6h later. Backup D/L 173T21:27; complexes G/G
---	168	168TI (nt)	TITAN	2012-179T06:58	Jun27	Wed	20	28	6	Inbound 458663.8 km flyby, v = 5.6 km/s, phase = 97 deg
---	168	Earth OCC	RING	2012-180T10:22	Jun28	Thu	21	26	8	Egress = 2012-180T15:16 (295 min)
---	168	Sun OCC	RING	2012-180T10:33	Jun28	Thu	21	26	8	Egress = 2012-180T14:12 (220 min)
---	168	Earth OCC	SATURN	2012-180T11:32	Jun28	Thu	21	26	8	Egress = 2012-180T15:01 (210 min)
---	168	Sun OCC	SATURN	2012-180T13:06	Jun28	Thu	22	26	8	Egress = 2012-180T15:30 (144 min)
---	168	Ring CRX	Ascending	2012-180T19:22	Jun28	Thu	22	26	8	r = 5.880 Rs (354713 km)
---	168	168TE (nt)	TETHYS	2012-180T21:09	Jun28	Thu	22	26	8	Inbound 68543.8 km flyby, v = 7.7 km/s, phase = 80 deg
---	168	Periapse		2012-181T00:27	Jun29	Fri	22	26	8	R = 5.063 Rs, lat = 15 deg, phase = 104 deg
---	168	Ring CRX	Descending	2012-182T23:34	Jun30	Sat	24	24	10	r = 20.670 Rs (1247047 km)
---	169	Apoapse		2012-192T23:45	Jul10	Tue	34	14	20	Per = 24.0 d, inc = 21.2 deg, r = 48.049 Rs, phase = 76 deg
---	169	OTM-328	169TI-4d	2012-203T01:38	Jul21	Sat	44	4	30	D/L start, burn ~6h later. Backup D/L 203T19:23; complexes C/G
---	169	Earth OCC	RING	2012-204T08:45	Jul22	Sun	45	2	1	Egress = 2012-204T13:41 (296 min)
---	169	Sun OCC	RING	2012-204T09:20	Jul22	Sun	45	2	1	Egress = 2012-204T12:29 (190 min)
---	169	Earth OCC	SATURN	2012-204T10:12	Jul22	Sun	45	2	1	Egress = 2012-204T13:38 (206 min)
---	169	Sun OCC	SATURN	2012-204T11:57	Jul22	Sun	45	2	1	Egress = 2012-204T14:03 (127 min)
---	169	Ring CRX	Ascending	2012-204T17:56	Jul22	Sun	46	2	2	r = 5.888 Rs (355207 km)
---	169	169HE (nt)	HELENE	2012-204T21:21	Jul22	Sun	46	2	2	Inbound 110960.2 km flyby, v = 6.0 km/s, phase = 91 deg
---	169	Periapse		2012-204T23:03	Jul22	Sun	46	2	2	R = 5.062 Rs, lat = 15 deg, phase = 104 deg
---	169	169TI (t) [T85]	TITAN	2012-206T20:03	Jul24	Tue	48	64	4	Outbound 1012.0 km flyby, v = 5.9 km/s, phase = 58 deg
---	169	Ring CRX	Descending	2012-206T21:04	Jul24	Tue	0	64	4	r = 20.359 Rs (1228276 km)
---	169	OTM-329	169TI+3d	2012-210T01:08	Jul28	Sat	3	61	7	D/L start, burn ~6h later. Backup D/L 211T01:08; complexes C/C
---	170	Apoapse		2012-215T10:52	Aug02	Thu	9	55	5	Per = 21.2 d, inc = 32.2 deg, r = 43.265 Rs, phase = 78 deg
---	170	OTM-330	~Apo	2012-220T00:36	Aug07	Tue	13	51	10	D/L start, burn ~6h later. Backup D/L 221T00:21; complexes C/C
---	170	Earth OCC	SATURN	2012-225T10:28	Aug12	Sun	19	45	5	Egress = 2012-225T13:31 (183 min)
---	170	Sun OCC	RING	2012-225T10:28	Aug12	Sun	19	45	5	Egress = 2012-225T14:41 (254 min)
---	170	Sun OCC	SATURN	2012-225T11:05	Aug12	Sun	19	45	5	Egress = 2012-225T14:37 (213 min)
---	170	Earth OCC	RING	2012-225T11:36	Aug12	Sun	19	45	5	Egress = 2012-225T15:03 (207 min)
---	170	Ring CRX	Ascending	2012-225T18:56	Aug12	Sun	19	45	6	r = 6.747 Rs (407023 km)
---	170	170TL (nt)	TELESTO	2012-225T21:46	Aug12	Sun	19	45	6	Inbound 92942.6 km flyby, v = 7.9 km/s, phase = 102 deg
---	170	Periapse		2012-226T01:32	Aug13	Mon	19	45	6	R = 5.746 Rs, lat = 24 deg, phase = 103 deg
---	170	Ring CRX	Descending	2012-228T02:30	Aug15	Wed	21	43	8	r = 20.424 Rs (1232151 km)
---	171	Apoapse		2012-236T16:39	Aug23	Thu	30	34	17	Per = 21.3 d, inc = 32.2 deg, r = 43.311 Rs, phase = 77 deg
---	171	Sun OCC	RING	2012-246T16:27	Sep02	Sun	40	24	27	Egress = 2012-246T20:41 (254 min)
---	171	Earth OCC	SATURN	2012-246T16:36	Sep02	Sun	40	24	27	Egress = 2012-246T20:05 (209 min)
---	171	Earth OCC	RING	2012-246T17:07	Sep02	Sun	40	24	27	Egress = 2012-246T21:00 (234 min)
---	171	Sun OCC	SATURN	2012-246T17:19	Sep02	Sun	40	24	27	Egress = 2012-246T20:47 (208 min)
---	171	171TI (nt)	TITAN	2012-246T19:44	Sep02	Sun	40	24	27	Inbound 784726.7 km flyby, v = 7.2 km/s, phase = 39 deg
---	171	Ring CRX	Ascending	2012-247T01:01	Sep03	Mon	40	24	27	r = 6.764 Rs (408053 km)
---	171	Periapse		2012-247T07:39	Sep03	Mon	40	23	27	R = 5.758 Rs, lat = 24 deg, phase = 103 deg
---	171	Ring CRX	Descending	2012-249T08:40	Sep05	Wed	43	21	29	r = 20.431 Rs (1232602 km)
---	172	Apoapse		2012-257T23:17	Sep13	Thu	51	13	38	Per = 21.3 d, inc = 32.2 deg, r = 43.383 Rs, phase = 77 deg
---	172	OTM-331	172TI-3d	2012-267T07:47	Sep23	Sun	60	3	47	D/L start, burn ~6h later. Backup D/L 268T07:47; complexes M/M
---	172	Sun OCC	RING	2012-267T23:42	Sep23	Sun	61	3	1	Egress = 2012-268T03:55 (254 min)
---	172	Earth OCC	RING	2012-267T23:55	Sep23	Sun	61	3	1	Egress = 2012-268T04:07 (253 min)
---	172	Earth OCC	SATURN	2012-268T00:12	Sep24	Mon	61	3	1	Egress = 2012-268T03:48 (217 min)
---	172	Sun OCC	SATURN	2012-268T00:48	Sep24	Mon	61	3	1	Egress = 2012-268T04:10 (202 min)
---	172	Ring CRX	Ascending	2012-268T08:20	Sep24	Mon	62	2	1	r = 6.764 Rs (408092 km)
---	172	Periapse		2012-268T14:59	Sep24	Mon	62	2	1	R = 5.753 Rs, lat = 24 deg, phase = 103 deg
---	172	172TI (t) [T86]	TITAN	2012-270T14:36	Sep26	Wed	64	48	3	Outbound 956.0 km flyby, v = 5.9 km/s, phase = 47 deg
---	172	Ring CRX	Descending	2012-270T15:19	Sep26	Wed	0	48	3	r = 20.262 Rs (1222383 km)
---	172	OTM-332	172TI+4d	2012-273T21:16	Sep29	Sat	3	45	7	D/L start, burn ~6h later. Backup D/L 274T21:16; complexes C/C
---	173	Apoapse		2012-280T10:19	Oct06	Sat	10	38	7	Per = 23.9 d, inc = 39.0 deg, r = 45.163 Rs, phase = 88 deg
---	173	OTM-333	~Apo	2012-283T07:01	Oct09	Tue	13	35	9	D/L start, burn ~6h later. Backup D/L 284T06:46; complexes M/M

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	173	Earth OCC	SATURN	2012-291T05:00	Oct17	Wed	21	27	8	Egress = 2012-291T09:32 (273 min)
---	173	Sun OCC	SATURN	2012-291T05:15	Oct17	Wed	21	27	8	Egress = 2012-291T09:51 (276 min)
---	173	Earth OCC	RING	2012-291T05:19	Oct17	Wed	21	27	8	Egress = 2012-291T10:12 (293 min)
---	173	Sun OCC	RING	2012-291T05:20	Oct17	Wed	21	27	8	Egress = 2012-291T10:14 (295 min)
---	173	Ring CRX	Ascending	2012-291T18:42	Oct17	Wed	21	27	8	r = 10.018 Rs (604397 km)
---	173	Periapse		2012-292T09:02	Oct18	Thu	22	26	9	R = 7.869 Rs, lat = 34 deg, phase = 92 deg
---	173	Ring CRX	Descending	2012-294T12:36	Oct20	Sat	24	24	11	r = 20.240 Rs (1221093 km)
---	173	SEP = 3.0 deg	Conjunction	2012-296T23:22	Oct22	Mon	26	21	14	Commanding must not be required below 3.0 deg
---	173	SEP = 3.0 deg	Conjunction	2012-301T16:24	Oct27	Sat	31	17	18	Commanding must not be required below 3.0 deg
---	174	Apoapse		2012-304T08:17	Oct30	Tue	34	14	21	Per = 24.0 d, inc = 39.0 deg, r = 45.233 Rs, phase = 88 deg
---	174	OTM-334	174TI-4d	2012-314T12:46	Nov09	Fri	44	4	31	D/L start, burn ~6h later. Backup D/L 315T12:31; complexes G/G
---	174	Earth OCC	RING	2012-315T03:05	Nov10	Sat	45	3	1	Egress = 2012-315T08:06 (302 min)
---	174	Sun OCC	RING	2012-315T03:32	Nov10	Sat	45	3	1	Egress = 2012-315T08:31 (300 min)
---	174	Sun OCC	SATURN	2012-315T03:52	Nov10	Sat	45	3	1	Egress = 2012-315T08:28 (276 min)
---	174	Earth OCC	SATURN	2012-315T04:18	Nov10	Sat	45	3	1	Egress = 2012-315T08:33 (255 min)
---	174	Ring CRX	Ascending	2012-315T17:08	Nov10	Sat	45	3	1	r = 10.028 Rs (604991 km)
---	174	Periapse		2012-316T07:30	Nov11	Sun	46	2	2	R = 7.870 Rs, lat = 34 deg, phase = 92 deg
---	174	174TI (t) [T87]	TITAN	2012-318T10:22	Nov13	Tue	48	16	4	Outbound 973.0 km flyby, v = 5.9 km/s, phase = 68 deg
---	174	Ring CRX	Descending	2012-318T10:49	Nov13	Tue	0	16	4	r = 20.185 Rs (1217732 km)
---	174	OTM-335	174TI+4d	2012-321T18:31	Nov16	Fri	3	13	7	D/L start, burn ~6h later. Backup D/L 322T18:31; complexes C/C
---	175	Apoapse		2012-324T02:54	Nov19	Mon	6	10	2	Per = 15.9 d, inc = 46.3 deg, r = 33.619 Rs, phase = 75 deg
---	175	OTM-336	~Apo	2012-326T18:16	Nov21	Wed	8	8	5	D/L start, burn ~6h later. Backup D/L 327T18:16; complexes C/C
---	175	OTM-337	175TI-3d	2012-330T18:01	Nov25	Sun	12	4	4	D/L start, burn ~6h later. Backup D/L 331T18:01; complexes C/C
---	175	Earth OCC	RING	2012-331T08:56	Nov26	Mon	13	3	1	Egress = 2012-331T12:41 (225 min)
---	175	Sun OCC	SATURN	2012-331T08:57	Nov26	Mon	13	3	1	Egress = 2012-331T12:16 (200 min)
---	175	Earth OCC	SATURN	2012-331T09:02	Nov26	Mon	13	3	1	Egress = 2012-331T12:40 (219 min)
---	175	Sun OCC	RING	2012-331T09:30	Nov26	Mon	13	3	1	Egress = 2012-331T12:56 (206 min)
---	175	Ring CRX	Ascending	2012-331T17:24	Nov26	Mon	13	3	1	r = 7.978 Rs (481311 km)
---	175	Periapse		2012-332T02:15	Nov27	Tue	14	2	1	R = 6.894 Rs, lat = 33 deg, phase = 105 deg
---	175	175TI (t) [T88]	TITAN	2012-334T08:57	Nov29	Thu	16	80	4	Outbound 1014.3 km flyby, v = 5.9 km/s, phase = 44 deg
---	175	Ring CRX	Descending	2012-334T09:25	Nov29	Thu	0	80	4	r = 20.179 Rs (1217422 km)
---	175	OTM-338	175TI+4d	2012-337T17:32	Dec02	Sun	3	76	7	D/L start, burn ~6h later. Backup D/L 338T17:32; complexes C/C
---	176	Apoapse		2012-338T10:59	Dec03	Mon	4	76	1	Per = 13.3 d, inc = 53.0 deg, r = 28.413 Rs, phase = 70 deg
---	176	Earth OCC	SATURN	2012-344T08:56	Dec09	Sun	10	70	7	Egress = 2012-344T12:26 (210 min)
---	176	Earth OCC	RING	2012-344T09:12	Dec09	Sun	10	70	7	Egress = 2012-344T12:37 (206 min)
---	176	Sun OCC	SATURN	2012-344T09:29	Dec09	Sun	10	70	7	Egress = 2012-344T11:28 (119 min)
---	176	Sun OCC	RING	2012-344T09:58	Dec09	Sun	10	70	7	Egress = 2012-344T12:51 (173 min)
---	176	Ring CRX	Ascending	2012-344T17:12	Dec09	Sun	10	69	7	r = 8.315 Rs (501617 km)
---	176	176TI (nt)	TITAN	2012-344T23:15	Dec09	Sun	11	69	7	Inbound 906833.8 km flyby, v = 8.1 km/s, phase = 85 deg
---	176	Periapse		2012-345T02:16	Dec10	Mon	11	69	7	R = 7.427 Rs, lat = 34 deg, phase = 110 deg
---	176	Ring CRX	Descending	2012-347T15:52	Dec12	Wed	13	66	10	r = 20.174 Rs (1217109 km)
---	176	176TI (nt)	TITAN	2012-350T13:50	Dec15	Sat	16	64	13	Outbound 899656.8 km flyby, v = 4.6 km/s, phase = 103 deg
---	177	Apoapse		2012-351T17:23	Dec16	Sun	17	62	14	Per = 13.3 d, inc = 53.0 deg, r = 28.405 Rs, phase = 70 deg
---	177	Earth OCC	SATURN	2012-357T15:19	Dec22	Sat	23	56	20	Egress = 2012-357T18:52 (214 min)
---	177	Earth OCC	RING	2012-357T15:20	Dec22	Sat	23	56	20	Egress = 2012-357T18:50 (210 min)
---	177	Sun OCC	SATURN	2012-357T15:37	Dec22	Sat	23	56	20	Egress = 2012-357T17:58 (141 min)
---	177	Sun OCC	RING	2012-357T16:10	Dec22	Sat	23	56	20	Egress = 2012-357T19:07 (178 min)
---	177	177TI (nt)	TITAN	2012-357T17:28	Dec22	Sat	23	56	20	Inbound 714541.4 km flyby, v = 8.1 km/s, phase = 31 deg
---	177	177RH (nt)	RHEA	2012-357T23:07	Dec22	Sat	24	56	20	Inbound 22871.8 km flyby, v = 9.1 km/s, phase = 43 deg
---	177	Ring CRX	Ascending	2012-357T23:29	Dec22	Sat	24	56	20	r = 8.307 Rs (501167 km)
---	177	Periapse		2012-358T08:30	Dec23	Sun	24	56	21	R = 7.424 Rs, lat = 34 deg, phase = 110 deg
---	177	Ring CRX	Descending	2012-360T22:14	Dec25	Tue	27	53	23	r = 20.197 Rs (1218465 km)
---	178	Apoapse		2012-364T23:50	Dec29	Sat	31	49	27	Per = 13.3 d, inc = 53.0 deg, r = 28.423 Rs, phase = 70 deg
---	178	Earth OCC	RING	2013-004T21:52	Jan04	Fri	37	43	33	Egress = 2013-005T01:24 (212 min)
---	178	Earth OCC	SATURN	2013-004T22:06	Jan04	Fri	37	43	33	Egress = 2013-005T01:35 (210 min)
---	178	Sun OCC	SATURN	2013-004T22:09	Jan04	Fri	37	43	33	Egress = 2013-005T00:47 (158 min)
---	178	Sun OCC	RING	2013-004T22:42	Jan04	Fri	37	43	33	Egress = 2013-005T01:45 (183 min)
---	178	Ring CRX	Ascending	2013-005T06:08	Jan05	Sat	37	43	34	r = 8.317 Rs (501769 km)
---	178	Periapse		2013-005T15:10	Jan05	Sat	37	42	34	R = 7.432 Rs, lat = 34 deg, phase = 111 deg
---	178	Ring CRX	Descending	2013-008T04:57	Jan08	Tue	40	40	36	r = 20.208 Rs (1219136 km)
---	179	Apoapse		2013-012T06:43	Jan12	Sat	44	36	41	Per = 13.3 d, inc = 53.0 deg, r = 28.445 Rs, phase = 69 deg
---	179	Earth OCC	RING	2013-018T04:54	Jan18	Fri	50	30	46	Egress = 2013-018T08:25 (212 min)
---	179	Sun OCC	SATURN	2013-018T05:08	Jan18	Fri	50	30	46	Egress = 2013-018T08:00 (172 min)
---	179	Earth OCC	SATURN	2013-018T05:20	Jan18	Fri	50	30	46	Egress = 2013-018T08:41 (201 min)
---	179	Sun OCC	RING	2013-018T05:41	Jan18	Fri	50	30	47	Egress = 2013-018T08:48 (187 min)
---	179	Ring CRX	Ascending	2013-018T13:12	Jan18	Fri	50	30	47	r = 8.327 Rs (502391 km)
---	179	Periapse		2013-018T22:16	Jan18	Fri	51	29	47	R = 7.441 Rs, lat = 34 deg, phase = 111 deg
---	179	Ring CRX	Descending	2013-021T12:07	Jan21	Mon	53	27	50	r = 20.220 Rs (1219860 km)
---	180	Apoapse		2013-025T13:55	Jan25	Fri	57	23	54	Per = 13.3 d, inc = 53.0 deg, r = 28.455 Rs, phase = 69 deg
---	180	OTM-339	~Peri	2013-030T14:09	Jan30	Wed	62	17	59	D/L start, burn ~6h later. Backup D/L 031T00:24; complexes C/M
---	180	Earth OCC	RING	2013-031T12:09	Jan31	Thu	63	17	1	Egress = 2013-031T15:39 (211 min)
---	180	Sun OCC	SATURN	2013-031T12:21	Jan31	Thu	63	17	1	Egress = 2013-031T15:24 (184 min)
---	180	Earth OCC	SATURN	2013-031T12:46	Jan31	Thu	63	17	1	Egress = 2013-031T15:58 (192 min)
---	180	Sun OCC	RING	2013-031T12:52	Jan31	Thu	63	17	1	Egress = 2013-031T16:03 (191 min)
---	180	Ring CRX	Ascending	2013-031T20:28	Jan31	Thu	63	16	1	r = 8.324 Rs (502209 km)
---	180	Periapse		2013-032T05:32	Feb01	Fri	64	16	2	R = 7.438 Rs, lat = 34 deg, phase = 111 deg
---	180	Ring CRX	Descending	2013-034T19:24	Feb03	Sun	66	13	4	r = 20.227 Rs (1220308 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	181	**Apoapse**		2013-038T21:12	Feb07	Thu	71	9	8	Per = 13.3 d, inc = 53.0 deg, r = 28.462 Rs, phase = 69 deg
---	181	**OTM-340**	181TI-4d	2013-043T23:26	Feb12	Tue	76	4	13	D/L start, burn ~6h later. Backup D/L 045T23:26; complexes M/M
---	181	**Earth OCC**	RING	2013-044T19:30	Feb13	Wed	76	3	1	Egress = 2013-044T22:59 (209 min)
---	181	**Sun OCC**	SATURN	2013-044T19:39	Feb13	Wed	76	3	1	Egress = 2013-044T22:51 (192 min)
---	181	**Sun OCC**	RING	2013-044T20:07	Feb13	Wed	76	3	1	Egress = 2013-044T23:21 (194 min)
---	181	**Earth OCC**	SATURN	2013-044T20:11	Feb13	Wed	76	3	1	Egress = 2013-044T23:18 (188 min)
---	181	Ring CRX	Ascending	2013-045T03:47	Feb14	Thu	77	3	1	r = 8.313 Rs (501526 km)
---	181	Periapse		2013-045T12:50	Feb14	Thu	77	3	2	R = 7.426 Rs, lat = 34 deg, phase = 111 deg
---	181	**181TI (t) [T89]**	**TITAN**	2013-048T01:57	Feb17	Sun	80	21	4	Outbound 1978.2 km flyby, v = 5.8 km/s, phase = 35 deg
---	181	Ring CRX	Descending	2013-048T02:23	Feb17	Sun	0	21	4	r = 20.171 Rs (1216906 km)
---	182	**Apoapse**		2013-051T05:02	Feb20	Wed	3	18	7	Per = 12.0 d, inc = 57.1 deg, r = 25.388 Rs, phase = 64 deg
---	182	**OTM-341**	~Peri	2013-055T06:12	Feb24	Sun	7	14	11	D/L start, burn ~6h later. Backup D/L 055T22:42; complexes G/M
---	182	**Earth OCC**	RING	2013-056T10:53	Feb25	Mon	8	12	1	Egress = 2013-056T14:18 (206 min)
---	182	**Earth OCC**	SATURN	2013-056T11:04	Feb25	Mon	8	12	1	Egress = 2013-056T14:32 (208 min)
---	182	**Sun OCC**	SATURN	2013-056T11:06	Feb25	Mon	8	12	1	Egress = 2013-056T13:46 (161 min)
---	182	**Sun OCC**	RING	2013-056T11:34	Feb25	Mon	8	12	1	Egress = 2013-056T14:36 (182 min)
---	182	182TI (nt)	**TITAN**	2013-056T17:27	Feb25	Mon	9	12	1	Inbound 707386.7 km flyby, v = 8.4 km/s, phase = 41 deg
---	182	Ring CRX	Ascending	2013-056T19:13	Feb25	Mon	9	12	2	r = 8.753 Rs (528090 km)
---	182	Periapse		2013-057T04:27	Feb26	Tue	9	12	2	R = 8.040 Rs, lat = 33 deg, phase = 116 deg
---	182	Ring CRX	Descending	2013-060T01:20	Mar01	Fri	12	9	5	r = 20.203 Rs (1218831 km)
---	182	**OTM-342**	~Apo	2013-060T22:28	Mar01	Fri	13	8	6	D/L start, burn ~6h later. Backup D/L 061T22:28; complexes M/M
---	183	**Apoapse**		2013-063T04:02	Mar04	Mon	15	6	2	Per = 12.0 d, inc = 57.1 deg, r = 25.408 Rs, phase = 64 deg
---	183	**OTM-343**	183RH-3d	2013-065T11:58	Mar06	Wed	17	3	5	D/L start, burn ~6h later. Backup D/L 066T11:59; complexes C/C
---	183	**Earth OCC**	RING	2013-068T10:05	Mar09	Sat	20	0	3	Egress = 2013-068T13:30 (205 min)
---	183	**Sun OCC**	SATURN	2013-068T10:11	Mar09	Sat	20	0	3	Egress = 2013-068T13:03 (173 min)
---	183	**Earth OCC**	SATURN	2013-068T10:13	Mar09	Sat	20	0	3	Egress = 2013-068T13:42 (209 min)
---	183	**Sun OCC**	RING	2013-068T10:39	Mar09	Sat	20	0	3	Egress = 2013-068T13:44 (185 min)
---	183	**183RH (t) [R4]**	**RHEA**	2013-068T18:17	Mar09	Sat	21	27	3	Inbound 997.2 km flyby, v = 9.3 km/s, phase = 138 deg
---	183	Ring CRX	Ascending	2013-068T18:22	Mar09	Sat	0	27	3	r = 8.756 Rs (528256 km)
---	183	Periapse		2013-069T03:40	Mar10	Sun	0	27	4	R = 8.032 Rs, lat = 33 deg, phase = 116 deg
---	183	**OTM-344**	183RH+2d	2013-070T21:44	Mar11	Mon	2	25	5	D/L start, burn ~6h later. Backup D/L 071T21:44; complexes M/M
---	183	Ring CRX	Descending	2013-072T00:04	Mar13	Wed	3	24	1	r = 20.133 Rs (1214648 km)
---	184	**Apoapse**		2013-075T03:13	Mar16	Sat	6	21	4	Per = 12.0 d, inc = 57.0 deg, r = 25.405 Rs, phase = 64 deg
---	184	**OTM-345**	~Apo	2013-075T21:29	Mar16	Sat	7	20	5	D/L start, burn ~6h later. Backup D/L 076T21:15; complexes M/M
---	184	**Sun OCC**	SATURN	2013-080T09:11	Mar21	Thu	12	16	4	Egress = 2013-080T12:14 (184 min)
---	184	**Earth OCC**	RING	2013-080T09:13	Mar21	Thu	12	16	4	Egress = 2013-080T12:36 (204 min)
---	184	**Earth OCC**	SATURN	2013-080T09:15	Mar21	Thu	12	16	4	Egress = 2013-080T12:46 (212 min)
---	184	**Sun OCC**	RING	2013-080T09:38	Mar21	Thu	12	16	5	Egress = 2013-080T12:47 (189 min)
---	184	Ring CRX	Ascending	2013-080T17:27	Mar21	Thu	12	15	5	r = 8.762 Rs (528607 km)
---	184	Periapse		2013-081T02:46	Mar22	Fri	12	15	5	R = 8.037 Rs, lat = 33 deg, phase = 116 deg
---	184	Ring CRX	Descending	2013-083T23:11	Mar24	Sun	15	12	8	r = 20.137 Rs (1214873 km)
---	185	**Apoapse**		2013-087T02:17	Mar28	Thu	18	9	11	Per = 12.0 d, inc = 57.0 deg, r = 25.401 Rs, phase = 64 deg
---	185	**OTM-346**	185TI-4d	2013-091T10:16	Apr01	Mon	23	4	16	D/L start, burn ~6h later. Backup D/L 093T10:01; complexes C/C
---	185	**Sun OCC**	SATURN	2013-092T08:11	Apr02	Tue	24	4	1	Egress = 2013-092T11:22 (192 min)
---	185	**Earth OCC**	SATURN	2013-092T08:15	Apr02	Tue	24	4	1	Egress = 2013-092T11:48 (214 min)
---	185	**Earth OCC**	RING	2013-092T08:21	Apr02	Tue	24	4	1	Egress = 2013-092T11:42 (202 min)
---	185	**Sun OCC**	RING	2013-092T08:37	Apr02	Tue	24	4	1	Egress = 2013-092T11:48 (192 min)
---	185	Ring CRX	Ascending	2013-092T16:28	Apr02	Tue	24	3	1	r = 8.750 Rs (527898 km)
---	185	Periapse		2013-093T01:46	Apr03	Wed	24	3	2	R = 8.025 Rs, lat = 33 deg, phase = 116 deg
---	185	**185TI (t) [T90]**	**TITAN**	2013-095T21:43	Apr05	Fri	27	48	4	Outbound 1400.0 km flyby, v = 5.8 km/s, phase = 69 deg
---	185	Ring CRX	Descending	2013-095T21:58	Apr05	Fri	0	48	4	r = 20.107 Rs (1213041 km)
---	186	**Apoapse**		2013-097T17:49	Apr07	Sun	2	46	6	Per = 9.6 d, inc = 61.7 deg, r = 22.565 Rs, phase = 48 deg
---	186	**OTM-347**	185TI+3d	2013-098T19:46	Apr08	Mon	3	45	7	D/L start, burn ~6h later. Backup D/L 099T19:46; complexes M/M
---	186	**Earth OCC**	SATURN	2013-102T03:45	Apr12	Fri	6	42	3	Egress = 2013-102T06:27 (162 min)
---	186	**Sun OCC**	SATURN	2013-102T03:47	Apr12	Fri	6	42	3	Egress = 2013-102T06:15 (148 min)
---	186	186TI (nt)	**TITAN**	2013-102T03:48	Apr12	Fri	6	42	3	Inbound 939359.2 km flyby, v = 10.1 km/s, phase = 50 deg
---	186	**Earth OCC**	RING	2013-102T04:02	Apr12	Fri	6	42	3	Egress = 2013-102T06:32 (151 min)
---	186	**Sun OCC**	RING	2013-102T04:07	Apr12	Fri	6	42	3	Egress = 2013-102T06:33 (146 min)
---	186	186PO (nt)	**POLYDEUCES**	2013-102T07:45	Apr12	Fri	6	41	3	Inbound 115442.5 km flyby, v = 11.5 km/s, phase = 126 deg
---	186	Ring CRX	Ascending	2013-102T08:52	Apr12	Fri	6	41	4	r = 6.462 Rs (389848 km)
---	186	Periapse		2013-102T12:35	Apr12	Fri	7	41	4	R = 6.244 Rs, lat = 22 deg, phase = 132 deg
---	186	Ring CRX	Descending	2013-105T11:28	Apr15	Mon	10	38	7	r = 20.116 Rs (1213574 km)
---	187	**Apoapse**		2013-107T07:26	Apr17	Wed	11	36	8	Per = 9.6 d, inc = 61.7 deg, r = 22.575 Rs, phase = 48 deg
---	187	**Earth OCC**	RHEA	2013-111T14:19	Apr21	Sun	16	32	13	Egress = 2013-111T14:19 (1 min)
---	187	**Sun OCC**	SATURN	2013-111T17:28	Apr21	Sun	16	32	13	Egress = 2013-111T20:00 (152 min)
---	187	**Earth OCC**	SATURN	2013-111T17:29	Apr21	Sun	16	32	13	Egress = 2013-111T20:07 (158 min)
---	187	**Earth OCC**	RING	2013-111T17:48	Apr21	Sun	16	32	13	Egress = 2013-111T20:16 (148 min)
---	187	**Sun OCC**	RING	2013-111T17:48	Apr21	Sun	16	32	13	Egress = 2013-111T20:15 (148 min)
---	187	Ring CRX	Ascending	2013-111T22:34	Apr21	Sun	16	32	13	r = 6.467 Rs (390140 km)
---	187	Periapse		2013-112T02:17	Apr22	Mon	16	32	13	R = 6.248 Rs, lat = 22 deg, phase = 132 deg
---	187	Ring CRX	Descending	2013-115T01:13	Apr25	Thu	19	29	16	r = 20.120 Rs (1213817 km)
---	188	**Apoapse**		2013-116T21:08	Apr26	Fri	21	27	18	Per = 9.6 d, inc = 61.7 deg, r = 22.574 Rs, phase = 48 deg
---	188	**OTM-348**	~Peri	2013-120T02:02	Apr30	Tue	24	24	21	D/L start, burn ~6h later. Backup D/L 120T18:17; complexes G/M
---	188	**Sun OCC**	SATURN	2013-121T07:09	May01	Wed	25	22	1	Egress = 2013-121T09:44 (155 min)
---	188	**Earth OCC**	SATURN	2013-121T07:14	May01	Wed	25	22	1	Egress = 2013-121T09:45 (151 min)
---	188	**Sun OCC**	RING	2013-121T07:28	May01	Wed	25	22	1	Egress = 2013-121T09:57 (149 min)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	188	Earth OCC	RING	2013-121T07:35	May01	Wed	25	22	1	Egress = 2013-121T09:59 (145 min)
---	188	Ring CRX	Ascending	2013-121T12:16	May01	Wed	26	22	1	r = 6.463 Rs (389896 km)
---	188	188TI (nt)	TITAN	2013-121T13:43	May01	Wed	26	22	1	Inbound 900098.8 km flyby, v = 10.7 km/s, phase = 59 deg
---	188	Periapse		2013-121T15:59	May01	Wed	26	22	2	R = 6.244 Rs, lat = 22 deg, phase = 132 deg
---	188	Ring CRX	Descending	2013-124T14:54	May04	Sat	29	19	5	r = 20.119 Rs (1213787 km)
---	189	Apoapse		2013-126T10:49	May06	Mon	31	17	6	Per = 9.6 d, inc = 61.7 deg, r = 22.575 Rs, phase = 48 deg
---	189	189TI (nt)	TITAN	2013-127T16:25	May07	Tue	32	16	8	Inbound 908120.9 km flyby, v = 6.3 km/s, phase = 127 deg
---	189	Sun OCC	SATURN	2013-130T20:47	May10	Fri	35	13	11	Egress = 2013-130T23:25 (158 min)
---	189	Earth OCC	SATURN	2013-130T20:56	May10	Fri	35	13	11	Egress = 2013-130T23:19 (143 min)
---	189	Sun OCC	RING	2013-130T21:05	May10	Fri	35	13	11	Egress = 2013-130T23:35 (150 min)
---	189	Earth OCC	RING	2013-130T21:17	May10	Fri	35	13	11	Egress = 2013-130T23:39 (142 min)
---	189	Ring CRX	Ascending	2013-131T01:55	May11	Sat	35	13	11	r = 6.459 Rs (389667 km)
---	189	Periapse		2013-131T05:37	May11	Sat	35	12	11	R = 6.241 Rs, lat = 22 deg, phase = 132 deg
---	189	Ring CRX	Descending	2013-134T04:33	May14	Tue	38	10	14	r = 20.125 Rs (1214144 km)
---	190	Apoapse		2013-136T00:30	May16	Thu	40	8	16	Per = 9.6 d, inc = 61.7 deg, r = 22.583 Rs, phase = 48 deg
---	190	OTM-349	190TI-4d	2013-139T06:46	May19	Sun	43	4	19	D/L start, burn ~6h later. Backup D/L 141T06:46; complexes C/C
---	190	Sun OCC	SATURN	2013-140T10:32	May20	Mon	45	3	1	Egress = 2013-140T13:12 (161 min)
---	190	Earth OCC	SATURN	2013-140T10:45	May20	Mon	45	3	1	Egress = 2013-140T12:59 (134 min)
---	190	Sun OCC	RING	2013-140T10:48	May20	Mon	45	3	1	Egress = 2013-140T13:19 (152 min)
---	190	Earth OCC	RING	2013-140T11:06	May20	Mon	45	3	1	Egress = 2013-140T13:25 (139 min)
---	190	Ring CRX	Ascending	2013-140T15:40	May20	Mon	45	3	1	r = 6.455 Rs (389442 km)
---	190	Periapse		2013-140T19:22	May20	Mon	45	3	2	R = 6.237 Rs, lat = 22 deg, phase = 132 deg
---	190	190TI (t) [T91]	TITAN	2013-143T17:33	May23	Thu	48	48	4	Outbound 970.0 km flyby, v = 5.9 km/s, phase = 50 deg
---	190	Ring CRX	Descending	2013-143T18:00	May23	Thu	0	48	4	r = 20.096 Rs (1212366 km)
---	191	Apoapse		2013-146T07:38	May26	Sun	3	45	7	Per = 12.0 d, inc = 59.4 deg, r = 23.188 Rs, phase = 63 deg
---	191	OTM-350	190TI+4d	2013-147T00:01	May27	Mon	3	45	8	D/L start, burn ~6h later. Backup D/L 148T00:01; complexes G/G
---	191	Sun OCC	SATURN	2013-151T06:00	May31	Fri	8	40	4	Egress = 2013-151T09:53 (233 min)
---	191	Sun OCC	RING	2013-151T06:21	May31	Fri	8	40	4	Egress = 2013-151T10:03 (222 min)
---	191	Earth OCC	RING	2013-151T07:18	May31	Fri	8	40	4	Egress = 2013-151T10:22 (184 min)
---	191	Earth OCC	SATURN	2013-151T07:20	May31	Fri	8	40	4	Egress = 2013-151T08:21 (61 min)
---	191	191TI (nt)	TITAN	2013-151T13:54	May31	Fri	8	40	5	Inbound 568230.7 km flyby, v = 7.6 km/s, phase = 27 deg
---	191	Ring CRX	Ascending	2013-151T17:10	May31	Fri	8	40	5	r = 10.980 Rs (662429 km)
---	191	Periapse		2013-152T07:01	Jun01	Sat	9	39	5	R = 10.237 Rs, lat = 34 deg, phase = 117 deg
---	191	Ring CRX	Descending	2013-155T16:53	Jun04	Tue	12	36	9	r = 20.110 Rs (1213218 km)
---	192	Apoapse		2013-158T06:29	Jun07	Fri	15	33	11	Per = 12.0 d, inc = 59.4 deg, r = 23.191 Rs, phase = 63 deg
---	192	OTM-351	~Peri	2013-162T15:14	Jun11	Tue	19	29	16	D/L start, burn ~6h later. Backup D/L 163T15:14; complexes M/M
---	192	Sun OCC	SATURN	2013-163T04:56	Jun12	Wed	19	28	1	Egress = 2013-163T08:53 (238 min)
---	192	Sun OCC	RING	2013-163T05:13	Jun12	Wed	19	28	1	Egress = 2013-163T08:57 (224 min)
---	192	Earth OCC	RING	2013-163T06:22	Jun12	Wed	20	28	1	Egress = 2013-163T09:18 (176 min)
---	192	Ring CRX	Ascending	2013-163T16:06	Jun12	Wed	20	28	1	r = 10.986 Rs (662791 km)
---	192	Periapse		2013-164T05:58	Jun13	Thu	21	27	2	R = 10.243 Rs, lat = 34 deg, phase = 117 deg
---	192	Ring CRX	Descending	2013-167T15:54	Jun16	Sun	24	24	5	r = 20.116 Rs (1213597 km)
---	193	Apoapse		2013-170T05:34	Jun19	Wed	27	21	8	Per = 12.0 d, inc = 59.4 deg, r = 23.200 Rs, phase = 63 deg
---	193	Sun OCC	SATURN	2013-175T04:06	Jun24	Mon	31	16	13	Egress = 2013-175T08:06 (240 min)
---	193	Sun OCC	RING	2013-175T04:18	Jun24	Mon	31	16	13	Egress = 2013-175T08:04 (226 min)
---	193	Earth OCC	RING	2013-175T05:37	Jun24	Mon	32	16	13	Egress = 2013-175T08:27 (171 min)
---	193	Ring CRX	Ascending	2013-175T15:16	Jun24	Mon	32	16	13	r = 10.990 Rs (663006 km)
---	193	Periapse		2013-176T05:09	Jun25	Tue	32	15	14	R = 10.246 Rs, lat = 34 deg, phase = 117 deg
---	193	Ring CRX	Descending	2013-179T15:04	Jun28	Fri	36	12	17	r = 20.115 Rs (1213518 km)
---	194	Apoapse		2013-182T04:39	Jul01	Mon	38	9	20	Per = 12.0 d, inc = 59.4 deg, r = 23.192 Rs, phase = 63 deg
---	194	Sun OCC	SATURN	2013-187T03:09	Jul06	Sat	43	4	24	Egress = 2013-187T07:10 (241 min)
---	194	Sun OCC	RING	2013-187T03:17	Jul06	Sat	43	4	25	Egress = 2013-187T07:03 (227 min)
---	194	Earth OCC	RING	2013-187T04:40	Jul06	Sat	43	4	25	Egress = 2013-187T07:28 (169 min)
---	194	Ring CRX	Ascending	2013-187T14:17	Jul06	Sat	44	4	25	r = 10.976 Rs (662193 km)
---	194	OTM-352	194TI-3d	2013-188T03:26	Jul07	Sun	44	3	26	D/L start, burn ~6h later. Backup D/L 189T03:26; complexes C/C
---	194	Periapse		2013-188T04:09	Jul07	Sun	44	3	0	R = 10.232 Rs, lat = 34 deg, phase = 117 deg
---	194	194TI (t) [T92]	TITAN	2013-191T13:22	Jul10	Wed	48	16	3	Outbound 964.0 km flyby, v = 5.9 km/s, phase = 46 deg
---	194	Ring CRX	Descending	2013-191T13:44	Jul10	Wed	0	16	3	r = 20.083 Rs (1211591 km)
---	194	OTM-353	194TI+4d	2013-194T20:40	Jul13	Sat	3	13	7	D/L start, burn ~6h later. Backup D/L 195T20:40; complexes G/G
---	195	Apoapse		2013-196T06:15	Jul15	Mon	5	11	1	Per = 16.0 d, inc = 56.7 deg, r = 25.136 Rs, phase = 100 deg
---	195	OTM-354	~Apo	2013-199T20:25	Jul18	Thu	8	8	5	D/L start, burn ~6h later. Backup D/L 200T12:40; complexes G/M
---	195	Sun OCC	RING	2013-200T21:03	Jul19	Fri	9	7	1	Egress = 2013-201T02:24 (322 min)
---	195	Sun OCC	SATURN	2013-200T22:09	Jul19	Fri	9	7	1	Egress = 2013-201T02:57 (289 min)
---	195	Earth OCC	RING	2013-201T00:06	Jul20	Sat	9	6	1	Egress = 2013-201T03:29 (204 min)
---	195	195TI (nt)	TITAN	2013-201T02:45	Jul20	Sat	10	6	1	Inbound 889495.4 km flyby, v = 6.2 km/s, phase = 111 deg
---	195	Ring CRX	Ascending	2013-201T22:04	Jul20	Sat	10	6	2	r = 18.176 Rs (1096531 km)
---	195	OTM-355	195TI-3d	2013-204T02:24	Jul23	Tue	13	3	4	D/L start, burn ~6h later. Backup D/L 205T02:24; complexes C/C
---	195	Periapse		2013-204T05:40	Jul23	Tue	13	3	0	R = 15.370 Rs, lat = 55 deg, phase = 80 deg
---	195	195TI (t) [T93]	TITAN	2013-207T11:56	Jul26	Fri	16	48	3	Outbound 1400.0 km flyby, v = 5.8 km/s, phase = 58 deg
---	195	Ring CRX	Descending	2013-207T12:16	Jul26	Fri	0	48	3	r = 20.066 Rs (1210559 km)
---	195	OTM-356	195TI+4d	2013-211T01:53	Jul30	Tue	4	44	7	D/L start, burn ~6h later. Backup D/L 212T01:53; complexes C/C
---	196	Apoapse		2013-217T16:38	Aug05	Mon	10	38	7	Per = 23.9 d, inc = 53.4 deg, r = 34.602 Rs, phase = 152 deg
---	196	OTM-357	~Apo	2013-219T01:22	Aug07	Wed	12	36	8	D/L start, burn ~6h later. Backup D/L 220T01:22; complexes C/C
---	196	Earth OCC	RING	2013-220T16:07	Aug08	Thu	13	35	2	Egress = 2013-220T22:19 (372 min)
---	196	Ring CRX	Ascending	2013-223T00:04	Aug11	Sun	16	32	4	r = 30.033 Rs (1811890 km)
---	196	Periapse		2013-229T15:35	Aug17	Sat	22	26	11	R = 18.474 Rs, lat = 38 deg, phase = 28 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	196	Ring CRX	Descending	2013-231T09:58	Aug19	Mon	24	24	12	r = 20.105 Rs (1212906 km)
---	197	**Apoapse**		2013-241T14:48	Aug29	Thu	34	14	23	Per = 23.9 d, inc = 53.4 deg, r = 34.610 Rs, phase = 152 deg
---	197	**Earth OCC**	SATURN	2013-244T10:47	Sep01	Sun	37	11	25	Egress = 2013-244T19:01 (494 min)
---	197	**Earth OCC**	RING	2013-244T11:34	Sep01	Sun	37	11	25	Egress = 2013-244T19:37 (484 min)
---	197	Ring CRX	Ascending	2013-246T22:11	Sep03	Tue	39	8	28	r = 30.063 Rs (1813671 km)
---	197	197TI (nt)	**TITAN**	2013-247T21:26	Sep04	Wed	40	7	29	Inbound 531389.0 km flyby, v = 4.7 km/s, phase = 126 deg
---	197	**OTM-358**	197TI-3d	2013-251T23:18	Sep08	Sun	44	3	33	D/L start, burn ~6h later. Backup D/L 252T23:18; complexes C/C
---	197	Periapse		2013-253T13:48	Sep10	Tue	46	2	2	R = 18.469 Rs, lat = 37 deg, phase = 28 deg
---	197	**197TI (t) [T94]**	**TITAN**	2013-255T07:44	Sep12	Thu	48	32	3	Outbound 1400.0 km flyby, v = 5.8 km/s, phase = 21 deg
---	197	Ring CRX	Descending	2013-255T08:01	Sep12	Thu	0	32	3	r = 20.087 Rs (1211869 km)
---	197	**OTM-359**	197TI+4d	2013-258T22:47	Sep15	Sun	4	28	7	D/L start, burn ~6h later. Backup D/L 259T22:47; complexes C/C
---	198	**Apoapse**		2013-270T15:02	Sep27	Fri	15	17	12	Per = 31.9 d, inc = 51.9 deg, r = 44.472 Rs, phase = 173 deg
---	198	**OTM-360**	~Apo	2013-272T22:01	Sep29	Sun	18	14	14	D/L start, burn ~6h later. Backup D/L 273T22:01; complexes C/C
---	198	Ring CRX	Ascending	2013-273T19:22	Sep30	Mon	18	13	1	r = 43.208 Rs (2606740 km)
---	198	**OTM-361**	198TI-3d	2013-283T21:15	Oct10	Thu	29	3	11	D/L start, burn ~6h later. Backup D/L 284T21:15; complexes C/C
---	198	Periapse		2013-286T13:40	Oct13	Sun	31	1	3	R = 19.819 Rs, lat = 14 deg, phase = 7 deg
---	198	**198TI (t) [T95]**	**TITAN**	2013-287T04:56	Oct14	Mon	32	48	3	Outbound 961.0 km flyby, v = 5.9 km/s, phase = 25 deg
---	198	Ring CRX	Descending	2013-287T05:11	Oct14	Mon	0	48	3	r = 20.069 Rs (1210774 km)
---	198	**OTM-362**	198TI+3d	2013-290T07:15	Oct17	Thu	3	45	6	D/L start, burn ~6h later. Backup D/L 291T07:15; complexes M/M
---	198	**OTM-363**	~Apo	2013-306T06:15	Nov02	Sat	19	29	16	D/L start, burn ~6h later. Backup D/L 307T06:15; complexes M/M
---	198	SEP = 3.0 deg	Conjunction	2013-308T03:22	Nov04	Mon	21	27	2	Commanding must not be required below 3.0 deg
---	198	Ring CRX	Ascending	2013-309T19:00	Nov05	Tue	23	25	4	r = 64.016 Rs (3862094 km)
---	199	**Apoapse**		2013-311T06:31	Nov07	Thu	24	24	5	Per = 47.9 d, inc = 49.7 deg, r = 64.196 Rs, phase = 155 deg
---	199	SEP = 3.0 deg	Conjunction	2013-312T20:15	Nov08	Fri	26	22	7	Commanding must not be required below 3.0 deg
---	199	**OTM-364**	199TI-3d	2013-331T18:45	Nov27	Wed	45	3	26	D/L start, burn ~6h later. Backup D/L 332T18:45; complexes C/C
---	199	**199TI (t) [T96]**	**TITAN**	2013-335T00:41	Dec01	Sun	48	32	3	Inbound 1400.0 km flyby, v = 5.8 km/s, phase = 50 deg
---	199	Ring CRX	Descending	2013-335T00:51	Dec01	Sun	0	32	3	r = 20.064 Rs (1210433 km)
---	199	Periapse		2013-335T22:43	Dec01	Sun	1	31	4	R = 19.523 Rs, lat = -19 deg, phase = 46 deg
---	199	**OTM-365**	199TI+4d	2013-338T11:45	Dec04	Wed	3	28	7	D/L start, burn ~6h later. Backup D/L 339T11:46; complexes G/G
---	199	Ring CRX	Ascending	2013-347T08:41	Dec13	Fri	12	20	9	r = 42.176 Rs (2544497 km)
---	199	**OTM-366**	~Apo	2013-351T17:32	Dec17	Tue	17	15	13	D/L start, burn ~6h later. Backup D/L 352T17:32; complexes C/C
---	200	**Apoapse**		2013-351T21:24	Dec17	Tue	17	15	0	Per = 31.9 d, inc = 51.3 deg, r = 44.775 Rs, phase = 134 deg
---	200	**OTM-367**	200TI-3d	2013-363T16:48	Dec29	Sun	29	3	12	D/L start, burn ~6h later. Backup D/L 364T16:48; complexes C/C
---	200	**200TI (t) [T97]**	**TITAN**	2014-001T22:00	Jan01	Wed	32	32	3	Inbound 1400.0 km flyby, v = 5.8 km/s, phase = 35 deg
---	200	Ring CRX	Descending	2014-001T22:10	Jan01	Wed	0	32	3	r = 20.064 Rs (1210453 km)
---	200	Periapse		2014-003T11:34	Jan03	Fri	2	30	5	R = 18.176 Rs, lat = -34 deg, phase = 66 deg
---	200	**OTM-368**	200TI+4d	2014-005T10:03	Jan05	Sun	4	28	7	D/L start, burn ~6h later. Backup D/L 006T10:03; complexes G/G
---	200	200TI (nt)	**TITAN**	2014-009T10:02	Jan09	Thu	8	24	4	Outbound 810304.7 km flyby, v = 4.3 km/s, phase = 169 deg
---	200	Ring CRX	Ascending	2014-011T07:04	Jan11	Sat	9	23	6	r = 37.259 Rs (2247841 km)
---	201	**Apoapse**		2014-019T10:10	Jan19	Sun	18	14	14	Per = 31.9 d, inc = 50.1 deg, r = 46.121 Rs, phase = 114 deg
---	201	**OTM-369**	~Apo	2014-025T08:51	Jan25	Sat	23	8	20	D/L start, burn ~6h later. Backup D/L 026T08:51; complexes G/G
---	201	**OTM-370**	201TI-3d	2014-030T14:51	Jan30	Thu	29	3	5	D/L start, burn ~6h later. Backup D/L 031T14:51; complexes C/C
---	201	**201TI (t) [T98]**	**TITAN**	2014-033T19:13	Feb02	Sun	32	32	3	Inbound 1235.7 km flyby, v = 5.9 km/s, phase = 44 deg
---	201	Ring CRX	Descending	2014-033T19:22	Feb02	Sun	0	32	3	r = 20.051 Rs (1209671 km)
---	201	Periapse		2014-035T18:12	Feb04	Tue	2	30	5	R = 16.244 Rs, lat = -42 deg, phase = 83 deg
---	201	**OTM-371**	201TI+3d	2014-036T08:07	Feb05	Wed	3	29	6	D/L start, burn ~6h later. Backup D/L 037T08:07; complexes G/G
---	201	Ring CRX	Ascending	2014-040T19:03	Feb09	Sun	7	25	4	r = 30.801 Rs (1858203 km)
---	201	201TI (nt)	**TITAN**	2014-041T13:05	Feb10	Mon	8	24	5	Outbound 887331.9 km flyby, v = 4.1 km/s, phase = 134 deg
---	201	**OTM-372**	~Apo	2014-048T07:24	Feb17	Mon	15	17	12	D/L start, burn ~6h later. Backup D/L 049T07:24; complexes G/G
---	202	**Apoapse**		2014-051T16:45	Feb20	Thu	18	14	3	Per = 31.9 d, inc = 48.1 deg, r = 48.048 Rs, phase = 97 deg
---	202	**OTM-373**	202TI-3d	2014-062T12:56	Mar03	Mon	29	3	14	D/L start, burn ~6h later. Backup D/L 063T12:56; complexes C/C
---	202	**202TI (t) [T99]**	**TITAN**	2014-065T16:27	Mar06	Thu	32	32	3	Inbound 1500.0 km flyby, v = 5.8 km/s, phase = 53 deg
---	202	Ring CRX	Descending	2014-065T16:34	Mar06	Thu	0	32	3	r = 20.047 Rs (1209438 km)
---	202	Periapse		2014-067T19:26	Mar08	Sat	2	30	5	R = 14.241 Rs, lat = -44 deg, phase = 96 deg
---	202	**OTM-374**	202TI+4d	2014-069T06:12	Mar10	Mon	4	28	7	D/L start, burn ~6h later. Backup D/L 070T06:12; complexes G/G
---	202	Ring CRX	Ascending	2014-071T01:27	Mar12	Wed	5	27	2	r = 24.817 Rs (1497191 km)
---	202	**OTM-375**	~Apo	2014-079T05:28	Mar20	Thu	14	18	10	D/L start, burn ~6h later. Backup D/L 080T05:28; complexes G/G
---	203	**Apoapse**		2014-083T18:02	Mar24	Mon	18	14	5	Per = 31.9 d, inc = 45.5 deg, r = 50.058 Rs, phase = 84 deg
---	203	**OTM-376**	203TI-3d	2014-094T04:29	Apr04	Fri	29	3	15	D/L start, burn ~6h later. Backup D/L 095T04:29; complexes G/G
---	203	**203TI (t) [T100]**	**TITAN**	2014-097T13:41	Apr07	Mon	32	40	3	Inbound 963.0 km flyby, v = 5.9 km/s, phase = 60 deg
---	203	Ring CRX	Descending	2014-097T13:50	Apr07	Mon	0	40	3	r = 20.025 Rs (1208105 km)
---	203	Periapse		2014-099T16:06	Apr09	Wed	2	38	5	R = 12.373 Rs, lat = -41 deg, phase = 104 deg
---	203	**OTM-377**	203TI+4d	2014-101T04:00	Apr11	Fri	4	37	7	D/L start, burn ~6h later. Backup D/L 102T04:00; complexes G/G
---	203	Ring CRX	Ascending	2014-101T21:26	Apr11	Fri	4	36	1	r = 20.664 Rs (1246650 km)
---	203	**OTM-378**	~Apo	2014-114T03:01	Apr24	Thu	17	24	13	D/L start, burn ~6h later. Backup D/L 115T03:01; complexes G/G
---	204	**Apoapse**		2014-117T13:47	Apr27	Sun	20	20	3	Per = 35.8 d, inc = 40.7 deg, r = 57.089 Rs, phase = 77 deg
---	204	Ring CRX	Descending	2014-133T09:11	May13	Tue	36	4	19	r = 20.018 Rs (1207684 km)
---	204	**OTM-379**	204TI-3d	2014-134T01:46	May14	Wed	37	4	20	D/L start, burn ~6h later. Backup D/L 135T01:31; complexes G/G
---	204	Periapse		2014-135T11:30	May15	Thu	38	2	1	R = 12.375 Rs, lat = -41 deg, phase = 103 deg
---	204	**Earth OCC**	TITAN	2014-137T16:10	May17	Sat	40	0	4	Egress = 2014-137T16:26 (16 min)
---	204	**Sun OCC**	TITAN	2014-137T16:10	May17	Sat	40	0	4	Egress = 2014-137T16:26 (16 min)
---	204	**204TI (t) [T101]**	**TITAN**	2014-137T16:12	May17	Sat	40	32	4	Outbound 2993.4 km flyby, v = 5.7 km/s, phase = 158 deg
---	204	Ring CRX	Ascending	2014-137T16:44	May17	Sat	0	32	4	r = 20.651 Rs (1245848 km)
---	204	**OTM-380**	204TI+4d	2014-141T01:16	May21	Wed	3	29	7	D/L start, burn ~6h later. Backup D/L 142T01:16; complexes G/G

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	205	**Apoapse**		2014-151T08:37	May31	Sat	14	18	10	Per = 31.9 d, inc = 44.3 deg, r = 50.833 Rs, phase = 72 deg
---	205	**OTM-381**	~Apo	2014-154T00:15	Jun03	Tue	16	16	13	D/L start, burn ~6h later. Backup D/L 155T00:15; complexes G/G
---	205	Ring CRX	Descending	2014-164T17:28	Jun13	Fri	27	5	11	r = 21.969 Rs (1325419 km)
---	205	**OTM-382**	205TI-3d	2014-166T05:44	Jun15	Sun	29	3	12	D/L start, burn ~6h later. Backup D/L 167T05:44; complexes C/C
---	205	Periapse		2014-167T07:15	Jun16	Mon	30	2	1	R = 13.470 Rs, lat = -44 deg, phase = 108 deg
---	205	**Earth OCC**	TITAN	2014-169T13:25	Jun18	Wed	32	0	3	Egress = 2014-169T13:42 (17 min)
---	205	**Sun OCC**	TITAN	2014-169T13:25	Jun18	Wed	32	0	3	Egress = 2014-169T13:41 (17 min)
---	205	**205TI (t) [T102]**	**TITAN**	2014-169T13:28	Jun18	Wed	32	32	3	Outbound 3658.3 km flyby, v = 5.6 km/s, phase = 165 deg
---	205	Ring CRX	Ascending	2014-169T13:59	Jun18	Wed	0	32	3	r = 20.649 Rs (1245766 km)
---	205	**OTM-383**	205TI+4d	2014-172T22:59	Jun21	Sat	3	28	7	D/L start, burn ~6h later. Backup D/L 173T22:59; complexes G/G
---	206	**Apoapse**		2014-183T06:52	Jul02	Wed	14	18	10	Per = 31.9 d, inc = 46.5 deg, r = 49.380 Rs, phase = 64 deg
---	206	**OTM-384**	~Apo	2014-185T21:58	Jul04	Fri	16	16	13	D/L start, burn ~6h later. Backup D/L 186T21:57; complexes G/G
---	206	Ring CRX	Descending	2014-195T17:19	Jul14	Mon	26	6	10	r = 25.716 Rs (1551455 km)
---	206	**OTM-385**	206TI-3d	2014-198T03:40	Jul17	Thu	29	3	12	D/L start, burn ~6h later. Backup D/L 199T03:41; complexes C/C
---	206	Periapse		2014-199T05:26	Jul18	Fri	30	2	1	R = 14.921 Rs, lat = -45 deg, phase = 116 deg
---	206	**Sun OCC**	TITAN	2014-201T10:36	Jul20	Sun	32	0	3	Egress = 2014-201T10:52 (17 min)
---	206	**Earth OCC**	TITAN	2014-201T10:37	Jul20	Sun	32	0	3	Egress = 2014-201T10:54 (17 min)
---	206	**206TI (t) [T103]**	**TITAN**	2014-201T10:41	Jul20	Sun	32	32	3	Outbound 5102.9 km flyby, v = 5.6 km/s, phase = 171 deg
---	206	Ring CRX	Ascending	2014-201T11:11	Jul20	Sun	0	32	3	r = 20.664 Rs (1246657 km)
---	206	**OTM-386**	206TI+4d	2014-204T20:40	Jul23	Wed	3	28	7	D/L start, burn ~6h later. Backup D/L 205T20:40; complexes G/G
---	207	**Apoapse**		2014-215T06:57	Aug03	Sun	14	18	10	Per = 31.9 d, inc = 48.0 deg, r = 48.279 Rs, phase = 56 deg
---	207	**OTM-387**	~Apo	2014-221T02:08	Aug09	Sat	20	12	16	D/L start, burn ~6h later. Backup D/L 222T02:08; complexes C/C
---	207	207TI (nt)	**TITAN**	2014-225T12:08	Aug13	Wed	24	8	4	Inbound 959106.5 km flyby, v = 4.2 km/s, phase = 34 deg
---	207	Ring CRX	Descending	2014-226T18:37	Aug14	Thu	25	7	6	r = 28.973 Rs (1747913 km)
---	207	**OTM-388**	207TI-3d	2014-230T01:37	Aug18	Mon	29	3	9	D/L start, burn ~6h later. Backup D/L 231T01:37; complexes C/C
---	207	Periapse		2014-231T05:47	Aug19	Tue	30	2	1	R = 15.941 Rs, lat = -44 deg, phase = 123 deg
---	207	**207TI (t) [T104]**	**TITAN**	2014-233T08:09	Aug21	Thu	32	32	3	Outbound 964.0 km flyby, v = 5.8 km/s, phase = 12 deg
---	207	Ring CRX	Ascending	2014-233T08:17	Aug21	Thu	0	32	3	r = 20.455 Rs (1234030 km)
---	207	**OTM-389**	207TI+4d	2014-237T01:06	Aug25	Mon	4	28	7	D/L start, burn ~6h later. Backup D/L 238T01:06; complexes C/C
---	208	**Apoapse**		2014-247T01:17	Sep04	Thu	14	18	10	Per = 31.9 d, inc = 44.6 deg, r = 50.809 Rs, phase = 73 deg
---	208	**OTM-390**	~Apo	2014-250T00:19	Sep07	Sun	17	15	13	D/L start, burn ~6h later. Backup D/L 251T00:19; complexes C/C
---	208	Ring CRX	Descending	2014-260T08:32	Sep17	Wed	27	5	10	r = 22.263 Rs (1343140 km)
---	208	**OTM-391**	208TI-3d	2014-261T23:33	Sep18	Thu	29	3	12	D/L start, burn ~6h later. Backup D/L 262T23:33; complexes C/C
---	208	Periapse		2014-262T23:53	Sep19	Fri	30	2	1	R = 13.483 Rs, lat = -44 deg, phase = 107 deg
---	208	**208TI (t) [T105]**	**TITAN**	2014-265T05:23	Sep22	Mon	32	32	3	Outbound 1400.0 km flyby, v = 5.8 km/s, phase = 21 deg
---	208	Ring CRX	Ascending	2014-265T05:30	Sep22	Mon	0	32	3	r = 20.455 Rs (1234022 km)
---	208	**OTM-392**	208TI+4d	2014-268T23:02	Sep25	Thu	4	28	7	D/L start, burn ~6h later. Backup D/L 269T23:02; complexes C/C
---	209	**Apoapse**		2014-278T22:43	Oct05	Sun	14	18	10	Per = 31.9 d, inc = 40.3 deg, r = 52.919 Rs, phase = 86 deg
---	209	**OTM-393**	~Apo	2014-281T22:16	Oct08	Wed	17	15	13	D/L start, burn ~6h later. Backup D/L 282T22:16; complexes C/C
---	209	Ring CRX	Descending	2014-293T06:10	Oct20	Mon	28	4	11	r = 17.281 Rs (1042536 km)
---	209	**OTM-394**	209TI-3d	2014-293T21:30	Oct20	Mon	29	3	12	D/L start, burn ~6h later. Backup D/L 294T21:30; complexes C/C
---	209	Periapse		2014-294T21:22	Oct21	Tue	30	2	1	R = 11.387 Rs, lat = -40 deg, phase = 94 deg
---	209	**209TI (t) [T106]**	**TITAN**	2014-297T02:40	Oct24	Fri	32	48	3	Outbound 1013.0 km flyby, v = 5.8 km/s, phase = 31 deg
---	209	Ring CRX	Ascending	2014-297T02:51	Oct24	Fri	0	48	3	r = 20.488 Rs (1236024 km)
---	209	**OTM-395**	209TI+4d	2014-300T14:45	Oct27	Mon	4	44	7	D/L start, burn ~6h later. Backup D/L 301T14:44; complexes G/G
---	210	**Apoapse**		2014-318T23:04	Nov14	Fri	22	26	18	Per = 47.8 d, inc = 33.1 deg, r = 73.341 Rs, phase = 88 deg
---	210	SEP = 3.0 deg	Conjunction	2014-320T00:04	Nov16	Sun	23	25	19	Commanding must not be required below 3.0 deg
---	210	SEP = 3.0 deg	Conjunction	2014-324T23:46	Nov20	Thu	28	20	24	Commanding must not be required below 3.0 deg
---	210	**OTM-396**	~Apo	2014-325T19:44	Nov21	Fri	29	19	25	D/L start, burn ~6h later. Backup D/L 326T19:44; complexes C/C
---	210	Ring CRX	Descending	2014-341T06:38	Dec07	Sun	44	4	15	r = 17.701 Rs (1067924 km)
---	210	**OTM-397**	210TI-3d	2014-341T12:30	Dec07	Sun	44	3	16	D/L start, burn ~6h later. Backup D/L 342T12:30; complexes G/G
---	210	Periapse		2014-342T21:09	Dec08	Mon	46	2	1	R = 10.910 Rs, lat = -33 deg, phase = 92 deg
---	210	**210TI (t) [T107]**	**TITAN**	2014-344T22:27	Dec10	Wed	48	32	3	Outbound 980.0 km flyby, v = 5.8 km/s, phase = 56 deg
---	210	Ring CRX	Ascending	2014-344T22:35	Dec10	Wed	0	32	3	r = 20.503 Rs (1236975 km)
---	210	**OTM-398**	210TI+3d	2014-347T18:15	Dec13	Sat	3	29	6	D/L start, burn ~6h later. Backup D/L 348T18:15; complexes C/C
---	211	**Apoapse**		2014-358T19:16	Dec24	Wed	14	18	11	Per = 31.9 d, inc = 28.6 deg, r = 56.152 Rs, phase = 107 deg
---	211	**OTM-399**	~Apo	2014-363T17:31	Dec29	Mon	19	13	16	D/L start, burn ~6h later. Backup D/L 364T17:31; complexes C/C
---	211	**OTM-400**	211TI-3d	2015-008T16:47	Jan08	Thu	29	3	10	D/L start, burn ~6h later. Backup D/L 009T16:47; complexes C/C
---	211	Ring CRX	Descending	2015-009T01:16	Jan09	Fri	29	3	0	r = 10.907 Rs (658026 km)
---	211	Periapse		2015-009T17:55	Jan09	Fri	30	2	1	R = 8.153 Rs, lat = -26 deg, phase = 72 deg
---	211	**211TI (t) [T108]**	**TITAN**	2015-011T19:49	Jan11	Sun	32	32	3	Outbound 970.0 km flyby, v = 5.8 km/s, phase = 46 deg
---	211	Ring CRX	Ascending	2015-011T19:57	Jan11	Sun	0	32	3	r = 20.503 Rs (1236973 km)
---	211	**OTM-401**	211TI+3d	2015-014T16:32	Jan14	Wed	3	29	6	D/L start, burn ~6h later. Backup D/L 015T16:32; complexes C/C
---	212	**Apoapse**		2015-025T18:42	Jan25	Sun	14	18	11	Per = 31.9 d, inc = 19.1 deg, r = 57.411 Rs, phase = 118 deg
---	212	**OTM-402**	~Apo	2015-031T15:34	Jan31	Sat	20	12	17	D/L start, burn ~6h later. Backup D/L 032T15:34; complexes C/C
---	212	**OTM-403**	212TI-3d	2015-040T15:06	Feb09	Mon	29	3	9	D/L start, burn ~6h later. Backup D/L 041T15:06; complexes C/C
---	212	Ring CRX	Descending	2015-041T06:10	Feb10	Tue	29	2	1	r = 8.783 Rs (529893 km)
---	212	212RH (nt)	**RHEA**	2015-041T06:46	Feb10	Tue	29	2	1	Inbound 46894.1 km flyby, v = 5.3 km/s, phase = 108 deg
---	212	Periapse		2015-041T17:17	Feb10	Tue	30	2	1	R = 6.887 Rs, lat = -16 deg, phase = 62 deg
---	212	**212TI (t) [T109]**	**TITAN**	2015-043T17:08	Feb12	Thu	32	32	3	Outbound 1200.0 km flyby, v = 5.8 km/s, phase = 55 deg
---	212	Ring CRX	Ascending	2015-043T17:17	Feb12	Thu	0	32	3	r = 20.517 Rs (1237797 km)
---	212	**OTM-404**	212TI+3d	2015-046T14:36	Feb15	Sun	3	29	6	D/L start, burn ~6h later. Backup D/L 047T14:36; complexes C/C
---	213	**Apoapse**		2015-057T17:15	Feb26	Thu	14	18	11	Per = 31.9 d, inc = 8.5 deg, r = 58.093 Rs, phase = 126 deg
---	213	**OTM-405**	~Apo	2015-063T13:39	Mar04	Wed	20	12	17	D/L start, burn ~6h later. Backup D/L 064T13:39; complexes C/C
---	213	**OTM-406**	213TI-3d	2015-072T13:10	Mar13	Fri	29	3	9	D/L start, burn ~6h later. Backup D/L 073T12:55; complexes C/C
---	213	Ring CRX	Descending	2015-073T07:04	Mar14	Sat	30	2	1	r = 7.721 Rs (465812 km)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	213	213HE (nt)	**HELENE**	2015-073T11:09	Mar14	Sat	30	2	1	Inbound 58312.7 km flyby, v = 5.4 km/s, phase = 80 deg
---	213	213CA (nt)	**CALYPSO**	2015-073T15:27	Mar14	Sat	30	2	1	Inbound 101492.6 km flyby, v = 3.3 km/s, phase = 84 deg
---	213	Periapse		2015-073T15:49	Mar14	Sat	30	2	1	R = 6.201 Rs, lat = -7 deg, phase = 54 deg
---	213	Ring CRX	Ascending	2015-075T14:12	Mar16	Mon	32	0	3	r = 20.393 Rs (1230336 km)
---	213	**213TI (t) [T110]**	**TITAN**	2015-075T14:30	Mar16	Mon	32	52	3	Outbound 2274.5 km flyby, v = 5.7 km/s, phase = 70 deg
---	213	Ring CRX	Descending	2015-076T04:15	Mar17	Tue	1	52	4	r = 24.366 Rs (1469974 km)
---	213	**OTM-407**	213TI+3d	2015-078T12:41	Mar19	Thu	3	49	6	D/L start, burn ~6h later. Backup D/L 079T12:41; complexes C/C
---	214	**Apoapse**		2015-087T16:41	Mar28	Sat	12	40	9	Per = 28.0 d, inc = 0.3 deg, r = 53.437 Rs, phase = 134 deg
---	214	Ring CRX	Ascending	2015-101T11:51	Apr11	Sat	26	26	23	r = 6.329 Rs (381833 km)
---	214	Periapse		2015-101T17:09	Apr11	Sat	26	26	23	R = 5.582 Rs, lat = 0 deg, phase = 45 deg
---	214	214TI (nt)	**TITAN**	2015-101T17:55	Apr11	Sat	26	26	23	Outbound 899009.2 km flyby, v = 8.8 km/s, phase = 132 deg
---	214	214TE (nt)	**TETHYS**	2015-101T17:58	Apr11	Sat	26	26	23	Outbound 52844.6 km flyby, v = 3.5 km/s, phase = 32 deg
---	214	214DI (nt)	**DIONE**	2015-101T23:22	Apr11	Sat	26	26	23	Outbound 110100.8 km flyby, v = 8.0 km/s, phase = 28 deg
---	214	Ring CRX	Descending	2015-104T07:59	Apr14	Tue	29	24	26	r = 25.047 Rs (1511101 km)
---	214	**OTM-408**	~Apo	2015-110T10:29	Apr20	Mon	35	18	32	D/L start, burn ~6h later. Backup D/L 111T10:29; complexes C/C
---	215	**Apoapse**		2015-115T17:46	Apr25	Sat	40	12	5	Per = 28.1 d, inc = 0.3 deg, r = 53.454 Rs, phase = 135 deg
---	215	**OTM-409**	215TI-3d	2015-124T09:30	May04	Mon	49	4	14	D/L start, burn ~6h later. Backup D/L 125T09:30; complexes C/C
---	215	**215TI (t) [T111]**	**TITAN**	2015-127T22:50	May07	Thu	52	40	4	Inbound 2721.5 km flyby, v = 5.7 km/s, phase = 34 deg
---	215	Ring CRX	Ascending	2015-129T09:44	May09	Sat	1	38	5	r = 6.629 Rs (399910 km)
---	215	215AE (nt)	**AEGAEON**	2015-129T16:51	May09	Sat	2	38	5	Inbound 92577.9 km flyby, v = 1.6 km/s, phase = 28 deg
---	215	215DA (nt)	**DAPHNIS**	2015-129T17:24	May09	Sat	2	38	5	Inbound 113743.9 km flyby, v = 0.1 km/s, phase = 43 deg
---	215	215TL (nt)	**TELESTO**	2015-129T17:53	May09	Sat	2	38	5	Inbound 46444.9 km flyby, v = 5.3 km/s, phase = 126 deg
---	215	Periapse		2015-129T18:03	May09	Sat	2	38	5	R = 4.113 Rs, lat = 0 deg, phase = 55 deg
---	215	215PO (nt)	**POLYDEUCES**	2015-130T01:12	May10	Sun	2	38	6	Outbound 33995.1 km flyby, v = 6.6 km/s, phase = 130 deg
---	215	Ring CRX	Descending	2015-130T06:41	May10	Sun	2	38	6	r = 8.550 Rs (515825 km)
---	215	**OTM-410**	215TI+4d	2015-131T09:00	May11	Mon	3	36	7	D/L start, burn ~6h later. Backup D/L 132T09:00; complexes C/C
---	216	**Apoapse**		2015-139T04:31	May19	Tue	11	29	8	Per = 18.9 d, inc = 0.3 deg, r = 41.198 Rs, phase = 125 deg
---	216	Ring CRX	Ascending	2015-148T06:33	May28	Thu	20	20	17	r = 6.641 Rs (400668 km)
---	216	216TI (nt)	**TITAN**	2015-148T10:24	May28	Thu	20	19	17	Inbound 942014.0 km flyby, v = 9.3 km/s, phase = 147 deg
---	216	216TL (nt)	**TELESTO**	2015-148T14:10	May28	Thu	21	19	17	Inbound 44717.9 km flyby, v = 5.3 km/s, phase = 127 deg
---	216	Periapse		2015-148T14:54	May28	Thu	21	19	17	R = 4.125 Rs, lat = 0 deg, phase = 55 deg
---	216	Ring CRX	Descending	2015-149T03:36	May29	Fri	21	19	18	r = 8.583 Rs (517804 km)
---	216	216HY (nt)	**HYPERION**	2015-151T13:36	May31	Sun	24	16	20	Outbound 34306.5 km flyby, v = 4.3 km/s, phase = 70 deg
---	217	**Apoapse**		2015-158T01:49	Jun07	Sun	30	10	27	Per = 18.9 d, inc = 0.3 deg, r = 41.261 Rs, phase = 126 deg
---	217	**OTM-411**	~Apo	2015-159T07:00	Jun08	Mon	31	9	28	D/L start, burn ~6h later. Backup D/L 160T07:00; complexes C/C
---	217	**OTM-412**	217DI-3d	2015-164T06:45	Jun13	Sat	36	4	5	D/L start, burn ~6h later. Backup D/L 165T06:45; complexes C/C
---	217	Ring CRX	Ascending	2015-167T04:24	Jun16	Tue	39	1	3	r = 6.637 Rs (400418 km)
---	217	217PO (nt)	**POLYDEUCES**	2015-167T05:25	Jun16	Tue	39	1	3	Inbound 34739.1 km flyby, v = 6.6 km/s, phase = 121 deg
---	217	217ME (nt)	**METHONE**	2015-167T11:40	Jun16	Tue	40	0	3	Inbound 109536.0 km flyby, v = 5.3 km/s, phase = 26 deg
---	217	217PN (nt)	**PAN**	2015-167T11:56	Jun16	Tue	40	0	3	Inbound 118214.9 km flyby, v = 0.3 km/s, phase = 41 deg
---	217	217AT (nt)	**ATLAS**	2015-167T12:30	Jun16	Tue	40	0	3	Inbound 111230.3 km flyby, v = 0.0 km/s, phase = 50 deg
---	217	Periapse		2015-167T12:44	Jun16	Tue	40	0	3	R = 4.124 Rs, lat = 0 deg, phase = 54 deg
---	217	217JA (nt)	**JANUS**	2015-167T13:38	Jun16	Tue	40	0	3	Outbound 101455.8 km flyby, v = 0.8 km/s, phase = 72 deg
---	217	217TL (nt)	**TELESTO**	2015-167T13:40	Jun16	Tue	40	0	3	Outbound 44143.3 km flyby, v = 5.3 km/s, phase = 127 deg
---	217	217TI (nt)	**TITAN**	2015-167T18:23	Jun16	Tue	40	0	3	Outbound 957876.7 km flyby, v = 8.8 km/s, phase = 97 deg
---	217	**217DI (t) [D4]**	**DIONE**	2015-167T20:12	Jun16	Tue	40	20	4	Outbound 516.2 km flyby, v = 7.3 km/s, phase = 58 deg
---	217	Ring CRX	Descending	2015-168T00:01	Jun17	Wed	0	20	4	r = 7.952 Rs (479748 km)
---	217	**OTM-413**	217DI+4d	2015-171T06:14	Jun20	Sat	3	17	7	D/L start, burn ~6h later. Backup D/L 172T06:14; complexes C/C
---	218	**Apoapse**		2015-176T23:56	Jun25	Thu	9	11	6	Per = 18.9 d, inc = 0.4 deg, r = 41.309 Rs, phase = 126 deg
---	218	**OTM-414**	~Apo	2015-177T05:44	Jun26	Fri	9	11	6	D/L start, burn ~6h later. Backup D/L 178T05:44; complexes C/C
---	218	**OTM-415**	218TI-3d	2015-185T05:13	Jul04	Sat	17	3	8	D/L start, burn ~6h later. Backup D/L 186T05:13; complexes C/C
---	218	Ring CRX	Ascending	2015-186T01:58	Jul05	Sun	18	2	1	r = 7.051 Rs (425415 km)
---	218	Periapse		2015-186T11:14	Jul05	Sun	19	2	1	R = 4.112 Rs, lat = 0 deg, phase = 54 deg
---	218	218PN (nt)	**PAN**	2015-186T11:47	Jul05	Sun	19	2	1	Outbound 115989.5 km flyby, v = 0.3 km/s, phase = 64 deg
---	218	218TL (nt)	**TELESTO**	2015-186T14:53	Jul05	Sun	19	2	1	Outbound 14193.2 km flyby, v = 5.8 km/s, phase = 127 deg
---	218	Ring CRX	Descending	2015-186T22:29	Jul05	Sun	19	1	2	r = 7.937 Rs (478817 km)
---	218	**218TI (t) [T112]**	**TITAN**	2015-188T08:10	Jul07	Tue	20	41	3	Outbound 10952.8 km flyby, v = 5.6 km/s, phase = 78 deg
---	218	**OTM-416**	218TI+3d	2015-191T04:58	Jul10	Fri	3	39	6	D/L start, burn ~6h later. Backup D/L 192T04:43; complexes C/C
---	219	**Apoapse**		2015-197T08:20	Jul16	Thu	9	32	6	Per = 21.8 d, inc = 0.5 deg, r = 45.173 Rs, phase = 122 deg
---	219	219TI (nt)	**TITAN**	2015-206T22:36	Jul25	Sat	19	23	16	Inbound 729282.8 km flyby, v = 9.2 km/s, phase = 31 deg
---	219	Ring CRX	Ascending	2015-207T18:05	Jul26	Sun	19	22	17	r = 8.207 Rs (495116 km)
---	219	219DI (nt)	**DIONE**	2015-208T00:56	Jul27	Mon	20	22	17	Inbound 60494.6 km flyby, v = 5.1 km/s, phase = 120 deg
---	219	Periapse		2015-208T05:49	Jul27	Mon	20	22	17	R = 4.698 Rs, lat = 0 deg, phase = 57 deg
---	219	219EN (nt)	**ENCELADUS**	2015-208T06:18	Jul27	Mon	20	22	17	Outbound 111410.3 km flyby, v = 6.7 km/s, phase = 24 deg
---	219	Ring CRX	Descending	2015-208T19:01	Jul27	Mon	20	21	18	r = 8.815 Rs (531835 km)
---	220	**Apoapse**		2015-219T03:41	Aug07	Fri	31	11	28	Per = 21.8 d, inc = 0.5 deg, r = 45.220 Rs, phase = 123 deg
---	220	**OTM-417**	~Apo	2015-221T02:54	Aug09	Sun	33	9	30	D/L start, burn ~6h later. Backup D/L 222T02:39; complexes C/C
---	220	**OTM-418**	220DI-3d	2015-226T02:23	Aug14	Fri	38	4	5	D/L start, burn ~6h later. Backup D/L 227T02:23; complexes C/C
---	220	Ring CRX	Ascending	2015-229T13:41	Aug17	Mon	41	0	3	r = 8.207 Rs (495102 km)
---	220	**220DI (t) [D5]**	**DIONE**	2015-229T18:33	Aug17	Mon	41	42	4	Inbound 473.4 km flyby, v = 6.4 km/s, phase = 67 deg
---	220	220TE (nt)	**TETHYS**	2015-229T21:56	Aug17	Mon	0	42	4	Inbound 41884.2 km flyby, v = 6.3 km/s, phase = 82 deg
---	220	220EN (nt)	**ENCELADUS**	2015-230T01:17	Aug18	Tue	0	42	4	Inbound 53156.5 km flyby, v = 3.3 km/s, phase = 78 deg
---	220	Periapse		2015-230T01:24	Aug18	Tue	0	42	4	R = 4.704 Rs, lat = 0 deg, phase = 57 deg
---	220	220ME (nt)	**METHONE**	2015-230T01:35	Aug18	Tue	0	42	4	Outbound 89815.5 km flyby, v = 1.6 km/s, phase = 61 deg
---	220	220TI (nt)	**TITAN**	2015-230T01:53	Aug18	Tue	0	42	4	Outbound 960414.9 km flyby, v = 10.1 km/s, phase = 120 deg
---	220	220HE (nt)	**HELENE**	2015-230T08:33	Aug18	Tue	1	42	4	Outbound 113514.7 km flyby, v = 9.4 km/s, phase = 46 deg
---	220	Ring CRX	Descending	2015-230T12:44	Aug18	Tue	1	41	4	r = 8.041 Rs (485099 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	220	**OTM-419**	220DI+4d	2015-233T01:52	Aug21	Fri	3	39	7	D/L start, burn ~6h later. Backup D/L 234T01:52; complexes C/C
---	221	**Apoapse**		2015-240T23:29	Aug28	Fri	11	31	8	Per = 21.9 d, inc = 0.4 deg, r = 45.265 Rs, phase = 123 deg
---	221	**OTM-420**	~Peri	2015-251T00:50	Sep08	Tue	21	21	18	D/L start, burn ~6h later. Backup D/L 252T00:50; complexes C/C
---	221	Ring CRX	Ascending	2015-251T08:03	Sep08	Tue	22	21	0	r = 8.998 Rs (542864 km)
---	221	221DI (nt)	**DIONE**	2015-251T14:33	Sep08	Tue	22	20	1	Inbound 41939.8 km flyby, v = 7.6 km/s, phase = 67 deg
---	221	Periapse		2015-251T21:41	Sep08	Tue	22	20	1	R = 4.696 Rs, lat = 0 deg, phase = 56 deg
---	221	221HE (nt)	**HELENE**	2015-252T04:53	Sep09	Wed	22	20	1	Outbound 82956.9 km flyby, v = 8.6 km/s, phase = 45 deg
---	221	Ring CRX	Descending	2015-252T09:03	Sep09	Wed	23	20	1	r = 8.049 Rs (485601 km)
---	221	221TI (nt)	**TITAN**	2015-253T06:21	Sep10	Thu	23	19	2	Outbound 706954.8 km flyby, v = 9.1 km/s, phase = 89 deg
---	222	**Apoapse**		2015-262T19:28	Sep19	Sat	33	9	12	Per = 21.8 d, inc = 0.4 deg, r = 45.211 Rs, phase = 124 deg
---	222	**OTM-421**	222TI-4d	2015-267T23:48	Sep24	Thu	38	4	17	D/L start, burn ~6h later. Backup D/L 268T23:33; complexes C/C
---	222	**222TI (t) [T113]**	**TITAN**	2015-271T21:37	Sep28	Mon	42	16	4	Inbound 1035.7 km flyby, v = 5.8 km/s, phase = 42 deg
---	222	Ring CRX	Ascending	2015-272T22:39	Sep29	Tue	1	15	5	r = 11.038 Rs (665939 km)
---	222	222DI (nt)	**DIONE**	2015-273T08:41	Sep30	Wed	1	14	5	Inbound 40811.5 km flyby, v = 10.1 km/s, phase = 63 deg
---	222	222CA (nt)	**CALYPSO**	2015-273T11:04	Sep30	Wed	2	14	5	Inbound 43869.4 km flyby, v = 10.8 km/s, phase = 72 deg
---	222	222MI (nt)	**MIMAS**	2015-273T14:13	Sep30	Wed	2	14	6	Inbound 64777.8 km flyby, v = 11.8 km/s, phase = 93 deg
---	222	Periapse		2015-273T16:17	Sep30	Wed	2	14	6	R = 2.899 Rs, lat = 1 deg, phase = 69 deg
---	222	222TE (nt)	**TETHYS**	2015-273T16:59	Sep30	Wed	2	14	6	Outbound 117351.1 km flyby, v = 8.5 km/s, phase = 106 deg
---	222	222AE (nt)	**AEGAEON**	2015-273T17:35	Sep30	Wed	2	14	6	Outbound 79840.3 km flyby, v = 12.1 km/s, phase = 40 deg
---	222	Pallene	**DUST HAZARD**	2015-273T18:43	Sep30	Wed	2	14	6	Egress at 2015-273T18:54 (0.0 min); protection = MEA cover closure.
---	222	Ring CRX	Descending	2015-273T18:48	Sep30	Wed	2	14	6	r = 3.514 Rs (211976 km)
---	222	E_ring_lg	**DUST HAZARD**	2015-273T18:53	Sep30	Wed	2	14	6	Egress at 2015-273T20:20 (0.1 min); protection = MEA cover closure.
---	222	222TL (nt)	**TELESTO**	2015-273T21:15	Sep30	Wed	2	14	6	Outbound 118009.2 km flyby, v = 13.7 km/s, phase = 68 deg
---	222	**OTM-423**	222TI+2d	2015-273T23:32	Sep30	Wed	2	13	6	D/L start, burn ~6h later. Backup D/L 274T23:17; complexes C/C
---	222	222RH (nt)	**RHEA**	2015-274T04:26	Oct01	Thu	2	13	0	Outbound 57919.0 km flyby, v = 9.3 km/s, phase = 85 deg
---	222	**OTM-424**	~Apo	2015-278T23:02	Oct05	Mon	7	8	5	D/L start, burn ~6h later. Backup D/L 279T23:02; complexes C/C
---	223	**Apoapse**		2015-280T15:12	Oct07	Wed	9	7	2	Per = 13.9 d, inc = 0.6 deg, r = 34.070 Rs, phase = 111 deg
---	223	**OTM-425**	223EN-3d	2015-283T22:46	Oct10	Sat	12	3	5	D/L start, burn ~6h later. Backup D/L 284T22:46; complexes C/C
---	223	223TI (nt)	**TITAN**	2015-286T12:43	Oct13	Tue	15	1	3	Inbound 826233.3 km flyby, v = 10.1 km/s, phase = 31 deg
---	223	Ring CRX	Ascending	2015-286T23:53	Oct13	Tue	15	0	3	r = 9.504 Rs (573405 km)
---	223	223PO (nt)	**POLYDEUCES**	2015-287T06:50	Oct14	Wed	15	0	3	Inbound 115888.4 km flyby, v = 12.3 km/s, phase = 61 deg
---	223	**223EN (t) [E20]**	**ENCELADUS**	2015-287T10:42	Oct14	Wed	16	14	3	Inbound 1838.8 km flyby, v = 8.5 km/s, phase = 63 deg
---	223	223ME (nt)	**METHONE**	2015-287T11:54	Oct14	Wed	0	14	4	Inbound 39921.8 km flyby, v = 9.9 km/s, phase = 89 deg
---	223	223PM (nt)	**PROMETHEUS**	2015-287T13:44	Oct14	Wed	0	14	4	Inbound 118633.1 km flyby, v = 12.4 km/s, phase = 25 deg
---	223	Periapse		2015-287T14:06	Oct14	Wed	0	14	4	R = 2.902 Rs, lat = 1 deg, phase = 69 deg
---	223	Pallene	**DUST HAZARD**	2015-287T16:31	Oct14	Wed	0	14	4	Egress at 2015-287T16:43 (0.0 min); protection = MEA cover closure.
---	223	E_ring_lg	**DUST HAZARD**	2015-287T16:41	Oct14	Wed	0	14	4	Egress at 2015-287T18:18 (0.1 min); protection = MEA cover closure.
---	223	Ring CRX	Descending	2015-287T17:02	Oct14	Wed	0	14	4	r = 3.700 Rs (223235 km)
---	223	223HE (nt)	**HELENE**	2015-287T20:59	Oct14	Wed	0	14	4	Outbound 117103.2 km flyby, v = 6.0 km/s, phase = 102 deg
---	223	**OTM-426**	~Apo	2015-292T22:15	Oct19	Mon	5	9	9	D/L start, burn ~6h later. Backup D/L 293T22:00; complexes C/C
---	224	**Apoapse**		2015-294T13:01	Oct21	Wed	7	7	2	Per = 13.9 d, inc = 0.6 deg, r = 34.070 Rs, phase = 111 deg
---	224	**OTM-427**	224EN-3d	2015-297T22:00	Oct24	Sat	10	4	5	D/L start, burn ~6h later. Backup D/L 298T21:45; complexes C/C
---	224	Ring CRX	Ascending	2015-300T21:22	Oct27	Tue	13	1	3	r = 9.662 Rs (582930 km)
---	224	224PO (nt)	**POLYDEUCES**	2015-301T04:12	Oct28	Wed	14	0	3	Inbound 84455.9 km flyby, v = 7.0 km/s, phase = 116 deg
---	224	224TL (nt)	**TELESTO**	2015-301T07:18	Oct28	Wed	14	0	3	Inbound 59688.5 km flyby, v = 6.9 km/s, phase = 113 deg
---	224	224DA (nt)	**DAPHNIS**	2015-301T11:25	Oct28	Wed	14	0	4	Inbound 73293.8 km flyby, v = 6.3 km/s, phase = 26 deg
---	224	Periapse		2015-301T11:56	Oct28	Wed	14	0	4	R = 2.902 Rs, lat = 1 deg, phase = 69 deg
---	224	224PN (nt)	**PAN**	2015-301T12:32	Oct28	Wed	14	0	4	Outbound 64881.5 km flyby, v = 4.7 km/s, phase = 116 deg
---	224	Pallene	**DUST HAZARD**	2015-301T14:21	Oct28	Wed	14	0	4	Egress at 2015-301T14:32 (0.0 min); protection = MEA cover closure.
---	224	E_ring_lg	**DUST HAZARD**	2015-301T14:31	Oct28	Wed	14	0	4	Egress at 2015-301T16:08 (0.1 min); protection = MEA cover closure.
---	224	Ring CRX	Descending	2015-301T14:50	Oct28	Wed	14	0	4	r = 3.685 Rs (222293 km)
---	224	**Earth OCC**	ENCELADUS	2015-301T15:23	Oct28	Wed	0	16	4	Egress = 2015-301T15:23 (1 min)
---	224	**Sun OCC**	ENCELADUS	2015-301T15:23	Oct28	Wed	0	16	4	Egress = 2015-301T15:23 (1 min)
---	224	**224EN (t) [E21]**	**ENCELADUS**	2015-301T15:23	Oct28	Wed	14	16	4	Outbound 49.0 km flyby, v = 8.5 km/s, phase = 118 deg
---	224	224HE (nt)	**HELENE**	2015-301T19:26	Oct28	Wed	0	15	4	Outbound 78771.6 km flyby, v = 11.1 km/s, phase = 77 deg
---	224	**OTM-428**	224EN+1d	2015-301T21:45	Oct28	Wed	0	15	4	D/L start, burn ~6h later. Backup D/L 302T21:45; complexes C/C
---	224	224TI (nt)	**TITAN**	2015-302T14:05	Oct29	Thu	1	15	1	Outbound 800182.2 km flyby, v = 10.0 km/s, phase = 112 deg
---	225	**Apoapse**		2015-308T10:43	Nov04	Wed	7	9	7	Per = 13.9 d, inc = 0.6 deg, r = 34.052 Rs, phase = 111 deg
---	225	**OTM-429**	~Apo	2015-308T21:14	Nov04	Wed	7	8	7	D/L start, burn ~6h later. Backup D/L 309T21:14; complexes C/C
---	225	**OTM-430**	225TI-3d	2015-313T20:59	Nov09	Mon	12	3	5	D/L start, burn ~6h later. Backup D/L 314T20:59; complexes C/C
---	225	Ring CRX	Ascending	2015-314T18:33	Nov10	Tue	13	2	1	r = 9.836 Rs (593424 km)
---	225	225CA (nt)	**CALYPSO**	2015-315T07:21	Nov11	Wed	14	2	1	Inbound 104487.7 km flyby, v = 7.5 km/s, phase = 122 deg
---	225	225PN (nt)	**PAN**	2015-315T07:22	Nov11	Wed	14	2	1	Inbound 111591.2 km flyby, v = 3.6 km/s, phase = 29 deg
---	225	Periapse		2015-315T09:30	Nov11	Wed	14	2	2	R = 2.898 Rs, lat = 0 deg, phase = 69 deg
---	225	225PA (nt)	**PANDORA**	2015-315T09:31	Nov11	Wed	14	2	2	Outbound 105771.8 km flyby, v = 12.1 km/s, phase = 118 deg
---	225	Pallene	**DUST HAZARD**	2015-315T11:56	Nov11	Wed	14	2	2	Egress at 2015-315T12:07 (0.0 min); protection = MEA cover closure.
---	225	E_ring_lg	**DUST HAZARD**	2015-315T12:05	Nov11	Wed	14	2	2	Egress at 2015-315T13:42 (0.1 min); protection = MEA cover closure.
---	225	Ring CRX	Descending	2015-315T12:20	Nov11	Wed	14	2	2	r = 3.653 Rs (220378 km)
---	225	225TE (nt)	**TETHYS**	2015-315T14:42	Nov11	Wed	14	2	2	Outbound 8363.4 km flyby, v = 8.8 km/s, phase = 100 deg
---	225	**225TI (t) [T114]**	**TITAN**	2015-317T05:47	Nov13	Fri	16	37	3	Outbound 11919.7 km flyby, v = 5.5 km/s, phase = 92 deg
---	225	**OTM-431**	225TI+3d	2015-319T20:29	Nov15	Sun	3	34	6	D/L start, burn ~6h later. Backup D/L 320T20:29; complexes C/C
---	225	Ring CRX	Ascending	2015-321T11:30	Nov17	Tue	4	32	2	r = 32.290 Rs (1948075 km)
---	226	**Apoapse**		2015-321T18:23	Nov17	Tue	5	32	2	Per = 12.7 d, inc = 1.3 deg, r = 32.334 Rs, phase = 114 deg
---	226	226TE (nt)	**TETHYS**	2015-327T21:57	Nov23	Mon	11	26	8	Inbound 17505.9 km flyby, v = 9.2 km/s, phase = 113 deg
---	226	226AT (nt)	**ATLAS**	2015-328T02:33	Nov24	Tue	11	26	8	Inbound 105213.8 km flyby, v = 16.6 km/s, phase = 102 deg
---	226	226DA (nt)	**DAPHNIS**	2015-328T02:35	Nov24	Tue	11	26	8	Inbound 77507.9 km flyby, v = 13.1 km/s, phase = 104 deg
---	226	JE_ring	**DUST HAZARD**	2015-328T02:38	Nov24	Tue	11	26	8	Egress at 2015-328T03:46 (0.0 min); protection = HGA to ram and MEA cover closure.
---	226	Ring CRX	Descending	2015-328T03:11	Nov24	Tue	11	26	8	r = 2.532 Rs (152756 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	226	Periapse		2015-328T03:14	Nov24	Tue	11	26	8	R = 2.532 Rs, lat = -0 deg, phase = 66 deg
---	226	226ME (nt)	**METHONE**	2015-328T05:46	Nov24	Tue	11	26	8	Outbound 37725.5 km flyby, v = 11.6 km/s, phase = 56 deg
---	226	**OTM-432**	~Apo	2015-329T20:13	Nov25	Wed	13	24	10	D/L start, burn ~6h later. Backup D/L 330T19:58; complexes C/C
---	226	SEP = 3.0 deg	Conjunction	2015-331T06:14	Nov27	Fri	14	22	1	Commanding must not be required below 3.0 deg
---	226	SEP = 2.0 deg	Conjunction	2015-332T19:43	Nov28	Sat	16	21	3	Plan no SSR playback below 2.0 deg
---	226	Ring CRX	Ascending	2015-334T00:32	Nov30	Mon	17	20	4	r = 32.198 Rs (1942495 km)
---	226	226TI (nt)	**TITAN**	2015-334T05:57	Nov30	Mon	17	19	4	Outbound 753131.1 km flyby, v = 4.0 km/s, phase = 113 deg
---	227	**Apoapse**		2015-334T11:53	Nov30	Mon	17	19	5	Per = 12.7 d, inc = 1.3 deg, r = 32.316 Rs, phase = 114 deg
---	227	SEP = 2.0 deg	Conjunction	2015-335T08:05	Dec01	Tue	18	18	5	Plan no SSR playback below 2.0 deg
---	227	SEP = 3.0 deg	Conjunction	2015-336T21:06	Dec02	Wed	20	17	7	Commanding must not be required below 3.0 deg
---	227	JE_ring	**DUST HAZARD**	2015-340T19:54	Dec06	Sun	24	13	11	Egress at 2015-340T21:02 (0.0 min); protection = HGA to ram and MEA cover closure.
---	227	Ring CRX	Descending	2015-340T20:26	Dec06	Sun	24	13	11	r = 2.531 Rs (152695 km)
---	227	Periapse		2015-340T20:31	Dec06	Sun	24	13	11	R = 2.530 Rs, lat = -0 deg, phase = 66 deg
---	227	227EP (nt)	**EPIMETHEUS**	2015-340T20:38	Dec06	Sun	24	13	11	Outbound 2616.0 km flyby, v = 5.5 km/s, phase = 65 deg
---	227	227AT (nt)	**ATLAS**	2015-340T20:55	Dec06	Sun	24	13	11	Outbound 20856.3 km flyby, v = 5.8 km/s, phase = 47 deg
---	227	227PM (nt)	**PROMETHEUS**	2015-340T21:02	Dec06	Sun	24	13	11	Outbound 20984.8 km flyby, v = 6.3 km/s, phase = 46 deg
---	227	Ring CRX	Ascending	2015-346T13:36	Dec12	Sat	29	7	17	r = 32.107 Rs (1936991 km)
---	228	**Apoapse**		2015-347T05:23	Dec13	Sun	30	7	17	Per = 12.7 d, inc = 1.3 deg, r = 32.335 Rs, phase = 114 deg
---	228	**OTM-433**	228EN-4d	2015-349T18:44	Dec15	Tue	33	4	20	D/L start, burn ~6h later. Backup D/L 350T18:44; complexes C/C
---	228	228TI (nt)	**TITAN**	2015-351T13:23	Dec17	Thu	34	2	2	Inbound 148671.3 km flyby, v = 4.7 km/s, phase = 130 deg
---	228	228AE (nt)	**AEGAEON**	2015-353T12:59	Dec19	Sat	36	0	4	Inbound 2556.8 km flyby, v = 6.9 km/s, phase = 84 deg
---	228	JE_ring	**DUST HAZARD**	2015-353T13:47	Dec19	Sat	36	0	4	Egress at 2015-353T14:31 (0.0 min); protection = MEA cover closure.
---	228	Ring CRX	Descending	2015-353T14:00	Dec19	Sat	36	0	4	r = 2.569 Rs (155018 km)
---	228	Periapse		2015-353T14:09	Dec19	Sat	36	0	4	R = 2.566 Rs, lat = -0 deg, phase = 66 deg
---	228	**228EN (t) [E22]**	**ENCELADUS**	2015-353T17:49	Dec19	Sat	37	27	4	Outbound 4999.0 km flyby, v = 9.5 km/s, phase = 114 deg
---	228	228CA (nt)	**CALYPSO**	2015-353T19:11	Dec19	Sat	0	27	4	Outbound 91778.1 km flyby, v = 13.4 km/s, phase = 69 deg
---	228	**OTM-434**	228EN+3d	2015-356T18:29	Dec22	Tue	3	24	7	D/L start, burn ~6h later. Backup D/L 357T18:29; complexes C/C
---	228	Ring CRX	Ascending	2015-358T21:55	Dec24	Thu	5	22	2	r = 31.871 Rs (1922768 km)
---	229	**Apoapse**		2015-360T00:24	Dec26	Sat	6	21	3	Per = 12.9 d, inc = 1.3 deg, r = 32.509 Rs, phase = 114 deg
---	229	**OTM-435**	~Peri	2015-363T18:00	Dec29	Tue	10	17	7	D/L start, burn ~6h later. Backup D/L 364T18:00; complexes C/C
---	229	229RH (nt)	**RHEA**	2015-365T22:31	Dec31	Thu	12	15	2	Inbound 24473.2 km flyby, v = 9.0 km/s, phase = 42 deg
---	229	229AE (nt)	**AEGAEON**	2016-001T08:59	Jan01	Fri	13	15	3	Inbound 93595.8 km flyby, v = 15.7 km/s, phase = 88 deg
---	229	229PN (nt)	**PAN**	2016-001T10:14	Jan01	Fri	13	15	3	Inbound 97034.4 km flyby, v = 14.7 km/s, phase = 107 deg
---	229	JE_ring	**DUST HAZARD**	2016-001T10:19	Jan01	Fri	13	15	3	Egress at 2016-001T10:59 (0.0 min); protection = MEA cover closure.
---	229	Ring CRX	Descending	2016-001T10:28	Jan01	Fri	13	15	3	r = 2.574 Rs (155312 km)
---	229	Periapse		2016-001T10:39	Jan01	Fri	13	15	3	R = 2.569 Rs, lat = -0 deg, phase = 66 deg
---	229	229PA (nt)	**PANDORA**	2016-001T11:27	Jan01	Fri	13	15	3	Outbound 109754.0 km flyby, v = 16.8 km/s, phase = 36 deg
---	229	229ME (nt)	**METHONE**	2016-001T12:10	Jan01	Fri	13	15	3	Outbound 27743.8 km flyby, v = 7.1 km/s, phase = 120 deg
---	229	229TE (nt)	**TETHYS**	2016-001T15:52	Jan01	Fri	13	14	3	Outbound 38843.0 km flyby, v = 11.3 km/s, phase = 68 deg
---	229	Ring CRX	Ascending	2016-006T13:14	Jan06	Wed	18	10	8	r = 31.585 Rs (1905549 km)
---	230	**Apoapse**		2016-007T20:49	Jan07	Thu	19	8	9	Per = 12.8 d, inc = 1.3 deg, r = 32.495 Rs, phase = 114 deg
---	230	**OTM-436**	230TI-3d	2016-012T17:16	Jan12	Tue	24	3	14	D/L start, burn ~6h later. Backup D/L 013T17:16; complexes C/C
---	230	230CA (nt)	**CALYPSO**	2016-014T02:03	Jan14	Thu	25	2	1	Inbound 115012.6 km flyby, v = 14.3 km/s, phase = 61 deg
---	230	230EN (nt)	**ENCELADUS**	2016-014T03:20	Jan14	Thu	25	2	1	Inbound 72194.9 km flyby, v = 13.4 km/s, phase = 70 deg
---	230	230MI (nt)	**MIMAS**	2016-014T06:07	Jan14	Thu	26	2	2	Inbound 27553.0 km flyby, v = 7.0 km/s, phase = 105 deg
---	230	230AT (nt)	**ATLAS**	2016-014T06:20	Jan14	Thu	26	2	2	Inbound 94471.6 km flyby, v = 14.9 km/s, phase = 103 deg
---	230	JE_ring	**DUST HAZARD**	2016-014T06:36	Jan14	Thu	26	2	2	Egress at 2016-014T07:20 (0.0 min); protection = MEA cover closure.
---	230	Ring CRX	Descending	2016-014T06:45	Jan14	Thu	26	2	2	r = 2.572 Rs (155196 km)
---	230	Periapse		2016-014T06:58	Jan14	Thu	26	2	2	R = 2.566 Rs, lat = -0 deg, phase = 66 deg
---	230	230DA (nt)	**DAPHNIS**	2016-014T07:15	Jan14	Thu	26	2	2	Outbound 23059.5 km flyby, v = 5.4 km/s, phase = 47 deg
---	230	230TL (nt)	**TELESTO**	2016-014T12:15	Jan14	Thu	26	2	2	Outbound 14052.5 km flyby, v = 9.3 km/s, phase = 115 deg
---	230	**230TI (t) [T115]**	**TITAN**	2016-016T02:21	Jan16	Sat	27	16	3	Outbound 3817.5 km flyby, v = 5.7 km/s, phase = 99 deg
---	230	Ring CRX	Ascending	2016-017T04:45	Jan17	Sun	1	15	4	r = 26.150 Rs (1577610 km)
---	230	**OTM-437**	230TI+3d	2016-018T17:01	Jan18	Mon	3	13	6	D/L start, burn ~6h later. Backup D/L 019T17:01; complexes C/C
---	231	**Apoapse**		2016-022T05:38	Jan22	Fri	6	10	4	Per = 15.9 d, inc = 2.4 deg, r = 37.003 Rs, phase = 107 deg
---	231	**OTM-438**	~Apo	2016-023T16:47	Jan23	Sat	8	8	5	D/L start, burn ~6h later. Backup D/L 024T16:32; complexes C/C
---	231	**OTM-439**	231TI-3d	2016-028T16:17	Jan28	Thu	13	3	5	D/L start, burn ~6h later. Backup D/L 029T16:17; complexes C/C
---	231	E_ring_lg	Dust Crossing	2016-030T03:23	Jan30	Sat	14	2	1	Egress at 2016-030T04:02 (0.0 min); protection = none.
---	231	Ring CRX	Descending	2016-030T03:43	Jan30	Sat	14	2	1	r = 3.597 Rs (216995 km)
---	231	Pallene	Dust Crossing	2016-030T03:57	Jan30	Sat	14	2	1	Egress at 2016-030T03:57 (0.0 min); protection = none.
---	231	Periapse		2016-030T04:56	Jan30	Sat	14	2	2	R = 3.487 Rs, lat = -1 deg, phase = 73 deg
---	231	231PN (nt)	**PAN**	2016-030T07:21	Jan30	Sat	14	2	2	Outbound 116655.3 km flyby, v = 0.6 km/s, phase = 129 deg
---	231	231PO (nt)	**POLYDEUCES**	2016-030T11:12	Jan30	Sat	14	2	2	Outbound 104755.6 km flyby, v = 5.9 km/s, phase = 105 deg
---	231	**231TI (t) [T116]**	**TITAN**	2016-032T01:01	Feb01	Mon	16	16	3	Outbound 1400.0 km flyby, v = 5.8 km/s, phase = 120 deg
---	231	**Earth OCC**	TITAN	2016-032T01:05	Feb01	Mon	0	16	3	Egress = 2016-032T01:33 (28 min)
---	231	**Sun OCC**	TITAN	2016-032T01:05	Feb01	Mon	0	16	3	Egress = 2016-032T01:31 (26 min)
---	231	Ring CRX	Ascending	2016-032T05:03	Feb01	Mon	0	16	4	r = 21.032 Rs (1268861 km)
---	231	**OTM-440**	231TI+3d	2016-034T16:03	Feb03	Wed	3	13	6	D/L start, burn ~6h later. Backup D/L 035T16:03; complexes C/C
---	232	**Apoapse**		2016-038T03:24	Feb07	Sun	6	10	3	Per = 16.0 d, inc = 16.0 deg, r = 36.724 Rs, phase = 103 deg
---	232	**OTM-441**	~Apo	2016-039T15:49	Feb08	Mon	8	8	5	D/L start, burn ~6h later. Backup D/L 040T15:34; complexes C/C
---	232	**OTM-442**	232TI-3d	2016-044T15:34	Feb13	Sat	13	3	5	D/L start, burn ~6h later. Backup D/L 045T15:19; complexes C/C
---	232	232PO (nt)	**POLYDEUCES**	2016-045T20:17	Feb14	Sun	14	2	1	Inbound 77361.3 km flyby, v = 7.8 km/s, phase = 96 deg
---	232	232TL (nt)	**TELESTO**	2016-046T00:15	Feb15	Mon	14	2	1	Inbound 48861.2 km flyby, v = 6.7 km/s, phase = 103 deg
---	232	232EP (nt)	**EPIMETHEUS**	2016-046T00:25	Feb15	Mon	14	2	1	Inbound 116051.9 km flyby, v = 4.5 km/s, phase = 26 deg
---	232	E_ring_lg	Dust Crossing	2016-046T00:34	Feb15	Mon	14	2	1	Egress at 2016-046T00:41 (0.0 min); protection = none.
---	232	Ring CRX	Descending	2016-046T00:38	Feb15	Mon	14	2	1	r = 4.122 Rs (248666 km)
---	232	Periapse		2016-046T02:57	Feb15	Mon	14	2	1	R = 3.801 Rs, lat = -9 deg, phase = 77 deg
---	232	232EN (nt)	**ENCELADUS**	2016-046T05:01	Feb15	Mon	14	2	2	Outbound 83953.8 km flyby, v = 8.0 km/s, phase = 74 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	232	**232TI (t) [T117]**	**TITAN**	2016-047T23:52	Feb16	Tue	16	48	3	Outbound 1018.0 km flyby, v = 5.9 km/s, phase = 122 deg
---	232	**Sun OCC**	TITAN	2016-047T23:54	Feb16	Tue	0	48	3	Egress = 2016-048T00:10 (17 min)
---	232	**Earth OCC**	TITAN	2016-047T23:55	Feb16	Tue	0	48	3	Egress = 2016-048T00:08 (13 min)
---	232	Ring CRX	Ascending	2016-048T01:58	Feb17	Wed	0	48	3	r = 20.528 Rs (1238466 km)
---	232	**OTM-443**	232TI+3d	2016-050T15:05	Feb19	Fri	3	45	6	D/L start, burn ~6h later. Backup D/L 051T15:05; complexes C/C
---	233	**Apoapse**		2016-058T00:09	Feb27	Sat	10	38	7	Per = 23.9 d, inc = 20.6 deg, r = 47.101 Rs, phase = 87 deg
---	233	233TI (nt)	**TITAN**	2016-068T08:01	Mar08	Tue	20	27	18	Inbound 425867.1 km flyby, v = 6.1 km/s, phase = 76 deg
---	233	Ring CRX	Descending	2016-069T15:35	Mar09	Wed	22	26	19	r = 7.126 Rs (429887 km)
---	233	Periapse		2016-069T23:00	Mar09	Wed	22	26	19	R = 5.949 Rs, lat = -16 deg, phase = 93 deg
---	233	Ring CRX	Ascending	2016-071T23:09	Mar11	Fri	24	24	21	r = 20.380 Rs (1229527 km)
---	234	**Apoapse**		2016-081T22:04	Mar21	Mon	34	14	31	Per = 23.9 d, inc = 20.6 deg, r = 47.133 Rs, phase = 87 deg
---	234	**OTM-444**	~Apo	2016-085T12:55	Mar25	Fri	38	10	35	D/L start, burn ~6h later. Backup D/L 086T12:55; complexes C/C
---	234	**OTM-445**	234TI-3d	2016-092T12:26	Apr01	Fri	45	3	7	D/L start, burn ~6h later. Backup D/L 093T12:26; complexes C/C
---	234	Ring CRX	Descending	2016-093T13:49	Apr02	Sat	46	2	1	r = 7.139 Rs (430714 km)
---	234	Periapse		2016-093T21:13	Apr02	Sat	46	2	1	R = 5.978 Rs, lat = -16 deg, phase = 92 deg
---	234	**Earth OCC**	TITAN	2016-095T19:47	Apr04	Mon	48	0	3	Egress = 2016-095T20:07 (21 min)
---	234	**Sun OCC**	TITAN	2016-095T19:47	Apr04	Mon	48	0	3	Egress = 2016-095T20:07 (21 min)
---	234	**234TI (t) [T118]**	**TITAN**	2016-095T19:48	Apr04	Mon	48	32	3	Outbound 990.0 km flyby, v = 5.9 km/s, phase = 138 deg
---	234	Ring CRX	Ascending	2016-095T21:17	Apr04	Mon	0	32	3	r = 20.363 Rs (1228519 km)
---	234	**OTM-446**	234TI+3d	2016-098T12:11	Apr07	Thu	3	29	6	D/L start, burn ~6h later. Backup D/L 099T11:11; complexes C/C
---	235	**Apoapse**		2016-109T18:23	Apr18	Mon	14	18	11	Per = 31.9 d, inc = 27.8 deg, r = 56.154 Rs, phase = 73 deg
---	235	**OTM-447**	~Apo	2016-113T11:13	Apr22	Fri	18	14	15	D/L start, burn ~6h later. Backup D/L 114T11:13; complexes C/C
---	235	**OTM-448**	235TI-3d	2016-124T10:29	May03	Tue	29	3	11	D/L start, burn ~6h later. Backup D/L 125T10:29; complexes C/C
---	235	Ring CRX	Descending	2016-125T00:14	May04	Wed	29	3	1	r = 10.917 Rs (658595 km)
---	235	Periapse		2016-125T16:56	May04	Wed	30	2	1	R = 8.142 Rs, lat = -25 deg, phase = 107 deg
---	235	**Earth OCC**	TITAN	2016-127T16:59	May06	Fri	32	0	3	Egress = 2016-127T17:15 (17 min)
---	235	**Sun OCC**	TITAN	2016-127T16:59	May06	Fri	32	0	3	Egress = 2016-127T17:14 (16 min)
---	235	**235TI (t) [T119]**	**TITAN**	2016-127T17:01	May06	Fri	32	32	3	Outbound 971.0 km flyby, v = 5.9 km/s, phase = 138 deg
---	235	Ring CRX	Ascending	2016-127T18:06	May06	Fri	0	32	3	r = 20.247 Rs (1221484 km)
---	235	**OTM-449**	235TI+3d	2016-130T09:59	May09	Mon	3	29	6	D/L start, burn ~6h later. Backup D/L 131T09:59; complexes C/C
---	236	**Apoapse**		2016-141T13:32	May20	Fri	14	18	11	Per = 31.9 d, inc = 35.3 deg, r = 54.493 Rs, phase = 64 deg
---	236	**OTM-450**	~Apo	2016-143T09:00	May22	Sun	16	16	13	D/L start, burn ~6h later. Backup D/L 144T09:00; complexes C/C
---	236	**OTM-451**	236TI-3d	2016-156T08:15	Jun04	Sat	29	3	13	D/L start, burn ~6h later. Backup D/L 157T08:15; complexes C/C
---	236	Ring CRX	Descending	2016-156T09:16	Jun04	Sat	29	3	0	r = 14.083 Rs (849638 km)
---	236	Periapse		2016-157T12:06	Jun05	Sun	30	2	1	R = 9.798 Rs, lat = -34 deg, phase = 115 deg
---	236	**Earth OCC**	RING	2016-158T05:05	Jun06	Mon	31	1	2	Egress = 2016-158T09:57 (292 min)
---	236	**Sun OCC**	RING	2016-158T05:11	Jun06	Mon	31	1	2	Egress = 2016-158T09:46 (275 min)
---	236	**236TI (t) [T120]**	**TITAN**	2016-159T14:13	Jun07	Tue	32	48	3	Outbound 975.0 km flyby, v = 5.9 km/s, phase = 124 deg
---	236	**Earth OCC**	TITAN	2016-159T14:15	Jun07	Tue	0	48	3	Egress = 2016-159T14:17 (2 min)
---	236	**Sun OCC**	TITAN	2016-159T14:15	Jun07	Tue	0	48	3	Egress = 2016-159T14:17 (3 min)
---	236	Ring CRX	Ascending	2016-159T15:03	Jun07	Tue	0	48	3	r = 20.167 Rs (1216673 km)
---	236	**OTM-452**	236TI+4d	2016-163T07:45	Jun11	Sat	4	44	7	D/L start, burn ~6h later. Backup D/L 164T07:45; complexes C/C
---	237	**Apoapse**		2016-169T06:13	Jun17	Fri	10	38	6	Per = 23.9 d, inc = 42.4 deg, r = 42.590 Rs, phase = 66 deg
---	237	Ring CRX	Descending	2016-180T00:02	Jun28	Tue	20	27	17	r = 14.362 Rs (866452 km)
---	237	Periapse		2016-181T04:56	Jun29	Wed	22	26	18	R = 10.447 Rs, lat = -40 deg, phase = 114 deg
---	237	**Earth OCC**	RING	2016-182T01:42	Jun30	Thu	22	25	19	Egress = 2016-182T08:18 (397 min)
---	237	**Sun OCC**	RING	2016-182T01:59	Jun30	Thu	22	25	19	Egress = 2016-182T08:33 (394 min)
---	237	**Earth OCC**	SATURN	2016-182T02:37	Jun30	Thu	23	25	19	Egress = 2016-182T06:44 (248 min)
---	237	**Sun OCC**	SATURN	2016-182T03:33	Jun30	Thu	23	25	19	Egress = 2016-182T07:42 (250 min)
---	237	Ring CRX	Ascending	2016-183T12:25	Jul01	Fri	24	24	20	r = 20.177 Rs (1217290 km)
---	238	**Apoapse**		2016-193T04:17	Jul11	Mon	34	14	30	Per = 24.0 d, inc = 42.3 deg, r = 42.665 Rs, phase = 67 deg
---	238	**OTM-453**	~Apo	2016-199T05:13	Jul17	Sun	40	8	36	D/L start, burn ~6h later. Backup D/L 200T05:13; complexes C/C
---	238	Ring CRX	Descending	2016-203T22:39	Jul21	Thu	44	3	5	r = 14.371 Rs (867017 km)
---	238	**OTM-454**	238TI-3d	2016-204T04:42	Jul22	Fri	45	3	5	D/L start, burn ~6h later. Backup D/L 205T04:57; complexes C/C
---	238	Periapse		2016-205T03:36	Jul23	Sat	46	2	1	R = 10.439 Rs, lat = -40 deg, phase = 113 deg
---	238	**Earth OCC**	RING	2016-206T00:07	Jul24	Sun	46	1	2	Egress = 2016-206T06:38 (392 min)
---	238	**Sun OCC**	RING	2016-206T00:42	Jul24	Sun	46	1	2	Egress = 2016-206T07:09 (387 min)
---	238	**Earth OCC**	SATURN	2016-206T00:51	Jul24	Sun	46	1	2	Egress = 2016-206T04:46 (235 min)
---	238	**Sun OCC**	SATURN	2016-206T02:28	Jul24	Sun	47	1	2	Egress = 2016-206T06:28 (240 min)
---	238	**238TI (t) [T121]**	**TITAN**	2016-207T10:05	Jul25	Mon	48	16	3	Outbound 976.0 km flyby, v = 5.9 km/s, phase = 101 deg
---	238	Ring CRX	Ascending	2016-207T10:45	Jul25	Mon	0	16	3	r = 20.108 Rs (1213115 km)
---	238	**OTM-455**	238TI+3d	2016-210T04:27	Jul28	Thu	3	13	6	D/L start, burn ~6h later. Backup D/L 211T04:27; complexes C/C
---	239	**Apoapse**		2016-212T20:06	Jul30	Sat	5	11	3	Per = 15.9 d, inc = 48.7 deg, r = 31.208 Rs, phase = 83 deg
---	239	**OTM-456**	~Apo	2016-215T04:11	Aug02	Tue	8	8	5	D/L start, burn ~6h later. Backup D/L 216T03:11; complexes C/C
---	239	Ring CRX	Descending	2016-220T01:58	Aug07	Sun	13	3	5	r = 11.124 Rs (671092 km)
---	239	**OTM-457**	239TI-3d	2016-220T03:40	Aug07	Sun	13	3	5	D/L start, burn ~6h later. Backup D/L 221T03:40; complexes C/C
---	239	Periapse		2016-220T19:27	Aug07	Sun	13	3	1	R = 9.293 Rs, lat = -40 deg, phase = 97 deg
---	239	**Sun OCC**	RING	2016-221T22:17	Aug08	Mon	15	1	2	Egress = 2016-222T04:47 (391 min)
---	239	**Sun OCC**	SATURN	2016-221T22:47	Aug08	Mon	15	1	2	Egress = 2016-222T02:45 (239 min)
---	239	**Earth OCC**	RING	2016-221T23:01	Aug08	Mon	15	1	2	Egress = 2016-222T02:54 (233 min)
---	239	**239TI (t) [T122]**	**TITAN**	2016-223T08:37	Aug10	Wed	16	48	3	Outbound 1599.5 km flyby, v = 5.8 km/s, phase = 92 deg
---	239	Ring CRX	Ascending	2016-223T09:13	Aug10	Wed	0	48	3	r = 20.083 Rs (1211623 km)
---	240	**Apoapse**		2016-226T15:02	Aug13	Sat	3	45	6	Per = 12.0 d, inc = 53.5 deg, r = 25.897 Rs, phase = 102 deg
---	240	**OTM-458**	239TI+4d	2016-227T03:10	Aug14	Sun	4	44	7	D/L start, burn ~6h later. Backup D/L 228T03:09; complexes C/C
---	240	**OTM-459**	~Peri	2016-232T02:54	Aug19	Fri	9	39	5	D/L start, burn ~6h later. Backup D/L 233T02:54; complexes C/C
---	240	Ring CRX	Descending	2016-232T06:10	Aug19	Fri	9	39	0	r = 8.213 Rs (495480 km)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	240	240TI (nt)	**TITAN**	2016-232T06:48	Aug19	Fri	9	39	0	Inbound 826346.9 km flyby, v = 8.5 km/s, phase = 110 deg
---	240	Periapse		2016-232T14:26	Aug19	Fri	9	39	0	R = 7.523 Rs, lat = -31 deg, phase = 78 deg
---	240	**Sun OCC**	RING	2016-233T19:46	Aug20	Sat	10	37	2	Egress = 2016-234T01:23 (338 min)
---	240	Ring CRX	Ascending	2016-235T07:58	Aug22	Mon	12	36	3	r = 20.087 Rs (1211832 km)
---	241	**Apoapse**		2016-238T13:51	Aug25	Thu	15	33	6	Per = 12.0 d, inc = 53.5 deg, r = 25.900 Rs, phase = 102 deg
---	241	241TI (nt)	**TITAN**	2016-243T14:13	Aug30	Tue	20	28	11	Inbound 985746.1 km flyby, v = 8.0 km/s, phase = 91 deg
---	241	Ring CRX	Descending	2016-244T05:00	Aug31	Wed	21	27	12	r = 8.214 Rs (495559 km)
---	241	Periapse		2016-244T13:15	Aug31	Wed	21	27	12	R = 7.524 Rs, lat = -31 deg, phase = 78 deg
---	241	**Sun OCC**	RING	2016-245T18:31	Sep01	Thu	22	25	14	Egress = 2016-246T00:21 (350 min)
---	241	Ring CRX	Ascending	2016-247T06:49	Sep03	Sat	24	24	15	r = 20.094 Rs (1212271 km)
---	242	**Apoapse**		2016-250T12:51	Sep06	Tue	27	21	18	Per = 12.0 d, inc = 53.5 deg, r = 25.920 Rs, phase = 102 deg
---	242	Ring CRX	Descending	2016-256T04:10	Sep12	Mon	33	15	24	r = 8.223 Rs (496068 km)
---	242	Periapse		2016-256T12:27	Sep12	Mon	33	15	24	R = 7.531 Rs, lat = -31 deg, phase = 78 deg
---	242	**Sun OCC**	RING	2016-257T17:41	Sep13	Tue	34	13	26	Egress = 2016-257T23:42 (362 min)
---	242	**Sun OCC**	SATURN	2016-257T18:35	Sep13	Tue	34	13	26	Egress = 2016-257T19:50 (76 min)
---	242	Ring CRX	Ascending	2016-259T06:03	Sep15	Thu	36	12	27	r = 20.097 Rs (1212469 km)
---	243	**Apoapse**		2016-262T12:02	Sep18	Sun	39	9	30	Per = 12.0 d, inc = 53.5 deg, r = 25.917 Rs, phase = 102 deg
---	243	**OTM-460**	243TI-4d	2016-267T00:34	Sep23	Fri	44	4	35	D/L start, burn ~6h later. Backup D/L 269T00:34; complexes C/C
---	243	Ring CRX	Descending	2016-268T03:19	Sep24	Sat	45	3	1	r = 8.212 Rs (495421 km)
---	243	243PO (nt)	**POLYDEUCES**	2016-268T03:54	Sep24	Sat	45	3	1	Inbound 110168.3 km flyby, v = 9.4 km/s, phase = 40 deg
---	243	Periapse		2016-268T11:35	Sep24	Sat	45	3	1	R = 7.519 Rs, lat = -31 deg, phase = 78 deg
---	243	**Sun OCC**	RING	2016-269T16:41	Sep25	Sun	46	1	3	Egress = 2016-269T22:52 (371 min)
---	243	**Sun OCC**	SATURN	2016-269T17:19	Sep25	Sun	46	1	3	Egress = 2016-269T19:31 (132 min)
---	243	**243TI (t) [T123]**	**TITAN**	2016-271T04:23	Sep27	Tue	48	48	4	Outbound 1736.5 km flyby, v = 5.8 km/s, phase = 88 deg
---	243	Ring CRX	Ascending	2016-271T04:58	Sep27	Tue	0	48	4	r = 20.067 Rs (1210642 km)
---	244	**Apoapse**		2016-273T05:54	Sep29	Thu	2	46	6	Per = 9.6 d, inc = 57.8 deg, r = 23.218 Rs, phase = 117 deg
---	244	**OTM-461**	243TI+3d	2016-274T00:03	Sep30	Fri	3	45	7	D/L start, burn ~6h later. Backup D/L 275T00:03; complexes C/C
---	244	244TI (nt)	**TITAN**	2016-277T17:31	Oct03	Mon	7	41	4	Inbound 876963.4 km flyby, v = 10.4 km/s, phase = 144 deg
---	244	Ring CRX	Descending	2016-277T21:26	Oct03	Mon	7	41	4	r = 5.806 Rs (350293 km)
---	244	244HE (nt)	**HELENE**	2016-277T21:56	Oct03	Mon	7	41	4	Inbound 106607.6 km flyby, v = 12.5 km/s, phase = 128 deg
---	244	Periapse		2016-278T00:39	Oct04	Tue	7	41	4	R = 5.587 Rs, lat = -22 deg, phase = 63 deg
---	244	**OTM-462**	~Peri	2016-278T23:48	Oct04	Tue	8	40	5	D/L start, burn ~6h later. Backup D/L 279T23:47; complexes C/C
---	244	**Sun OCC**	RING	2016-279T04:06	Oct05	Wed	8	40	0	Egress = 2016-279T09:19 (313 min)
---	244	Ring CRX	Ascending	2016-280T18:25	Oct06	Thu	10	38	2	r = 20.076 Rs (1211155 km)
---	245	**Apoapse**		2016-282T19:29	Oct08	Sat	12	36	4	Per = 9.6 d, inc = 57.8 deg, r = 23.229 Rs, phase = 118 deg
---	245	Ring CRX	Descending	2016-287T11:07	Oct13	Thu	16	32	8	r = 5.812 Rs (350623 km)
---	245	Periapse		2016-287T14:20	Oct13	Thu	16	31	9	R = 5.592 Rs, lat = -22 deg, phase = 62 deg
---	245	**Sun OCC**	RING	2016-288T17:42	Oct14	Fri	18	30	10	Egress = 2016-288T23:11 (329 min)
---	245	Ring CRX	Ascending	2016-290T08:08	Oct16	Sun	19	29	11	r = 20.077 Rs (1211243 km)
---	246	**Apoapse**		2016-292T09:10	Oct18	Tue	21	27	13	Per = 9.6 d, inc = 57.8 deg, r = 23.228 Rs, phase = 118 deg
---	246	Ring CRX	Descending	2016-297T00:47	Oct23	Sun	26	22	18	r = 5.806 Rs (350279 km)
---	246	246TL (nt)	**TELESTO**	2016-297T01:32	Oct23	Sun	26	22	18	Inbound 58966.4 km flyby, v = 12.1 km/s, phase = 71 deg
---	246	Periapse		2016-297T04:00	Oct23	Sun	26	22	18	R = 5.586 Rs, lat = -22 deg, phase = 62 deg
---	246	**Sun OCC**	RING	2016-298T07:15	Oct24	Mon	27	21	19	Egress = 2016-298T12:56 (342 min)
---	246	Ring CRX	Ascending	2016-299T21:41	Oct25	Tue	29	19	21	r = 20.063 Rs (1210400 km)
---	247	**Apoapse**		2016-301T22:45	Oct27	Thu	31	17	23	Per = 9.6 d, inc = 57.8 deg, r = 23.221 Rs, phase = 118 deg
---	247	247TI (nt)	**TITAN**	2016-303T08:56	Oct29	Sat	32	16	24	Inbound 860392.2 km flyby, v = 5.6 km/s, phase = 41 deg
---	247	247TL (nt)	**TELESTO**	2016-306T13:35	Nov01	Tue	35	12	28	Inbound 103930.5 km flyby, v = 11.7 km/s, phase = 22 deg
---	247	247PO (nt)	**POLYDEUCES**	2016-306T13:38	Nov01	Tue	35	12	28	Inbound 106888.2 km flyby, v = 11.3 km/s, phase = 71 deg
---	247	Ring CRX	Descending	2016-306T14:16	Nov01	Tue	35	12	28	r = 5.801 Rs (349952 km)
---	247	Periapse		2016-306T17:29	Nov01	Tue	36	12	28	R = 5.581 Rs, lat = -22 deg, phase = 62 deg
---	247	**Sun OCC**	RING	2016-307T20:37	Nov02	Wed	37	11	29	Egress = 2016-308T02:31 (354 min)
---	247	**Earth OCC**	RING	2016-307T21:33	Nov02	Wed	37	11	29	Egress = 2016-308T00:43 (190 min)
---	247	Ring CRX	Ascending	2016-309T11:07	Nov04	Fri	38	10	30	r = 20.062 Rs (1210330 km)
---	248	**Apoapse**		2016-311T12:17	Nov06	Sun	40	7	33	Per = 9.6 d, inc = 57.8 deg, r = 23.230 Rs, phase = 118 deg
---	248	**OTM-463**	248TI-4d	2016-314T21:29	Nov09	Wed	44	4	36	D/L start, burn ~6h later. Backup D/L 315T21:29; complexes C/C
---	248	Ring CRX	Descending	2016-316T03:52	Nov11	Fri	45	3	1	r = 5.799 Rs (349836 km)
---	248	248TE (nt)	**TETHYS**	2016-316T05:17	Nov11	Fri	45	3	1	Inbound 102468.7 km flyby, v = 12.8 km/s, phase = 101 deg
---	248	Periapse		2016-316T07:05	Nov11	Fri	45	3	1	R = 5.578 Rs, lat = -22 deg, phase = 62 deg
---	248	**Sun OCC**	RING	2016-317T10:06	Nov12	Sat	46	2	3	Egress = 2016-317T16:11 (365 min)
---	248	**Earth OCC**	RING	2016-317T10:31	Nov12	Sat	46	2	3	Egress = 2016-317T15:03 (272 min)
---	248	**248TI (t) [T124]**	**TITAN**	2016-319T00:02	Nov14	Mon	48	16	4	Outbound 1581.6 km flyby, v = 5.8 km/s, phase = 80 deg
---	248	Ring CRX	Ascending	2016-319T00:36	Nov14	Mon	0	16	4	r = 20.050 Rs (1209597 km)
---	249	**Apoapse**		2016-320T09:11	Nov15	Tue	1	15	5	Per = 8.0 d, inc = 61.4 deg, r = 21.875 Rs, phase = 129 deg
---	249	**OTM-464**	248TI+3d	2016-321T21:14	Nov16	Wed	3	13	7	D/L start, burn ~6h later. Backup D/L 322T20:14; complexes C/C
---	249	249MI (nt)	**MIMAS**	2016-324T07:36	Nov19	Sat	5	11	2	Inbound 47041.8 km flyby, v = 15.7 km/s, phase = 18 deg
---	249	E_ring_lg	Dust Crossing	2016-324T07:45	Nov19	Sat	5	11	2	Egress at 2016-324T07:47 (0.0 min); protection = none.
---	249	Ring CRX	Descending	2016-324T07:46	Nov19	Sat	5	11	2	r = 3.677 Rs (221824 km)
---	249	249DA (nt)	**DAPHNIS**	2016-324T08:16	Nov19	Sat	5	11	2	Inbound 86072.1 km flyby, v = 17.5 km/s, phase = 56 deg
---	249	249ME (nt)	**METHONE**	2016-324T08:24	Nov19	Sat	5	11	2	Inbound 72613.7 km flyby, v = 17.2 km/s, phase = 103 deg
---	249	249PA (nt)	**PANDORA**	2016-324T08:41	Nov19	Sat	5	11	2	Inbound 93417.5 km flyby, v = 17.8 km/s, phase = 77 deg
---	249	Periapse		2016-324T08:44	Nov19	Sat	5	11	2	R = 3.621 Rs, lat = -14 deg, phase = 51 deg
---	249	**Sun OCC**	RING	2016-325T05:37	Nov20	Sun	6	10	3	Egress = 2016-325T11:22 (345 min)
---	249	**Earth OCC**	RING	2016-325T06:01	Nov20	Sun	6	10	3	Egress = 2016-325T10:23 (263 min)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	249	OTM-465	~Apo	2016-326T20:59	Nov21	Mon	8	8	5	D/L start, burn ~6h later. Backup D/L 327T20:43; complexes C/C
---	249	Ring CRX	Ascending	2016-326T23:40	Nov21	Mon	8	8	0	r = 20.055 Rs (1209938 km)
---	250	Apoapse		2016-328T08:21	Nov23	Wed	9	7	1	Per = 8.0 d, inc = 61.4 deg, r = 21.884 Rs, phase = 129 deg
---	250	OTM-466	250TI-3d	2016-331T20:28	Nov26	Sat	13	3	5	D/L start, burn ~6h later. Backup D/L 332T19:15; complexes C/C
---	250	250EP (nt)	EPIMETHEUS	2016-332T06:32	Nov27	Sun	13	3	0	Inbound 85999.0 km flyby, v = 16.7 km/s, phase = 10 deg
---	250	250EN (nt)	ENCELADUS	2016-332T06:55	Nov27	Sun	13	3	0	Inbound 21897.3 km flyby, v = 15.6 km/s, phase = 103 deg
---	250	E_ring_lg	Dust Crossing	2016-332T07:00	Nov27	Sun	13	3	0	Egress at 2016-332T07:01 (0.0 min); protection = none.
---	250	Ring CRX	Descending	2016-332T07:01	Nov27	Sun	13	3	0	r = 3.678 Rs (221876 km)
---	250	Periapse		2016-332T07:58	Nov27	Sun	13	3	0	R = 3.622 Rs, lat = -14 deg, phase = 51 deg
---	250	Sun OCC	RING	2016-333T04:46	Nov28	Mon	14	2	1	Egress = 2016-333T10:45 (359 min)
---	250	Earth OCC	RING	2016-333T04:52	Nov28	Mon	14	2	1	Egress = 2016-333T10:12 (321 min)
---	250	250TI (t) [T125]	TITAN	2016-334T22:20	Nov29	Tue	16	143	3	Outbound 3223.1 km flyby, v = 5.7 km/s, phase = 82 deg
---	250	Ring CRX	Ascending	2016-334T22:51	Nov29	Tue	0	143	3	r = 20.052 Rs (1209721 km)
---	251	Apoapse		2016-335T23:33	Nov30	Wed	1	142	4	Per = 7.2 d, inc = 63.8 deg, r = 21.268 Rs, phase = 135 deg
---	251	START OF F RING ORBITS		2016-335T23:33	Nov30	Wed	1	142	4	Series of 20 orbits with periapses / ring crossings just outside F ring
---	251	OTM-467	250TI+5d	2016-339T06:58	Dec04	Sun	4	139	7	D/L start, burn ~6h later. Backup D/L 340T06:58; complexes M/M
---	251	251ME (nt)	METHONE	2016-339T13:10	Dec04	Sun	5	139	0	Inbound 50465.5 km flyby, v = 19.4 km/s, phase = 111 deg
---	251	JE_ring	DUST HAZARD	2016-339T13:14	Dec04	Sun	5	139	0	Egress at 2016-339T13:16 (0.0 min); protection = HGA to ram and MEA cover closure.
---	251	Ring CRX	Descending	2016-339T13:15	Dec04	Sun	5	139	0	r = 2.533 Rs (152809 km)
---	251	Periapse		2016-339T13:37	Dec04	Sun	5	139	0	R = 2.515 Rs, lat = -9 deg, phase = 45 deg
---	251	251PN (nt)	PAN	2016-339T13:39	Dec04	Sun	5	139	0	Outbound 56595.5 km flyby, v = 21.6 km/s, phase = 103 deg
---	251	Earth OCC	RING	2016-340T05:17	Dec05	Mon	5	138	1	Egress = 2016-340T11:16 (360 min)
---	251	Sun OCC	RING	2016-340T05:20	Dec05	Mon	5	138	1	Egress = 2016-340T11:31 (372 min)
---	251	Ring CRX	Ascending	2016-342T03:00	Dec07	Wed	7	136	3	r = 20.062 Rs (1210320 km)
---	251	SEP = 3.0 deg	Conjunction	2016-342T12:21	Dec07	Wed	8	136	3	Commanding must not be required below 3.0 deg
---	252	Apoapse		2016-343T03:45	Dec08	Thu	8	135	4	Per = 7.2 d, inc = 63.8 deg, r = 21.276 Rs, phase = 135 deg
---	252	SEP = 2.0 deg	Conjunction	2016-343T20:25	Dec08	Thu	9	134	5	Plan no SSR playback below 2.0 deg
---	252	252PM (nt)	PROMETHEUS	2016-346T16:55	Dec11	Sun	12	132	7	Inbound 101474.2 km flyby, v = 21.5 km/s, phase = 43 deg
---	252	JE_ring	DUST HAZARD	2016-346T17:31	Dec11	Sun	12	132	7	Egress at 2016-346T17:32 (0.0 min); protection = HGA to ram and MEA cover closure.
---	252	252PL (nt)	PALLENE	2016-346T17:31	Dec11	Sun	12	132	7	Inbound 94789.1 km flyby, v = 20.3 km/s, phase = 132 deg
---	252	Ring CRX	Descending	2016-346T17:32	Dec11	Sun	12	132	7	r = 2.533 Rs (152788 km)
---	252	Periapse		2016-346T17:54	Dec11	Sun	12	132	7	R = 2.514 Rs, lat = -9 deg, phase = 45 deg
---	252	SEP = 2.0 deg	Conjunction	2016-347T04:43	Dec12	Mon	12	131	8	Plan no SSR playback below 2.0 deg
---	252	Earth OCC	RING	2016-347T09:22	Dec12	Mon	12	131	8	Egress = 2016-347T15:57 (396 min)
---	252	Sun OCC	RING	2016-347T09:31	Dec12	Mon	12	131	8	Egress = 2016-347T15:56 (386 min)
---	252	SEP = 3.0 deg	Conjunction	2016-348T12:24	Dec13	Tue	14	130	9	Commanding must not be required below 3.0 deg
---	252	Ring CRX	Ascending	2016-349T07:15	Dec14	Wed	14	129	10	r = 20.057 Rs (1210060 km)
---	253	Apoapse		2016-350T07:58	Dec15	Thu	15	128	11	Per = 7.2 d, inc = 63.8 deg, r = 21.270 Rs, phase = 135 deg
---	253	253TI (nt)	TITAN	2016-350T21:53	Dec15	Thu	16	127	12	Inbound 343227.7 km flyby, v = 5.3 km/s, phase = 58 deg
---	253	253AE (nt)	AEGAEON	2016-353T21:15	Dec18	Sun	19	124	15	Inbound 98157.6 km flyby, v = 20.7 km/s, phase = 60 deg
---	253	JE_ring	DUST HAZARD	2016-353T21:34	Dec18	Sun	19	124	15	Egress at 2016-353T21:35 (0.0 min); protection = MEA cover closure.
---	253	Ring CRX	Descending	2016-353T21:35	Dec18	Sun	19	124	15	r = 2.514 Rs (151642 km)
---	253	253PA (nt)	PANDORA	2016-353T21:39	Dec18	Sun	19	124	15	Inbound 13837.8 km flyby, v = 20.4 km/s, phase = 81 deg
---	253	Periapse		2016-353T21:56	Dec18	Sun	19	124	15	R = 2.495 Rs, lat = -9 deg, phase = 45 deg
---	253	253JA (nt)	JANUS	2016-353T22:09	Dec18	Sun	19	124	15	Outbound 110222.2 km flyby, v = 23.5 km/s, phase = 109 deg
---	253	253PN (nt)	PAN	2016-353T22:13	Dec18	Sun	19	124	15	Outbound 93273.6 km flyby, v = 23.2 km/s, phase = 107 deg
---	253	Earth OCC	RING	2016-354T13:08	Dec19	Mon	20	124	15	Egress = 2016-354T20:13 (425 min)
---	253	Sun OCC	RING	2016-354T13:21	Dec19	Mon	20	124	15	Egress = 2016-354T20:00 (400 min)
---	253	Ring CRX	Ascending	2016-356T11:08	Dec21	Wed	22	122	17	r = 20.049 Rs (1209535 km)
---	254	Apoapse		2016-357T11:57	Dec22	Thu	23	121	18	Per = 7.2 d, inc = 63.8 deg, r = 21.273 Rs, phase = 135 deg
---	254	OTM-468	~Apo	2016-358T18:58	Dec23	Fri	24	119	20	D/L start, burn ~6h later. Backup D/L 359T18:58; complexes C/C
---	254	254EN (nt)	ENCELADUS	2016-361T01:32	Dec26	Mon	26	117	2	Inbound 88314.0 km flyby, v = 19.2 km/s, phase = 135 deg
---	254	JE_ring	DUST HAZARD	2016-361T01:35	Dec26	Mon	26	117	2	Egress at 2016-361T01:37 (0.0 min); protection = MEA cover closure.
---	254	254MI (nt)	MIMAS	2016-361T01:35	Dec26	Mon	26	117	2	Inbound 40700.6 km flyby, v = 19.4 km/s, phase = 114 deg
---	254	Ring CRX	Descending	2016-361T01:36	Dec26	Mon	26	117	2	r = 2.512 Rs (151539 km)
---	254	254AE (nt)	AEGAEON	2016-361T01:41	Dec26	Mon	26	117	2	Inbound 32482.9 km flyby, v = 20.2 km/s, phase = 138 deg
---	254	Periapse		2016-361T01:58	Dec26	Mon	26	117	2	R = 2.494 Rs, lat = -9 deg, phase = 45 deg
---	254	Earth OCC	RING	2016-361T17:05	Dec26	Mon	27	117	3	Egress = 2016-362T00:31 (447 min)
---	254	Sun OCC	RING	2016-361T17:18	Dec26	Mon	27	117	3	Egress = 2016-362T00:08 (411 min)
---	254	Earth OCC	SATURN	2016-361T18:04	Dec26	Mon	27	117	3	Egress = 2016-361T20:41 (158 min)
---	254	Ring CRX	Ascending	2016-363T15:07	Dec28	Wed	29	115	5	r = 20.044 Rs (1209237 km)
---	255	Apoapse		2016-364T15:56	Dec29	Thu	30	114	6	Per = 7.2 d, inc = 63.8 deg, r = 21.269 Rs, phase = 136 deg
---	255	255TI (nt)	TITAN	2016-366T12:17	Dec31	Sat	32	112	8	Inbound 681818.6 km flyby, v = 7.2 km/s, phase = 29 deg
---	255	JE_ring	DUST HAZARD	2017-002T05:31	Jan02	Mon	33	110	9	Egress at 2017-002T05:32 (0.0 min); protection = MEA cover closure.
---	255	Ring CRX	Descending	2017-002T05:32	Jan02	Mon	33	110	9	r = 2.508 Rs (151319 km)
---	255	Periapse		2017-002T05:53	Jan02	Mon	33	110	9	R = 2.490 Rs, lat = -9 deg, phase = 44 deg
---	255	255DA (nt)	DAPHNIS	2017-002T06:10	Jan02	Mon	33	110	9	Outbound 98983.2 km flyby, v = 23.5 km/s, phase = 107 deg
---	255	Earth OCC	RING	2017-002T20:57	Jan02	Mon	34	109	10	Egress = 2017-003T04:41 (464 min)
---	255	Sun OCC	RING	2017-002T21:09	Jan02	Mon	34	109	10	Egress = 2017-003T04:10 (421 min)
---	255	Earth OCC	SATURN	2017-002T21:35	Jan02	Mon	34	109	10	Egress = 2017-003T01:43 (248 min)
---	255	Ring CRX	Ascending	2017-004T19:00	Jan04	Wed	36	107	12	r = 20.041 Rs (1209059 km)
---	256	Apoapse		2017-005T19:50	Jan05	Thu	37	106	13	Per = 7.2 d, inc = 63.8 deg, r = 21.267 Rs, phase = 136 deg
---	256	256AT (nt)	ATLAS	2017-009T08:48	Jan09	Mon	40	103	17	Inbound 100863.3 km flyby, v = 21.7 km/s, phase = 44 deg
---	256	256EP (nt)	EPIMETHEUS	2017-009T09:06	Jan09	Mon	40	103	17	Inbound 70798.9 km flyby, v = 20.6 km/s, phase = 55 deg
---	256	JE_ring	DUST HAZARD	2017-009T09:24	Jan09	Mon	40	103	17	Egress at 2017-009T09:26 (0.0 min); protection = MEA cover closure.
---	256	Ring CRX	Descending	2017-009T09:25	Jan09	Mon	40	103	17	r = 2.506 Rs (151192 km)
---	256	256DA (nt)	DAPHNIS	2017-009T09:47	Jan09	Mon	40	103	17	Outbound 55957.1 km flyby, v = 21.6 km/s, phase = 104 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	256	Periapse		2017-009T09:47	Jan09	Mon	40	103	17	R = 2.488 Rs, lat = -9 deg, phase = 44 deg
---	256	256PM (nt)	**PROMETHEUS**	2017-009T10:01	Jan09	Mon	40	103	17	Outbound 95524.0 km flyby, v = 23.2 km/s, phase = 107 deg
---	256	**Earth OCC**	RING	2017-010T00:49	Jan10	Tue	41	102	17	Egress = 2017-010T08:46 (477 min)
---	256	**Sun OCC**	RING	2017-010T00:59	Jan10	Tue	41	102	17	Egress = 2017-010T08:09 (430 min)
---	256	**Earth OCC**	SATURN	2017-010T01:22	Jan10	Tue	41	102	17	Egress = 2017-010T06:26 (305 min)
---	256	Ring CRX	Ascending	2017-011T22:52	Jan11	Wed	43	100	19	r = 20.038 Rs (1208896 km)
---	257	**Apoapse**		2017-012T23:43	Jan12	Thu	44	99	20	Per = 7.2 d, inc = 63.8 deg, r = 21.266 Rs, phase = 136 deg
---	257	257JA (nt)	**JANUS**	2017-016T12:42	Jan16	Mon	48	96	24	Inbound 117179.0 km flyby, v = 21.9 km/s, phase = 50 deg
---	257	257AT (nt)	**ATLAS**	2017-016T13:10	Jan16	Mon	48	96	24	Inbound 24557.0 km flyby, v = 20.3 km/s, phase = 28 deg
---	257	257PL (nt)	**PALLENE**	2017-016T13:14	Jan16	Mon	48	96	24	Inbound 62882.4 km flyby, v = 19.6 km/s, phase = 149 deg
---	257	JE_ring	**DUST HAZARD**	2017-016T13:16	Jan16	Mon	48	96	24	Egress at 2017-016T13:17 (0.0 min); protection = MEA cover closure.
---	257	Ring CRX	Descending	2017-016T13:17	Jan16	Mon	48	96	24	r = 2.507 Rs (151248 km)
---	257	257DA (nt)	**DAPHNIS**	2017-016T13:22	Jan16	Mon	48	96	24	Inbound 17845.0 km flyby, v = 20.6 km/s, phase = 69 deg
---	257	Periapse		2017-016T13:38	Jan16	Mon	48	96	24	R = 2.489 Rs, lat = -9 deg, phase = 44 deg
---	257	**Earth OCC**	RING	2017-017T04:44	Jan17	Tue	48	95	24	Egress = 2017-017T12:50 (487 min)
---	257	**Sun OCC**	RING	2017-017T04:49	Jan17	Tue	48	95	24	Egress = 2017-017T12:08 (440 min)
---	257	**Earth OCC**	SATURN	2017-017T05:15	Jan17	Tue	48	95	24	Egress = 2017-017T10:59 (345 min)
---	257	**Sun OCC**	SATURN	2017-017T06:04	Jan17	Tue	48	95	24	Egress = 2017-017T07:49 (106 min)
---	257	257TI (nt)	**TITAN**	2017-017T18:30	Jan17	Tue	49	94	25	Outbound 724242.3 km flyby, v = 9.3 km/s, phase = 96 deg
---	257	Ring CRX	Ascending	2017-019T02:46	Jan19	Thu	50	93	26	r = 20.042 Rs (1209140 km)
---	258	**Apoapse**		2017-020T03:35	Jan20	Fri	51	92	27	Per = 7.2 d, inc = 63.8 deg, r = 21.268 Rs, phase = 136 deg
---	258	258DA (nt)	**DAPHNIS**	2017-023T16:58	Jan23	Mon	55	89	31	Inbound 38531.2 km flyby, v = 20.4 km/s, phase = 36 deg
---	258	258EN (nt)	**ENCELADUS**	2017-023T17:01	Jan23	Mon	55	89	31	Inbound 124361.1 km flyby, v = 20.2 km/s, phase = 129 deg
---	258	JE_ring	**DUST HAZARD**	2017-023T17:09	Jan23	Mon	55	89	31	Egress at 2017-023T17:11 (0.0 min); protection = MEA cover closure.
---	258	Ring CRX	Descending	2017-023T17:10	Jan23	Mon	55	89	31	r = 2.507 Rs (151266 km)
---	258	258PA (nt)	**PANDORA**	2017-023T17:17	Jan23	Mon	55	89	31	Inbound 19443.5 km flyby, v = 20.5 km/s, phase = 92 deg
---	258	Periapse		2017-023T17:32	Jan23	Mon	55	89	31	R = 2.489 Rs, lat = -9 deg, phase = 44 deg
---	258	258AT (nt)	**ATLAS**	2017-023T17:33	Jan23	Mon	55	89	31	Outbound 60042.1 km flyby, v = 21.8 km/s, phase = 105 deg
---	258	**Earth OCC**	RING	2017-024T08:40	Jan24	Tue	55	88	32	Egress = 2017-024T16:53 (493 min)
---	258	**Sun OCC**	RING	2017-024T08:41	Jan24	Tue	55	88	32	Egress = 2017-024T16:08 (448 min)
---	258	**Earth OCC**	SATURN	2017-024T09:14	Jan24	Tue	55	88	32	Egress = 2017-024T15:26 (373 min)
---	258	**Sun OCC**	SATURN	2017-024T09:38	Jan24	Tue	55	88	32	Egress = 2017-024T12:20 (162 min)
---	258	Ring CRX	Ascending	2017-026T06:40	Jan26	Thu	57	86	33	r = 20.046 Rs (1209347 km)
---	259	**Apoapse**		2017-027T07:30	Jan27	Fri	58	85	35	Per = 7.2 d, inc = 63.8 deg, r = 21.272 Rs, phase = 137 deg
---	259	259DA (nt)	**DAPHNIS**	2017-030T20:36	Jan30	Mon	62	81	38	Inbound 85527.1 km flyby, v = 21.2 km/s, phase = 44 deg
---	259	JE_ring	**DUST HAZARD**	2017-030T21:06	Jan30	Mon	62	81	38	Egress at 2017-030T21:08 (0.0 min); protection = MEA cover closure.
---	259	259MI (nt)	**MIMAS**	2017-030T21:06	Jan30	Mon	62	81	38	Inbound 40385.1 km flyby, v = 19.7 km/s, phase = 112 deg
---	259	259EP (nt)	**EPIMETHEUS**	2017-030T21:06	Jan30	Mon	62	81	38	Inbound 5838.8 km flyby, v = 20.1 km/s, phase = 76 deg
---	259	Ring CRX	Descending	2017-030T21:07	Jan30	Mon	62	81	38	r = 2.509 Rs (151349 km)
---	259	259PM (nt)	**PROMETHEUS**	2017-030T21:27	Jan30	Mon	62	81	38	Inbound 52584.6 km flyby, v = 21.5 km/s, phase = 104 deg
---	259	Periapse		2017-030T21:29	Jan30	Mon	62	81	38	R = 2.491 Rs, lat = -9 deg, phase = 43 deg
---	259	**Sun OCC**	RING	2017-031T12:37	Jan31	Tue	63	81	39	Egress = 2017-031T20:12 (455 min)
---	259	**Earth OCC**	RING	2017-031T12:42	Jan31	Tue	63	81	39	Egress = 2017-031T20:59 (497 min)
---	259	**Earth OCC**	SATURN	2017-031T13:20	Jan31	Tue	63	81	39	Egress = 2017-031T19:51 (392 min)
---	259	**Sun OCC**	SATURN	2017-031T13:23	Jan31	Tue	63	81	39	Egress = 2017-031T16:49 (206 min)
---	259	259TI (nt)	**TITAN**	2017-032T20:04	Feb01	Wed	64	79	40	Outbound 220952.7 km flyby, v = 6.3 km/s, phase = 91 deg
---	259	Ring CRX	Ascending	2017-033T10:36	Feb02	Thu	65	79	41	r = 20.053 Rs (1209780 km)
---	260	**Apoapse**		2017-034T11:39	Feb03	Fri	66	78	42	Per = 7.2 d, inc = 63.7 deg, r = 21.297 Rs, phase = 137 deg
---	260	260PN (nt)	**PAN**	2017-038T00:51	Feb07	Tue	69	74	45	Inbound 95495.7 km flyby, v = 21.4 km/s, phase = 43 deg
---	260	260JA (nt)	**JANUS**	2017-038T01:02	Feb07	Tue	69	74	45	Inbound 90893.4 km flyby, v = 21.0 km/s, phase = 53 deg
---	260	JE_ring	**DUST HAZARD**	2017-038T01:28	Feb07	Tue	69	74	45	Egress at 2017-038T01:30 (0.0 min); protection = HGA to ram and MEA cover closure.
---	260	Ring CRX	Descending	2017-038T01:29	Feb07	Tue	69	74	45	r = 2.528 Rs (152524 km)
---	260	260ME (nt)	**METHONE**	2017-038T01:33	Feb07	Tue	69	74	45	Inbound 84478.7 km flyby, v = 20.7 km/s, phase = 125 deg
---	260	Periapse		2017-038T01:51	Feb07	Tue	69	74	45	R = 2.510 Rs, lat = -9 deg, phase = 43 deg
---	260	**Sun OCC**	RING	2017-038T17:03	Feb07	Tue	70	74	46	Egress = 2017-039T00:45 (463 min)
---	260	**Earth OCC**	RING	2017-038T17:15	Feb07	Tue	70	74	46	Egress = 2017-039T01:32 (498 min)
---	260	**Sun OCC**	SATURN	2017-038T17:42	Feb07	Tue	70	74	46	Egress = 2017-038T21:45 (243 min)
---	260	**Earth OCC**	SATURN	2017-038T18:01	Feb07	Tue	70	74	46	Egress = 2017-039T00:41 (401 min)
---	260	Ring CRX	Ascending	2017-040T15:03	Feb09	Thu	72	72	48	r = 20.061 Rs (1210275 km)
---	261	**Apoapse**		2017-041T16:07	Feb10	Fri	73	71	49	Per = 7.2 d, inc = 63.7 deg, r = 21.304 Rs, phase = 137 deg
---	261	261AE (nt)	**AEGAEON**	2017-045T05:45	Feb14	Tue	76	67	52	Inbound 74803.5 km flyby, v = 20.1 km/s, phase = 68 deg
---	261	261ME (nt)	**METHONE**	2017-045T05:54	Feb14	Tue	76	67	52	Inbound 69193.2 km flyby, v = 19.5 km/s, phase = 95 deg
---	261	JE_ring	**DUST HAZARD**	2017-045T06:00	Feb14	Tue	76	67	52	Egress at 2017-045T06:01 (0.0 min); protection = HGA to ram and MEA cover closure.
---	261	Ring CRX	Descending	2017-045T06:01	Feb14	Tue	76	67	52	r = 2.529 Rs (152546 km)
---	261	Periapse		2017-045T06:23	Feb14	Tue	76	67	52	R = 2.510 Rs, lat = -9 deg, phase = 43 deg
---	261	**Sun OCC**	RING	2017-045T21:33	Feb14	Tue	77	66	53	Egress = 2017-046T05:22 (469 min)
---	261	**Earth OCC**	RING	2017-045T21:52	Feb14	Tue	77	66	53	Egress = 2017-046T06:08 (497 min)
---	261	**Sun OCC**	SATURN	2017-045T22:08	Feb14	Tue	77	66	53	Egress = 2017-046T02:39 (271 min)
---	261	**Earth OCC**	SATURN	2017-045T22:46	Feb14	Tue	77	66	53	Egress = 2017-046T05:28 (402 min)
---	261	Ring CRX	Ascending	2017-047T19:34	Feb16	Thu	79	64	55	r = 20.058 Rs (1210128 km)
---	261	261TI (nt)	**TITAN**	2017-048T13:17	Feb17	Fri	80	64	56	Outbound 186447.5 km flyby, v = 5.1 km/s, phase = 77 deg
---	262	**Apoapse**		2017-048T20:29	Feb17	Fri	80	63	56	Per = 7.2 d, inc = 63.8 deg, r = 21.296 Rs, phase = 137 deg
---	262	262PN (nt)	**PAN**	2017-052T09:39	Feb21	Tue	83	60	60	Inbound 73575.0 km flyby, v = 21.0 km/s, phase = 43 deg
---	262	262PM (nt)	**PROMETHEUS**	2017-052T09:43	Feb21	Tue	83	60	60	Inbound 67919.5 km flyby, v = 20.8 km/s, phase = 47 deg
---	262	262PL (nt)	**PALLENE**	2017-052T10:04	Feb21	Tue	83	60	60	Inbound 106030.1 km flyby, v = 19.8 km/s, phase = 91 deg
---	262	262EP (nt)	**EPIMETHEUS**	2017-052T10:04	Feb21	Tue	83	60	60	Inbound 10481.4 km flyby, v = 20.1 km/s, phase = 75 deg
---	262	2.47_Rs_gap	**DUST HAZARD**	2017-052T10:06	Feb21	Tue	83	60	60	Egress at 2017-052T10:07 (0.0 min); protection = MEA cover closure.
---	262	Ring CRX	Descending	2017-052T10:07	Feb21	Tue	83	60	60	r = 2.493 Rs (150398 km)

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	262	262AE (nt)	**AEGAEON**	2017-052T10:15	Feb21	Tue	83	60	60	Inbound 51344.4 km flyby, v = 20.7 km/s, phase = 124 deg
---	262	Periapse		2017-052T10:28	Feb21	Tue	84	60	60	R = 2.475 Rs, lat = -9 deg, phase = 43 deg
---	262	**Sun OCC**	RING	2017-053T01:24	Feb22	Wed	84	59	60	Egress = 2017-053T09:18 (475 min)
---	262	**Earth OCC**	RING	2017-053T01:49	Feb22	Wed	84	59	60	Egress = 2017-053T10:03 (494 min)
---	262	**Sun OCC**	SATURN	2017-053T01:56	Feb22	Wed	84	59	60	Egress = 2017-053T06:53 (297 min)
---	262	**Earth OCC**	SATURN	2017-053T02:53	Feb22	Wed	84	59	60	Egress = 2017-053T09:31 (398 min)
---	262	Ring CRX	Ascending	2017-054T23:30	Feb23	Thu	86	57	62	r = 20.057 Rs (1210062 km)
---	263	**Apoapse**		2017-056T00:31	Feb25	Sat	87	56	63	Per = 7.2 d, inc = 63.7 deg, r = 21.302 Rs, phase = 137 deg
---	263	263JA (nt)	**JANUS**	2017-059T13:47	Feb28	Tue	91	53	67	Inbound 92457.8 km flyby, v = 21.1 km/s, phase = 56 deg
---	263	263PA (nt)	**PANDORA**	2017-059T13:55	Feb28	Tue	91	53	67	Inbound 57213.1 km flyby, v = 20.5 km/s, phase = 49 deg
---	263	2.47_Rs_gap	**DUST HAZARD**	2017-059T14:13	Feb28	Tue	91	53	67	Egress at 2017-059T14:14 (0.0 min); protection = MEA cover closure.
---	263	Ring CRX	Descending	2017-059T14:14	Feb28	Tue	91	53	67	r = 2.492 Rs (150334 km)
---	263	Periapse		2017-059T14:35	Feb28	Tue	91	53	67	R = 2.474 Rs, lat = -9 deg, phase = 42 deg
---	263	**Sun OCC**	RING	2017-060T05:30	Mar01	Wed	91	52	67	Egress = 2017-060T13:29 (480 min)
---	263	**Sun OCC**	SATURN	2017-060T06:01	Mar01	Wed	91	52	67	Egress = 2017-060T11:17 (317 min)
---	263	**Earth OCC**	RING	2017-060T06:02	Mar01	Wed	91	52	67	Egress = 2017-060T14:12 (491 min)
---	263	**Earth OCC**	SATURN	2017-060T07:16	Mar01	Wed	91	52	68	Egress = 2017-060T13:43 (388 min)
---	263	Ring CRX	Ascending	2017-062T03:35	Mar03	Fri	93	50	69	r = 20.053 Rs (1209824 km)
---	264	**Apoapse**		2017-063T04:35	Mar04	Sat	94	49	70	Per = 7.2 d, inc = 63.7 deg, r = 21.297 Rs, phase = 138 deg
---	264	264TI (nt)	**TITAN**	2017-064T11:53	Mar05	Sun	96	48	72	Inbound 492397.4 km flyby, v = 5.9 km/s, phase = 42 deg
---	264	264MI (nt)	**MIMAS**	2017-066T18:02	Mar07	Tue	98	46	74	Inbound 101805.5 km flyby, v = 20.7 km/s, phase = 74 deg
---	264	264PN (nt)	**PAN**	2017-066T18:07	Mar07	Tue	98	46	74	Inbound 24886.8 km flyby, v = 20.4 km/s, phase = 26 deg
---	264	2.47_Rs_gap	**DUST HAZARD**	2017-066T18:12	Mar07	Tue	98	46	74	Egress at 2017-066T18:13 (0.0 min); protection = MEA cover closure.
---	264	Ring CRX	Descending	2017-066T18:13	Mar07	Tue	98	46	74	r = 2.482 Rs (149733 km)
---	264	Periapse		2017-066T18:34	Mar07	Tue	98	45	74	R = 2.464 Rs, lat = -9 deg, phase = 42 deg
---	264	**Sun OCC**	RING	2017-067T09:23	Mar08	Wed	98	45	75	Egress = 2017-067T17:27 (484 min)
---	264	**Sun OCC**	SATURN	2017-067T09:54	Mar08	Wed	98	45	75	Egress = 2017-067T15:28 (334 min)
---	264	**Earth OCC**	RING	2017-067T10:01	Mar08	Wed	98	45	75	Egress = 2017-067T18:07 (486 min)
---	264	**Earth OCC**	SATURN	2017-067T11:26	Mar08	Wed	99	45	75	Egress = 2017-067T17:39 (374 min)
---	264	Ring CRX	Ascending	2017-069T07:26	Mar10	Fri	100	43	77	r = 20.042 Rs (1209130 km)
---	265	**Apoapse**		2017-070T08:31	Mar11	Sat	101	42	78	Per = 7.2 d, inc = 63.7 deg, r = 21.295 Rs, phase = 138 deg
---	265	2.47_Rs_gap	**DUST HAZARD**	2017-073T22:07	Mar14	Tue	105	38	81	Egress at 2017-073T22:08 (0.0 min); protection = MEA cover closure.
---	265	Ring CRX	Descending	2017-073T22:07	Mar14	Tue	105	38	81	r = 2.480 Rs (149612 km)
---	265	265EP (nt)	**EPIMETHEUS**	2017-073T22:17	Mar14	Tue	105	38	81	Inbound 38332.8 km flyby, v = 21.0 km/s, phase = 113 deg
---	265	Periapse		2017-073T22:29	Mar14	Tue	105	38	81	R = 2.462 Rs, lat = -9 deg, phase = 42 deg
---	265	265PA (nt)	**PANDORA**	2017-073T22:46	Mar14	Tue	105	38	81	Outbound 110590.1 km flyby, v = 24.0 km/s, phase = 104 deg
---	265	**Sun OCC**	RING	2017-074T13:18	Mar15	Wed	106	38	82	Egress = 2017-074T21:24 (487 min)
---	265	**Sun OCC**	SATURN	2017-074T13:49	Mar15	Wed	106	38	82	Egress = 2017-074T19:37 (348 min)
---	265	**Earth OCC**	RING	2017-074T14:01	Mar15	Wed	106	38	82	Egress = 2017-074T22:02 (482 min)
---	265	**Earth OCC**	SATURN	2017-074T15:35	Mar15	Wed	106	38	82	Egress = 2017-074T21:33 (358 min)
---	265	Ring CRX	Ascending	2017-076T11:19	Mar17	Fri	108	36	84	r = 20.038 Rs (1208910 km)
---	266	**Apoapse**		2017-077T12:24	Mar18	Sat	109	35	85	Per = 7.2 d, inc = 63.7 deg, r = 21.293 Rs, phase = 138 deg
---	266	266TI (nt)	**TITAN**	2017-079T17:40	Mar20	Mon	111	33	87	Inbound 874286.0 km flyby, v = 8.7 km/s, phase = 32 deg
---	266	266JA (nt)	**JANUS**	2017-081T01:47	Mar22	Wed	112	31	88	Inbound 43770.0 km flyby, v = 20.2 km/s, phase = 65 deg
---	266	266PL (nt)	**PALLENE**	2017-081T01:56	Mar22	Wed	112	31	88	Inbound 106896.1 km flyby, v = 20.7 km/s, phase = 122 deg
---	266	2.47_Rs_gap	**DUST HAZARD**	2017-081T01:58	Mar22	Wed	112	31	88	Egress at 2017-081T01:59 (0.0 min); protection = MEA cover closure.
---	266	Ring CRX	Descending	2017-081T01:59	Mar22	Wed	112	31	88	r = 2.479 Rs (149575 km)
---	266	266PN (nt)	**PAN**	2017-081T02:20	Mar22	Wed	112	31	88	Outbound 55136.5 km flyby, v = 21.8 km/s, phase = 99 deg
---	266	Periapse		2017-081T02:20	Mar22	Wed	112	31	88	R = 2.461 Rs, lat = -9 deg, phase = 42 deg
---	266	**Sun OCC**	RING	2017-081T17:10	Mar22	Wed	113	31	89	Egress = 2017-082T01:19 (490 min)
---	266	**Sun OCC**	SATURN	2017-081T17:42	Mar22	Wed	113	31	89	Egress = 2017-081T23:42 (361 min)
---	266	**Earth OCC**	RING	2017-081T17:57	Mar22	Wed	113	31	89	Egress = 2017-082T01:54 (478 min)
---	266	**Earth OCC**	SATURN	2017-081T19:41	Mar22	Wed	113	30	89	Egress = 2017-082T01:22 (341 min)
---	266	266TI (nt)	**TITAN**	2017-082T00:23	Mar23	Thu	113	30	89	Outbound 954986.1 km flyby, v = 11.3 km/s, phase = 97 deg
---	266	Ring CRX	Ascending	2017-083T15:11	Mar24	Fri	115	29	91	r = 20.039 Rs (1208981 km)
---	267	**Apoapse**		2017-084T16:15	Mar25	Sat	116	28	92	Per = 7.2 d, inc = 63.7 deg, r = 21.292 Rs, phase = 138 deg
---	267	267EN (nt)	**ENCELADUS**	2017-088T05:46	Mar29	Wed	119	24	95	Inbound 93096.0 km flyby, v = 19.3 km/s, phase = 134 deg
---	267	2.47_Rs_gap	**DUST HAZARD**	2017-088T05:49	Mar29	Wed	119	24	95	Egress at 2017-088T05:50 (0.0 min); protection = MEA cover closure.
---	267	Ring CRX	Descending	2017-088T05:50	Mar29	Wed	119	24	95	r = 2.478 Rs (149489 km)
---	267	267MI (nt)	**MIMAS**	2017-088T06:07	Mar29	Wed	119	24	95	Inbound 121337.7 km flyby, v = 22.8 km/s, phase = 108 deg
---	267	Periapse		2017-088T06:11	Mar29	Wed	119	24	95	R = 2.460 Rs, lat = -9 deg, phase = 42 deg
---	267	**Sun OCC**	RING	2017-088T21:01	Mar29	Wed	120	23	96	Egress = 2017-089T05:13 (493 min)
---	267	**Sun OCC**	SATURN	2017-088T21:35	Mar29	Wed	120	23	96	Egress = 2017-089T03:46 (372 min)
---	267	**Earth OCC**	RING	2017-088T21:51	Mar29	Wed	120	23	96	Egress = 2017-089T05:45 (475 min)
---	267	**Earth OCC**	SATURN	2017-088T23:44	Mar29	Wed	120	23	96	Egress = 2017-089T05:10 (326 min)
---	267	Ring CRX	Ascending	2017-090T19:01	Mar31	Fri	122	21	98	r = 20.038 Rs (1208903 km)
---	268	**Apoapse**		2017-091T20:06	Apr01	Sat	123	20	99	Per = 7.2 d, inc = 63.7 deg, r = 21.292 Rs, phase = 139 deg
---	268	268AT (nt)	**ATLAS**	2017-095T09:18	Apr05	Wed	126	17	103	Inbound 66051.0 km flyby, v = 20.8 km/s, phase = 49 deg
---	268	268PA (nt)	**PANDORA**	2017-095T09:28	Apr05	Wed	126	17	103	Inbound 39664.4 km flyby, v = 20.4 km/s, phase = 50 deg
---	268	2.47_Rs_gap	**DUST HAZARD**	2017-095T09:40	Apr05	Wed	126	17	103	Egress at 2017-095T09:41 (0.0 min); protection = MEA cover closure.
---	268	Ring CRX	Descending	2017-095T09:41	Apr05	Wed	126	17	103	r = 2.479 Rs (149573 km)
---	268	Periapse		2017-095T10:02	Apr05	Wed	126	17	103	R = 2.461 Rs, lat = -9 deg, phase = 41 deg
---	268	268EP (nt)	**EPIMETHEUS**	2017-095T10:12	Apr05	Wed	126	17	103	Outbound 110979.9 km flyby, v = 23.8 km/s, phase = 103 deg
---	268	**Sun OCC**	RING	2017-096T00:53	Apr06	Thu	127	16	103	Egress = 2017-096T09:08 (495 min)
---	268	**Sun OCC**	SATURN	2017-096T01:29	Apr06	Thu	127	16	103	Egress = 2017-096T07:50 (381 min)
---	268	**Earth OCC**	RING	2017-096T01:45	Apr06	Thu	127	16	103	Egress = 2017-096T09:37 (472 min)
---	268	**Earth OCC**	SATURN	2017-096T03:46	Apr06	Thu	127	16	103	Egress = 2017-096T08:58 (312 min)
---	268	268TI (nt)	**TITAN**	2017-096T22:18	Apr06	Thu	128	15	104	Outbound 477715.9 km flyby, v = 7.7 km/s, phase = 89 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	268	Ring CRX	Ascending	2017-097T22:53	Apr07	Fri	129	14	105	r = 20.045 Rs (1209325 km)
---	269	**Apoapse**		2017-099T00:01	Apr09	Sun	130	13	106	Per = 7.2 d, inc = 63.6 deg, r = 21.302 Rs, phase = 139 deg
---	269	269MI (nt)	**MIMAS**	2017-102T13:27	Apr12	Wed	134	10	110	Inbound 97740.9 km flyby, v = 20.8 km/s, phase = 76 deg
---	269	269AE (nt)	**AEGAEON**	2017-102T13:37	Apr12	Wed	134	10	110	Inbound 20653.5 km flyby, v = 19.9 km/s, phase = 120 deg
---	269	2.47_Rs_gap	**DUST HAZARD**	2017-102T13:39	Apr12	Wed	134	10	110	Egress at 2017-102T13:40 (0.0 min); protection = MEA cover closure.
---	269	Ring CRX	Descending	2017-102T13:40	Apr12	Wed	134	10	110	r = 2.484 Rs (149839 km)
---	269	269JA (nt)	**JANUS**	2017-102T13:42	Apr12	Wed	134	10	110	Inbound 8962.0 km flyby, v = 20.2 km/s, phase = 126 deg
---	269	269AT (nt)	**ATLAS**	2017-102T13:44	Apr12	Wed	134	10	110	Inbound 14833.6 km flyby, v = 20.5 km/s, phase = 65 deg
---	269	Periapse		2017-102T14:01	Apr12	Wed	134	10	110	R = 2.466 Rs, lat = -9 deg, phase = 41 deg
---	269	269PM (nt)	**PROMETHEUS**	2017-102T14:11	Apr12	Wed	134	10	110	Outbound 88144.7 km flyby, v = 22.9 km/s, phase = 103 deg
---	269	**Sun OCC**	RING	2017-103T04:55	Apr13	Thu	134	9	110	Egress = 2017-103T13:11 (496 min)
---	269	**Sun OCC**	SATURN	2017-103T05:34	Apr13	Thu	134	9	110	Egress = 2017-103T12:02 (389 min)
---	269	**Earth OCC**	RING	2017-103T05:48	Apr13	Thu	134	9	110	Egress = 2017-103T13:37 (470 min)
---	269	**Earth OCC**	SATURN	2017-103T07:54	Apr13	Thu	134	9	111	Egress = 2017-103T12:56 (302 min)
---	269	Ring CRX	Ascending	2017-105T02:54	Apr15	Sat	136	7	112	r = 20.049 Rs (1209571 km)
---	270	**Apoapse**		2017-106T04:03	Apr16	Sun	137	6	113	Per = 7.2 d, inc = 63.6 deg, r = 21.306 Rs, phase = 139 deg
---	270	**OTM-469**	270TI-4d	2017-108T12:11	Apr18	Tue	140	4	116	D/L start, burn ~6h later. Backup D/L 109T12:11; complexes C/C
---	270	2.47_Rs_gap	**DUST HAZARD**	2017-109T17:43	Apr19	Wed	141	3	1	Egress at 2017-109T17:44 (0.0 min); protection = MEA cover closure.
---	270	Ring CRX	Descending	2017-109T17:44	Apr19	Wed	141	3	1	r = 2.485 Rs (149902 km)
---	270	270ME (nt)	**METHONE**	2017-109T17:47	Apr19	Wed	141	3	1	Inbound 89296.5 km flyby, v = 20.9 km/s, phase = 120 deg
---	270	270AE (nt)	**AEGAEON**	2017-109T18:05	Apr19	Wed	141	3	1	Inbound 111259.9 km flyby, v = 22.8 km/s, phase = 105 deg
---	270	Periapse		2017-109T18:05	Apr19	Wed	141	3	1	R = 2.467 Rs, lat = -9 deg, phase = 41 deg
---	270	270AT (nt)	**ATLAS**	2017-109T18:13	Apr19	Wed	141	3	1	Outbound 77237.5 km flyby, v = 22.5 km/s, phase = 102 deg
---	270	270PA (nt)	**PANDORA**	2017-109T18:24	Apr19	Wed	141	2	1	Outbound 117315.7 km flyby, v = 24.3 km/s, phase = 101 deg
---	270	270DA (nt)	**DAPHNIS**	2017-109T18:28	Apr19	Wed	141	2	1	Outbound 117728.7 km flyby, v = 24.5 km/s, phase = 101 deg
---	270	**Sun OCC**	RING	2017-110T09:01	Apr20	Thu	141	2	2	Egress = 2017-110T17:18 (497 min)
---	270	**Sun OCC**	SATURN	2017-110T09:43	Apr20	Thu	141	2	2	Egress = 2017-110T16:18 (395 min)
---	270	**Earth OCC**	RING	2017-110T09:53	Apr20	Thu	141	2	2	Egress = 2017-110T17:41 (469 min)
---	270	**Earth OCC**	SATURN	2017-110T12:02	Apr20	Thu	142	2	2	Egress = 2017-110T16:58 (297 min)
---	270	**270TI (t) [T126]**	**TITAN**	2017-112T06:20	Apr22	Sat	143	---	4	Outbound 979.0 km flyby, v = 5.8 km/s, phase = 61 deg
---	270	Ring CRX	Ascending	2017-112T07:10	Apr22	Sat	0	---	4	r = 20.081 Rs (1211482 km)
---	271	**Apoapse**		2017-113T04:17	Apr23	Sun	1	---	5	Per = 6.5 d, inc = 62.2 deg, r = 21.114 Rs, phase = 144 deg
---	271	**START OF PROXIMAL ORBITS**		2017-113T04:17	Apr23	Sun	1	---	5	Series of 22 orbits with periapses / ring crossings between rings and planet
---	271	271JA (nt)	**JANUS**	2017-116T09:32	Apr26	Wed	4	---	8	Inbound 99468.6 km flyby, v = 29.6 km/s, phase = 129 deg
---	271	271AT (nt)	**ATLAS**	2017-116T09:39	Apr26	Wed	4	---	8	Inbound 73643.7 km flyby, v = 29.8 km/s, phase = 149 deg
---	271	Ring CRX	Descending	2017-116T09:42	Apr26	Wed	4	---	8	r = 1.063 Rs (64148 km)
---	271	Periapse		2017-116T09:45	Apr26	Wed	4	---	8	R = 1.060 Rs, lat = -5 deg, phase = 36 deg
---	271	271DA (nt)	**DAPHNIS**	2017-116T09:46	Apr26	Wed	4	---	8	Outbound 83075.8 km flyby, v = 29.8 km/s, phase = 119 deg
---	271	271EP (nt)	**EPIMETHEUS**	2017-116T09:50	Apr26	Wed	4	---	8	Outbound 103995.5 km flyby, v = 29.5 km/s, phase = 110 deg
---	271	**Sun OCC**	RING	2017-116T10:01	Apr26	Wed	4	---	8	Egress = 2017-116T10:10 (10 min)
---	271	**Earth OCC**	RING	2017-116T10:07	Apr26	Wed	4	---	8	Egress = 2017-116T10:08 (2 min)
---	271	**Sun OCC**	RING	2017-116T14:13	Apr26	Wed	4	---	8	Egress = 2017-116T22:34 (501 min)
---	271	**Earth OCC**	RING	2017-116T14:45	Apr26	Wed	4	---	8	Egress = 2017-116T22:45 (480 min)
---	271	**Sun OCC**	SATURN	2017-116T15:18	Apr26	Wed	4	---	8	Egress = 2017-116T21:57 (400 min)
---	271	**Earth OCC**	SATURN	2017-116T16:45	Apr26	Wed	4	---	8	Egress = 2017-116T21:58 (313 min)
---	271	Ring CRX	Ascending	2017-118T17:52	Apr28	Fri	6	---	10	r = 20.062 Rs (1210351 km)
---	272	**Apoapse**		2017-119T15:16	Apr29	Sat	7	---	11	Per = 6.5 d, inc = 62.2 deg, r = 21.118 Rs, phase = 144 deg
---	272	272DA (nt)	**DAPHNIS**	2017-122T20:38	May02	Tue	11	---	14	Inbound 77478.7 km flyby, v = 29.9 km/s, phase = 140 deg
---	272	272PM (nt)	**PROMETHEUS**	2017-122T20:43	May02	Tue	11	---	14	Inbound 75494.6 km flyby, v = 29.8 km/s, phase = 148 deg
---	272	Ring CRX	Descending	2017-122T20:44	May02	Tue	11	---	14	r = 1.063 Rs (64128 km)
---	272	Periapse		2017-122T20:47	May02	Tue	11	---	14	R = 1.060 Rs, lat = -5 deg, phase = 36 deg
---	272	**Sun OCC**	RING	2017-122T21:07	May02	Tue	11	---	14	Egress = 2017-122T21:11 (4 min)
---	272	**Earth OCC**	RING	2017-122T21:09	May02	Tue	11	---	14	Egress = 2017-122T21:10 (1 min)
---	272	**Sun OCC**	RING	2017-123T01:18	May03	Wed	11	---	15	Egress = 2017-123T09:35 (497 min)
---	272	**Earth OCC**	RING	2017-123T01:49	May03	Wed	11	---	15	Egress = 2017-123T09:41 (473 min)
---	272	**Sun OCC**	SATURN	2017-123T02:32	May03	Wed	11	---	15	Egress = 2017-123T09:04 (393 min)
---	272	**Earth OCC**	SATURN	2017-123T04:05	May03	Wed	11	---	15	Egress = 2017-123T08:40 (275 min)
---	272	Ring CRX	Ascending	2017-125T04:38	May05	Fri	13	---	17	r = 20.034 Rs (1208661 km)
---	273	**Apoapse**		2017-126T02:16	May06	Sat	14	---	18	Per = 6.5 d, inc = 62.1 deg, r = 21.115 Rs, phase = 145 deg
---	273	273TI (nt)	**TITAN**	2017-127T21:25	May07	Sun	16	---	19	Inbound 487541.0 km flyby, v = 7.1 km/s, phase = 30 deg
---	273	273DA (nt)	**DAPHNIS**	2017-129T07:22	May09	Tue	17	---	21	Inbound 119539.8 km flyby, v = 30.1 km/s, phase = 85 deg
---	273	273PN (nt)	**PAN**	2017-129T07:24	May09	Tue	17	---	21	Inbound 114475.5 km flyby, v = 30.5 km/s, phase = 87 deg
---	273	273PA (nt)	**PANDORA**	2017-129T07:42	May09	Tue	17	---	21	Inbound 79278.1 km flyby, v = 29.9 km/s, phase = 144 deg
---	273	Ring CRX	Descending	2017-129T07:42	May09	Tue	17	---	21	r = 1.059 Rs (63896 km)
---	273	Periapse		2017-129T07:45	May09	Tue	17	---	21	R = 1.056 Rs, lat = -6 deg, phase = 35 deg
---	273	**Sun OCC**	RING	2017-129T07:49	May09	Tue	17	---	21	Egress = 2017-129T08:09 (21 min)
---	273	**Earth OCC**	RING	2017-129T08:07	May09	Tue	17	---	21	Egress = 2017-129T08:08 (1 min)
---	273	**Sun OCC**	RING	2017-129T12:18	May09	Tue	17	---	21	Egress = 2017-129T20:28 (491 min)
---	273	**Earth OCC**	RING	2017-129T12:46	May09	Tue	17	---	21	Egress = 2017-129T20:31 (466 min)
---	273	**Sun OCC**	SATURN	2017-129T13:41	May09	Tue	17	---	21	Egress = 2017-129T19:58 (377 min)
---	273	**Earth OCC**	SATURN	2017-129T15:21	May09	Tue	17	---	21	Egress = 2017-129T19:11 (231 min)
---	273	273TI (nt)	**TITAN**	2017-129T22:18	May09	Tue	18	---	21	Outbound 979235.1 km flyby, v = 12.6 km/s, phase = 90 deg
---	273	Ring CRX	Ascending	2017-131T15:14	May11	Thu	19	---	23	r = 19.993 Rs (1206185 km)
---	274	**Apoapse**		2017-132T13:11	May12	Fri	20	---	24	Per = 6.5 d, inc = 62.2 deg, r = 21.108 Rs, phase = 145 deg
---	274	274EP (nt)	**EPIMETHEUS**	2017-135T18:30	May15	Mon	24	---	27	Inbound 86877.2 km flyby, v = 29.8 km/s, phase = 149 deg
---	274	Ring CRX	Descending	2017-135T18:33	May15	Mon	24	---	27	r = 1.058 Rs (63844 km)
---	274	274PN (nt)	**PAN**	2017-135T18:36	May15	Mon	24	---	27	Inbound 75472.1 km flyby, v = 29.9 km/s, phase = 131 deg
---	274	Periapse		2017-135T18:37	May15	Mon	24	---	27	R = 1.055 Rs, lat = -6 deg, phase = 35 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	274	274PM (nt)	PROMETHEUS	2017-135T18:38	May15	Mon	24	---	27	Outbound 85766.9 km flyby, v = 29.9 km/s, phase = 123 deg
---	274	274AE (nt)	AEGAEON	2017-135T18:45	May15	Mon	24	---	27	Outbound 123076.0 km flyby, v = 29.2 km/s, phase = 111 deg
---	274	274AT (nt)	ATLAS	2017-135T18:57	May15	Mon	24	---	27	Outbound 121039.7 km flyby, v = 27.9 km/s, phase = 81 deg
---	274	Sun OCC	RING	2017-135T18:58	May15	Mon	24	---	27	Egress = 2017-135T19:01 (3 min)
---	274	Earth OCC	RING	2017-135T18:59	May15	Mon	24	---	27	Egress = 2017-135T19:00 (1 min)
---	274	Sun OCC	RING	2017-135T23:13	May15	Mon	24	---	27	Egress = 2017-136T07:17 (484 min)
---	274	Earth OCC	RING	2017-135T23:39	May15	Mon	24	---	27	Egress = 2017-136T07:18 (459 min)
---	274	Sun OCC	SATURN	2017-136T00:51	May16	Tue	24	---	28	Egress = 2017-136T06:42 (351 min)
---	274	Earth OCC	SATURN	2017-136T02:39	May16	Tue	24	---	28	Egress = 2017-136T05:32 (174 min)
---	274	Ring CRX	Ascending	2017-138T01:51	May18	Thu	26	---	30	r = 19.968 Rs (1204647 km)
---	275	Apoapse		2017-139T00:03	May19	Fri	27	---	30	Per = 6.5 d, inc = 62.2 deg, r = 21.108 Rs, phase = 145 deg
---	275	275AT (nt)	ATLAS	2017-142T05:17	May22	Mon	30	---	34	Inbound 88066.0 km flyby, v = 30.1 km/s, phase = 122 deg
---	275	Ring CRX	Descending	2017-142T05:26	May22	Mon	30	---	34	r = 1.059 Rs (63894 km)
---	275	Periapse		2017-142T05:29	May22	Mon	30	---	34	R = 1.056 Rs, lat = -6 deg, phase = 35 deg
---	275	275JA (nt)	JANUS	2017-142T05:35	May22	Mon	30	---	34	Outbound 106741.2 km flyby, v = 29.5 km/s, phase = 112 deg
---	275	275AE (nt)	AEGAEON	2017-142T05:35	May22	Mon	30	---	34	Outbound 119686.4 km flyby, v = 29.4 km/s, phase = 117 deg
---	275	Earth OCC	RING	2017-142T05:50	May22	Mon	30	---	34	Egress = 2017-142T05:52 (2 min)
---	275	Sun OCC	RING	2017-142T05:50	May22	Mon	30	---	34	Egress = 2017-142T05:53 (3 min)
---	275	Sun OCC	RING	2017-142T10:11	May22	Mon	30	---	34	Egress = 2017-142T18:05 (475 min)
---	275	Earth OCC	RING	2017-142T10:33	May22	Mon	30	---	34	Egress = 2017-142T18:04 (451 min)
---	275	Sun OCC	SATURN	2017-142T12:09	May22	Mon	30	---	34	Egress = 2017-142T17:17 (309 min)
---	275	Earth OCC	SATURN	2017-142T14:21	May22	Mon	30	---	34	Egress = 2017-142T15:29 (69 min)
---	275	275TI (nt)	TITAN	2017-144T01:09	May24	Wed	32	---	36	Outbound 138533.1 km flyby, v = 6.1 km/s, phase = 69 deg
---	275	Ring CRX	Ascending	2017-144T12:25	May24	Wed	32	---	36	r = 19.953 Rs (1203762 km)
---	275	NORTHERN SUMM. SOLSTICE		2017-144T21:31	May24	Wed	33	---	36	Northern summer solstice (maximum Solar declination)
---	276	Apoapse		2017-145T11:14	May25	Thu	33	---	37	Per = 6.5 d, inc = 61.9 deg, r = 21.151 Rs, phase = 146 deg
---	276	276PA (nt)	PANDORA	2017-148T16:41	May28	Sun	36	---	40	Inbound 103923.8 km flyby, v = 29.7 km/s, phase = 103 deg
---	276	Ring CRX	Descending	2017-148T16:54	May28	Sun	36	---	40	r = 1.076 Rs (64901 km)
---	276	276MI (nt)	MIMAS	2017-148T16:58	May28	Sun	36	---	40	Inbound 123285.7 km flyby, v = 29.8 km/s, phase = 137 deg
---	276	Periapse		2017-148T16:58	May28	Sun	36	---	40	R = 1.072 Rs, lat = -6 deg, phase = 34 deg
---	276	276AE (nt)	AEGAEON	2017-148T17:07	May28	Sun	36	---	40	Outbound 124823.5 km flyby, v = 28.7 km/s, phase = 110 deg
---	276	276PM (nt)	PROMETHEUS	2017-148T17:15	May28	Sun	36	---	40	Outbound 118219.6 km flyby, v = 28.1 km/s, phase = 87 deg
---	276	Earth OCC	RING	2017-148T17:16	May28	Sun	36	---	40	Egress = 2017-148T17:20 (5 min)
---	276	Sun OCC	RING	2017-148T17:20	May28	Sun	36	---	40	Egress = 2017-148T17:21 (1 min)
---	276	Sun OCC	RING	2017-148T21:51	May28	Sun	37	---	40	Egress = 2017-149T05:31 (461 min)
---	276	Earth OCC	RING	2017-148T22:10	May28	Sun	37	---	40	Egress = 2017-149T05:30 (441 min)
---	276	Sun OCC	SATURN	2017-149T00:25	May29	Mon	37	---	41	Egress = 2017-149T04:11 (227 min)
---	276	Ring CRX	Ascending	2017-150T23:38	May30	Tue	39	---	42	r = 19.927 Rs (1202173 km)
---	277	Apoapse		2017-151T22:45	May31	Wed	40	---	43	Per = 6.5 d, inc = 61.9 deg, r = 21.156 Rs, phase = 146 deg
---	277	277PN (nt)	PAN	2017-155T04:09	Jun04	Sun	43	---	47	Inbound 118237.9 km flyby, v = 30.4 km/s, phase = 79 deg
---	277	277EP (nt)	EPIMETHEUS	2017-155T04:19	Jun04	Sun	43	---	47	Inbound 96602.0 km flyby, v = 29.3 km/s, phase = 128 deg
---	277	Ring CRX	Descending	2017-155T04:28	Jun04	Sun	43	---	47	r = 1.076 Rs (64898 km)
---	277	Periapse		2017-155T04:31	Jun04	Sun	43	---	47	R = 1.072 Rs, lat = -6 deg, phase = 34 deg
---	277	Earth OCC	RING	2017-155T04:46	Jun04	Sun	43	---	47	Egress = 2017-155T04:54 (8 min)
---	277	Sun OCC	RING	2017-155T04:54	Jun04	Sun	43	---	47	Egress = 2017-155T04:54 (1 min)
---	277	Sun OCC	RING	2017-155T09:30	Jun04	Sun	43	---	47	Egress = 2017-155T16:57 (448 min)
---	277	Earth OCC	RING	2017-155T09:44	Jun04	Sun	43	---	47	Egress = 2017-155T16:57 (433 min)
---	277	Sun OCC	SATURN	2017-155T13:14	Jun04	Sun	43	---	47	Egress = 2017-155T14:25 (72 min)
---	277	Ring CRX	Ascending	2017-157T10:53	Jun06	Tue	45	---	49	r = 19.891 Rs (1200035 km)
---	278	Apoapse		2017-158T10:16	Jun07	Wed	46	---	50	Per = 6.5 d, inc = 61.9 deg, r = 21.153 Rs, phase = 146 deg
---	278	278TI (nt)	TITAN	2017-159T19:58	Jun08	Thu	48	---	51	Inbound 348599.5 km flyby, v = 6.0 km/s, phase = 40 deg
---	278	Ring CRX	Descending	2017-161T15:55	Jun10	Sat	49	---	53	r = 1.067 Rs (64366 km)
---	278	Periapse		2017-161T15:59	Jun10	Sat	49	---	53	R = 1.063 Rs, lat = -6 deg, phase = 34 deg
---	278	278PN (nt)	PAN	2017-161T16:01	Jun10	Sat	49	---	53	Outbound 83862.3 km flyby, v = 29.7 km/s, phase = 120 deg
---	278	278JA (nt)	JANUS	2017-161T16:04	Jun10	Sat	49	---	53	Outbound 104366.9 km flyby, v = 29.4 km/s, phase = 117 deg
---	278	Earth OCC	RING	2017-161T16:22	Jun10	Sat	49	---	53	Egress = 2017-161T16:22 (0 min)
---	278	Sun OCC	RING	2017-161T16:22	Jun10	Sat	49	---	53	Egress = 2017-161T16:22 (1 min)
---	278	Sun OCC	RING	2017-161T20:58	Jun10	Sat	50	---	53	Egress = 2017-162T04:09 (432 min)
---	278	Earth OCC	RING	2017-161T21:06	Jun10	Sat	50	---	53	Egress = 2017-162T04:11 (426 min)
---	278	Ring CRX	Ascending	2017-163T21:45	Jun12	Mon	52	---	55	r = 19.825 Rs (1196017 km)
---	279	Apoapse		2017-164T21:39	Jun13	Tue	53	---	56	Per = 6.5 d, inc = 61.9 deg, r = 21.144 Rs, phase = 146 deg
---	279	279PM (nt)	PROMETHEUS	2017-168T03:02	Jun17	Sat	56	---	60	Inbound 100073.8 km flyby, v = 30.0 km/s, phase = 103 deg
---	279	279AT (nt)	ATLAS	2017-168T03:12	Jun17	Sat	56	---	60	Inbound 73808.6 km flyby, v = 29.7 km/s, phase = 149 deg
---	279	Ring CRX	Descending	2017-168T03:15	Jun17	Sat	56	---	60	r = 1.066 Rs (64309 km)
---	279	Periapse		2017-168T03:19	Jun17	Sat	56	---	60	R = 1.062 Rs, lat = -6 deg, phase = 34 deg
---	279	279DA (nt)	DAPHNIS	2017-168T03:35	Jun17	Sat	56	---	60	Outbound 116091.8 km flyby, v = 28.3 km/s, phase = 89 deg
---	279	Earth OCC	RING	2017-168T03:41	Jun17	Sat	56	---	60	Egress = 2017-168T03:41 (0 min)
---	279	Sun OCC	RING	2017-168T03:42	Jun17	Sat	56	---	60	Egress = 2017-168T03:41 (0 min)
---	279	Sun OCC	RING	2017-168T08:24	Jun17	Sat	56	---	60	Egress = 2017-168T15:18 (415 min)
---	279	Earth OCC	RING	2017-168T08:25	Jun17	Sat	56	---	60	Egress = 2017-168T15:24 (419 min)
---	279	Ring CRX	Ascending	2017-170T08:46	Jun19	Mon	58	---	62	r = 19.789 Rs (1193892 km)
---	280	Apoapse		2017-171T08:58	Jun20	Tue	59	---	63	Per = 6.5 d, inc = 61.9 deg, r = 21.144 Rs, phase = 146 deg
---	280	280EP (nt)	EPIMETHEUS	2017-174T14:20	Jun23	Fri	62	---	66	Inbound 109935.7 km flyby, v = 29.2 km/s, phase = 106 deg
---	280	Ring CRX	Descending	2017-174T14:34	Jun23	Fri	62	---	66	r = 1.067 Rs (64352 km)
---	280	Periapse		2017-174T14:37	Jun23	Fri	62	---	66	R = 1.063 Rs, lat = -6 deg, phase = 34 deg
---	280	280DA (nt)	DAPHNIS	2017-174T14:38	Jun23	Fri	62	---	66	Outbound 81942.5 km flyby, v = 29.7 km/s, phase = 130 deg
---	280	280PA (nt)	PANDORA	2017-174T14:41	Jun23	Fri	62	---	66	Outbound 94837.4 km flyby, v = 29.6 km/s, phase = 119 deg
---	280	Earth OCC	RING	2017-174T15:00	Jun23	Fri	62	---	66	Egress = 2017-174T15:00 (0 min)

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	280	Earth OCC	RING	2017-174T19:45	Jun23	Fri	63	---	66	Egress = 2017-175T02:36 (411 min)
---	280	Sun OCC	RING	2017-174T19:51	Jun23	Fri	63	---	66	Egress = 2017-175T02:25 (395 min)
---	280	280TI (nt)	TITAN	2017-176T00:29	Jun25	Sun	64	---	68	Outbound 337923.6 km flyby, v = 7.2 km/s, phase = 68 deg
---	280	Ring CRX	Ascending	2017-176T19:47	Jun25	Sun	65	---	68	r = 19.763 Rs (1192277 km)
---	281	Apoapse		2017-177T20:22	Jun26	Mon	66	---	69	Per = 6.5 d, inc = 61.8 deg, r = 21.160 Rs, phase = 147 deg
---	281	281PN (nt)	PAN	2017-181T01:52	Jun30	Fri	69	---	73	Inbound 98544.6 km flyby, v = 30.2 km/s, phase = 95 deg
---	281	281DA (nt)	DAPHNIS	2017-181T02:01	Jun30	Fri	69	---	73	Inbound 72636.1 km flyby, v = 29.6 km/s, phase = 146 deg
---	281	281PM (nt)	PROMETHEUS	2017-181T02:03	Jun30	Fri	69	---	73	Inbound 74642.4 km flyby, v = 29.6 km/s, phase = 151 deg
---	281	Ring CRX	Descending	2017-181T02:04	Jun30	Fri	69	---	73	r = 1.071 Rs (64622 km)
---	281	Periapse		2017-181T02:08	Jun30	Fri	69	---	73	R = 1.067 Rs, lat = -6 deg, phase = 33 deg
---	281	281JA (nt)	JANUS	2017-181T02:10	Jun30	Fri	69	---	73	Outbound 96083.3 km flyby, v = 29.5 km/s, phase = 131 deg
---	281	Earth OCC	RING	2017-181T07:19	Jun30	Fri	69	---	73	Egress = 2017-181T14:00 (402 min)
---	281	Sun OCC	RING	2017-181T07:32	Jun30	Fri	69	---	73	Egress = 2017-181T13:43 (371 min)
---	281	Ring CRX	Ascending	2017-183T07:01	Jul02	Sun	71	---	75	r = 19.730 Rs (1190320 km)
---	282	Apoapse		2017-184T07:56	Jul03	Mon	72	---	76	Per = 6.5 d, inc = 61.8 deg, r = 21.164 Rs, phase = 147 deg
---	282	282DA (nt)	DAPHNIS	2017-187T13:29	Jul06	Thu	75	---	79	Inbound 92376.1 km flyby, v = 29.9 km/s, phase = 106 deg
---	282	Ring CRX	Descending	2017-187T13:39	Jul06	Thu	75	---	79	r = 1.071 Rs (64638 km)
---	282	Periapse		2017-187T13:43	Jul06	Thu	75	---	79	R = 1.067 Rs, lat = -6 deg, phase = 33 deg
---	282	282PN (nt)	PAN	2017-187T13:53	Jul06	Thu	75	---	79	Outbound 104099.8 km flyby, v = 29.2 km/s, phase = 100 deg
---	282	Earth OCC	RING	2017-187T18:55	Jul06	Thu	76	---	79	Egress = 2017-188T01:27 (392 min)
---	282	Sun OCC	RING	2017-187T19:18	Jul06	Thu	76	---	79	Egress = 2017-188T01:02 (344 min)
---	282	Ring CRX	Ascending	2017-189T18:16	Jul08	Sat	77	---	81	r = 19.690 Rs (1187897 km)
---	283	Apoapse		2017-190T19:28	Jul09	Sun	79	---	82	Per = 6.5 d, inc = 61.9 deg, r = 21.160 Rs, phase = 147 deg
---	283	283TI (nt)	TITAN	2017-191T14:24	Jul10	Mon	79	---	83	Inbound 247911.0 km flyby, v = 5.3 km/s, phase = 62 deg
---	283	283EP (nt)	EPIMETHEUS	2017-194T00:49	Jul13	Thu	82	---	86	Inbound 114813.5 km flyby, v = 29.1 km/s, phase = 97 deg
---	283	Ring CRX	Descending	2017-194T01:05	Jul13	Thu	82	---	86	r = 1.056 Rs (63681 km)
---	283	283PA (nt)	PANDORA	2017-194T01:06	Jul13	Thu	82	---	86	Inbound 79805.5 km flyby, v = 29.8 km/s, phase = 148 deg
---	283	283AT (nt)	ATLAS	2017-194T01:08	Jul13	Thu	82	---	86	Inbound 80994.5 km flyby, v = 29.8 km/s, phase = 138 deg
---	283	Periapse		2017-194T01:09	Jul13	Thu	82	---	86	R = 1.051 Rs, lat = -7 deg, phase = 33 deg
---	283	283PM (nt)	PROMETHEUS	2017-194T01:22	Jul13	Thu	82	---	86	Outbound 112688.9 km flyby, v = 28.7 km/s, phase = 99 deg
---	283	Earth OCC	RING	2017-194T01:32	Jul13	Thu	82	---	86	Egress = 2017-194T01:32 (1 min)
---	283	Earth OCC	RING	2017-194T06:12	Jul13	Thu	82	---	86	Egress = 2017-194T12:33 (382 min)
---	283	Sun OCC	RING	2017-194T06:44	Jul13	Thu	82	---	86	Egress = 2017-194T11:58 (315 min)
---	283	Ring CRX	Ascending	2017-196T04:58	Jul15	Sat	84	---	88	r = 19.608 Rs (1182935 km)
---	284	Apoapse		2017-197T06:50	Jul16	Sun	85	---	89	Per = 6.5 d, inc = 61.8 deg, r = 21.158 Rs, phase = 147 deg
---	284	284AT (nt)	ATLAS	2017-200T12:06	Jul19	Wed	88	---	92	Inbound 120654.8 km flyby, v = 30.0 km/s, phase = 72 deg
---	284	Ring CRX	Descending	2017-200T12:26	Jul19	Wed	88	---	92	r = 1.055 Rs (63625 km)
---	284	284JA (nt)	JANUS	2017-200T12:30	Jul19	Wed	88	---	92	Inbound 93742.3 km flyby, v = 29.8 km/s, phase = 140 deg
---	284	Periapse		2017-200T12:30	Jul19	Wed	88	---	92	R = 1.050 Rs, lat = -7 deg, phase = 33 deg
---	284	Earth OCC	RING	2017-200T12:53	Jul19	Wed	88	---	92	Egress = 2017-200T12:53 (1 min)
---	284	Earth OCC	RING	2017-200T17:35	Jul19	Wed	88	---	92	Egress = 2017-200T23:45 (370 min)
---	284	Sun OCC	RING	2017-200T18:19	Jul19	Wed	88	---	92	Egress = 2017-200T22:58 (280 min)
---	284	Ring CRX	Ascending	2017-202T15:57	Jul21	Fri	90	---	94	r = 19.563 Rs (1180211 km)
---	285	Apoapse		2017-203T18:10	Jul22	Sat	91	---	95	Per = 6.5 d, inc = 61.7 deg, r = 21.156 Rs, phase = 147 deg
---	285	285TI (nt)	TITAN	2017-206T03:33	Jul25	Tue	94	---	98	Inbound 907880.1 km flyby, v = 10.3 km/s, phase = 56 deg
---	285	285PN (nt)	PAN	2017-206T23:37	Jul25	Tue	95	---	98	Inbound 80302.0 km flyby, v = 30.0 km/s, phase = 118 deg
---	285	Ring CRX	Descending	2017-206T23:45	Jul25	Tue	95	---	98	r = 1.055 Rs (63641 km)
---	285	Periapse		2017-206T23:49	Jul25	Tue	95	---	98	R = 1.050 Rs, lat = -7 deg, phase = 33 deg
---	285	Earth OCC	RING	2017-207T00:12	Jul26	Wed	95	---	99	Egress = 2017-207T00:12 (0 min)
---	285	Earth OCC	RING	2017-207T04:57	Jul26	Wed	95	---	99	Egress = 2017-207T10:54 (357 min)
---	285	Sun OCC	RING	2017-207T05:55	Jul26	Wed	95	---	99	Egress = 2017-207T09:52 (237 min)
---	285	285TI (nt)	TITAN	2017-208T02:28	Jul27	Thu	96	---	100	Outbound 548630.8 km flyby, v = 8.6 km/s, phase = 68 deg
---	285	Ring CRX	Ascending	2017-209T02:56	Jul28	Fri	97	---	101	r = 19.524 Rs (1177871 km)
---	286	Apoapse		2017-210T05:30	Jul29	Sat	98	---	102	Per = 6.5 d, inc = 61.7 deg, r = 21.162 Rs, phase = 147 deg
---	286	286PM (nt)	PROMETHEUS	2017-213T10:51	Aug01	Tue	101	---	105	Inbound 112748.3 km flyby, v = 29.9 km/s, phase = 81 deg
---	286	286PA (nt)	PANDORA	2017-213T11:00	Aug01	Tue	101	---	105	Inbound 84973.3 km flyby, v = 29.8 km/s, phase = 124 deg
---	286	Ring CRX	Descending	2017-213T11:08	Aug01	Tue	101	---	105	r = 1.056 Rs (63719 km)
---	286	Periapse		2017-213T11:12	Aug01	Tue	101	---	105	R = 1.052 Rs, lat = -7 deg, phase = 34 deg
---	286	286PN (nt)	PAN	2017-213T11:32	Aug01	Tue	101	---	105	Outbound 118341.4 km flyby, v = 27.6 km/s, phase = 92 deg
---	286	Earth OCC	RING	2017-213T11:35	Aug01	Tue	101	---	105	Egress = 2017-213T11:34 (0 min)
---	286	Earth OCC	RING	2017-213T16:25	Aug01	Tue	101	---	105	Egress = 2017-213T22:05 (341 min)
---	286	Sun OCC	RING	2017-213T17:43	Aug01	Tue	101	---	105	Egress = 2017-213T20:42 (180 min)
---	286	Ring CRX	Ascending	2017-215T14:00	Aug03	Thu	103	---	107	r = 19.484 Rs (1175476 km)
---	287	Apoapse		2017-216T16:55	Aug04	Fri	104	---	108	Per = 6.5 d, inc = 61.7 deg, r = 21.166 Rs, phase = 147 deg
---	287	Ring CRX	Descending	2017-219T22:34	Aug07	Mon	108	---	111	r = 1.057 Rs (63753 km)
---	287	287JA (nt)	JANUS	2017-219T22:35	Aug07	Mon	108	---	111	Inbound 89749.2 km flyby, v = 29.8 km/s, phase = 151 deg
---	287	Periapse		2017-219T22:39	Aug07	Mon	108	---	111	R = 1.052 Rs, lat = -7 deg, phase = 34 deg
---	287	287AT (nt)	ATLAS	2017-219T22:43	Aug07	Mon	108	---	111	Outbound 94580.2 km flyby, v = 29.6 km/s, phase = 122 deg
---	287	287EP (nt)	EPIMETHEUS	2017-219T22:51	Aug07	Mon	108	---	111	Outbound 120137.7 km flyby, v = 28.5 km/s, phase = 111 deg
---	287	Earth OCC	RING	2017-220T03:57	Aug08	Tue	108	---	112	Egress = 2017-220T09:19 (323 min)
---	287	Sun OCC	RING	2017-220T05:52	Aug08	Tue	108	---	112	Egress = 2017-220T07:17 (86 min)
---	287	Ring CRX	Ascending	2017-222T01:06	Aug10	Thu	110	---	114	r = 19.438 Rs (1172711 km)
---	288	Apoapse		2017-223T04:17	Aug11	Fri	111	---	115	Per = 6.5 d, inc = 61.9 deg, r = 21.161 Rs, phase = 146 deg
---	288	288TI (nt)	TITAN	2017-223T04:55	Aug11	Fri	111	---	115	Inbound 175225.5 km flyby, v = 5.0 km/s, phase = 82 deg
---	288	288AT (nt)	ATLAS	2017-226T09:29	Aug14	Mon	114	---	118	Inbound 107173.9 km flyby, v = 30.1 km/s, phase = 83 deg
---	288	288PM (nt)	PROMETHEUS	2017-226T09:39	Aug14	Mon	114	---	118	Inbound 79456.0 km flyby, v = 30.1 km/s, phase = 135 deg

CASSINI TOUR EVENT SUMMARY | 090721 Reference Trajectory | v1 *DST / DTT = days since / until Targ. Flyby, DSM = days since OTM*

Seq	Rev	Name	Event	Epoch (SCET)	Date	DOW	DST	DTT	DSM	Comment
---	288	Ring CRX	Descending	2017-226T09:44	Aug14	Mon	114	---	118	r = 1.034 Rs (62401 km)
---	288	Periapse		2017-226T09:48	Aug14	Mon	114	---	118	R = 1.030 Rs, lat = -7 deg, phase = 34 deg
---	288	**Earth OCC**	RING	2017-226T10:10	Aug14	Mon	114	---	118	Egress = 2017-226T10:11 (1 min)
---	288	**Earth OCC**	RING	2017-226T14:59	Aug14	Mon	114	---	118	Egress = 2017-226T20:01 (303 min)
---	288	Ring CRX	Ascending	2017-228T11:34	Aug16	Wed	116	---	120	r = 19.359 Rs (1167918 km)
---	289	**Apoapse**		2017-229T15:25	Aug17	Thu	117	---	121	Per = 6.5 d, inc = 61.6 deg, r = 21.167 Rs, phase = 146 deg
---	289	289PA (nt)	**PANDORA**	2017-232T20:36	Aug20	Sun	121	---	124	Inbound 120848.3 km flyby, v = 29.5 km/s, phase = 71 deg
---	289	289PN (nt)	**PAN**	2017-232T20:51	Aug20	Sun	121	---	124	Inbound 76005.0 km flyby, v = 30.2 km/s, phase = 126 deg
---	289	Ring CRX	Descending	2017-232T20:57	Aug20	Sun	121	---	124	r = 1.034 Rs (62357 km)
---	289	Periapse		2017-232T21:01	Aug20	Sun	121	---	124	R = 1.029 Rs, lat = -7 deg, phase = 34 deg
---	289	289DA (nt)	**DAPHNIS**	2017-232T21:18	Aug20	Sun	121	---	124	Outbound 115598.6 km flyby, v = 28.1 km/s, phase = 102 deg
---	289	**Earth OCC**	RING	2017-232T21:23	Aug20	Sun	121	---	124	Egress = 2017-232T21:24 (1 min)
---	289	Ring CRX	Ascending	2017-234T22:22	Aug22	Tue	123	---	126	r = 19.303 Rs (1164537 km)
---	290	**Apoapse**		2017-236T02:36	Aug24	Thu	124	---	128	Per = 6.5 d, inc = 61.6 deg, r = 21.163 Rs, phase = 146 deg
---	290	290TI (nt)	**TITAN**	2017-238T06:27	Aug26	Sat	126	---	130	Inbound 736276.7 km flyby, v = 8.8 km/s, phase = 54 deg
---	290	290JA (nt)	**JANUS**	2017-239T08:01	Aug27	Sun	127	---	131	Inbound 90912.3 km flyby, v = 30.1 km/s, phase = 139 deg
---	290	Ring CRX	Descending	2017-239T08:06	Aug27	Sun	127	---	131	r = 1.033 Rs (62314 km)
---	290	290DA (nt)	**DAPHNIS**	2017-239T08:09	Aug27	Sun	127	---	131	Inbound 80871.6 km flyby, v = 30.1 km/s, phase = 145 deg
---	290	290PM (nt)	**PROMETHEUS**	2017-239T08:10	Aug27	Sun	127	---	131	Inbound 85497.8 km flyby, v = 30.0 km/s, phase = 143 deg
---	290	Periapse		2017-239T08:10	Aug27	Sun	127	---	131	R = 1.028 Rs, lat = -7 deg, phase = 34 deg
---	290	290EP (nt)	**EPIMETHEUS**	2017-239T08:11	Aug27	Sun	127	---	131	Outbound 99654.2 km flyby, v = 30.1 km/s, phase = 142 deg
---	290	290PA (nt)	**PANDORA**	2017-239T08:16	Aug27	Sun	127	---	131	Outbound 100942.7 km flyby, v = 29.7 km/s, phase = 124 deg
---	290	290PN (nt)	**PAN**	2017-239T08:33	Aug27	Sun	127	---	131	Outbound 119571.4 km flyby, v = 26.8 km/s, phase = 96 deg
---	290	**Earth OCC**	RING	2017-239T08:33	Aug27	Sun	127	---	131	Egress = 2017-239T08:33 (1 min)
---	290	290TI (nt)	**TITAN**	2017-240T05:12	Aug28	Mon	128	---	132	Outbound 743747.8 km flyby, v = 10.2 km/s, phase = 67 deg
---	290	Ring CRX	Ascending	2017-241T09:07	Aug29	Tue	129	---	133	r = 19.248 Rs (1161247 km)
---	291	**Apoapse**		2017-242T13:44	Aug30	Wed	130	---	134	Per = 6.5 d, inc = 61.6 deg, r = 21.163 Rs, phase = 146 deg
---	291	291DA (nt)	**DAPHNIS**	2017-245T19:08	Sep02	Sat	134	---	137	Inbound 80143.2 km flyby, v = 30.2 km/s, phase = 121 deg
---	291	Ring CRX	Descending	2017-245T19:14	Sep02	Sat	134	---	137	r = 1.033 Rs (62313 km)
---	291	Periapse		2017-245T19:19	Sep02	Sat	134	---	137	R = 1.028 Rs, lat = -7 deg, phase = 35 deg
---	291	291AT (nt)	**ATLAS**	2017-245T19:22	Sep02	Sat	134	---	137	Outbound 92826.6 km flyby, v = 29.9 km/s, phase = 130 deg
---	291	**Earth OCC**	RING	2017-245T19:41	Sep02	Sat	134	---	137	Egress = 2017-245T19:41 (0 min)
---	291	Ring CRX	Ascending	2017-247T19:53	Sep04	Mon	136	---	139	r = 19.198 Rs (1158196 km)
---	292	**Apoapse**		2017-249T00:54	Sep06	Wed	137	---	141	Per = 6.5 d, inc = 61.6 deg, r = 21.165 Rs, phase = 145 deg
---	292	292DA (nt)	**DAPHNIS**	2017-252T06:07	Sep09	Sat	140	---	144	Inbound 115809.5 km flyby, v = 30.2 km/s, phase = 67 deg
---	292	292AT (nt)	**ATLAS**	2017-252T06:09	Sep09	Sat	140	---	144	Inbound 109526.2 km flyby, v = 30.2 km/s, phase = 76 deg
---	292	Ring CRX	Descending	2017-252T06:25	Sep09	Sat	140	---	144	r = 1.034 Rs (62359 km)
---	292	Periapse		2017-252T06:30	Sep09	Sat	140	---	144	R = 1.028 Rs, lat = -7 deg, phase = 35 deg
---	292	292PM (nt)	**PROMETHEUS**	2017-252T06:44	Sep09	Sat	140	---	144	Outbound 115378.0 km flyby, v = 28.3 km/s, phase = 108 deg
---	292	**Earth OCC**	RING	2017-252T12:42	Sep09	Sat	140	---	144	Egress = 2017-252T15:03 (142 min)
---	292	Ring CRX	Ascending	2017-254T06:42	Sep11	Mon	142	---	146	r = 19.145 Rs (1155046 km)
---	292	292TI (nt)	**TITAN**	2017-254T19:14	Sep11	Mon	143	---	146	Outbound 84137.2 km flyby, v = 5.2 km/s, phase = 80 deg
---	293	**Apoapse**		2017-255T11:53	Sep12	Tue	143	---	147	Per = 6.4 d, inc = 61.5 deg, r = 21.157 Rs, phase = 145 deg
---	293	293JA (nt)	**JANUS**	2017-258T16:51	Sep15	Fri	146	---	150	Inbound 111325.1 km flyby, v = 29.9 km/s, phase = 93 deg
---	293	293PN (nt)	**PAN**	2017-258T16:55	Sep15	Fri	146	---	150	Inbound 91103.9 km flyby, v = 30.7 km/s, phase = 96 deg
---	293	293PA (nt)	**PANDORA**	2017-258T17:00	Sep15	Fri	146	---	150	Inbound 85950.7 km flyby, v = 30.6 km/s, phase = 126 deg
---	293	293EP (nt)	**EPIMETHEUS**	2017-258T17:00	Sep15	Fri	146	---	150	Inbound 92227.9 km flyby, v = 30.7 km/s, phase = 134 deg
---	293	**END OF MISSION**		2017-258T17:03	Sep15	Fri	146	---	150	End of mission, atmospheric entry into Saturn (1.0 Rs)
---	293	Ring CRX	Descending	2017-258T17:06	Sep15	Fri	146	---	150	r = 0.988 Rs (59588 km)
---	293	Periapse		2017-258T17:10	Sep15	Fri	146	---	150	R = 0.983 Rs, lat = -7 deg, phase = 35 deg